AICTE FACULTY RECRUITMENT

Assistant Professor Screening Test — Computer Science & Applications

SUBJECT CODE: 12

COMPREHENSIVE QUESTION BANK — 3,250 MCQs

In-Line Correct Answers & Consolidated Topic Answer Keys

Target Examination	AICTE Faculty Recruitment / University Assistant Professor Screening
Standard / Level	UGC-NET, GATE, AP State Eligibility & Screening Test Standard
Coverage Depth	All 12 Syllabus Units (17 Sub-Modules / Modules)
Question Distribution	200 MCQs per Unit (50 MCQs for Unit 12a Parallel Computing)
Total Volume	**3,250 Multiple Choice Questions with Full Answer Keys**

Table of Contents & Topic Breakdown

#	Topic Name / Syllabus Domain	MCQs
1	Topic 1: Discrete Structures	200
2	Topic 2: Computer Arithmetic	200
3	Topic 3: Programming in C and C++	200
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	Total Questions Across All 12 Topics	**3,250**

Topic 1: Discrete Structures

Syllabus: *Set Theory: Sets, Relations, Functions, Pigeonhole Principle, Inclusion-Exclusion Principle, Equivalence and Partial Orderings, Elementary Counting Techniques, Probability, Measure(s) for information and Mutual information; Graph: Definition, walks, paths, trails, connected graphs, regular and bipartite graphs, cycles and circuits, Tree and rooted tree, Spanning trees, Eccentricity of a vertex, radius and diameter of a graph, Central Graphs, Centres of a tree, Hamiltonian and Eulerian graphs, Planar graphs; Groups: Finite fields and Error correcting / detecting codes.*

Q1. Which of the following is a set?

A. All tall people
B. All prime numbers less than 10
C. All beautiful paintings
D. All rich people
Correct Answer: B

Q2. The set {x | x is an even prime number} contains

A. 0 elements
B. 1 element
C. 2 elements
D. Infinite elements
Correct Answer: B

Q3. If A={1,2,3} and B={2,3,4}, then A∪B=

A. {1,2,3}
B. {2,3}
C. {1,2,3,4}
D. {1,4}
Correct Answer: C

Q4. The power set of {a,b} has

A. 2 elements
B. 3 elements
C. 4 elements
D. 8 elements
Correct Answer: C

Q5. |A×B| where |A|=3, |B|=4 equals

A. 7
B. 12
C. 1
D. 24
Correct Answer: B

Q6. Which relation is an equivalence relation?

A. R={(1,2),(2,3)}
B. R={(1,1),(2,2),(1,2),(2,1)}
C. R={(1,2)}
D. R={(1,2),(2,3),(1,3)}
Correct Answer: B

Q7. A function f:A→B is onto if

A. Every element of A maps to unique element of B
B. Every element of B has preimage in A
C. No two elements map same
D. Domain equals codomain
Correct Answer: B

Q8. The pigeonhole principle states if n items are put into m containers with n>m, then

A. Each container has exactly 1 item
B. At least one container has ≥2 items
C. No container has items
D. All containers have equal items
Correct Answer: B

Q9. Inclusion-exclusion for $|A \cup B|$ equals

A. $|A|+|B|$
B. $|A|+|B|-|A \cap B|$
C. $|A|-|B|$
D. $|A| \times |B|$
Correct Answer: B

Q10. The number of ways to arrange 5 books on a shelf is

A. 10
B. 25
C. 120
D. 5
Correct Answer: C

Q11. C(5,2) equals

A. 10
B. 20
C. 5
D. 25
Correct Answer: A

Q12. The probability of getting heads on a fair coin toss is

A. 0
B. 0.5
C. 1
D. 0.25
Correct Answer: B

Q13. If P(A)=0.3 and P(B)=0.5 with A,B independent, $P(A \cap B)$=

A. 0.8
B. 0.15
C. 0.2
D. 0.5
Correct Answer: B

Q14. The entropy of a fair coin is

A. 0 bits
B. 1 bit
C. 2 bits
D. 0.5 bits
Correct Answer: B

Q15. Which is a partial order?

A. Divisibility on positive integers
B. Equality on real numbers
C. Subset relation on power set
D. All of the above
Correct Answer: D

Q16. The number of equivalence relations on {1,2,3} is

A. 3
B. 5
C. 6
D. 9
Correct Answer: B

Q17. A relation that is reflexive, symmetric, and transitive is called

A. Partial order
B. Equivalence relation
C. Function
D. Relation
Correct Answer: B

Q18. The number of functions from set A (m elements) to set B (n elements) is

A. m+n
B. m×n
C. n^m
D. m^n
Correct Answer: C

Q19. Which is NOT a set?

A. The set of all solutions to $x^2=4$
B. The set of all tall people
C. The set of vowels
D. The set of prime numbers
Correct Answer: B

Q20. $A \cap B$ when $A \subseteq B$ equals

A. A
B. B
C. ∅
D. U
Correct Answer: A

Q21. The complement of set A (relative to universal set U) is

A. U–A
B. A–U
C. $A \cap U$
D. ∅
Correct Answer: A

Q22. How many subsets does {1,2,3,4} have?

A. 8
B. 16
C. 32
D. 4
Correct Answer: B

Q23. The Cartesian product A×B has |A|×|B| elements

A. True
B. False
C. Sometimes
D. Depends
Correct Answer: A

Q24. In modular arithmetic, 7 mod 3 =

A. 1
B. 2
C. 3
D. 4
Correct Answer: B

Q25. The number of permutations of n distinct objects is

A. n
B. n^2
C. n!
D. 2^n
Correct Answer: C

Q26. P(n,r) = n!/(n–r)! gives

A. Combinations
B. Permutations
C. Probability
D. Power set
Correct Answer: B

Q27. The number of ways to choose 3 from 10 is

A. 120
B. 720
C. 30
D. 1000
Correct Answer: A

Q28. If events A and B are mutually exclusive, P(A∪B) =

A. P(A)P(B)
B. P(A)+P(B)
C. P(A)−P(B)
D. 1
Correct Answer: B

Q29. The information content of a message with probability 1/8 is

A. 1 bit
B. 2 bits
C. 3 bits
D. 8 bits
Correct Answer: C

Q30. Which relation is antisymmetric?

A. Equality
B. Divisibility on integers
C. Subset relation
D. Both B and C
Correct Answer: D

Q31. A total order requires

A. Reflexivity and symmetry
B. Reflexivity, antisymmetry, transitivity, and comparability
C. Only transitivity
D. Only reflexivity
Correct Answer: B

Q32. The number of onto functions from {1,...,5} to {1,...,3} is

A. 150
B. 243
C. 25
D. 3
Correct Answer: A

Q33. Bayes' theorem states P(A|B) =

A. P(B|A)P(A)/P(B)
B. P(A|B)P(B)/P(A)
C. P(A)P(B)
D. P(B)/P(A)
Correct Answer: A

Q34. The expected value of a fair die roll is

A. 3
B. 3.5
C. 4
D. 6
Correct Answer: B

Q35. Variance of a constant is

A. 0
B. 1
C. The constant
D. Undefined
Correct Answer: A

Q36. The number of derangements of 4 objects is

A. 6
B. 9
C. 12
D. 24
Correct Answer: B

Q37. Which is a valid partial order?

A. Strict less than
B. Non-strict less than or equal
C. Divisibility on positive integers
D. Both B and C
Correct Answer: D

Q38. The number of lattice points in {1,...,n}■ is

A. n
B. n^2
C. n^n
D. 2^n
Correct Answer: C

Q39. If f is injective and surjective, f is

A. Neither
B. Bijective
C. Just injective
D. Just surjective
Correct Answer: B

Q40. The transitive closure of a relation R is

A. R itself
B. Smallest transitive relation containing R
C. R's complement
D. R's inverse
Correct Answer: B

Q41. A partition of a set S is

A. Any collection of subsets
B. Collection of disjoint non-empty subsets whose union is S
C. Collection of all subsets
D. A single subset
Correct Answer: B

Q42. In how many ways can 5 identical balls be placed in 3 distinct boxes?

A. C(7,2)
B. 3^5
C. 5^3
D. 15
Correct Answer: A

Q43. The probability of drawing a red card from a standard deck is

A. 1/4
B. 1/2
C. 1/3
D. 1/13
Correct Answer: B

Q44. If A={1,2}, B={3,4}, then A×B =

A. {(1,3),(1,4),(2,3),(2,4)}
B. {(1,2),(3,4)}
C. {(1,3),(2,4)}
D. {(1,1),(2,2)}
Correct Answer: A

Q45. The symmetric difference $A \oplus B$ equals

A. $A \cap B$
B. $(A-B) \cup (B-A)$
C. $A \cup B$
D. $A-B$
Correct Answer: B

Q46. The number of reflexive relations on {1,...,n} is

A. 2^n
B. $2^{(n^2-n)}$
C. $2^{(n^2)}$
D. n^2
Correct Answer: B

Q47. The number of binary relations on {1,...,n} is

A. 2^n
B. $2^{(n^2)}$
C. n^2
D. n!
Correct Answer: B

Q48. Conditional probability P(A|B) is defined when

A. $P(A)>0$
B. $P(B)>0$
C. $P(A \cap B)>0$
D. $P(A)>0$ and $P(B)>0$
Correct Answer: B

Q49. The variance of a binomial distribution B(n,p) is

A. np
B. $np(1-p)$
C. np^2
D. $n(1-p)$
Correct Answer: B

Q50. A relation R on set A is irreflexive if

A. $(a,a) \in R$ for all a
B. $(a,a) \notin R$ for all $a \in A$
C. Some $(a,a) \in R$
D. No elements exist
Correct Answer: B

Q51. The number of surjections from {1,...,n} to itself is

A. n
B. n!
C. 2^n
D. n^n
Correct Answer: B

Q52. If $P(A)=0.4$, $P(B)=0.5$, $P(A \cup B)=0.7$, then $P(A \cap B)=$

A. 0.2
B. 0.9
C. 0
D. 0.35
Correct Answer: A

Q53. The mutual information I(X;Y) is always

A. Negative
B. Non-negative
C. Zero
D. Greater than 1
Correct Answer: B

Q54. A well-ordered set is one where every non-empty subset has

A. A maximum
B. A minimum
C. No elements
D. A supremum
Correct Answer: B

Q55. The number of total orders on {1,...,n} is

A. n
B. n!
C. 2^n
D. C(2n,n)
Correct Answer: B

Q56. The union of two equivalence relations is

A. Always an equivalence relation
B. Never an equivalence relation
C. Sometimes an equivalence relation
D. Always symmetric
Correct Answer: C

Q57. In the inclusion-exclusion formula for 3 sets, we subtract

A. |A∩B|+|A∩C|+|B∩C|
B. |A∩B∩C|
C. |A|+|B|+|C|
D. Nothing
Correct Answer: A

Q58. The number of permutations of n objects taken r at a time with repetition is

A. n^r
B. n!/(n–r)!
C. C(n,r)
D. n!/r!
Correct Answer: A

Q59. If two events are independent, P(A∩B) =

A. P(A)+P(B)
B. P(A)P(B)
C. P(A)–P(B)
D. 1
Correct Answer: B

Q60. The Shannon entropy of a source with uniform distribution over n symbols is

A. 0
B. log■(n)
C. n
D. 1/log■(n)
Correct Answer: B

Q61. A linearly ordered set is also called a

A. Lattice
B. Chain
C. Antichain
D. Boolean algebra
Correct Answer: B

Q62. The number of functions that are both one-to-one and onto from {1,...,n} to {1,...,n} is

A. n^2
B. n!
C. 2^n
D. n^n
Correct Answer: B

Q63. If R is symmetric and transitive, R is

A. Always reflexive
B. Not necessarily reflexive
C. Never reflexive
D. Always antisymmetric
Correct Answer: B

Q64. The number of antichains in {1,...,n} ordered by divisibility

A. 2^n
B. The Dedekind number
C. n!
D. n^2
Correct Answer: B

Q65. The number of onto functions from {1,...,4} to {1,...,2} is

A. 14
B. 16
C. 8
D. 12
Correct Answer: A

Q66. In a group of 367 people, at least two share a birthday by

A. Bayes' theorem
B. Pigeonhole principle
C. Central limit theorem
D. Multiplication principle
Correct Answer: B

Q67. The probability of getting exactly 3 heads in 5 fair coin flips is

A. 5/16
B. 10/32
C. 5/32
D. 1/32
Correct Answer: B

Q68. A complete relation on a set A is one where

A. Every pair is related
B. For any $a \neq b$, either $(a,b) \in R$ or $(b,a) \in R$
C. No pairs are related
D. All elements map to themselves
Correct Answer: B

Q69. The number of strings of length n over alphabet {0,1} is

A. n
B. 2n
C. 2^n
D. n^2
Correct Answer: C

Q70. If X~Bernoulli(p), E(X) =

A. 0
B. p
C. 1–p
D. 1
Correct Answer: B

Q71. A walk in a graph is

A. A sequence of vertices and edges where edges connect consecutive vertices
B. A closed path
C. A tree structure
D. A cycle
Correct Answer: A

Q72. A path is a walk that

A. Can repeat vertices
B. Does not repeat vertices
C. Is always closed
D. Has no edges
Correct Answer: B

Q73. A trail is a walk that

A. Does not repeat vertices
B. Does not repeat edges
C. Is a cycle
D. Is a tree
Correct Answer: B

Q74. A graph is connected if

A. It has no edges
B. There is a path between every pair of vertices
C. All vertices have same degree
D. It is a tree
Correct Answer: B

Q75. A regular graph of degree k means

A. Every vertex has degree k
B. Every vertex has degree 1
C. All edges are equal
D. It is bipartite
Correct Answer: A

Q76. A bipartite graph has

A. All vertices connected
B. Vertices divided into two sets with edges only between sets
C. No edges
D. Only odd cycles
Correct Answer: B

Q77. A cycle of length n has

A. n vertices and n edges
B. n vertices and n–1 edges
C. n+1 edges
D. n–1 vertices
Correct Answer: A

Q78. A tree with n vertices has

A. n edges
B. n–1 edges
C. n+1 edges
D. 2n edges
Correct Answer: B

Q79. A rooted tree has

A. All vertices with degree 1
B. One designated root vertex
C. No edges
D. Only leaf nodes
Correct Answer: B

Q80. A spanning tree of a graph G is

A. A subgraph that is a tree and includes all vertices of G
B. A cycle in G
C. A complete subgraph
D. A bipartite subgraph
Correct Answer: A

Q81. The eccentricity of a vertex is

A. The number of edges incident to it
B. The maximum distance from that vertex to any other vertex
C. The degree of the vertex
D. The number of paths through it
Correct Answer: B

Q82. The radius of a graph is

A. The maximum eccentricity
B. The minimum eccentricity among all vertices
C. The number of vertices
D. The diameter
Correct Answer: B

Q83. The diameter of a graph is

A. The minimum eccentricity
B. The maximum eccentricity among all vertices
C. The average distance
D. The number of edges
Correct Answer: B

Q84. The centre of a graph consists of vertices with

A. Maximum eccentricity
B. Minimum eccentricity
C. Degree 1
D. Maximum degree
Correct Answer: B

Q85. A graph has a Hamiltonian path if

A. It visits every vertex exactly once
B. It visits every edge exactly once
C. It is bipartite
D. It is a tree
Correct Answer: A

Q86. A graph has an Eulerian circuit if

A. All vertices have even degree and it is connected
B. All vertices have odd degree
C. It has a Hamiltonian cycle
D. It is a tree
Correct Answer: A

Q87. A connected planar graph satisfies

A. V−E+F=2
B. V+E=F
C. E=V+F
D. V=E=F
Correct Answer: A

Q88. Kuratowski's theorem states a graph is planar iff it does not contain

A. K■ or C■ as subgraph
B. K■ or K■,■ as subdivision
C. Any cycle
D. Trees
Correct Answer: B

Q89. A bipartite graph is

A. Always planar
B. 2-colorable
C. Always complete
D. Always a tree
Correct Answer: B

Q90. The number of edges in a complete graph K■ is

A. n
B. n(n–1)/2
C. n²
D. n(n+1)/2
Correct Answer: B

Q91. In a binary tree with n internal nodes, the number of leaves is

A. n
B. n+1
C. 2n
D. n–1
Correct Answer: B

Q92. A binary search tree has

A. The left child key < parent key < right child key
B. All keys equal
C. No ordering
D. Only left children
Correct Answer: A

Q93. The height of a balanced binary tree with n nodes is

A. n
B. ■log■(n+1)■
C. n/2
D. √n
Correct Answer: B

Q94. A spanning tree of a connected graph with n vertices has

A. n edges
B. n–1 edges
C. n+1 edges
D. 2n edges
Correct Answer: B

Q95. Kruskal's algorithm finds

A. Shortest path
B. Minimum spanning tree
C. Maximum flow
D. Eulerian circuit
Correct Answer: B

Q96. Dijkstra's algorithm solves

A. Minimum spanning tree
B. Single-source shortest path
C. Maximum flow
D. Topological sort
Correct Answer: B

Q97. The number of spanning trees in K■ is

A. 4
B. 16
C. 64
D. 256
Correct Answer: B

Q98. A graph is Eulerian if

A. It has a Hamiltonian cycle
B. It has an Eulerian circuit
C. It is bipartite
D. It is planar
Correct Answer: B

Q99. The chromatic number of a bipartite graph is

A. 1
B. 2
C. 3
D. 4
Correct Answer: B

Q100. A Hamiltonian cycle visits

A. Every edge exactly once
B. Every vertex exactly once and returns to start
C. Some vertices
D. Only degree-2 vertices
Correct Answer: B

Q101. The shortest path in an unweighted graph can be found by

A. Dijkstra
B. BFS
C. DFS
D. Kruskal
Correct Answer: B

Q102. A directed acyclic graph (DAG) is used for

A. Finding cycles
B. Topological sorting
C. Shortest paths with negative weights
D. Network flow
Correct Answer: B

Q103. The adjacency matrix of a graph is

A. A tree
B. A square matrix where entry (i,j) indicates edge from i to j
C. A list of edges
D. A binary search tree
Correct Answer: B

Q104. In a complete bipartite graph K■,■ the number of edges is

A. m+n
B. m×n
C. m+n−1
D. m×n−1
Correct Answer: B

Q105. A graph G is isomorphic to G' if

A. They have same number of vertices
B. There exists a bijection preserving adjacency
C. They have same edges
D. They are identical
Correct Answer: B

Q106. The number of edges in a tree with n vertices is

A. n
B. n−1
C. n+1
D. 2n
Correct Answer: B

Q107. A forest is a graph that is

A. Connected and has cycles
B. Acyclic
C. Complete
D. Bipartite
Correct Answer: B

Q108. The number of leaves in a full binary tree with n internal nodes is

A. n
B. n+1
C. 2n
D. n–1
Correct Answer: B

Q109. An Eulerian path exists if

A. All vertices have even degree
B. Exactly 0 or 2 vertices have odd degree
C. All vertices have odd degree
D. The graph is complete
Correct Answer: B

Q110. The maximum independent set problem is

A. Solvable in polynomial time for all graphs
B. NP-hard
C. O(n) solvable
D. Trivial
Correct Answer: B

Q111. A graph with n vertices where every pair is connected by an edge is called

A. Bipartite
B. Complete
C. Planar
D. Sparse
Correct Answer: B

Q112. The girth of a graph is

A. The number of vertices
B. The length of the shortest cycle
C. The diameter
D. The number of edges
Correct Answer: B

Q113. A multigraph allows

A. Only simple edges
B. Multiple edges between same pair of vertices and/or loops
C. Only directed edges
D. No edges
Correct Answer: B

Q114. In a weighted graph, the weight of a spanning tree is

A. The number of edges
B. The sum of weights of all its edges
C. The maximum edge weight
D. The product of weights
Correct Answer: B

Q115. The number of vertices in a path graph P■ is

A. n–1
B. n
C. n+1
D. 2n
Correct Answer: B

Q116. A wheel graph W■ has

A. n vertices
B. n+1 vertices
C. 2n vertices
D. n–1 vertices
Correct Answer: B

Q117. The adjacency list representation uses

A. A matrix
B. A list of lists for each vertex
C. A tree
D. A hash table only
Correct Answer: B

Q118. Breadth-first search (BFS) uses

A. A stack
B. A queue
C. A priority queue
D. Recursion only
Correct Answer: B

Q119. Depth-first search (DFS) uses

A. A queue
B. A stack
C. A priority queue
D. A heap
Correct Answer: B

Q120. The topological sort of a DAG produces

A. A cycle
B. A linear ordering of vertices
C. A tree
D. A bipartite graph
Correct Answer: B

Q121. The strongly connected components of a directed graph can be found by

A. BFS
B. Kosaraju's or Tarjan's algorithm
C. Kruskal
D. Dijkstra
Correct Answer: B

Q122. In a flow network, the max-flow min-cut theorem states

A. Max flow equals min cut capacity
B. Max flow equals number of vertices
C. Min cut equals number of edges
D. Max flow equals shortest path
Correct Answer: A

Q123. The Bellman-Ford algorithm handles

A. Only positive weights
B. Graphs with negative weight edges
C. Only bipartite graphs
D. Only trees
Correct Answer: B

Q124. The Floyd-Warshall algorithm finds

A. MST
B. All-pairs shortest paths
C. Maximum flow
D. Eulerian path
Correct Answer: B

Q125. A graph is a tree iff

A. It is connected and has n–1 edges
B. It has cycles
C. It is complete
D. It is bipartite
Correct Answer: A

Q126. A tree has exactly one

A. Cycle
B. Path between any two vertices
C. Eulerian circuit
D. Hamiltonian cycle
Correct Answer: B

Q127. A binary heap is based on a

A. Binary search tree
B. Complete binary tree
C. AVL tree
D. Red-black tree
Correct Answer: B

Q128. The in-order traversal of a binary search tree produces

A. A random sequence
B. A sorted sequence
C. A reverse-sorted sequence
D. A level-order sequence
Correct Answer: B

Q129. The post-order traversal of a binary tree visits

A. Root first, then left and right subtrees
B. Left subtree, right subtree, then root
C. Right subtree, left subtree, then root
D. Root, left, right
Correct Answer: B

Q130. A graph with n vertices and n edges has

A. No cycles
B. At least one cycle (if connected)
C. Exactly one spanning tree
D. More than one component
Correct Answer: B

Q131. The complement of a graph G has

A. Same vertices, edges where G had no edge
B. Different vertices
C. Same edges
D. No vertices
Correct Answer: A

Q132. A Eulerian graph with 6 vertices each of degree 4 has

A. 6 edges
B. 12 edges
C. 24 edges
D. 48 edges
Correct Answer: B

Q133. The number of edges in a complete bipartite graph K■,■ is

A. 3
B. 6
C. 9
D. 12
Correct Answer: C

Q134. A tree with 10 vertices has

A. 8 edges
B. 9 edges
C. 10 edges
D. 11 edges
Correct Answer: B

Q135. The radius of a path graph P■ is

A. n
B. ■n/2■
C. ■n/2■
D. n–1
Correct Answer: C

Q136. A graph with maximum degree Δ has chromatic number at most

A. Δ–1
B. Δ
C. Δ+1
D. 2Δ
Correct Answer: C

Q137. A matching in a graph is

A. A set of edges with no shared vertices
B. A cycle
C. A spanning tree
D. A clique
Correct Answer: A

Q138. A perfect matching exists in a graph with

A. Odd number of vertices
B. Even number of vertices and perfect pairing
C. Any number of vertices
D. Only trees
Correct Answer: B

Q139. A bridge (cut-edge) is an edge whose removal

A. Increases the number of components
B. Decreases the number of components
C. Has no effect
D. Creates a cycle
Correct Answer: A

Q140. A cut vertex (articulation point) is

A. A vertex whose removal increases number of components
B. A vertex with maximum degree
C. A leaf vertex
D. A vertex in a cycle
Correct Answer: A

Q141. The number of edges in a planar graph with n≥3 vertices is at most

A. n
B. 3n–6
C. n^2
D. 2n
Correct Answer: B

Q142. A graph is bipartite iff it contains

A. No cycles
B. No odd cycles
C. No even cycles
D. At least one cycle
Correct Answer: B

Q143. The Ramsey number R(3,3) equals

A. 3
B. 5
C. 6
D. 9
Correct Answer: C

Q144. A graph is Hamiltonian if

A. It has an Eulerian circuit
B. It has a Hamiltonian cycle
C. It is bipartite
D. It is planar
Correct Answer: B

Q145. The number of edges in an n-vertex complete graph is

A. n(n–1)/2
B. n^2
C. n(n+1)/2
D. 2n
Correct Answer: A

Q146. A rooted tree where each node has at most 2 children is a

A. General tree
B. Binary tree
C. B-tree
D. AVL tree
Correct Answer: B

Q147. The level of a node in a rooted tree is

A. The number of children
B. The number of edges from root to that node
C. The degree
D. The height
Correct Answer: B

Q148. The height of a rooted tree is

A. The number of leaves
B. The maximum level of any node
C. The number of edges
D. The number of internal nodes
Correct Answer: B

Q149. A preorder traversal of a tree visits

A. Root, then left subtree, then right subtree
B. Left, root, right
C. Left, right, root
D. Right, root, left
Correct Answer: A

Q150. A graph with n vertices and 0 edges has

A. Chromatic number 0
B. Chromatic number 1
C. Chromatic number n
D. Infinite chromatic number
Correct Answer: B

Q151. The Petersen graph is

A. Planar
B. Non-planar
C. Bipartite
D. A tree
Correct Answer: B

Q152. In a bipartite graph K■,■, the number of vertices is

A. 2
B. 3
C. 5
D. 6
Correct Answer: C

Q153. A dense graph has

A. Few edges relative to vertices
B. Many edges close to the maximum
C. Exactly n–1 edges
D. No edges
Correct Answer: B

Q154. The number of different spanning trees in a cycle C■ is

A. n
B. n–1
C. n+1
D. 2n
Correct Answer: A

Q155. An adjacency list for a graph with n vertices and m edges uses

A. O(n²) space
B. O(n+m) space
C. O(m) space
D. O(n) space
Correct Answer: B

Q156. The chromatic polynomial of a tree with n vertices is

A. k(k–1)^(n–1)
B. k^n
C. k(k–1)^n
D. k+n
Correct Answer: A

Q157. A graph is k-chromatic if

A. It can be colored with k colors and no fewer
B. It has k vertices
C. It has k edges
D. It has k components
Correct Answer: A

Q158. The handshake lemma states the sum of all vertex degrees equals

A. Number of edges
B. Twice the number of edges
C. Number of vertices squared
D. Number of edges divided by 2
Correct Answer: B

Q159. A graph with all vertices having degree 2 consists of

A. A single cycle
B. Disjoint cycles
C. A tree
D. A path
Correct Answer: B

Q160. The graph K■ has chromatic number

A. 2
B. 3
C. 4
D. 5
Correct Answer: D

Q161. A graph is an interval graph if

A. Vertices correspond to intervals on a line and edges represent overlaps
B. It is bipartite
C. It is planar
D. It has no edges
Correct Answer: A

Q162. The shortest path problem in an undirected unweighted graph is solved by

A. DFS
B. BFS
C. Dijkstra
D. Prim
Correct Answer: B

Q163. A directed tree is a directed graph that is

A. Connected and acyclic
B. A complete graph
C. A bipartite graph
D. A cycle
Correct Answer: A

Q164. In a directed graph, the indegree of a vertex is

A. The number of edges leaving it
B. The number of edges entering it
C. The number of paths to it
D. The distance from root
Correct Answer: B

Q165. A tournament is a directed graph where

A. Every pair of vertices has exactly one directed edge between them
B. All edges are bidirectional
C. There are no edges
D. It is a tree
Correct Answer: A

Q166. The number of edges in a tournament on n vertices is

A. n
B. n(n–1)/2
C. n^2
D. n(n+1)/2
Correct Answer: B

Q167. An Eulerian graph must be

A. Connected and all vertices have even degree
B. All vertices have odd degree
C. Bipartite
D. Planar
Correct Answer: A

Q168. The vertex cover number is

A. The maximum independent set size
B. The minimum number of vertices that cover all edges
C. The number of edges
D. The maximum matching size
Correct Answer: B

Q169. A Hamiltonian path in a complete graph K■ exists

A. Only if n is even
B. For all n≥2
C. Only if n≥4
D. Never
Correct Answer: B

Q170. A graph with n vertices and at least $n^2/4$ edges must contain

A. A tree
B. A triangle (by Turán's theorem)
C. A bipartite graph
D. An Eulerian circuit
Correct Answer: B

Q171. A group (G,*) is a set G with a binary operation * that is

A. Closed, associative, has identity, and every element has inverse
B. Commutative and closed
C. Associative and commutative
D. Closed and has identity only
Correct Answer: A

Q172. The order of a group is

A. The number of elements
B. The largest element
C. The number of subgroups
D. The order of its generator
Correct Answer: A

Q173. The order of an element a in a group is

A. The number of elements in G
B. The smallest positive integer n such that a^n = e
C. The number of subgroups
D. The identity element
Correct Answer: B

Q174. Z■ under addition is a group of order

A. 4
B. 5
C. 10
D. 25
Correct Answer: B

Q175. The integers under multiplication form

A. A group
B. A monoid but not a group
C. A ring
D. A field
Correct Answer: B

Q176. A ring is a set with two operations satisfying

A. Addition forms abelian group, multiplication is associative, and distributive laws hold
B. Both operations form groups
C. Only addition is closed
D. Only multiplication has identity
Correct Answer: A

Q177. A field is a ring where

A. Multiplication is commutative and every nonzero element has a multiplicative inverse
B. Addition is not commutative
C. There are no elements
D. Multiplication is not associative
Correct Answer: A

Q178. GF(2) has elements

A. {0,1,2}
B. {0,1}
C. {0}
D. {1,2}
Correct Answer: B

Q179. A cyclic group of order n is isomorphic to

A. S_n
B. Z_n
C. Z■
D. R
Correct Answer: B

Q180. The number of generators of Z■■ is

A. 4
B. 12
C. 2
D. 6
Correct Answer: A

Q181. Lagrange's theorem states that the order of a subgroup divides

A. The order of the group
B. The order of the element
C. The number of cosets
D. The characteristic
Correct Answer: A

Q182. A subgroup of Z■■ generated by 4 has order

A. 3
B. 4
C. 6
D. 12
Correct Answer: A

Q183. The symmetric group S■ has order

A. 3
B. 6
C. 9
D. 12
Correct Answer: B

Q184. An abelian group is one where

A. Every element is its own inverse
B. The operation is commutative
C. It has no subgroups
D. It is cyclic
Correct Answer: B

Q185. The alternating group A■ has order

A. 4
B. 8
C. 12
D. 24
Correct Answer: C

Q186. Z_p where p is prime is

A. Not a field
B. A field
C. A ring but not a field
D. An integral domain only
Correct Answer: B

Q187. A finite field GF(p■) has

A. p elements
B. p■ elements
C. n elements
D. pn elements
Correct Answer: B

Q188. The characteristic of GF(p■) is

A. 0
B. p
C. p■
D. n
Correct Answer: B

Q189. In GF(2^8) used in AES, the polynomial is irreducible of degree

A. 2
B. 4
C. 8
D. 16
Correct Answer: C

Q190. A Hamming code can

A. Only detect errors
B. Detect and correct single-bit errors
C. Only correct errors
D. Compress data
Correct Answer: B

Q191. The Hamming distance between 0110 and 1100 is

A. 1
B. 2
C. 3
D. 4
Correct Answer: C

Q192. A code with minimum distance d can detect up to

A. d errors
B. d–1 errors
C. d+1 errors
D. 2d errors
Correct Answer: B

Q193. A code with minimum distance d can correct up to

A. d errors
B. ■(d–1)/2■ errors
C. d–1 errors
D. 2d errors
Correct Answer: B

Q194. The parity check matrix of a linear code is used to

A. Encode messages
B. Detect errors by checking syndrome
C. Decode without error
D. Compress data
Correct Answer: B

Q195. A cyclic code of length n over GF(q) is an ideal in

A. GF(q)[x]/(x^n–1)
B. GF(q)[x]/(x^n+1)
C. Z[x]/(x^n)
D. R[x]
Correct Answer: A

Q196. The generator polynomial g(x) of a cyclic code divides

A. x^n+1
B. x^n–1
C. x^n
D. x+1
Correct Answer: B

Q197. Reed-Solomon codes are

A. Binary codes only
B. Non-binary cyclic codes
C. Convolutional codes
D. Block codes that are not cyclic
Correct Answer: B

Q198. A CRC (cyclic redundancy check) uses

A. Polynomial division over GF(2)
B. Matrix multiplication
C. Fourier transforms
D. Linear programming
Correct Answer: A

Q199. The Hamming(7,4) code has

A. 7 data bits and 4 parity bits
B. 4 data bits and 3 parity bits
C. 7 data bits and 3 parity bits
D. 4 data bits and 4 parity bits
Correct Answer: B

Q200. A linear block code [n,k] has

A. n information bits and k coded bits
B. k information bits and n coded bits (n total)
C. n+k bits total
D. k parity bits
Correct Answer: B

ANSWER KEY — Topic 1: Discrete Structures

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1-B 2-B 3-C 4-C 5-B 6-B 7-B 8-B 9-B 10-C
11-A 12-B 13-B 14-B 15-D 16-B 17-B 18-C 19-B 20-A
21-A 22-B 23-A 24-B 25-C 26-B 27-A 28-B 29-C 30-D
31-B 32-A 33-A 34-B 35-A 36-B 37-D 38-C 39-B 40-B
41-B 42-A 43-B 44-A 45-B 46-B 47-B 48-B 49-B 50-B
51-B 52-A 53-B 54-B 55-B 56-C 57-A 58-A 59-B 60-B
61-B 62-B 63-B 64-B 65-A 66-B 67-B 68-B 69-C 70-B
71-A 72-B 73-B 74-B 75-A 76-B 77-A 78-B 79-B 80-A
81-B 82-B 83-B 84-B 85-A 86-A 87-A 88-B 89-B 90-B
91-B 92-A 93-B 94-B 95-B 96-B 97-B 98-B 99-B 100-B
101-B 102-B 103-B 104-B 105-B 106-B 107-B 108-B 109-B 110-B
111-B 112-B 113-B 114-B 115-B 116-B 117-B 118-B 119-B 120-B
121-B 122-A 123-B 124-B 125-A 126-B 127-B 128-B 129-B 130-B
131-A 132-B 133-C 134-B 135-C 136-C 137-A 138-B 139-A 140-A
141-B 142-B 143-C 144-B 145-A 146-B 147-B 148-B 149-A 150-B
151-B 152-C 153-B 154-A 155-B 156-A 157-A 158-B 159-B 160-D
161-A 162-B 163-A 164-B 165-A 166-B 167-A 168-B 169-B 170-B
171-A 172-A 173-B 174-B 175-B 176-A 177-A 178-B 179-B 180-A
181-A 182-A 183-B 184-B 185-C 186-B 187-B 188-B 189-C 190-B
191-C 192-B 193-B 194-B 195-A 196-B 197-B 198-A 199-B 200-B
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Topic 2: Computer Arithmetic

Syllabus: *Propositional (Boolean) Logic, Predicate Logic, Well-formed-formulae (WFF), Satisfiability and Tautology; Logic Families: TTL, ECL and C-MOS gates, Boolean algebra and Minimization of Boolean functions, Flip-flops - types, race condition and comparison, Design of combinational and sequential circuits; Representation of Integers: Octal, Hex, Decimal, and Binary, 2's complement and 1's complement arithmetic, Floating point representation.*

Q1. Which of the following is a tautology?

A. $p \vee \neg p$
B. $p \wedge \neg p$
C. $p \rightarrow \neg p$
D. $(p \wedge q) \wedge \neg q$
Correct Answer: A

Q2. Which formula is unsatisfiable (contradiction)?

A. $p \vee \neg p$
B. $p \rightarrow p$
C. $p \wedge \neg p$
D. True
Correct Answer: C

Q3. A wff is satisfiable iff?

A. It is false in all interpretations
B. It is true in all interpretations
C. There exists an interpretation making it true
D. It has no models
Correct Answer: C

Q4. De Morgan's law states $\neg(p \vee q)$ equals?

A. $p \wedge q$
B. $\neg p \wedge \neg q$
C. $\neg p \vee \neg q$
D. $p \vee q$
Correct Answer: B

Q5. $p \rightarrow q$ is logically equivalent to?

A. $\neg q \vee p$
B. $p \vee \neg q$
C. $\neg p \vee q$
D. $p \wedge q$
Correct Answer: C

Q6. Which logic family has the highest speed?

A. TTL
B. ECL
C. CMOS
D. RTL
Correct Answer: B

Q7. Which logic family has lowest static power dissipation?

A. CMOS
B. ECL
C. TTL
D. DTL
Correct Answer: A

Q8. TTL stands for?

A. Transistor-Transistor Logic
B. Time-Triggered Logic
C. Transistor-Timed Loop
D. Triode-Transistor Logic
Correct Answer: A

Q9. ECL's key characteristic is?

A. Saturated operation, low speed
B. Highest noise margin of all
C. Zero power
D. Non-saturated operation, very high speed but high power
Correct Answer: D

Q10. CMOS inverter in steady state draws?

A. Near-zero static current
B. Negative current
C. Large static current
D. Maximum current always
Correct Answer: A

Q11. Predicate $\forall x\ P(x)$ is true when?

A. P holds for no x
B. P(x) holds for every x in domain
C. P holds for at least one x
D. P holds for finitely many x
Correct Answer: B

Q12. $\exists x\ \forall y\ R(x,y)$ versus $\forall y\ \exists x\ R(x,y)$: the former?

A. Is always false
B. Implies the latter but not conversely
C. Is weaker
D. Is equivalent
Correct Answer: B

Q13. Skolemization is used to?

A. Minimize gates
B. Convert to tautology
C. Eliminate universal quantifiers
D. Eliminate existential quantifiers preserving satisfiability
Correct Answer: D

Q14. A valid argument's conclusion must be true when?

A. At least one premise is true
B. Premises are false
C. Conclusion is satisfiable
D. All premises are true
Correct Answer: D

Q15. Resolution principle applies to formulas in?

A. Horn form only
B. Arithmetic form
C. Conjunctive normal form (clausal form)
D. Disjunctive normal form only
Correct Answer: C

Q16. Which of the following is logically equivalent to $\neg(p \wedge q)$?

A. $p \rightarrow q$
B. $p \wedge q$
C. $\neg p \vee \neg q$
D. $\neg p \wedge \neg q$
Correct Answer: C

Q17. The dual of Boolean expression A·(B+C) is?

A. A+B·C
B. A·B+C
C. A+(B·C)
D. (A+B)·C
Correct Answer: C

Q18. Fan-out of a logic gate refers to?

A. Power supply voltage
B. Number of outputs it has
C. Number of gate inputs it can drive
D. Propagation delay
Correct Answer: C

Q19. Noise margin is highest (among classic families) in?

A. CMOS
B. Schottky TTL
C. ECL
D. I2L
Correct Answer: A

Q20. A tautology with n variables is true in how many rows of truth table?

A. 4-1 rows
B. 2 rows
C. All 4 rows
D. Exactly 1 row
Correct Answer: C

Q21. Which of the following is logically equivalent to ¬(p ∧ q)?

A. ¬p ∨ ¬q
B. p ∧ q
C. ¬p ∧ ¬q
D. p → q
Correct Answer: A

Q22. The dual of Boolean expression A·(B+C) is?

A. (A+B)·C
B. A+(B·C)
C. A·B+C
D. A+B·C
Correct Answer: B

Q23. Fan-out of a logic gate refers to?

A. Number of gate inputs it can drive
B. Power supply voltage
C. Propagation delay
D. Number of outputs it has
Correct Answer: A

Q24. Noise margin is highest (among classic families) in?

A. ECL
B. Schottky TTL
C. CMOS
D. I2L
Correct Answer: C

Q25. A tautology with n variables is true in how many rows of truth table?

A. 16-1 rows
B. 4 rows
C. All 16 rows
D. Exactly 1 row
Correct Answer: C

Q26. Which of the following is logically equivalent to ¬(p ∧ q)?

A. ¬p ∨ ¬q
B. ¬p ∧ ¬q
C. p ∧ q
D. p → q
Correct Answer: A

Q27. The dual of Boolean expression A·(B+C) is?

A. A+B·C
B. (A+B)·C
C. A·B+C
D. A+(B·C)
Correct Answer: D

Q28. Fan-out of a logic gate refers to?

A. Number of gate inputs it can drive
B. Number of outputs it has
C. Power supply voltage
D. Propagation delay
Correct Answer: A

Q29. Noise margin is highest (among classic families) in?

A. ECL
B. CMOS
C. Schottky TTL
D. I2L
Correct Answer: B

Q30. A tautology with n variables is true in how many rows of truth table?

A. All 16 rows
B. 4 rows
C. Exactly 1 row
D. 16-1 rows
Correct Answer: A

Q31. Which of the following is logically equivalent to ¬(p ∧ q)?

A. p ∧ q
B. ¬p ∨ ¬q
C. p → q
D. ¬p ∧ ¬q
Correct Answer: B

Q32. The dual of Boolean expression A·(B+C) is?

A. A+B·C
B. A·B+C
C. (A+B)·C
D. A+(B·C)
Correct Answer: D

Q33. Fan-out of a logic gate refers to?

A. Propagation delay
B. Number of gate inputs it can drive
C. Number of outputs it has
D. Power supply voltage
Correct Answer: B

Q34. Noise margin is highest (among classic families) in?

A. I2L
B. Schottky TTL
C. ECL
D. CMOS
Correct Answer: D

Q35. A tautology with n variables is true in how many rows of truth table?

A. 2 rows
B. All 4 rows
C. Exactly 1 row
D. 4-1 rows
Correct Answer: B

Q36. Which of the following is logically equivalent to ¬(p ∧ q)?

A. ¬p ∧ ¬q
B. p → q
C. p ∧ q
D. ¬p ∨ ¬q
Correct Answer: D

Q37. The dual of Boolean expression A·(B+C) is?

A. A+B·C
B. A+(B·C)
C. A·B+C
D. (A+B)·C
Correct Answer: B

Q38. Fan-out of a logic gate refers to?

A. Number of outputs it has
B. Power supply voltage
C. Number of gate inputs it can drive
D. Propagation delay
Correct Answer: C

Q39. Noise margin is highest (among classic families) in?

A. CMOS
B. I2L
C. Schottky TTL
D. ECL
Correct Answer: A

Q40. A tautology with n variables is true in how many rows of truth table?

A. All 4 rows
B. 2 rows
C. 4-1 rows
D. Exactly 1 row
Correct Answer: A

Q41. Which of the following is logically equivalent to ¬(p ∧ q)?

A. p → q
B. ¬p ∨ ¬q
C. p ∧ q
D. ¬p ∧ ¬q
Correct Answer: B

Q42. The dual of Boolean expression A·(B+C) is?

A. A·B+C
B. A+(B·C)
C. A+B·C
D. (A+B)·C
Correct Answer: B

Q43. Fan-out of a logic gate refers to?

A. Propagation delay
B. Number of outputs it has
C. Power supply voltage
D. Number of gate inputs it can drive
Correct Answer: D

Q44. Noise margin is highest (among classic families) in?

A. ECL
B. CMOS
C. I2L
D. Schottky TTL
Correct Answer: B

Q45. A tautology with n variables is true in how many rows of truth table?

A. 2 rows
B. Exactly 1 row
C. All 4 rows
D. 4-1 rows
Correct Answer: C

Q46. Which of the following is logically equivalent to ¬(p ∧ q)?

A. p ∧ q
B. ¬p ∧ ¬q
C. ¬p ∨ ¬q
D. p → q
Correct Answer: C

Q47. The dual of Boolean expression A·(B+C) is?

A. A+(B·C)
B. A+B·C
C. A·B+C
D. (A+B)·C
Correct Answer: A

Q48. Fan-out of a logic gate refers to?

A. Power supply voltage
B. Number of gate inputs it can drive
C. Number of outputs it has
D. Propagation delay
Correct Answer: B

Q49. Noise margin is highest (among classic families) in?

A. Schottky TTL
B. I2L
C. CMOS
D. ECL
Correct Answer: C

Q50. A tautology with n variables is true in how many rows of truth table?

A. All 16 rows
B. Exactly 1 row
C. 16-1 rows
D. 4 rows
Correct Answer: A

Q51. An n-variable K-map (n=4) has how many cells?

A. 8
B. 17
C. 16
D. 16
Correct Answer: C

Q52. In a K-map, groups must contain cells in powers of?

A. 2 (1,2,4,8,...)
B. 3
C. 10
D. 5
Correct Answer: A

Q53. Quine-McCluskey method is used for?

A. Floating-point rounding
B. Encoding flip-flops
C. Designing ALUs
D. Tabular minimization of Boolean functions, finding prime implicants
Correct Answer: D

Q54. An essential prime implicant is one that?

A. Covers a minterm covered by no other prime implicant
B. Is always redundant
C. Has the largest cost
D. Covers all minterms
Correct Answer: A

Q55. A 4:1 multiplexer has how many select lines?

A. 4
B. 5
C. 2
D. 3
Correct Answer: C

Q56. A 4-to-16 decoder has how many output lines?

A. 16
B. 8
C. 5
D. 4
Correct Answer: A

Q57. Race-around condition occurs in?

A. D flip-flop always
B. ROM
C. JK flip-flop with J=K=1 and long clock pulse
D. Combinational gates only
Correct Answer: C

Q58. Master-slave JK flip-flop was introduced to overcome?

A. Fan-in limit
B. Power dissipation
C. Race-around condition
D. Clock skew only
Correct Answer: C

Q59. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = JQ'+K'Q
B. Q+ = D
C. Q+ = Q'
D. Q+ = T xor Q
Correct Answer: B

Q60. Characteristic equation of JK flip-flop is?

A. Q+ = TQ'+T'Q
B. Q+ = JQ' + K'Q
C. Q+ = SQ+R'Q
D. Q+ = D
Correct Answer: B

Q61. A mod-8 ripple counter requires at least how many flip-flops?

A. 8
B. 7
C. 3
D. 4
Correct Answer: C

Q62. A combinational circuit's output depends on?

A. Present inputs only
B. Previous output only
C. Present + past inputs
D. Clock only
Correct Answer: A

Q63. A sequential circuit's output depends on?

A. Temperature only
B. Power supply only
C. Present inputs and present state (past inputs)
D. Present inputs only
Correct Answer: C

Q64. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1010
B. 1101
C. 1110
D. 0100
Correct Answer: C

Q65. The Boolean function F=A+A'B equals (absorption)?

A. A
B. A'B'
C. A'B
D. A+B
Correct Answer: D

Q66. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 1,1,1
B. 3,2,1
C. 1, 2, 3
D. 2,2,2
Correct Answer: C

Q67. A half-adder has inputs/outputs?

A. 2 inputs, 1 output
B. 3 inputs, 2 outputs
C. 2 inputs, 2 outputs (Sum, Carry)
D. 1 input, 2 outputs
Correct Answer: C

Q68. A full-adder can be built from how many half-adders + OR?

A. 1 half-adder
B. 2 half-adders + 1 OR gate
C. 4 half-adders
D. 3 half-adders
Correct Answer: B

Q69. Propagation delay in ripple-carry adder grows as?

A. O(n) with number of bits
B. O(n^2)
C. O(1)
D. O(log n) always
Correct Answer: A

Q70. Edge-triggered flip-flop samples input on?

A. Entire HIGH level
B. Both levels continuously
C. Active clock edge only
D. Entire LOW level
Correct Answer: C

Q71. An n-variable K-map (n=3) has how many cells?

A. 8
B. 6
C. 9
D. 9
Correct Answer: A

Q72. In a K-map, groups must contain cells in powers of?

A. 3
B. 2 (1,2,4,8,...)
C. 5
D. 10
Correct Answer: B

Q73. Quine-McCluskey method is used for?

A. Encoding flip-flops
B. Floating-point rounding
C. Tabular minimization of Boolean functions, finding prime implicants
D. Designing ALUs
Correct Answer: C

Q74. An essential prime implicant is one that?

A. Covers all minterms
B. Covers a minterm covered by no other prime implicant
C. Is always redundant
D. Has the largest cost
Correct Answer: B

Q75. A 2:1 multiplexer has how many select lines?

A. 2
B. 3
C. 2
D. 1
Correct Answer: D

Q76. A 4-to-16 decoder has how many output lines?

A. 5
B. 8
C. 16
D. 4
Correct Answer: C

Q77. Race-around condition occurs in?

A. ROM
B. JK flip-flop with J=K=1 and long clock pulse
C. D flip-flop always
D. Combinational gates only
Correct Answer: B

Q78. Master-slave JK flip-flop was introduced to overcome?

A. Race-around condition
B. Clock skew only
C. Power dissipation
D. Fan-in limit
Correct Answer: A

Q79. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = D
B. Q+ = T xor Q
C. Q+ = Q'
D. Q+ = JQ'+K'Q
Correct Answer: A

Q80. Characteristic equation of JK flip-flop is?

A. Q+ = JQ' + K'Q
B. Q+ = SQ+R'Q
C. Q+ = D
D. Q+ = TQ'+T'Q
Correct Answer: A

Q81. A mod-10 ripple counter requires at least how many flip-flops?

A. 5
B. 9
C. 4
D. 10
Correct Answer: C

Q82. A combinational circuit's output depends on?

A. Present + past inputs
B. Previous output only
C. Present inputs only
D. Clock only
Correct Answer: C

Q83. A sequential circuit's output depends on?

A. Temperature only
B. Present inputs and present state (past inputs)
C. Power supply only
D. Present inputs only
Correct Answer: B

Q84. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1110
B. 1101
C. 0100
D. 1010
Correct Answer: A

Q85. The Boolean function F=A+A'B equals (absorption)?

A. A'B
B. A'B'
C. A+B
D. A
Correct Answer: C

Q86. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 2,2,2
B. 1,1,1
C. 3,2,1
D. 1, 2, 3
Correct Answer: D

Q87. A half-adder has inputs/outputs?

A. 2 inputs, 1 output
B. 3 inputs, 2 outputs
C. 1 input, 2 outputs
D. 2 inputs, 2 outputs (Sum, Carry)
Correct Answer: D

Q88. A full-adder can be built from how many half-adders + OR?

A. 2 half-adders + 1 OR gate
B. 3 half-adders
C. 1 half-adder
D. 4 half-adders
Correct Answer: A

Q89. Propagation delay in ripple-carry adder grows as?

A. O(log n) always
B. O(n) with number of bits
C. O(n^2)
D. O(1)
Correct Answer: B

Q90. Edge-triggered flip-flop samples input on?

A. Entire HIGH level
B. Active clock edge only
C. Both levels continuously
D. Entire LOW level
Correct Answer: B

Q91. An n-variable K-map (n=2) has how many cells?

A. 4
B. 4
C. 5
D. 4
Correct Answer: A

Q92. In a K-map, groups must contain cells in powers of?

A. 2 (1,2,4,8,...)
B. 10
C. 5
D. 3
Correct Answer: A

Q93. Quine-McCluskey method is used for?

A. Designing ALUs
B. Floating-point rounding
C. Tabular minimization of Boolean functions, finding prime implicants
D. Encoding flip-flops
Correct Answer: C

Q94. An essential prime implicant is one that?

A. Covers all minterms
B. Is always redundant
C. Covers a minterm covered by no other prime implicant
D. Has the largest cost
Correct Answer: C

Q95. A 2:1 multiplexer has how many select lines?

A. 2
B. 3
C. 2
D. 1
Correct Answer: D

Q96. A 3-to-8 decoder has how many output lines?

A. 8
B. 3
C. 6
D. 4
Correct Answer: A

Q97. Race-around condition occurs in?

A. JK flip-flop with J=K=1 and long clock pulse
B. ROM
C. D flip-flop always
D. Combinational gates only
Correct Answer: A

Q98. Master-slave JK flip-flop was introduced to overcome?

A. Clock skew only
B. Power dissipation
C. Race-around condition
D. Fan-in limit
Correct Answer: C

Q99. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = Q'
B. Q+ = D
C. Q+ = JQ'+K'Q
D. Q+ = T xor Q
Correct Answer: B

Q100. Characteristic equation of JK flip-flop is?

A. Q+ = SQ+R'Q
B. Q+ = JQ' + K'Q
C. Q+ = D
D. Q+ = TQ'+T'Q
Correct Answer: B

Q101. A mod-16 ripple counter requires at least how many flip-flops?

A. 5
B. 16
C. 15
D. 4
Correct Answer: D

Q102. A combinational circuit's output depends on?

A. Clock only
B. Present inputs only
C. Present + past inputs
D. Previous output only
Correct Answer: B

Q103. A sequential circuit's output depends on?

A. Power supply only
B. Temperature only
C. Present inputs and present state (past inputs)
D. Present inputs only
Correct Answer: C

Q104. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 0100
B. 1010
C. 1110
D. 1101
Correct Answer: C

Q105. The Boolean function F=A+A'B equals (absorption)?

A. A'B
B. A
C. A'B'
D. A+B
Correct Answer: D

Q106. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 3,2,1
B. 1, 2, 3
C. 1,1,1
D. 2,2,2
Correct Answer: B

Q107. A half-adder has inputs/outputs?

A. 3 inputs, 2 outputs
B. 1 input, 2 outputs
C. 2 inputs, 2 outputs (Sum, Carry)
D. 2 inputs, 1 output
Correct Answer: C

Q108. A full-adder can be built from how many half-adders + OR?

A. 4 half-adders
B. 2 half-adders + 1 OR gate
C. 3 half-adders
D. 1 half-adder
Correct Answer: B

Q109. Propagation delay in ripple-carry adder grows as?

A. O(log n) always
B. O(n^2)
C. O(1)
D. O(n) with number of bits
Correct Answer: D

Q110. Edge-triggered flip-flop samples input on?

A. Entire LOW level
B. Entire HIGH level
C. Both levels continuously
D. Active clock edge only
Correct Answer: D

Q111. An n-variable K-map (n=4) has how many cells?

A. 16
B. 16
C. 8
D. 17
Correct Answer: A

Q112. In a K-map, groups must contain cells in powers of?

A. 3
B. 2 (1,2,4,8,...)
C. 5
D. 10
Correct Answer: B

Q113. Quine-McCluskey method is used for?

A. Floating-point rounding
B. Designing ALUs
C. Encoding flip-flops
D. Tabular minimization of Boolean functions, finding prime implicants
Correct Answer: D

Q114. An essential prime implicant is one that?

A. Is always redundant
B. Covers all minterms
C. Has the largest cost
D. Covers a minterm covered by no other prime implicant
Correct Answer: D

Q115. A 2:1 multiplexer has how many select lines?

A. 2
B. 3
C. 1
D. 2
Correct Answer: C

Q116. A 4-to-16 decoder has how many output lines?

A. 8
B. 5
C. 16
D. 4
Correct Answer: C

Q117. Race-around condition occurs in?

A. D flip-flop always
B. Combinational gates only
C. JK flip-flop with J=K=1 and long clock pulse
D. ROM
Correct Answer: C

Q118. Master-slave JK flip-flop was introduced to overcome?

A. Power dissipation
B. Race-around condition
C. Clock skew only
D. Fan-in limit
Correct Answer: B

Q119. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = Q'
B. Q+ = D
C. Q+ = JQ'+K'Q
D. Q+ = T xor Q
Correct Answer: B

Q120. Characteristic equation of JK flip-flop is?

A. Q+ = D
B. Q+ = JQ' + K'Q
C. Q+ = SQ+R'Q
D. Q+ = TQ'+T'Q
Correct Answer: B

Q121. A mod-8 ripple counter requires at least how many flip-flops?

A. 8
B. 3
C. 4
D. 7
Correct Answer: B

Q122. A combinational circuit's output depends on?

A. Present inputs only
B. Previous output only
C. Clock only
D. Present + past inputs
Correct Answer: A

Q123. A sequential circuit's output depends on?

A. Present inputs only
B. Power supply only
C. Present inputs and present state (past inputs)
D. Temperature only
Correct Answer: C

Q124. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1010
B. 1110
C. 0100
D. 1101
Correct Answer: B

Q125. The Boolean function F=A+A'B equals (absorption)?

A. A
B. A'B'
C. A'B
D. A+B
Correct Answer: D

Q126. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 2,2,2
B. 1,1,1
C. 1, 2, 3
D. 3,2,1
Correct Answer: C

Q127. A half-adder has inputs/outputs?

A. 3 inputs, 2 outputs
B. 2 inputs, 2 outputs (Sum, Carry)
C. 1 input, 2 outputs
D. 2 inputs, 1 output
Correct Answer: B

Q128. A full-adder can be built from how many half-adders + OR?

A. 3 half-adders
B. 4 half-adders
C. 1 half-adder
D. 2 half-adders + 1 OR gate
Correct Answer: D

Q129. Propagation delay in ripple-carry adder grows as?

A. O(log n) always
B. O(1)
C. O(n^2)
D. O(n) with number of bits
Correct Answer: D

Q130. Edge-triggered flip-flop samples input on?

A. Active clock edge only
B. Entire LOW level
C. Both levels continuously
D. Entire HIGH level
Correct Answer: A

Q131. An n-variable K-map (n=3) has how many cells?

A. 8
B. 9
C. 9
D. 6
Correct Answer: A

Q132. In a K-map, groups must contain cells in powers of?

A. 3
B. 2 (1,2,4,8,...)
C. 5
D. 10
Correct Answer: B

Q133. Quine-McCluskey method is used for?

A. Designing ALUs
B. Encoding flip-flops
C. Tabular minimization of Boolean functions, finding prime implicants
D. Floating-point rounding
Correct Answer: C

Q134. An essential prime implicant is one that?

A. Is always redundant
B. Covers all minterms
C. Has the largest cost
D. Covers a minterm covered by no other prime implicant
Correct Answer: D

Q135. A 2:1 multiplexer has how many select lines?

A. 1
B. 2
C. 2
D. 3
Correct Answer: A

Q136. A 3-to-8 decoder has how many output lines?

A. 8
B. 3
C. 4
D. 6
Correct Answer: A

Q137. Race-around condition occurs in?

A. JK flip-flop with J=K=1 and long clock pulse
B. ROM
C. Combinational gates only
D. D flip-flop always
Correct Answer: A

Q138. Master-slave JK flip-flop was introduced to overcome?

A. Fan-in limit
B. Power dissipation
C. Clock skew only
D. Race-around condition
Correct Answer: D

Q139. Characteristic equation of D flip-flop (next state Q+) is?

A. Q+ = JQ'+K'Q
B. Q+ = D
C. Q+ = T xor Q
D. Q+ = Q'
Correct Answer: B

Q140. Characteristic equation of JK flip-flop is?

A. Q+ = SQ+R'Q
B. Q+ = TQ'+T'Q
C. Q+ = D
D. Q+ = JQ' + K'Q
Correct Answer: D

Q141. A mod-8 ripple counter requires at least how many flip-flops?

A. 8
B. 7
C. 4
D. 3
Correct Answer: D

Q142. A combinational circuit's output depends on?

A. Previous output only
B. Present inputs only
C. Present + past inputs
D. Clock only
Correct Answer: B

Q143. A sequential circuit's output depends on?

A. Present inputs only
B. Power supply only
C. Present inputs and present state (past inputs)
D. Temperature only
Correct Answer: C

Q144. Gray code of binary 1011 is? (adjacent codes differ by 1 bit)

A. 1110
B. 0100
C. 1101
D. 1010
Correct Answer: A

Q145. The Boolean function F=A+A'B equals (absorption)?

A. A
B. A+B
C. A'B'
D. A'B
Correct Answer: B

Q146. Minimum number of NAND gates to implement NOT, AND, OR respectively (using NAND only)?

A. 2,2,2
B. 1, 2, 3
C. 1,1,1
D. 3,2,1
Correct Answer: B

Q147. A half-adder has inputs/outputs?

A. 2 inputs, 2 outputs (Sum, Carry)
B. 2 inputs, 1 output
C. 3 inputs, 2 outputs
D. 1 input, 2 outputs
Correct Answer: A

Q148. A full-adder can be built from how many half-adders + OR?

A. 4 half-adders
B. 2 half-adders + 1 OR gate
C. 3 half-adders
D. 1 half-adder
Correct Answer: B

Q149. Propagation delay in ripple-carry adder grows as?

A. O(n) with number of bits
B. O(n^2)
C. O(1)
D. O(log n) always
Correct Answer: A

Q150. Edge-triggered flip-flop samples input on?

A. Both levels continuously
B. Entire HIGH level
C. Entire LOW level
D. Active clock edge only
Correct Answer: D

Q151. Decimal 109 in 8-bit binary is?

A. 1101101
B. 01101110
C. 01101101
D. 01101100
Correct Answer: C

Q152. Decimal 222 in hexadecimal (2 digits) is?

A. 0xDF
B. 0o336
C. 0xDD
D. 0xDE
Correct Answer: D

Q153. Decimal 86 in octal is?

A. 127
B. 126
C. 125
D. 56
Correct Answer: B

Q154. Range of signed integers in 3-bit 2's complement is?

A. -4 to +3
B. 0 to 7
C. -7 to 7
D. 0 to 8
Correct Answer: A

Q155. 1's complement of binary 01101 is?

A. 01101
B. 10010
C. 00000
D. 11111
Correct Answer: B

Q156. In IEEE 754 single precision, exponent bias is?

A. 1023
B. 128
C. 127
D. 15
Correct Answer: C

Q157. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1+7+24
B. 1+11+20
C. 8+8+16
D. 1 sign + 8 exponent + 23 mantissa
Correct Answer: D

Q158. In IEEE 754 double precision, exponent bias is?

A. 1023
B. 127
C. 63
D. 2047
Correct Answer: A

Q159. Binary 1100 + 0111 equals decimal?

A. 18
B. 19
C. 84
D. 20
Correct Answer: B

Q160. Overflow in 2's complement addition occurs when?

A. There is any carry
B. Operands have opposite signs
C. Carry into sign bit differs from carry out
D. Result is zero
Correct Answer: C

Q161. Decimal 103 in 8-bit binary is?

A. 01100111
B. 01100110
C. 1100111
D. 01101000
Correct Answer: A

Q162. Decimal 55 in hexadecimal (2 digits) is?

A. 0x38
B. 0o067
C. 0x36
D. 0x37
Correct Answer: D

Q163. Decimal 87 in octal is?

A. 126
B. 130
C. 127
D. 57
Correct Answer: C

Q164. Range of signed integers in 5-bit 2's complement is?

A. -31 to 31
B. -16 to +15
C. 0 to 32
D. 0 to 31
Correct Answer: B

Q165. 1's complement of binary 10110 is?

A. 01001
B. 10110
C. 11111
D. 00000
Correct Answer: A

Q166. In IEEE 754 single precision, exponent bias is?

A. 128
B. 15
C. 127
D. 1023
Correct Answer: C

Q167. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1 sign + 8 exponent + 23 mantissa
B. 8+8+16
C. 1+7+24
D. 1+11+20
Correct Answer: A

Q168. In IEEE 754 double precision, exponent bias is?

A. 1023
B. 63
C. 2047
D. 127
Correct Answer: A

Q169. Binary 1110 + 0010 equals decimal?

A. 17
B. 16
C. 28
D. 15
Correct Answer: B

Q170. Overflow in 2's complement addition occurs when?

A. Result is zero
B. There is any carry
C. Operands have opposite signs
D. Carry into sign bit differs from carry out
Correct Answer: D

Q171. Decimal 14 in 8-bit binary is?

A. 0001110
B. 00001110
C. 00001111
D. 00001101
Correct Answer: B

Q172. Decimal 194 in hexadecimal (2 digits) is?

A. 0xC2
B. 0xC3
C. 0xC1
D. 0o302
Correct Answer: A

Q173. Decimal 80 in octal is?

A. 117
B. 121
C. 50
D. 120
Correct Answer: D

Q174. Range of signed integers in 7-bit 2's complement is?

A. 0 to 127
B. -64 to +63
C. 0 to 128
D. -127 to 127
Correct Answer: B

Q175. 1's complement of binary 10011 is?

A. 11111
B. 01100
C. 10011
D. 00000
Correct Answer: B

Q176. In IEEE 754 single precision, exponent bias is?

A. 127
B. 128
C. 15
D. 1023
Correct Answer: A

Q177. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1 sign + 8 exponent + 23 mantissa
B. 1+7+24
C. 8+8+16
D. 1+11+20
Correct Answer: A

Q178. In IEEE 754 double precision, exponent bias is?

A. 63
B. 1023
C. 2047
D. 127
Correct Answer: B

Q179. Binary 0010 + 0011 equals decimal?

A. 6
B. 6
C. 4
D. 5
Correct Answer: D

Q180. Overflow in 2's complement addition occurs when?

A. Operands have opposite signs
B. There is any carry
C. Result is zero
D. Carry into sign bit differs from carry out
Correct Answer: D

Q181. Decimal 79 in 8-bit binary is?

A. 01001111
B. 1001111
C. 01001110
D. 01010000
Correct Answer: A

Q182. Decimal 73 in hexadecimal (2 digits) is?

A. 0x4A
B. 0x48
C. 0x49
D. 0o111
Correct Answer: C

Q183. Decimal 38 in octal is?

A. 26
B. 46
C. 45
D. 47
Correct Answer: B

Q184. Range of signed integers in 6-bit 2's complement is?

A. -32 to +31
B. 0 to 64
C. -63 to 63
D. 0 to 63
Correct Answer: A

Q185. 1's complement of binary 01101 is?

A. 00000
B. 11111
C. 01101
D. 10010
Correct Answer: D

Q186. In IEEE 754 single precision, exponent bias is?

A. 1023
B. 15
C. 127
D. 128
Correct Answer: C

Q187. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1+7+24
B. 8+8+16
C. 1 sign + 8 exponent + 23 mantissa
D. 1+11+20
Correct Answer: C

Q188. In IEEE 754 double precision, exponent bias is?

A. 127
B. 63
C. 2047
D. 1023
Correct Answer: D

Q189. Binary 0011 + 0111 equals decimal?

A. 21
B. 11
C. 9
D. 10
Correct Answer: D

Q190. Overflow in 2's complement addition occurs when?

A. Result is zero
B. Carry into sign bit differs from carry out
C. There is any carry
D. Operands have opposite signs
Correct Answer: B

Q191. Decimal 145 in 8-bit binary is?

A. 10010001
B. 10010000
C. 10010001
D. 10010010
Correct Answer: A

Q192. Decimal 25 in hexadecimal (2 digits) is?

A. 0x19
B. 0x18
C. 0x1A
D. 0o031
Correct Answer: A

Q193. Decimal 18 in octal is?

A. 23
B. 21
C. 12
D. 22
Correct Answer: D

Q194. Range of signed integers in 7-bit 2's complement is?

A. 0 to 128
B. -64 to +63
C. 0 to 127
D. -127 to 127
Correct Answer: B

Q195. 1's complement of binary 11001 is?

A. 11111
B. 00000
C. 11001
D. 00110
Correct Answer: D

Q196. In IEEE 754 single precision, exponent bias is?

A. 1023
B. 15
C. 128
D. 127
Correct Answer: D

Q197. In IEEE 754 single precision (32-bit), bit-fields are?

A. 1+11+20
B. 1+7+24
C. 8+8+16
D. 1 sign + 8 exponent + 23 mantissa
Correct Answer: D

Q198. In IEEE 754 double precision, exponent bias is?

A. 2047
B. 63
C. 1023
D. 127
Correct Answer: C

Q199. Binary 0110 + 0110 equals decimal?

A. 12
B. 11
C. 13
D. 36
Correct Answer: A

Q200. Overflow in 2's complement addition occurs when?

A. Carry into sign bit differs from carry out
B. Operands have opposite signs
C. There is any carry
D. Result is zero
Correct Answer: A

ANSWER KEY — Topic 2: Computer Arithmetic

```
1-A 2-C 3-C 4-B 5-C 6-B 7-A 8-A 9-D 10-A
11-B 12-B 13-D 14-D 15-C 16-C 17-C 18-C 19-A 20-C
21-A 22-B 23-A 24-C 25-C 26-A 27-D 28-A 29-B 30-A
31-B 32-D 33-B 34-D 35-B 36-D 37-B 38-C 39-A 40-A
41-B 42-B 43-D 44-B 45-C 46-C 47-A 48-B 49-C 50-A
51-C 52-A 53-D 54-A 55-C 56-A 57-C 58-C 59-B 60-B
61-C 62-A 63-C 64-C 65-D 66-C 67-C 68-B 69-A 70-C
71-A 72-B 73-C 74-B 75-D 76-C 77-B 78-A 79-A 80-A
81-C 82-C 83-B 84-A 85-C 86-D 87-D 88-A 89-B 90-B
91-A 92-A 93-C 94-C 95-D 96-A 97-A 98-C 99-B 100-B
101-D 102-B 103-C 104-C 105-D 106-B 107-C 108-B 109-D 110-D
111-A 112-B 113-D 114-D 115-C 116-C 117-C 118-B 119-B 120-B
121-B 122-A 123-C 124-B 125-D 126-C 127-B 128-D 129-D 130-A
131-A 132-B 133-C 134-D 135-A 136-A 137-A 138-D 139-B 140-D
141-D 142-B 143-C 144-A 145-B 146-B 147-A 148-B 149-A 150-D
151-C 152-D 153-B 154-A 155-B 156-C 157-D 158-A 159-B 160-C
161-A 162-D 163-C 164-B 165-A 166-C 167-A 168-A 169-B 170-D
171-B 172-A 173-D 174-B 175-B 176-A 177-A 178-B 179-D 180-D
181-A 182-C 183-B 184-A 185-D 186-C 187-C 188-D 189-D 190-B
191-A 192-A 193-D 194-B 195-D 196-D 197-D 198-C 199-A 200-A
```

Topic 3: Programming in C and C++

Syllabus: *Elements of C: Tokens, identifiers, data types in C, Control structures in C, Sequence, selection and iteration(s), Structured data types in C: arrays, struct, union, string, and pointers; Object Oriented Programming Concepts: Class, object, instantiation, Inheritance, polymorphism and overloading; C++ Programming: Elements of C++, Tokens, identifiers, Variables and constants, Datatypes, Operators, Control statements, Functions parameter passing, Class and objects, Constructors and destructors, Overloading, Inheritance, Templates, Exception handling.*

Q1. What is the output of: int a=5; printf("%d", a++ + ++a); assuming evaluation left-to-right with gcc giving 12? In strict ANSI C the behaviour is?

A. 12 guaranteed
B. 11 guaranteed
C. 0 guaranteed
D. Undefined behaviour (sequence point violation)
Correct Answer: D

Q2. What is output of: int x=10; printf("%d %d", x++, ++x); ?

A. 10 11 guaranteed
B. 10 12
C. 11 11
D. Undefined / compiler-dependent (sequence-point violation)
Correct Answer: D

Q3. Output of: for(i=0;i<5;i++); printf("%d",i); with i declared int i; ?

A. 0
B. 4
C. 5
D. 6
Correct Answer: C

Q4. Output of: int a[]={1,2,3,4}; printf("%d", *(a+2)); ?

A. 2
B. 3
C. 1
D. 4
Correct Answer: B

Q5. Output of: char s[]="hello"; printf("%lu", (unsigned long)strlen(s)); ?

A. 6
B. 5
C. 0
D. 4
Correct Answer: B

Q6. Output of: int x=5; if(x=0) printf("hi"); else printf("bye"); ?

A. hi bye
B. hi
C. Compile error
D. bye
Correct Answer: D

Q7. Output of: printf("%d", 5/2); in C (both operands int)?

A. 2
B. 2.0
C. 3
D. 2.5
Correct Answer: A

Q8. Output of: printf("%f", 5/2); ?

A. 0.000000
B. 2.500000
C. 3.000000
D. 2.000000 (integer division then converted)
Correct Answer: D

Q9. Output of: int n=0; while(n++<3) printf("%d",n); ?

A. 012
B. 123
C. 321
D. 0123
Correct Answer: B

Q10. Output of: char c='A'; printf("%d", c+1); ?

A. B
B. 67
C. 66
D. 65
Correct Answer: C

Q11. sizeof(int) on typical 32/64-bit gcc/Linux is?

A. 1 byte
B. 8 bytes
C. 4 bytes
D. 2 bytes
Correct Answer: C

Q12. sizeof(char) in C is by definition?

A. 2 bytes
B. Implementation defined, never 1
C. 1 byte
D. 4 bytes
Correct Answer: C

Q13. Which storage class retains value between function calls?

A. auto
B. static
C. extern only
D. register
Correct Answer: B

Q14. Default storage class of local variables is?

A. register
B. extern
C. static
D. auto
Correct Answer: D

Q15. Which operator has highest precedence?

A. * over + (multiplicative over additive)
B. || over &&
C. = over *
D. + over *
Correct Answer: A

Q16. Output of: int a=10,b=20; printf("%d", a^b); (bitwise XOR)?

A. 0
B. 10
C. 20
D. 30
Correct Answer: D

Q17. Output of: printf("%d", ~5); on 32-bit 2's complement?

A. -5
B. -6
C. 6
D. 5
Correct Answer: B

Q18. Output of: int p=5; printf("%d", p<<2); ?

A. 10
B. 25
C. 7
D. 20
Correct Answer: D

Q19. Output of: union {int i; char c[4];} u; u.i=1; printf("%d", u.c[0]); on little-endian?

A. 0
B. 256
C. 1
D. Undefined always 0
Correct Answer: C

Q20. Dereferencing a NULL pointer causes?

A. Always returns NULL
B. Always returns 0
C. Undefined behaviour / segmentation fault
D. Compiles to no-op
Correct Answer: C

Q21. C tokens include?

A. Only functions
B. Only headers
C. Keywords, identifiers, constants, strings, operators, special symbols
D. Only keywords
Correct Answer: C

Q22. Which is NOT a valid C identifier?

A. _abc
B. abc2
C. 2abc (starts with digit)
D. _2abc
Correct Answer: C

Q23. Which is a valid constant in C?

A. Only 123
B. Only keywords
C. All of 123, 1.23, 'a', "abc"
D. Only variables
Correct Answer: C

Q24. Difference between struct and union?

A. No difference
B. Union sums sizes
C. Union shares memory; size = largest member
D. Struct shares memory
Correct Answer: C

Q25. Arrays in C are?

A. Indexed from 1
B. Dynamic by default
C. Linked lists
D. Contiguous memory with fixed size, indexed from 0
Correct Answer: D

Q26. Pointer arithmetic p+1 advances by?

A. Always 4 bytes
B. 1 bit
C. Always 1 byte
D. sizeof(*p) bytes
Correct Answer: D

Q27. String in C is terminated by?

A. Null character '\0'
B. Space
C. EOF
D. Newline
Correct Answer: A

Q28. Which function copies strings?

A. strlen
B. strcmp only
C. strcat only
D. strcpy
Correct Answer: D

Q29. Recursion requires?

A. Global variables only
B. Base case + recursive case, uses call stack
C. No stack
D. Only loops
Correct Answer: B

Q30. Passing array to function in C passes?

A. By value deep copy
B. Only length
C. Whole copy
D. Pointer to first element (call by address)
Correct Answer: D

Q31. Output of: int a=4,b=5; printf("%d", a*b + a/b); ?

A. 20
B. 21
C. 19
D. 9
Correct Answer: A

Q32. Output of: int i,sum=0; for(i=1;i<=7;i++) sum+=i; printf("%d",sum); ?

A. 27
B. 28
C. 29
D. 49
Correct Answer: B

Q33. Output of: printf("%d", 8 & 8); (bitwise AND)?

A. 0
B. 8
C. 8
D. 16
Correct Answer: B

Q34. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. while
B. for
C. do-while
D. none
Correct Answer: C

Q35. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 20
B. 4
C. 9
D. 5
Correct Answer: A

Q36. Which header declares malloc/free?

A. string.h
B. math.h
C. stdio.h
D. stdlib.h
Correct Answer: D

Q37. Output of: int a=4,b=7; printf("%d", a*b + a/b); ?

A. 28
B. 29
C. 11
D. 27
Correct Answer: A

Q38. Output of: int i,sum=0; for(i=1;i<=4;i++) sum+=i; printf("%d",sum); ?

A. 10
B. 16
C. 11
D. 9
Correct Answer: A

Q39. Output of: printf("%d", 1 & 10); (bitwise AND)?

A. 11
B. 11
C. 0
D. 11
Correct Answer: C

Q40. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. while
B. do-while
C. none
D. for
Correct Answer: B

Q41. sizeof(int arr[4]) on machine with sizeof(int)=4 is?

A. 8
B. 16
C. 4
D. 4
Correct Answer: B

Q42. Which header declares malloc/free?

A. math.h
B. string.h
C. stdlib.h
D. stdio.h
Correct Answer: C

Q43. Output of: int a=6,b=6; printf("%d", a*b + a/b); ?

A. 12
B. 36
C. 38
D. 37
Correct Answer: D

Q44. Output of: int i,sum=0; for(i=1;i<=4;i++) sum+=i; printf("%d",sum); ?

A. 9
B. 16
C. 11
D. 10
Correct Answer: D

Q45. Output of: printf("%d", 5 & 9); (bitwise AND)?

A. 1
B. 12
C. 14
D. 13
Correct Answer: A

Q46. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-while
B. none
C. for
D. while
Correct Answer: A

Q47. sizeof(int arr[4]) on machine with sizeof(int)=4 is?

A. 4
B. 16
C. 8
D. 4
Correct Answer: B

Q48. Which header declares malloc/free?

A. string.h
B. stdlib.h
C. stdio.h
D. math.h
Correct Answer: B

Q49. Output of: int a=8,b=3; printf("%d", a*b + a/b); ?

A. 27
B. 25
C. 11
D. 26
Correct Answer: D

Q50. Output of: int i,sum=0; for(i=1;i<=6;i++) sum+=i; printf("%d",sum); ?

A. 21
B. 22
C. 36
D. 20
Correct Answer: A

Q51. Output of: printf("%d", 9 & 9); (bitwise AND)?

A. 0
B. 9
C. 18
D. 9
Correct Answer: B

Q52. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. none
B. do-while
C. while
D. for
Correct Answer: B

Q53. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 4
B. 20
C. 5
D. 9
Correct Answer: B

Q54. Which header declares malloc/free?

A. math.h
B. stdlib.h
C. string.h
D. stdio.h
Correct Answer: B

Q55. Output of: int a=2,b=3; printf("%d", a*b + a/b); ?

A. 7
B. 5
C. 6
D. 5
Correct Answer: C

Q56. Output of: int i,sum=0; for(i=1;i<=3;i++) sum+=i; printf("%d",sum); ?

A. 6
B. 5
C. 7
D. 9
Correct Answer: A

Q57. Output of: printf("%d", 6 & 1); (bitwise AND)?

A. 7
B. 7
C. 7
D. 0
Correct Answer: D

Q58. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. for
B. none
C. do-while
D. while
Correct Answer: C

Q59. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 9
B. 5
C. 20
D. 4
Correct Answer: C

Q60. Which header declares malloc/free?

A. stdlib.h
B. string.h
C. math.h
D. stdio.h
Correct Answer: A

Q61. Output of: int a=7,b=3; printf("%d", a*b + a/b); ?

A. 24
B. 22
C. 10
D. 23
Correct Answer: D

Q62. Output of: int i,sum=0; for(i=1;i<=5;i++) sum+=i; printf("%d",sum); ?

A. 25
B. 14
C. 15
D. 16
Correct Answer: C

Q63. Output of: printf("%d", 10 & 4); (bitwise AND)?

A. 14
B. 14
C. 0
D. 14
Correct Answer: C

Q64. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-while
B. while
C. none
D. for
Correct Answer: A

Q65. sizeof(int arr[5]) on machine with sizeof(int)=4 is?

A. 5
B. 4
C. 20
D. 9
Correct Answer: C

Q66. Which header declares malloc/free?

A. stdlib.h
B. math.h
C. stdio.h
D. string.h
Correct Answer: A

Q67. Output of: int a=4,b=9; printf("%d", a*b + a/b); ?

A. 36
B. 35
C. 37
D. 13
Correct Answer: A

Q68. Output of: int i,sum=0; for(i=1;i<=4;i++) sum+=i; printf("%d",sum); ?

A. 10
B. 11
C. 16
D. 9
Correct Answer: A

Q69. Output of: printf("%d", 6 & 8); (bitwise AND)?

A. 14
B. 14
C. 0
D. 14
Correct Answer: C

Q70. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-while
B. none
C. while
D. for
Correct Answer: A

Q71. sizeof(int arr[2]) on machine with sizeof(int)=4 is?

A. 8
B. 6
C. 2
D. 4
Correct Answer: A

Q72. Which header declares malloc/free?

A. stdlib.h
B. math.h
C. string.h
D. stdio.h
Correct Answer: A

Q73. Output of: int a=2,b=9; printf("%d", a*b + a/b); ?

A. 11
B. 18
C. 17
D. 19
Correct Answer: B

Q74. Output of: int i,sum=0; for(i=1;i<=3;i++) sum+=i; printf("%d",sum); ?

A. 9
B. 6
C. 7
D. 5
Correct Answer: B

Q75. Output of: printf("%d", 7 & 3); (bitwise AND)?

A. 7
B. 10
C. 3
D. 4
Correct Answer: C

Q76. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. for
B. while
C. none
D. do-while
Correct Answer: D

Q77. sizeof(int arr[3]) on machine with sizeof(int)=4 is?

A. 7
B. 12
C. 3
D. 4
Correct Answer: B

Q78. Which header declares malloc/free?

A. stdlib.h
B. math.h
C. string.h
D. stdio.h
Correct Answer: A

Q79. Output of: int a=3,b=6; printf("%d", a*b + a/b); ?

A. 18
B. 9
C. 17
D. 19
Correct Answer: A

Q80. Output of: int i,sum=0; for(i=1;i<=5;i++) sum+=i; printf("%d",sum); ?

A. 14
B. 25
C. 15
D. 16
Correct Answer: C

Q81. Output of: printf("%d", 7 & 1); (bitwise AND)?

A. 1
B. 7
C. 6
D. 8
Correct Answer: A

Q82. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. for
B. do-while
C. none
D. while
Correct Answer: B

Q83. sizeof(int arr[2]) on machine with sizeof(int)=4 is?

A. 6
B. 2
C. 4
D. 8
Correct Answer: D

Q84. Which header declares malloc/free?

A. math.h
B. string.h
C. stdio.h
D. stdlib.h
Correct Answer: D

Q85. Output of: int a=9,b=2; printf("%d", a*b + a/b); ?

A. 23
B. 11
C. 21
D. 22
Correct Answer: D

Q86. Output of: int i,sum=0; for(i=1;i<=3;i++) sum+=i; printf("%d",sum); ?

A. 9
B. 7
C. 5
D. 6
Correct Answer: D

Q87. Output of: printf("%d", 5 & 5); (bitwise AND)?

A. 5
B. 0
C. 10
D. 5
Correct Answer: A

Q88. Which loop guarantees at least one execution? (among while / do-while / for with typical conditions)

A. do-while
B. while
C. for
D. none
Correct Answer: A

Q89. sizeof(int arr[3]) on machine with sizeof(int)=4 is?

A. 3
B. 12
C. 4
D. 7
Correct Answer: B

Q90. Which header declares malloc/free?

A. math.h
B. stdio.h
C. stdlib.h
D. string.h
Correct Answer: C

Q91. Encapsulation primarily achieves?

A. Data hiding + bundling data and methods
B. More memory
C. Multiple inheritance
D. Faster code
Correct Answer: A

Q92. Inheritance represents?

A. IS-A relationship, code reuse
B. Only polymorphism
C. No reuse
D. HAS-A only
Correct Answer: A

Q93. Polymorphism in OOP means?

A. Only data hiding
B. Same code always
C. Same interface, different behaviour (compile-time + run-time)
D. No overloading
Correct Answer: C

Q94. Abstraction focuses on?

A. Hiding interface
B. Essential features, hiding implementation
C. Showing all details
D. Memory layout
Correct Answer: B

Q95. A class vs object?

A. Class is instance
B. Object is blueprint
C. Same thing
D. Class is blueprint; object is instance
Correct Answer: D

Q96. Constructor is called?

A. At object creation, same name as class, no return type
B. At destruction
C. Manually always
D. Never automatically
Correct Answer: A

Q97. Destructor in C++ is denoted by?

A. Hash (#)
B. Tilde (~) prefix, no args/return
C. At (@)
D. Dollar ($)
Correct Answer: B

Q98. Virtual function enables?

A. Static binding only
B. Run-time polymorphism via dynamic dispatch
C. Compile-time only
D. No polymorphism
Correct Answer: B

Q99. Pure virtual function makes class?

A. Concrete
B. Static only
C. Final
D. Abstract (cannot instantiate)
Correct Answer: D

Q100. Function overloading requires?

A. Different name
B. Same name, different signature
C. Different class only
D. Same signature
Correct Answer: B

Q101. Operator overloading cannot overload?

A. ::, ., .*, ?:, sizeof among non-overloadable
B. [], ()
C. +, -, *, /
D. ==, !=
Correct Answer: A

Q102. Friend function in C++?

A. Cannot access private
B. Only public
C. Non-member with access to private members
D. Member always
Correct Answer: C

Q103. Multiple inheritance ambiguity is resolved via?

A. Macros
B. No solution
C. Virtual base class / scope resolution
D. Templates only
Correct Answer: C

Q104. HAS-A relationship is implemented via?

A. Inheritance
B. Composition / aggregation
C. Templating
D. Overloading
Correct Answer: B

Q105. Overriding vs overloading?

A. Overloading is run-time
B. Overriding is compile-time
C. Same thing
D. Overriding is run-time (same signature in derived); overloading is compile-time
Correct Answer: D

Q106. Which OOP pillar binds code and data together?

A. Abstraction
B. Polymorphism
C. Inheritance
D. Encapsulation
Correct Answer: D

Q107. Which OOP pillar enables code reuse via hierarchy?

A. Abstraction
B. Encapsulation
C. Polymorphism
D. Inheritance
Correct Answer: D

Q108. Dynamic binding is associated with?

A. Macros
B. Static functions
C. Inline functions
D. Virtual functions
Correct Answer: D

Q109. An abstract class in C++ has?

A. Only static members
B. At least one pure virtual function
C. No constructor
D. No members
Correct Answer: B

Q110. Which OOP pillar binds code and data together?

A. Inheritance
B. Abstraction
C. Polymorphism
D. Encapsulation
Correct Answer: D

Q111. Which OOP pillar enables code reuse via hierarchy?

A. Abstraction
B. Polymorphism
C. Encapsulation
D. Inheritance
Correct Answer: D

Q112. Dynamic binding is associated with?

A. Virtual functions
B. Macros
C. Static functions
D. Inline functions
Correct Answer: A

Q113. An abstract class in C++ has?

A. At least one pure virtual function
B. Only static members
C. No constructor
D. No members
Correct Answer: A

Q114. Which OOP pillar binds code and data together?

A. Abstraction
B. Inheritance
C. Polymorphism
D. Encapsulation
Correct Answer: D

Q115. Which OOP pillar enables code reuse via hierarchy?

A. Encapsulation
B. Abstraction
C. Inheritance
D. Polymorphism
Correct Answer: C

Q116. Dynamic binding is associated with?

A. Inline functions
B. Virtual functions
C. Macros
D. Static functions
Correct Answer: B

Q117. An abstract class in C++ has?

A. No members
B. Only static members
C. No constructor
D. At least one pure virtual function
Correct Answer: D

Q118. Which OOP pillar binds code and data together?

A. Inheritance
B. Polymorphism
C. Encapsulation
D. Abstraction
Correct Answer: C

Q119. Which OOP pillar enables code reuse via hierarchy?

A. Inheritance
B. Encapsulation
C. Polymorphism
D. Abstraction
Correct Answer: A

Q120. Dynamic binding is associated with?

A. Virtual functions
B. Macros
C. Inline functions
D. Static functions
Correct Answer: A

Q121. Which is correct C++ input/output?

A. cin >> x; cout << x;
B. in>>x
C. cout>>x
D. scanf/printf only
Correct Answer: A

Q122. Scope resolution operator is?

A. ->
B. :
C. .
D. ::
Correct Answer: D

Q123. Which header for C++ strings?

A. <string.h> only
B. <stdio.h>
C. <string>
D. <stdlib.h>
Correct Answer: C

Q124. new/delete in C++ differ from malloc/free because?

A. They are faster always
B. They never allocate
C. They are C-only
D. They call constructors/destructors + type-safe
Correct Answer: D

Q125. Default access in C++ class is?

A. public
B. private
C. global
D. protected
Correct Answer: B

Q126. Default access in C++ struct is?

A. none
B. public
C. private
D. protected
Correct Answer: B

Q127. Copy constructor is called when?

A. Every assignment
B. Never
C. Object initialized from another object of same class
D. Only on destruction
Correct Answer: C

Q128. Which constructor has no parameters?

A. Default constructor
B. Parameterized
C. Virtual
D. Copy constructor
Correct Answer: A

Q129. Function template allows?

A. No reuse
B. Generic code for any type
C. Only classes
D. Only int
Correct Answer: B

Q130. Class template parameter is declared with?

A. overload<>
B. template <int>
C. template <typename T> / <class T>
D. generic<>
Correct Answer: C

Q131. Exception handling keywords are?

A. do, while, for
B. try, catch, throw
C. new, delete, this
D. if, else, switch
Correct Answer: B

Q132. An uncaught exception causes?

A. terminate() / abnormal termination
B. Return 0 always
C. Silent ignore
D. Compile error only
Correct Answer: A

Q133. STL vector provides?

A. Associative map
B. Dynamic array with random access
C. Fixed array
D. Linked list only
Correct Answer: B

Q134. Which STL container is LIFO?

A. set
B. queue
C. map
D. stack
Correct Answer: D

Q135. Inline function suggests?

A. No return
B. Always recursion
C. Code substitution to avoid call overhead
D. Virtual dispatch
Correct Answer: C

Q136. 'this' pointer refers to?

A. Current object invoking member function
B. Global object
C. Parent class
D. Null always
Correct Answer: A

Q137. Pass by reference in C++ uses?

A. * only
B. Macros
C. No such feature
D. & in parameter (aliases actual argument)
Correct Answer: D

Q138. Function overloading resolution is?

A. Link-time only
B. Compile-time based on signature
C. Never done
D. Run-time
Correct Answer: B

Q139. Virtual destructor is needed when?

A. Only for templates
B. Only for static objects
C. Deleting derived object via base pointer
D. Never
Correct Answer: C

Q140. Namespace in C++ is used to?

A. Avoid name collisions
B. Increase speed
C. Handle exceptions
D. Allocate memory
Correct Answer: A

Q141. Output of C++: int x=6; cout << (x>10 ? "big" : "small"); ?

A. error
B. small
C. 0
D. big
Correct Answer: B

Q142. Output of C++: for(int i=0;i<5;i++) cout<<i; ?

A. 43210
B. 5
C. 012340
D. 01234
Correct Answer: D

Q143. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. try always throws
B. catch always runs
C. C++ has no finally; RAII/destructors handle cleanup
D. C++ has finally like Java
Correct Answer: C

Q144. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Reference error
B. 0
C. 5
D. NULL
Correct Answer: C

Q145. Which cast is safest for polymorphism in C++?

A. const_cast for polymorphism
B. dynamic_cast
C. reinterpret_cast always
D. C-style cast
Correct Answer: B

Q146. Template specialization allows?

A. Custom implementation for specific type
B. Faster compilation always
C. No generics
D. Only macros
Correct Answer: A

Q147. Output of C++: int x=19; cout << (x>10 ? "big" : "small"); ?

A. error
B. small
C. big
D. 0
Correct Answer: C

Q148. Output of C++: for(int i=0;i<3;i++) cout<<i; ?

A. 0120
B. 012
C. 3
D. 210
Correct Answer: B

Q149. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. catch always runs
B. C++ has finally like Java
C. try always throws
D. C++ has no finally; RAII/destructors handle cleanup
Correct Answer: D

Q150. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. NULL
B. Reference error
C. 5
D. 0
Correct Answer: C

Q151. Which cast is safest for polymorphism in C++?

A. dynamic_cast
B. C-style cast
C. reinterpret_cast always
D. const_cast for polymorphism
Correct Answer: A

Q152. Template specialization allows?

A. Custom implementation for specific type
B. Faster compilation always
C. No generics
D. Only macros
Correct Answer: A

Q153. Output of C++: int x=7; cout << (x>10 ? "big" : "small"); ?

A. 0
B. small
C. big
D. error
Correct Answer: B

Q154. Output of C++: for(int i=0;i<2;i++) cout<<i; ?

A. 2
B. 01
C. 10
D. 010
Correct Answer: B

Q155. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ has no finally; RAII/destructors handle cleanup
B. catch always runs
C. C++ has finally like Java
D. try always throws
Correct Answer: A

Q156. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Reference error
B. NULL
C. 0
D. 5
Correct Answer: D

Q157. Which cast is safest for polymorphism in C++?

A. const_cast for polymorphism
B. dynamic_cast
C. reinterpret_cast always
D. C-style cast
Correct Answer: B

Q158. Template specialization allows?

A. Faster compilation always
B. Only macros
C. No generics
D. Custom implementation for specific type
Correct Answer: D

Q159. Output of C++: int x=13; cout << (x>10 ? "big" : "small"); ?

A. big
B. 0
C. error
D. small
Correct Answer: A

Q160. Output of C++: for(int i=0;i<6;i++) cout<<i; ?

A. 012345
B. 543210
C. 6
D. 0123450
Correct Answer: A

Q161. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. catch always runs
B. try always throws
C. C++ has finally like Java
D. C++ has no finally; RAII/destructors handle cleanup
Correct Answer: D

Q162. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 0
B. NULL
C. Reference error
D. 5
Correct Answer: D

Q163. Which cast is safest for polymorphism in C++?

A. C-style cast
B. reinterpret_cast always
C. const_cast for polymorphism
D. dynamic_cast
Correct Answer: D

Q164. Template specialization allows?

A. No generics
B. Only macros
C. Custom implementation for specific type
D. Faster compilation always
Correct Answer: C

Q165. Output of C++: int x=9; cout << (x>10 ? "big" : "small"); ?

A. error
B. small
C. 0
D. big
Correct Answer: B

Q166. Output of C++: for(int i=0;i<3;i++) cout<<i; ?

A. 3
B. 0120
C. 012
D. 210
Correct Answer: C

Q167. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ has no finally; RAII/destructors handle cleanup
B. try always throws
C. catch always runs
D. C++ has finally like Java
Correct Answer: A

Q168. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. NULL
B. Reference error
C. 5
D. 0
Correct Answer: C

Q169. Which cast is safest for polymorphism in C++?

A. dynamic_cast
B. reinterpret_cast always
C. const_cast for polymorphism
D. C-style cast
Correct Answer: A

Q170. Template specialization allows?

A. Only macros
B. No generics
C. Custom implementation for specific type
D. Faster compilation always
Correct Answer: C

Q171. Output of C++: int x=6; cout << (x>10 ? "big" : "small"); ?

A. 0
B. big
C. error
D. small
Correct Answer: D

Q172. Output of C++: for(int i=0;i<2;i++) cout<<i; ?

A. 01
B. 10
C. 2
D. 010
Correct Answer: A

Q173. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ has no finally; RAII/destructors handle cleanup
B. try always throws
C. catch always runs
D. C++ has finally like Java
Correct Answer: A

Q174. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Reference error
B. NULL
C. 0
D. 5
Correct Answer: D

Q175. Which cast is safest for polymorphism in C++?

A. const_cast for polymorphism
B. C-style cast
C. reinterpret_cast always
D. dynamic_cast
Correct Answer: D

Q176. Template specialization allows?

A. No generics
B. Faster compilation always
C. Custom implementation for specific type
D. Only macros
Correct Answer: C

Q177. Output of C++: int x=11; cout << (x>10 ? "big" : "small"); ?

A. small
B. 0
C. error
D. big
Correct Answer: D

Q178. Output of C++: for(int i=0;i<4;i++) cout<<i; ?

A. 4
B. 3210
C. 0123
D. 01230
Correct Answer: C

Q179. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. try always throws
B. catch always runs
C. C++ has finally like Java
D. C++ has no finally; RAII/destructors handle cleanup
Correct Answer: D

Q180. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 5
B. Reference error
C. 0
D. NULL
Correct Answer: A

Q181. Which cast is safest for polymorphism in C++?

A. dynamic_cast
B. const_cast for polymorphism
C. C-style cast
D. reinterpret_cast always
Correct Answer: A

Q182. Template specialization allows?

A. No generics
B. Custom implementation for specific type
C. Only macros
D. Faster compilation always
Correct Answer: B

Q183. Output of C++: int x=9; cout << (x>10 ? "big" : "small"); ?

A. big
B. small
C. 0
D. error
Correct Answer: B

Q184. Output of C++: for(int i=0;i<6;i++) cout<<i; ?

A. 543210
B. 012345
C. 0123450
D. 6
Correct Answer: B

Q185. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. C++ has finally like Java
B. try always throws
C. catch always runs
D. C++ has no finally; RAII/destructors handle cleanup
Correct Answer: D

Q186. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 0
B. 5
C. Reference error
D. NULL
Correct Answer: B

Q187. Which cast is safest for polymorphism in C++?

A. reinterpret_cast always
B. dynamic_cast
C. const_cast for polymorphism
D. C-style cast
Correct Answer: B

Q188. Template specialization allows?

A. Faster compilation always
B. No generics
C. Only macros
D. Custom implementation for specific type
Correct Answer: D

Q189. Output of C++: int x=2; cout << (x>10 ? "big" : "small"); ?

A. error
B. big
C. 0
D. small
Correct Answer: D

Q190. Output of C++: for(int i=0;i<5;i++) cout<<i; ?

A. 43210
B. 5
C. 012340
D. 01234
Correct Answer: D

Q191. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. catch always runs
B. try always throws
C. C++ has finally like Java
D. C++ has no finally; RAII/destructors handle cleanup
Correct Answer: D

Q192. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. Reference error
B. 0
C. 5
D. NULL
Correct Answer: C

Q193. Which cast is safest for polymorphism in C++?

A. const_cast for polymorphism
B. dynamic_cast
C. C-style cast
D. reinterpret_cast always
Correct Answer: B

Q194. Template specialization allows?

A. Only macros
B. No generics
C. Faster compilation always
D. Custom implementation for specific type
Correct Answer: D

Q195. Output of C++: int x=15; cout << (x>10 ? "big" : "small"); ?

A. big
B. small
C. error
D. 0
Correct Answer: A

Q196. Output of C++: for(int i=0;i<3;i++) cout<<i; ?

A. 0120
B. 210
C. 012
D. 3
Correct Answer: C

Q197. Which exception block always executes (in Java-like analogy)? In C++ RAII/destructors ensure cleanup; finally is?

A. try always throws
B. C++ has no finally; RAII/destructors handle cleanup
C. C++ has finally like Java
D. catch always runs
Correct Answer: B

Q198. Output of C++: int &r=x; r=5; (x initially 0). Value of x?

A. 0
B. 5
C. Reference error
D. NULL
Correct Answer: B

Q199. Which cast is safest for polymorphism in C++?

A. const_cast for polymorphism
B. dynamic_cast
C. reinterpret_cast always
D. C-style cast
Correct Answer: B

Q200. Template specialization allows?

A. No generics
B. Custom implementation for specific type
C. Only macros
D. Faster compilation always
Correct Answer: B

ANSWER KEY — Topic 3: Programming in C and C++

```
1-D 2-D 3-C 4-B 5-B 6-D 7-A 8-D 9-B 10-C
11-C 12-C 13-B 14-D 15-A 16-D 17-B 18-D 19-C 20-C
21-C 22-C 23-C 24-C 25-D 26-D 27-A 28-D 29-B 30-D
31-A 32-B 33-B 34-C 35-A 36-D 37-A 38-A 39-C 40-B
41-B 42-C 43-D 44-D 45-A 46-A 47-B 48-B 49-D 50-A
51-B 52-B 53-B 54-B 55-C 56-A 57-D 58-C 59-C 60-A
61-D 62-C 63-C 64-A 65-C 66-A 67-A 68-A 69-C 70-A
71-A 72-A 73-B 74-B 75-C 76-D 77-B 78-A 79-A 80-C
81-A 82-B 83-D 84-D 85-D 86-D 87-A 88-A 89-B 90-C
91-A 92-A 93-C 94-B 95-D 96-A 97-B 98-B 99-D 100-B
101-A 102-C 103-C 104-B 105-D 106-D 107-D 108-D 109-B 110-D
111-D 112-A 113-A 114-D 115-C 116-B 117-D 118-C 119-A 120-A
121-A 122-D 123-C 124-D 125-B 126-B 127-C 128-A 129-B 130-C
131-B 132-A 133-B 134-D 135-C 136-A 137-D 138-B 139-C 140-A
141-B 142-D 143-C 144-C 145-B 146-A 147-C 148-B 149-D 150-C
151-A 152-A 153-B 154-B 155-A 156-D 157-B 158-D 159-A 160-A
161-D 162-D 163-D 164-C 165-B 166-C 167-A 168-C 169-A 170-C
171-D 172-A 173-A 174-D 175-D 176-C 177-D 178-C 179-D 180-A
181-A 182-B 183-B 184-B 185-D 186-B 187-B 188-D 189-D 190-D
191-D 192-C 193-B 194-D 195-A 196-C 197-B 198-B 199-B 200-B
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Topic 4: Theory of Computation

Syllabus: *Computability: Models of computation - Finite Automata, Pushdown Automata, Non-determinism and NFA, DPDA and PDAs, and Languages accepted by these structures, Grammars, Languages, Non-computability and Examples of non-computable problems, DFA/NFA conversions, Pumping lemma, Chomsky hierarchy, Turing machines, Decidability and Halting problem.*

Q1. A DFA is formally a 5-tuple (Q, Σ, δ, q0, F) where δ is?

A. Q × Σ → Q × stack
B. Q → Q
C. Q × Σ → 2^Q
D. Q × Σ → Q (total transition function)
Correct Answer: D

Q2. An NFA transition function maps to?

A. Single state
B. Tape symbol
C. Stack symbol
D. 2^Q (set of states, possibly empty)
Correct Answer: D

Q3. Every NFA can be converted to an equivalent DFA with at most?

A. n! states
B. n states
C. 2^n states (subset construction)
D. n^2 states
Correct Answer: C

Q4. ε-closure of a state includes?

A. Only final states
B. Only itself
C. Only start state
D. State itself + all reachable via ε moves
Correct Answer: D

Q5. Language accepted by DFA is called?

A. Context-free
B. Decidable only non-regular
C. Recursive enumerable
D. Regular language
Correct Answer: D

Q6. Pumping lemma for regular languages states: for long enough string xyz with |y|>0, |xy|≤p, then?

A. p must be infinite
B. xy^iz ∈ L for all i≥0
C. y must be empty
D. xy^iz ∉ L
Correct Answer: B

Q7. Which language is NOT regular?

A. (01)*
B. Finite language
C. {a^n b^n | n≥0}
D. {a^n | n even}
Correct Answer: C

Q8. Myhill-Nerode theorem characterizes regularity via?

A. Finite index of right-invariant equivalence
B. Tape length
C. Infinite states
D. Stack depth
Correct Answer: A

Q9. Minimal DFA is unique up to isomorphism and found by?

A. Removing alphabet
B. Pumping
C. Adding states
D. Partition refinement / table-filling (state minimization)
Correct Answer: D

Q10. Regular expressions, NFA, DFA are?

A. Only RE = PDA
B. Unrelated
C. Equivalent in expressive power
D. Strictly hierarchical
Correct Answer: C

Q11. Context-free grammars are recognized by?

A. Finite automata
B. DFA
C. Pushdown automata (PDA)
D. Turing machine only
Correct Answer: C

Q12. A DPDA is strictly weaker than?

A. NFA
B. NPDA (non-deterministic PDA)
C. DFA
D. TM
Correct Answer: B

Q13. Example of deterministic CFL but ambiguous grammar?

A. a^n b^n c^n
B. WW
C. Balanced parentheses can be deterministic
D. Halting set
Correct Answer: C

Q14. Chomsky Normal Form productions are of type?

A. A→ε only
B. A→aB
C. A→BC or A→a
D. A→a*
Correct Answer: C

Q15. Greibach Normal Form productions start with?

A. Terminal followed by variables
B. Variable only
C. Epsilon only
D. Two terminals
Correct Answer: A

Q16. Chomsky hierarchy from most restricted: Regular ⊂ CFL ⊂ CSL ⊂ ?

A. Finite only
B. Recursively enumerable
C. Undecidable only
D. Regular again
Correct Answer: B

Q17. Type-0 grammars correspond to?

A. PDA
B. DFA
C. Turing machines / RE languages
D. LBA only regular
Correct Answer: C

Q18. Type-1 (context-sensitive) correspond to?

A. Linear bounded automata
B. Unrestricted TM with infinite tape, same as Type-0 but bounded
C. PDA
D. DFA
Correct Answer: A

Q19. Pumping lemma for CFL uses?

A. No bound
B. uvxyz with |vy|>0, |vxy|≤p
C. Single pump
D. xyz only
Correct Answer: B

Q20. Which language is CFL but not regular?

A. {ww | w∈{0,1}*} over binary with copy
B. {a^n b^n}
C. {a^p | p prime} is not even CFL
D. {a^n b^n c^n}
Correct Answer: B

Q21. Ambiguous grammar means?

A. Infinite language
B. No parse tree
C. Some string has ≥2 parse trees
D. No terminals
Correct Answer: C

Q22. Inherent ambiguity example?

A. (a+b)*
B. a*b*
C. {a^i b^j c^k | i=j or j=k}
D. Finite sets
Correct Answer: C

Q23. CYK algorithm decides membership for CFG in CNF in?

A. O(n)
B. O(n^3) time
C. O(n!)
D. O(2^n)
Correct Answer: B

Q24. Pushdown automaton uses?

A. Stack (LIFO) + finite control + input
B. Queue
C. Tape only
D. Two stacks always
Correct Answer: A

Q25. Turing machine adds over PDA?

A. Only stack
B. Infinite read/write tape + arbitrary movement
C. Only finite control
D. No input
Correct Answer: B

Q26. Church-Turing thesis states?

A. P=NP
B. All problems decidable
C. TM captures effective computability
D. NFA=DFA only
Correct Answer: C

Q27. Halting problem is?

A. Regular
B. Decidable
C. Context-free
D. Undecidable / non-computable (no TM decides it)
Correct Answer: D

Q28. An example non-computable / undecidable problem?

A. Sorting numbers
B. Determining if arbitrary program halts
C. Checking if DFA accepts empty string
D. Checking if string in finite set
Correct Answer: B

Q29. Rice's theorem says?

A. TMs always halt
B. All regular properties decidable implies all decidable
C. All non-trivial semantic properties of RE languages are undecidable
D. P=NP
Correct Answer: C

Q30. Decidable problems for DFA include?

A. Halting
B. Ambiguity of CFGs
C. Emptiness, finiteness, equivalence, membership
D. Universality of CFLs
Correct Answer: C

Q31. Which CFL problem is undecidable?

A. Membership
B. Ambiguity / universality / equivalence of CFGs
C. Emptiness
D. Finiteness
Correct Answer: B

Q32. Post Correspondence Problem (PCP) is?

A. Finite
B. Undecidable in general
C. Regular
D. Decidable always
Correct Answer: B

Q33. Linear bounded automata decide?

A. Regular only
B. RE only
C. Context-sensitive languages
D. All languages
Correct Answer: C

Q34. A universal Turing machine?

A. Cannot simulate
B. Accepts only regular
C. Decides halting
D. Simulates any TM given its description
Correct Answer: D

Q35. Recursively enumerable vs recursive?

A. Recursive = decidable (TM halts always); RE = semi-decidable
B. Recursive $\subset$ regular
C. RE $\subset$ finite
D. Same thing
Correct Answer: A

Q36. Complement of halting language is?

A. CFL
B. Recursive
C. Regular
D. Not RE (non-RE)
Correct Answer: D

Q37. If L and its complement are both RE then L is?

A. Recursive (decidable)
B. Non-RE
C. Regular
D. Context-free
Correct Answer: A

Q38. Kleene closure L* is regular if L is regular: proof uses?

A. ε-NFA construction
B. Pumping only
C. TM
D. PDA
Correct Answer: A

Q39. Arden's theorem solves?

A. Equations R = Q + RP for regular expressions
B. PCP
C. Halting
D. CFGs
Correct Answer: A

Q40. Mealy vs Moore machines differ in?

A. Output depends on transition vs state
B. Input alphabet
C. No difference
D. Acceptance by stack
Correct Answer: A

Q41. The number of states in minimal DFA for language (0+1)*0(0+1){2} (k fixed suffix length) is at least?

A. 8
B. 9
C. 4
D. 4
Correct Answer: A

Q42. Consider the standard result: An NFA transition function maps to?

A. Stack symbol
B. Single state
C. Tape symbol
D. 2^Q (set of states, possibly empty)
Correct Answer: D

Q43. L = {a^n b^n c^n} is?

A. Regular
B. Context-sensitive but not CFL
C. Finite
D. CFL
Correct Answer: B

Q44. If NFA has 3 states, worst-case DFA via subset construction has?

A. 7 states
B. 3 states
C. 3^2 states
D. 2^3 = 8 states
Correct Answer: D

Q45. Consider the standard result: Language accepted by DFA is called?

A. Recursive enumerable
B. Context-free
C. Decidable only non-regular
D. Regular language
Correct Answer: D

Q46. Intersection of CFL with regular is?

A. Regular always
B. Not CFL
C. Undecidable
D. CFL
Correct Answer: D

Q47. Pumping length p for DFA with 3 states can be taken as?

A. 4
B. 2
C. 2k
D. 3
Correct Answer: D

Q48. Consider the standard result: Myhill-Nerode theorem characterizes regularity via?

A. Stack depth
B. Finite index of right-invariant equivalence
C. Tape length
D. Infinite states
Correct Answer: B

Q49. L = {a^p | p prime} is?

A. Star-free only regular
B. Regular
C. Finite
D. Not regular (and not CFL)
Correct Answer: D

Q50. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Constant 1
B. Exponential always required
C. Polynomial in size
D. Undecidable
Correct Answer: C

Q51. Consider the standard result: Context-free grammars are recognized by?

A. Turing machine only
B. Finite automata
C. DFA
D. Pushdown automata (PDA)
Correct Answer: D

Q52. Empty-string membership for RE language is?

A. Always decidable in O(1)
B. Semi-decidable
C. Regular
D. Never RE
Correct Answer: B

Q53. For alphabet size 3, number of strings of length n is?

A. 3^n
B. 3*n
C. n^3
D. 3+n
Correct Answer: A

Q54. Consider the standard result: Chomsky Normal Form productions are of type?

A. A→a*
B. A→ε only
C. A→aB
D. A→BC or A→a
Correct Answer: D

Q55. Complement of CFL is?

A. Not necessarily CFL
B. Always CFL
C. Always regular
D. Always finite
Correct Answer: A

Q56. The number of states in minimal DFA for language (0+1)*0(0+1){2} (k fixed suffix length) is at least?

A. 8
B. 4
C. 9
D. 4
Correct Answer: A

Q57. Consider the standard result: Type-0 grammars correspond to?

A. Turing machines / RE languages
B. LBA only regular
C. DFA
D. PDA
Correct Answer: A

Q58. L = {ww^R | w∈{a,b}*} is?

A. Deterministic CFL (palindromes even length)
B. Regular
C. Not CFL
D. Recursive only non-CFL
Correct Answer: A

Q59. If NFA has 5 states, worst-case DFA via subset construction has?

A. 5^2 states
B. 31 states
C. 5 states
D. 2^5 = 32 states
Correct Answer: D

Q60. Consider the standard result: Which language is CFL but not regular?

A. {a^n b^n c^n}
B. {a^n b^n}
C. {ww | w∈{0,1}*} over binary with copy
D. {a^p | p prime} is not even CFL
Correct Answer: B

Q61. Equivalence of two DFAs is decidable by?

A. Minimization / product construction checking symmetric difference emptiness
B. Halting reduction
C. Pumping
D. Rice's theorem
Correct Answer: A

Q62. Pumping length p for DFA with 3 states can be taken as?

A. 3
B. 4
C. 2
D. 2k
Correct Answer: A

Q63. Consider the standard result: CYK algorithm decides membership for CFG in CNF in?

A. O(2^n)
B. O(n^3) time
C. O(n!)
D. O(n)
Correct Answer: B

Q64. Every finite language is?

A. Not regular
B. Regular
C. Undecidable
D. CFL only non-regular
Correct Answer: B

Q65. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Exponential always required
B. Constant 1
C. Undecidable
D. Polynomial in size
Correct Answer: D

Q66. Consider the standard result: Church-Turing thesis states?

A. P=NP
B. NFA=DFA only
C. TM captures effective computability
D. All problems decidable
Correct Answer: C

Q67. L = {a^n b^n c^n} is?

A. CFL
B. Regular
C. Context-sensitive but not CFL
D. Finite
Correct Answer: C

Q68. For alphabet size 3, number of strings of length n is?

A. 3+n
B. n^3
C. 3^n
D. 3*n
Correct Answer: C

Q69. Consider the standard result: Rice's theorem says?

A. P=NP
B. All regular properties decidable implies all decidable
C. All non-trivial semantic properties of RE languages are undecidable
D. TMs always halt
Correct Answer: C

Q70. Intersection of CFL with regular is?

A. Undecidable
B. Not CFL
C. CFL
D. Regular always
Correct Answer: C

Q71. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 17
B. 16
C. 8
D. 5
Correct Answer: B

Q72. Consider the standard result: Post Correspondence Problem (PCP) is?

A. Decidable always
B. Regular
C. Undecidable in general
D. Finite
Correct Answer: C

Q73. L = {a^p | p prime} is?

A. Star-free only regular
B. Not regular (and not CFL)
C. Finite
D. Regular
Correct Answer: B

Q74. If NFA has 4 states, worst-case DFA via subset construction has?

A. 2^4 = 16 states
B. 4 states
C. 4^2 states
D. 15 states
Correct Answer: A

Q75. Consider the standard result: Recursively enumerable vs recursive?

A. Recursive = decidable (TM halts always); RE = semi-decidable
B. Recursive ⊂ regular
C. RE ⊂ finite
D. Same thing
Correct Answer: A

Q76. Empty-string membership for RE language is?

A. Semi-decidable
B. Never RE
C. Always decidable in O(1)
D. Regular
Correct Answer: A

Q77. Pumping length p for DFA with 4 states can be taken as?

A. 3
B. 5
C. 2k
D. 4
Correct Answer: D

Q78. Consider the standard result: Kleene closure L* is regular if L is regular: proof uses?

A. PDA
B. ε-NFA construction
C. Pumping only
D. TM
Correct Answer: B

Q79. Complement of CFL is?

A. Always regular
B. Always finite
C. Always CFL
D. Not necessarily CFL
Correct Answer: D

Q80. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Constant 1
B. Undecidable
C. Exponential always required
D. Polynomial in size
Correct Answer: D

Q81. Consider the standard result: A DFA is formally a 5-tuple (Q, Σ, δ, q0, F) where δ is?

A. Q → Q
B. Q × Σ → 2^Q
C. Q × Σ → Q × stack
D. Q × Σ → Q (total transition function)
Correct Answer: D

Q82. L = {ww^R | w∈{a,b}*} is?

A. Regular
B. Deterministic CFL (palindromes even length)
C. Recursive only non-CFL
D. Not CFL
Correct Answer: B

Q83. For alphabet size 5, number of strings of length n is?

A. 5*n
B. 5^n
C. 5+n
D. n^5
Correct Answer: B

Q84. Consider the standard result: ε-closure of a state includes?

A. Only start state
B. Only itself
C. State itself + all reachable via ε moves
D. Only final states
Correct Answer: C

Q85. Equivalence of two DFAs is decidable by?

A. Halting reduction
B. Minimization / product construction checking symmetric difference emptiness
C. Rice's theorem
D. Pumping
Correct Answer: B

Q86. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 17
B. 16
C. 8
D. 5
Correct Answer: B

Q87. Consider the standard result: Which language is NOT regular?

A. {a^n | n even}
B. Finite language
C. (01)*
D. {a^n b^n | n≥0}
Correct Answer: D

Q88. Every finite language is?

A. Undecidable
B. Regular
C. Not regular
D. CFL only non-regular
Correct Answer: B

Q89. If NFA has 2 states, worst-case DFA via subset construction has?

A. 2^2 states
B. 2^2 = 4 states
C. 2 states
D. 3 states
Correct Answer: B

Q90. Consider the standard result: Regular expressions, NFA, DFA are?

A. Equivalent in expressive power
B. Strictly hierarchical
C. Unrelated
D. Only RE = PDA
Correct Answer: A

Q91. L = {a^n b^n c^n} is?

A. Context-sensitive but not CFL
B. Finite
C. CFL
D. Regular
Correct Answer: A

Q92. Pumping length p for DFA with 5 states can be taken as?

A. 5
B. 2k
C. 4
D. 6
Correct Answer: A

Q93. Consider the standard result: Example of deterministic CFL but ambiguous grammar?

A. Halting set
B. Balanced parentheses can be deterministic
C. WW
D. a^n b^n c^n
Correct Answer: B

Q94. Intersection of CFL with regular is?

A. Undecidable
B. Regular always
C. CFL
D. Not CFL
Correct Answer: C

Q95. Number of productions to convert CFG with 4 productions to CNF is bounded by?

A. Exponential always required
B. Constant 1
C. Polynomial in size
D. Undecidable
Correct Answer: C

Q96. Consider the standard result: Chomsky hierarchy from most restricted: Regular $\subset$ CFL $\subset$ CSL $\subset$?

A. Finite only
B. Undecidable only
C. Regular again
D. Recursively enumerable
Correct Answer: D

Q97. L = {a^p | p prime} is?

A. Star-free only regular
B. Finite
C. Not regular (and not CFL)
D. Regular
Correct Answer: C

Q98. For alphabet size 5, number of strings of length n is?

A. 5+n
B. n^5
C. 5*n
D. 5^n
Correct Answer: D

Q99. Consider the standard result: Pumping lemma for CFL uses?

A. xyz only
B. Single pump
C. No bound
D. uvxyz with |vy|>0, |vxy|≤p
Correct Answer: D

Q100. Empty-string membership for RE language is?

A. Regular
B. Always decidable in O(1)
C. Semi-decidable
D. Never RE
Correct Answer: C

Q101. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 32
B. 64
C. 7
D. 65
Correct Answer: B

Q102. Consider the standard result: Inherent ambiguity example?

A. (a+b)*
B. a*b*
C. {a^i b^j c^k | i=j or j=k}
D. Finite sets
Correct Answer: C

Q103. Complement of CFL is?

A. Not necessarily CFL
B. Always CFL
C. Always regular
D. Always finite
Correct Answer: A

Q104. If NFA has 4 states, worst-case DFA via subset construction has?

A. 4^2 states
B. 2^4 = 16 states
C. 4 states
D. 15 states
Correct Answer: B

Q105. Consider the standard result: Turing machine adds over PDA?

A. Only stack
B. Infinite read/write tape + arbitrary movement
C. Only finite control
D. No input
Correct Answer: B

Q106. L = {ww^R | w∈{a,b}*} is?

A. Not CFL
B. Deterministic CFL (palindromes even length)
C. Recursive only non-CFL
D. Regular
Correct Answer: B

Q107. Pumping length p for DFA with 3 states can be taken as?

A. 3
B. 4
C. 2k
D. 2
Correct Answer: A

Q108. Consider the standard result: An example non-computable / undecidable problem?

A. Determining if arbitrary program halts
B. Checking if DFA accepts empty string
C. Checking if string in finite set
D. Sorting numbers
Correct Answer: A

Q109. Equivalence of two DFAs is decidable by?

A. Halting reduction
B. Pumping
C. Minimization / product construction checking symmetric difference emptiness
D. Rice's theorem
Correct Answer: C

Q110. Number of productions to convert CFG with 3 productions to CNF is bounded by?

A. Polynomial in size
B. Undecidable
C. Exponential always required
D. Constant 1
Correct Answer: A

Q111. Consider the standard result: Which CFL problem is undecidable?

A. Emptiness
B. Ambiguity / universality / equivalence of CFGs
C. Finiteness
D. Membership
Correct Answer: B

Q112. Every finite language is?

A. Not regular
B. CFL only non-regular
C. Regular
D. Undecidable
Correct Answer: C

Q113. For alphabet size 5, number of strings of length n is?

A. 5*n
B. n^5
C. 5^n
D. 5+n
Correct Answer: C

Q114. Consider the standard result: A universal Turing machine?

A. Simulates any TM given its description
B. Cannot simulate
C. Decides halting
D. Accepts only regular
Correct Answer: A

Q115. L = {a^n b^n c^n} is?

A. CFL
B. Regular
C. Context-sensitive but not CFL
D. Finite
Correct Answer: C

Q116. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 7
B. 32
C. 65
D. 64
Correct Answer: D

Q117. Consider the standard result: If L and its complement are both RE then L is?

A. Context-free
B. Non-RE
C. Regular
D. Recursive (decidable)
Correct Answer: D

Q118. Intersection of CFL with regular is?

A. Not CFL
B. Regular always
C. CFL
D. Undecidable
Correct Answer: C

Q119. If NFA has 5 states, worst-case DFA via subset construction has?

A. 5 states
B. 31 states
C. 2^5 = 32 states
D. 5^2 states
Correct Answer: C

Q120. Consider the standard result: Mealy vs Moore machines differ in?

A. Input alphabet
B. Output depends on transition vs state
C. No difference
D. Acceptance by stack
Correct Answer: B

Q121. L = {a^p | p prime} is?

A. Regular
B. Not regular (and not CFL)
C. Finite
D. Star-free only regular
Correct Answer: B

Q122. Pumping length p for DFA with 4 states can be taken as?

A. 5
B. 3
C. 4
D. 2k
Correct Answer: C

Q123. Consider the standard result: Every NFA can be converted to an equivalent DFA with at most?

A. 2^n states (subset construction)
B. n states
C. n! states
D. n^2 states
Correct Answer: A

Q124. Empty-string membership for RE language is?

A. Always decidable in O(1)
B. Never RE
C. Semi-decidable
D. Regular
Correct Answer: C

Q125. Number of productions to convert CFG with 5 productions to CNF is bounded by?

A. Polynomial in size
B. Constant 1
C. Undecidable
D. Exponential always required
Correct Answer: A

Q126. Consider the standard result: Pumping lemma for regular languages states: for long enough string xyz with |y|>0, |xy|≤p, then?

A. xy^iz ∉ L
B. p must be infinite
C. xy^iz ∈ L for all i≥0
D. y must be empty
Correct Answer: C

Q127. Complement of CFL is?

A. Always finite
B. Not necessarily CFL
C. Always regular
D. Always CFL
Correct Answer: B

Q128. For alphabet size 2, number of strings of length n is?

A. 2^n
B. 2+n
C. n^2
D. 2*n
Correct Answer: A

Q129. Consider the standard result: Minimal DFA is unique up to isomorphism and found by?

A. Partition refinement / table-filling (state minimization)
B. Pumping
C. Removing alphabet
D. Adding states
Correct Answer: A

Q130. L = {ww^R | w∈{a,b}*} is?

A. Not CFL
B. Regular
C. Recursive only non-CFL
D. Deterministic CFL (palindromes even length)
Correct Answer: D

Q131. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 65
B. 32
C. 64
D. 7
Correct Answer: C

Q132. Consider the standard result: A DPDA is strictly weaker than?

A. DFA
B. NPDA (non-deterministic PDA)
C. TM
D. NFA
Correct Answer: B

Q133. Equivalence of two DFAs is decidable by?

A. Halting reduction
B. Rice's theorem
C. Minimization / product construction checking symmetric difference emptiness
D. Pumping
Correct Answer: C

Q134. If NFA has 4 states, worst-case DFA via subset construction has?

A. 4^2 states
B. 4 states
C. 15 states
D. 2^4 = 16 states
Correct Answer: D

Q135. Consider the standard result: Greibach Normal Form productions start with?

A. Variable only
B. Epsilon only
C. Two terminals
D. Terminal followed by variables
Correct Answer: D

Q136. Every finite language is?

A. CFL only non-regular
B. Regular
C. Not regular
D. Undecidable
Correct Answer: B

Q137. Pumping length p for DFA with 2 states can be taken as?

A. 3
B. 1
C. 2k
D. 2
Correct Answer: D

Q138. Consider the standard result: Type-1 (context-sensitive) correspond to?

A. Linear bounded automata
B. PDA
C. Unrestricted TM with infinite tape, same as Type-0 but bounded
D. DFA
Correct Answer: A

Q139. L = {a^n b^n c^n} is?

A. Finite
B. Regular
C. CFL
D. Context-sensitive but not CFL
Correct Answer: D

Q140. Number of productions to convert CFG with 4 productions to CNF is bounded by?

A. Undecidable
B. Polynomial in size
C. Constant 1
D. Exponential always required
Correct Answer: B

Q141. Consider the standard result: Ambiguous grammar means?

A. No terminals
B. No parse tree
C. Infinite language
D. Some string has ≥2 parse trees
Correct Answer: D

Q142. Intersection of CFL with regular is?

A. Undecidable
B. CFL
C. Regular always
D. Not CFL
Correct Answer: B

Q143. For alphabet size 3, number of strings of length n is?

A. n^3
B. 3+n
C. 3^n
D. 3*n
Correct Answer: C

Q144. Consider the standard result: Pushdown automaton uses?

A. Tape only
B. Two stacks always
C. Stack (LIFO) + finite control + input
D. Queue
Correct Answer: C

Q145. L = {a^p | p prime} is?

A. Finite
B. Regular
C. Star-free only regular
D. Not regular (and not CFL)
Correct Answer: D

Q146. The number of states in minimal DFA for language (0+1)*0(0+1){4} (k fixed suffix length) is at least?

A. 6
B. 32
C. 33
D. 16
Correct Answer: B

Q147. Consider the standard result: Halting problem is?

A. Decidable
B. Undecidable / non-computable (no TM decides it)
C. Context-free
D. Regular
Correct Answer: B

Q148. Empty-string membership for RE language is?

A. Regular
B. Never RE
C. Semi-decidable
D. Always decidable in O(1)
Correct Answer: C

Q149. If NFA has 5 states, worst-case DFA via subset construction has?

A. 5 states
B. 31 states
C. 5^2 states
D. 2^5 = 32 states
Correct Answer: D

Q150. Consider the standard result: Decidable problems for DFA include?

A. Ambiguity of CFGs
B. Universality of CFLs
C. Emptiness, finiteness, equivalence, membership
D. Halting
Correct Answer: C

Q151. Complement of CFL is?

A. Always finite
B. Always regular
C. Not necessarily CFL
D. Always CFL
Correct Answer: C

Q152. Pumping length p for DFA with 4 states can be taken as?

A. 3
B. 5
C. 4
D. 2k
Correct Answer: C

Q153. Consider the standard result: Linear bounded automata decide?

A. Regular only
B. RE only
C. All languages
D. Context-sensitive languages
Correct Answer: D

Q154. L = {ww^R | w∈{a,b}*} is?

A. Regular
B. Not CFL
C. Deterministic CFL (palindromes even length)
D. Recursive only non-CFL
Correct Answer: C

Q155. Number of productions to convert CFG with 2 productions to CNF is bounded by?

A. Exponential always required
B. Polynomial in size
C. Constant 1
D. Undecidable
Correct Answer: B

Q156. Consider the standard result: Complement of halting language is?

A. Regular
B. CFL
C. Not RE (non-RE)
D. Recursive
Correct Answer: C

Q157. Equivalence of two DFAs is decidable by?

A. Minimization / product construction checking symmetric difference emptiness
B. Rice's theorem
C. Halting reduction
D. Pumping
Correct Answer: A

Q158. For alphabet size 2, number of strings of length n is?

A. 2*n
B. 2+n
C. 2^n
D. n^2
Correct Answer: C

Q159. Consider the standard result: Arden's theorem solves?

A. CFGs
B. PCP
C. Halting
D. Equations R = Q + RP for regular expressions
Correct Answer: D

Q160. Every finite language is?

A. CFL only non-regular
B. Regular
C. Undecidable
D. Not regular
Correct Answer: B

Q161. The number of states in minimal DFA for language (0+1)*0(0+1){5} (k fixed suffix length) is at least?

A. 7
B. 65
C. 32
D. 64
Correct Answer: D

Q162. Consider the standard result: An NFA transition function maps to?

A. Single state
B. 2^Q (set of states, possibly empty)
C. Stack symbol
D. Tape symbol
Correct Answer: B

Q163. L = {a^n b^n c^n} is?

A. Context-sensitive but not CFL
B. Finite
C. Regular
D. CFL
Correct Answer: A

Q164. If NFA has 2 states, worst-case DFA via subset construction has?

A. 2^2 = 4 states
B. 3 states
C. 2^2 states
D. 2 states
Correct Answer: A

Q165. Consider the standard result: Language accepted by DFA is called?

A. Regular language
B. Context-free
C. Recursive enumerable
D. Decidable only non-regular
Correct Answer: A

Q166. Intersection of CFL with regular is?

A. Regular always
B. CFL
C. Not CFL
D. Undecidable
Correct Answer: B

Q167. Pumping length p for DFA with 4 states can be taken as?

A. 3
B. 5
C. 2k
D. 4
Correct Answer: D

Q168. Consider the standard result: Myhill-Nerode theorem characterizes regularity via?

A. Infinite states
B. Finite index of right-invariant equivalence
C. Tape length
D. Stack depth
Correct Answer: B

Q169. L = {a^p | p prime} is?

A. Not regular (and not CFL)
B. Star-free only regular
C. Finite
D. Regular
Correct Answer: A

Q170. Number of productions to convert CFG with 3 productions to CNF is bounded by?

A. Undecidable
B. Constant 1
C. Polynomial in size
D. Exponential always required
Correct Answer: C

Q171. Consider the standard result: Context-free grammars are recognized by?

A. Pushdown automata (PDA)
B. Turing machine only
C. Finite automata
D. DFA
Correct Answer: A

Q172. Empty-string membership for RE language is?

A. Semi-decidable
B. Always decidable in O(1)
C. Regular
D. Never RE
Correct Answer: A

Q173. For alphabet size 5, number of strings of length n is?

A. 5+n
B. 5*n
C. 5^n
D. n^5
Correct Answer: C

Q174. Consider the standard result: Chomsky Normal Form productions are of type?

A. A→BC or A→a
B. A→a*
C. A→ε only
D. A→aB
Correct Answer: A

Q175. Complement of CFL is?

A. Always regular
B. Always CFL
C. Always finite
D. Not necessarily CFL
Correct Answer: D

Q176. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 8
B. 5
C. 16
D. 17
Correct Answer: C

Q177. Consider the standard result: Type-0 grammars correspond to?

A. DFA
B. Turing machines / RE languages
C. PDA
D. LBA only regular
Correct Answer: B

Q178. L = {ww^R | w∈{a,b}*} is?

A. Not CFL
B. Regular
C. Recursive only non-CFL
D. Deterministic CFL (palindromes even length)
Correct Answer: D

Q179. If NFA has 2 states, worst-case DFA via subset construction has?

A. 2^2 states
B. 2^2 = 4 states
C. 2 states
D. 3 states
Correct Answer: B

Q180. Consider the standard result: Which language is CFL but not regular?

A. {a^n b^n c^n}
B. {a^n b^n}
C. {ww | w∈{0,1}*} over binary with copy
D. {a^p | p prime} is not even CFL
Correct Answer: B

Q181. Equivalence of two DFAs is decidable by?

A. Pumping
B. Minimization / product construction checking symmetric difference emptiness
C. Halting reduction
D. Rice's theorem
Correct Answer: B

Q182. Pumping length p for DFA with 3 states can be taken as?

A. 4
B. 3
C. 2
D. 2k
Correct Answer: B

Q183. Consider the standard result: CYK algorithm decides membership for CFG in CNF in?

A. O(n^3) time
B. O(n)
C. O(2^n)
D. O(n!)
Correct Answer: A

Q184. Every finite language is?

A. Undecidable
B. Regular
C. Not regular
D. CFL only non-regular
Correct Answer: B

Q185. Number of productions to convert CFG with 2 productions to CNF is bounded by?

A. Exponential always required
B. Undecidable
C. Polynomial in size
D. Constant 1
Correct Answer: C

Q186. Consider the standard result: Church-Turing thesis states?

A. NFA=DFA only
B. All problems decidable
C. P=NP
D. TM captures effective computability
Correct Answer: D

Q187. L = {a^n b^n c^n} is?

A. Regular
B. Finite
C. CFL
D. Context-sensitive but not CFL
Correct Answer: D

Q188. For alphabet size 2, number of strings of length n is?

A. 2^n
B. 2+n
C. n^2
D. 2*n
Correct Answer: A

Q189. Consider the standard result: Rice's theorem says?

A. P=NP
B. All regular properties decidable implies all decidable
C. TMs always halt
D. All non-trivial semantic properties of RE languages are undecidable
Correct Answer: D

Q190. Intersection of CFL with regular is?

A. Undecidable
B. CFL
C. Not CFL
D. Regular always
Correct Answer: B

Q191. The number of states in minimal DFA for language (0+1)*0(0+1){3} (k fixed suffix length) is at least?

A. 5
B. 8
C. 17
D. 16
Correct Answer: D

Q192. Consider the standard result: Post Correspondence Problem (PCP) is?

A. Finite
B. Undecidable in general
C. Regular
D. Decidable always
Correct Answer: B

Q193. L = {a^p | p prime} is?

A. Star-free only regular
B. Not regular (and not CFL)
C. Finite
D. Regular
Correct Answer: B

Q194. If NFA has 3 states, worst-case DFA via subset construction has?

A. 3^2 states
B. 2^3 = 8 states
C. 7 states
D. 3 states
Correct Answer: B

Q195. Consider the standard result: Recursively enumerable vs recursive?

A. Recursive = decidable (TM halts always); RE = semi-decidable
B. RE $\subset$ finite
C. Recursive $\subset$ regular
D. Same thing
Correct Answer: A

Q196. Empty-string membership for RE language is?

A. Semi-decidable
B. Regular
C. Always decidable in O(1)
D. Never RE
Correct Answer: A

Q197. Pumping length p for DFA with 4 states can be taken as?

A. 2k
B. 4
C. 5
D. 3
Correct Answer: B

Q198. Consider the standard result: Kleene closure L* is regular if L is regular: proof uses?

A. Pumping only
B. PDA
C. ε-NFA construction
D. TM
Correct Answer: C

Q199. Complement of CFL is?

A. Always finite
B. Not necessarily CFL
C. Always CFL
D. Always regular
Correct Answer: B

Q200. Number of productions to convert CFG with 2 productions to CNF is bounded by?

A. Exponential always required
B. Constant 1
C. Polynomial in size
D. Undecidable
Correct Answer: C

ANSWER KEY — Topic 4: Theory of Computation

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1-D 2-D 3-C 4-D 5-D 6-B 7-C 8-A 9-D 10-C
11-C 12-B 13-C 14-C 15-A 16-B 17-C 18-A 19-B 20-B
21-C 22-C 23-B 24-A 25-B 26-C 27-D 28-B 29-C 30-C
31-B 32-B 33-C 34-D 35-A 36-D 37-A 38-A 39-A 40-A
41-A 42-D 43-B 44-D 45-D 46-D 47-D 48-B 49-D 50-C
51-D 52-B 53-A 54-D 55-A 56-A 57-A 58-A 59-D 60-B
61-A 62-A 63-B 64-B 65-D 66-C 67-C 68-C 69-C 70-C
71-B 72-C 73-B 74-A 75-A 76-A 77-D 78-B 79-D 80-D
81-D 82-B 83-B 84-C 85-B 86-B 87-D 88-B 89-B 90-A
91-A 92-A 93-B 94-C 95-C 96-D 97-C 98-D 99-D 100-C
101-B 102-C 103-A 104-B 105-B 106-B 107-A 108-A 109-C 110-A
111-B 112-C 113-C 114-A 115-C 116-D 117-D 118-C 119-C 120-B
121-B 122-C 123-A 124-C 125-A 126-C 127-B 128-A 129-A 130-D
131-C 132-B 133-C 134-D 135-D 136-B 137-D 138-A 139-D 140-B
141-D 142-B 143-C 144-C 145-D 146-B 147-B 148-C 149-D 150-C
151-C 152-C 153-D 154-C 155-B 156-C 157-A 158-C 159-D 160-B
161-D 162-B 163-A 164-A 165-A 166-B 167-D 168-B 169-A 170-C
171-A 172-A 173-C 174-A 175-D 176-C 177-B 178-D 179-B 180-B
181-B 182-B 183-A 184-B 185-C 186-D 187-D 188-A 189-D 190-B
191-D 192-B 193-B 194-B 195-A 196-A 197-B 198-C 199-B 200-C
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Topic 5: Computer Organization

Syllabus: *Machine instructions and addressing modes, ALU, data path and control unit, Instruction pipelining, pipeline hazards, Memory hierarchy: cache, main memory and secondary storage; I/O interface (interrupt and DMA mode).*

Q1. Which addressing mode uses PC-relative offset for branch target?

A. Immediate
B. Direct
C. Relative
D. Indirect
Correct Answer: C

Q2. In zero-address machine, operands are stored in?

A. Registers
B. Stack
C. Memory
D. Cache
Correct Answer: B

Q3. Opcode length n bits allows maximum how many distinct operations?

A. n
B. 2n
C. 2^n
D. n^2
Correct Answer: C

Q4. Which addressing mode is best for array traversal with auto-increment?

A. Immediate
B. Auto-increment
C. Relative
D. Implicit
Correct Answer: B

Q5. MOV R1, (R2) is an example of?

A. Register
B. Register-indirect
C. Direct
D. Immediate
Correct Answer: B

Q6. Effective address in indexed addressing is?

A. PC+offset
B. Index+base
C. Register content
D. Immediate value
Correct Answer: B

Q7. Stack addressing for ADD implicitly uses?

A. TOS+next
B. PC
C. MAR
D. IR
Correct Answer: B

Q8. Immediate addressing is suitable for?

A. Constants
B. Pointers
C. Arrays
D. Subroutines
Correct Answer: A

Q9. Number of address fields in three-address instruction for A=B+C?

A. 1
B. 2
C. 3
D. 4
Correct Answer: C

Q10. Indirect addressing requires how many memory accesses minimum?

A. 0
B. 1
C. 2
D. 3
Correct Answer: C

Q11. Base-register addressing is used for?

A. Relocation
B. Constants
C. Stack
D. I/O
Correct Answer: A

Q12. Which mode supports position-independent code?

A. Absolute
B. PC-relative
C. Indirect
D. Immediate
Correct Answer: B

Q13. Accumulator machine uses how many addresses?

A. 0
B. 1
C. 2
D. 3
Correct Answer: A

Q14. In RISC, addressing modes are typically?

A. Many
B. Few and simple
C. Microcoded
D. Variable
Correct Answer: B

Q15. Displacement = base 1000 + offset 250 gives EA?

A. 750
B. 1000
C. 1250
D. 250000
Correct Answer: A

Q16. Register-indirect saves what compared to direct?

A. Bits
B. Time
C. Power
D. Nothing
Correct Answer: B

Q17. Immediate field in MIPS I-type is how many bits?

A. 5
B. 16
C. 26
D. 32
Correct Answer: A

Q18. J-type instruction is used for?

A. ALU
B. Jump
C. Load
D. Branch conditional
Correct Answer: B

Q19. Big-endian stores MSB at?

A. Highest address
B. Lowest address
C. Middle
D. Register
Correct Answer: A

Q20. Little-endian example processor?

A. IBM 370
B. Intel x86
C. Motorola 68000
D. SPARC
Correct Answer: B

Q21. Stack overflow occurs when?

A. POP empty
B. PUSH full
C. ADD zero
D. JMP fail
Correct Answer: B

Q22. Which instruction changes PC conditionally?

A. MOV
B. Branch
C. ADD
D. NOP
Correct Answer: B

Q23. Addressing mode of loop counter update is often?

A. Register
B. Immediate
C. Indirect
D. Relative
Correct Answer: A

Q24. Memory-indirect vs cache: extra access causes?

A. Speedup
B. Penalty
C. Hit
D. Coherence
Correct Answer: C

Q25. Auto-decrement is used for?

A. Stack push
B. Queue
C. Heap
D. Array read
Correct Answer: B

Q26. `LOAD M[X]` where X is address of address is?

A. Direct
B. Indirect
C. Immediate
D. Indexed
Correct Answer: B

Q27. Which is not an addressing mode?

A. Implied
B. Paging
C. Indexed
D. Autoincrement
Correct Answer: B

Q28. Instruction format contains?

A. Opcode+operands
B. Data only
C. Control only
D. Clock
Correct Answer: A

Q29. Fixed-length instructions simplify?

A. Decoding
B. Storage
C. Pipelining both fetch/decode
D. All
Correct Answer: C

Q30. Variable-length instruction example ISA?

A. MIPS
B. RISC-V
C. x86
D. ARM fixed
Correct Answer: C

Q31. Effective address of `ADD #5,R1` source is?

A. R1
B. 5 itself
C. M[5]
D. PC
Correct Answer: B

Q32. Post-indexed addressing updates register?

A. Before use
B. After use
C. Never
D. On interrupt
Correct Answer: B

Q33. Which mode needs no EA calculation?

A. Immediate
B. Indexed
C. Based
D. Indirect
Correct Answer: A

Q34. Stack machine expression A+B requires?

A. PUSH A,PUSH B,ADD
B. ADD A,B
C. LOAD both
D. MOV
Correct Answer: A

Q35. One-address instruction needs implicit?

A. Accumulator
B. Cache
C. Bus
D. ALU flag
Correct Answer: B

Q36. Two-address `ADD R1,R2` means?

A. R1=R1+R2
B. R2=R1+R2
C. R1=R2
D. MEM=R1+R2
Correct Answer: A

Q37. Endianness affects?

A. ALU speed
B. Byte order
C. Clock
D. Opcode
Correct Answer: B

Q38. Alignment requirement improves?

A. Memory bandwidth
B. Power
C. Code size
D. Interrupts
Correct Answer: B

Q39. Unaligned access may cause?

A. Extra cycles/trap
B. Faster
C. No effect
D. Halt always
Correct Answer: A

Q40. Instruction cycle order is?

A. Decode-fetch-execute
B. Fetch-decode-execute
C. Execute-fetch-decode
D. Fetch-execute-decode
Correct Answer: B

Q41. PC holds?

A. Next instruction addr
B. Current data
C. Opcode
D. Flag
Correct Answer: A

Q42. IR holds?

A. Current instruction
B. Next address
C. Data
D. Status
Correct Answer: A

Q43. MAR holds?

A. Address for memory access
B. Data
C. Instruction
D. Flag
Correct Answer: A

Q44. MDR holds?

A. Data to/from memory
B. Address
C. PC
D. Opcode
Correct Answer: A

Q45. Expanding opcode allows?

A. Variable op lengths
B. Fixed ops
C. No operands
D. Larger memory
Correct Answer: A

Q46. `BEQ offset` uses which mode?

A. PC-relative
B. Absolute
C. Indirect
D. Stack
Correct Answer: A

Q47. Register file with 32 regs needs bits per specifier?

A. 3
B. 4
C. 5
D. 6
Correct Answer: B

Q48. Memory address 16-bit, word-addressable, max words?

A. 16K
B. 32K
C. 64K
D. 1M
Correct Answer: C

Q49. Byte-addressable 24-bit address space size?

A. 16MB
B. 8MB
C. 4MB
D. 2MB
Correct Answer: A

Q50. Which supports subroutine linkage best?

A. Implied
B. Register-indirect + stack
C. Immediate
D. Direct only
Correct Answer: B

Q51. ALU performs?

A. Arithmetic/logic
B. Control sequencing
C. I/O transfer
D. Decoding
Correct Answer: A

Q52. Control unit function is?

A. Execute ALU
B. Generate control signals
C. Store data
D. Fetch operands only
Correct Answer: B

Q53. Hardwired control is?

A. Slow flexible
B. Fast inflexible
C. Microprogrammed
D. Software
Correct Answer: B

Q54. Microprogrammed control uses?

A. Control memory
B. PLA only
C. Random logic
D. Cache
Correct Answer: A

Q55. Datapath includes?

A. ALU+regs+buses
B. Only memory
C. Only CU
D. I/O devices
Correct Answer: A

Q56. Single-bus organization bottleneck is?

A. Bus contention
B. ALU speed
C. Clock
D. Memory size
Correct Answer: A

Q57. Three-bus datapath allows?

A. Parallel operand fetch
B. Slower
C. Serial only
D. No pipelining
Correct Answer: A

Q58. Register transfer `R1<-[R2]+[R3]` needs ALU op?

A. ADD
B. SUB
C. AND
D. OR
Correct Answer: A

Q59. Flags Z,N,C,V are set by?

A. CU
B. ALU outcome
C. Memory
D. Bus
Correct Answer: B

Q60. Carry flag indicates?

A. Zero
B. Unsigned overflow
C. Sign
D. Parity
Correct Answer: B

Q61. Overflow flag indicates?

A. Signed overflow
B. Zero
C. Carry
D. Interrupt
Correct Answer: A

Q62. Shifter in datapath does?

A. Multiply only
B. Shift/rotate
C. Add
D. Store
Correct Answer: B

Q63. Single-cycle datapath CPI is?

A. 1
B. >1 variable
C. <1
D. 0
Correct Answer: A

Q64. Multi-cycle datapath advantage?

A. Shorter cycle, resource sharing
B. Higher CPI always bad
C. No control
D. Larger clock only
Correct Answer: B

Q65. Pipelining improves?

A. Latency
B. Throughput
C. Clock period
D. Code size
Correct Answer: B

Q66. 5-stage MIPS pipeline stages order?

A. IF-ID-EX-MEM-WB
B. ID-IF-EX-MEM-WB
C. IF-EX-ID-MEM-WB
D. FE-DE-ME-WB
Correct Answer: A

Q67. Pipeline speedup ideal with k stages?

A. k
B. 1/k
C. k^2
D. log k
Correct Answer: B

Q68. Data hazard occurs due to?

A. Branch
B. Data dependency
C. Structural
D. Control signal
Correct Answer: B

Q69. RAW hazard example?

A. ADD R1,R2; SUB R3,R1
B. Two loads same cycle
C. Branch mispredict
D. Cache miss
Correct Answer: A

Q70. WAR hazard occurs in?

A. In-order 5-stage
B. Out-of-order
C. Single-cycle
D. Unpipelined
Correct Answer: B

Q71. Structural hazard is due to?

A. Resource conflict
B. Data dependency
C. Branch
D. Interrupt
Correct Answer: A

Q72. Control hazard is due to?

A. Branch/jump
B. ALU delay
C. Memory conflict
D. Register file
Correct Answer: A

Q73. Data forwarding resolves?

A. Structural
B. Data hazard stall
C. Control
D. Cache miss
Correct Answer: B

Q74. Branch prediction reduces?

A. Data stalls
B. Control stalls
C. Structural
D. TLB miss
Correct Answer: B

Q75. Delayed branch uses?

A. NOP/branch slot
B. Flush always
C. Predictor
D. Cache
Correct Answer: A

Q76. Pipeline flush penalty equals?

A. Stages remaining
B. Depth of pipe
C. 1 always
D. 0
Correct Answer: B

Q77. 4-stage pipe, 100 instr, no stalls, cycles?

A. 100
B. 103
C. 400
D. 104
Correct Answer: B

Q78. Throughput with cycle 2ns, ideal CPI 1?

A. 500 MIPS
B. 2 MIPS
C. 1000 MIPS
D. 250 MIPS
Correct Answer: A

Q79. Speedup with branch penalty 3, 20% branches?

A. Depends CPI
B. 1.6 degraded
C. 3
D. Ideal
Correct Answer: B

Q80. Superscalar means?

A. Multiple issue per cycle
B. Deeper pipe
C. Vector
D. Multithread
Correct Answer: A

Q81. Superpipelined means?

A. Many narrow stages
B. Multiple pipes
C. VLIW
D. SIMD
Correct Answer: A

Q82. VLIW relies on?

A. Hardware scheduling
B. Compiler scheduling
C. Branch predictor
D. Cache
Correct Answer: B

Q83. Out-of-order needs?

A. Scoreboard/Tomasulo
B. In-order only
C. No rename
D. No buffer
Correct Answer: B

Q84. Register renaming removes?

A. True dep
B. Name dep (WAR/WAW)
C. Control
D. Structural
Correct Answer: A

Q85. Tomasulo uses?

A. Reservation stations
B. Branch stack
C. Victim cache
D. Write buffer
Correct Answer: B

Q86. Speculative execution needs?

A. Reorder buffer
B. Larger cache
C. Slower clock
D. No predictor
Correct Answer: A

Q87. Precise interrupt requires?

A. In-order commit
B. Out-of-order commit
C. No pipeline
D. Flush cache
Correct Answer: A

Q88. MIPS load-use hazard needs?

A. 0 stalls
B. 1 stall/bubble
C. 2 stalls
D. Flush
Correct Answer: A

Q89. Compiler avoids load-use by?

A. Instruction scheduling
B. Loop unroll only
C. Inlining
D. Dead code
Correct Answer: B

Q90. Loop unrolling helps?

A. Reduce branch/data hazards
B. Increase code
C. Both
D. Neither
Correct Answer: A

Q91. CPI pipelined = base +?

A. Stall cycles/instr
B. Clock
C. Depth
D. Cache size
Correct Answer: A

Q92. Amdahl's law for pipeline?

A. Limited by sequential part
B. Unlimited
C. Linear always
D. Exponential
Correct Answer: A

Q93. RISC pipeline friendly because?

A. Fixed length, load-store
B. Complex modes
C. Microcode
D. Variable length
Correct Answer: A

Q94. CISC pipelining difficult due to?

A. Simple ops
B. Variable length/complex
C. Load-store
D. Many regs
Correct Answer: B

Q95. Control memory in Wilkes design holds?

A. Microinstructions
B. Macro ops
C. Data
D. Addresses only
Correct Answer: B

Q96. Horizontal microcode is?

A. Compact
B. Wide, parallel control
C. Vertical
D. Nanocode
Correct Answer: A

Q97. Datapath cycle time set by?

A. Fastest stage
B. Slowest stage
C. Average
D. First stage
Correct Answer: B

Q98. Pipeline register between stages holds?

A. Control only
B. Data+control
C. Clock
D. PC only
Correct Answer: B

Q99. Hazard detection unit in ID stage detects?

A. Load-use
B. Structural
C. WAW
D. Branch taken
Correct Answer: A

Q100. Forwarding paths needed EX-MEM and?

A. MEM-WB
B. ID-IF
C. WB-IF
D. IF-ID
Correct Answer: A

Q101. Memory hierarchy goal?

A. Speed of SRAM cost of disk
B. Speed of cache at cost of SRAM?
C. Average near fastest, cost near cheapest
D. Largest size only
Correct Answer: C

Q102. Locality exploited by hierarchy?

A. Temporal+spatial
B. Only temporal
C. Only spatial
D. None
Correct Answer: A

Q103. SRAM vs DRAM: SRAM is?

A. Dense cheap
B. Fast, no refresh
C. Slow refresh
D. Nonvolatile
Correct Answer: B

Q104. DRAM needs?

A. Refresh
B. No refresh
C. Battery
D. Larger transistors only
Correct Answer: B

Q105. Cache between?

A. CPU-registers
B. CPU-main memory
C. Main-disk
D. I/O-CPU
Correct Answer: A

Q106. Cache hit means?

A. Found in cache
B. Miss
C. Page fault
D. TLB miss
Correct Answer: B

Q107. Hit ratio 0.95, Tc=10ns, Tm=100ns, AMAT?

A. 14.5ns
B. 100ns
C. 10ns
D. 19ns
Correct Answer: A

Q108. AMAT formula?

A. Hit+Miss*Penalty
B. Hit*Miss
C. Miss/Hit
D. Penalty-Hit
Correct Answer: A

Q109. Direct-mapped cache index uses?

A. Tag only
B. Middle bits
C. High bits
D. Offset only
Correct Answer: B

Q110. Fully associative search uses?

A. Index
B. Parallel comparators
C. Decoder
D. Hash only
Correct Answer: B

Q111. Set-associative balances?

A. Cost/miss rate
B. Size only
C. Power only
D. Speed only
Correct Answer: A

Q112. Block size large exploits?

A. Temporal
B. Spatial
C. Both
D. Neither
Correct Answer: B

Q113. Larger block may increase?

A. Miss penalty
B. Hit time always down
C. Tag size up
D. Associativity
Correct Answer: B

Q114. Write-through policy writes to?

A. Cache only
B. Cache+memory
C. Memory only
D. Buffer only
Correct Answer: A

Q115. Write-back uses bit?

A. Valid
B. Dirty
C. Tag
D. LRU
Correct Answer: B

Q116. Write allocate on miss brings block for?

A. Write-through no-allocate
B. Write-back typically
C. Never
D. ROM
Correct Answer: B

Q117. No-write-allocate is common with?

A. Write-back
B. Write-through
C. LRU
D. FIFO
Correct Answer: B

Q118. Valid bit indicates?

A. Dirty
B. Block has valid data
C. LRU
D. Shared
Correct Answer: B

Q119. Tag bits = Addr - (index+offset) example 32b, 1K sets, 32B block?

A. 17
B. 15
C. 20
D. 22
Correct Answer: A

Q120. Cache size = sets*assoc*block. 512 sets,4-way,64B?

A. 64KB
B. 128KB
C. 256KB
D. 32KB
Correct Answer: B

Q121. LRU replacement needs bits for 4-way?

A. 1
B. 2
C. 4
D. 8
Correct Answer: B

Q122. FIFO vs LRU: Belady anomaly in?

A. LRU
B. FIFO
C. Both
D. Neither
Correct Answer: B

Q123. Cold miss is?

A. Compulsory
B. Capacity
C. Conflict
D. Coherence
Correct Answer: B

Q124. Conflict miss reduced by?

A. Larger block
B. Higher associativity
C. Smaller cache
D. Write-through
Correct Answer: B

Q125. Capacity miss reduced by?

A. Larger cache
B. Associativity only
C. Victim always
D. Prefetch only
Correct Answer: A

Q126. Thrashing due to?

A. Conflict/capacity
B. Compulsory
C. Hit
D. Write buffer
Correct Answer: A

Q127. Victim cache holds?

A. Recently evicted
B. Prefetched
C. Write buffer
D. TLB
Correct Answer: A

Q128. Prefetching exploits?

A. Spatial/stride
B. Random
C. LRU
D. Write
Correct Answer: B

Q129. Split I/D cache avoids?

A. Capacity
B. Structural hazard
C. Conflict
D. Coherence
Correct Answer: B

Q130. Unified cache advantage?

A. No interference
B. Flexible allocation
C. Faster hit
D. Less tag
Correct Answer: B

Q131. Main memory typically?

A. SRAM
B. DRAM
C. Flash
D. Magnetic
Correct Answer: B

Q132. ROM is?

A. Volatile
B. Nonvolatile
C. Cache
D. Register
Correct Answer: B

Q133. Virtual memory maps?

A. Physical-virtual
B. Virtual-physical
C. Cache-memory
D. Disk-cache
Correct Answer: B

Q134. Page size typical?

A. 32B
B. 4KB
C. 4MB cache
D. 1B
Correct Answer: B

Q135. Page fault handled by?

A. Hardware only
B. OS
C. Compiler
D. Cache controller
Correct Answer: B

Q136. TLB caches?

A. Data
B. Page table entries
C. Instructions
D. Disk blocks
Correct Answer: B

Q137. TLB miss slower than?

A. Page fault
B. Cache miss? Actually TLB miss < page fault
C. Hit
D. Equal
Correct Answer: B

Q138. Two-level page table saves?

A. Time
B. Space
C. TLB
D. Cache
Correct Answer: B

Q139. Inverted page table size depends on?

A. Virtual
B. Physical frames
C. Page size
D. Process
Correct Answer: B

Q140. Thrashing in VM due to?

A. High locality
B. Over-allocation, low frames
C. Large RAM
D. Fast disk
Correct Answer: B

Q141. Working set model ensures?

A. Locality retained
B. FIFO
C. Belady
D. Segmentation
Correct Answer: A

Q142. Secondary memory example?

A. Registers
B. HDD/SSD
C. SRAM
D. DRAM
Correct Answer: B

Q143. SSD vs HDD: SSD has?

A. Seek time
B. No mechanical seek
C. Slower
D. Magnetic
Correct Answer: B

Q144. Disk access time =?

A. Seek+rotational+transfer
B. Hit+miss
C. CAS only
D. Cycle only
Correct Answer: B

Q145. Average rotational latency 7200 RPM?

A. 8.33ms
B. 4.17ms
C. 2ms
D. 16ms
Correct Answer: B

Q146. RAID 0 provides?

A. Redundancy
B. Striping, no redundancy
C. Mirror
D. Parity
Correct Answer: A

Q147. RAID 1 provides?

A. Striping
B. Mirroring
C. Parity
D. Speed only
Correct Answer: B

Q148. RAID 5 tolerates how many disk failures?

A. 0
B. 1
C. 2
D. All
Correct Answer: B

Q149. Optical vs magnetic: optical is?

A. Faster
B. Removable, slower
C. Volatile
D. SRAM
Correct Answer: B

Q150. Memory interleaving improves?

A. Latency
B. Bandwidth
C. Capacity
D. Reliability
Correct Answer: B

Q151. Low-order interleave uses?

A. High bits
B. Low bits for module
C. Tag
D. Row only
Correct Answer: B

Q152. Burst mode in DRAM (FPM/EDO/SDRAM) exploits?

A. Locality sequential
B. Random
C. Refresh
D. ECC
Correct Answer: A

Q153. ECC can?

A. Detect only
B. Correct single-bit
C. No detection
D. Replace disk
Correct Answer: B

Q154. Parity detects?

A. Even errors
B. Odd/single-bit
C. All
D. Burst
Correct Answer: B

Q155. Cache coherence problem in?

A. Uniprocessor
B. Multiprocessor
C. Unpipelined
D. VM
Correct Answer: B

Q156. MSI/MESI are?

A. Coherence protocols
B. Replacement
C. Bus arb
D. Page replacement
Correct Answer: A

Q157. Write-invalidate vs update: invalidate?

A. Broadcast data
B. Invalidate others
C. Update all
D. No bus
Correct Answer: B

Q158. Snooping uses?

A. Shared bus monitoring
B. Directory
C. OS
D. Compiler
Correct Answer: B

Q159. NUMA vs UMA: NUMA has?

A. Uniform latency
B. Non-uniform
C. No shared
D. Single bus
Correct Answer: B

Q160. Memory-mapped I/O maps devices to?

A. I/O ports
B. Memory addresses
C. Cache
D. Registers only
Correct Answer: B

Q161. Isolated I/O uses?

A. Same addresses
B. Separate instructions/ports
C. Memory
D. DMA only
Correct Answer: B

Q162. Polling vs interrupt: polling?

A. CPU waits/busy
B. Async
C. Efficient
D. DMA
Correct Answer: B

Q163. Interrupt allows?

A. Busy wait
B. Overlap, async service
C. No context switch
D. Sync only
Correct Answer: B

Q164. ISR address via?

A. Polling
B. Vector table
C. Stack
D. Cache
Correct Answer: B

Q165. Interrupt priority handled by?

A. Daisy chain/controller
B. ALU
C. Cache
D. Bus data
Correct Answer: A

Q166. INTR vs NMI: NMI is?

A. Maskable
B. Non-maskable
C. Software
D. Poll
Correct Answer: B

Q167. DMA transfers between?

A. CPU-register
B. Memory-peripheral w/o CPU
C. Cache-CPU
D. ALU-IO
Correct Answer: B

Q168. DMA modes: burst vs cycle-steal?

A. Burst blocks bus
B. Steal interleaves
C. Both
D. Neither
Correct Answer: B

Q169. DMA controller needs?

A. Address+count+control
B. ALU
C. Cache
D. TLB
Correct Answer: A

Q170. Cycle stealing means?

A. CPU halted fully
B. One cycle borrowed
C. Burst
D. Interrupt only
Correct Answer: B

Q171. I/O processor (channel) is?

A. Simple DMA
B. Dedicated I/O CPU
C. ALU
D. Bus
Correct Answer: B

Q172. Synchronous vs async bus: async uses?

A. Clock
B. Handshake
C. Fixed timing
D. Burst only
Correct Answer: B

Q173. Bus arbitration decides?

A. Priority
B. Master
C. Slave speed
D. Address
Correct Answer: B

Q174. Daisy chaining priority depends on?

A. Position
B. Request time
C. Device speed
D. Random
Correct Answer: A

Q175. Programmed I/O uses?

A. Interrupts
B. CPU instructions
C. DMA
D. Channel
Correct Answer: B

Q176. Interrupt-driven I/O saves CPU vs?

A. DMA
B. Programmed
C. Channel
D. Cache
Correct Answer: B

Q177. Which fastest for bulk transfer?

A. Programmed
B. Interrupt
C. DMA
D. Polling
Correct Answer: C

Q178. Device status checked via?

A. Data reg
B. Status reg
C. Address
D. Cache
Correct Answer: B

Q179. UART is for?

A. Parallel
B. Serial
C. DMA
D. Disk
Correct Answer: B

Q180. SCSI/USB are?

A. CPU buses
B. I/O buses/interfaces
C. Memory
D. ALU
Correct Answer: B

Q181. Throughput of bus = width*freq example 32b 100MHz?

A. 400MB/s
B. 100MB/s
C. 3200MB/s
D. 32MB/s
Correct Answer: A

Q182. Bus width 64b, 66MHz throughput?

A. 528MB/s
B. 66MB/s
C. 4224MB/s
D. 132MB/s
Correct Answer: A

Q183. Daisy chain fails if device?

A. Fast
B. Fails/powered off breaks chain
C. Slow
D. Multiple
Correct Answer: B

Q184. Polling overhead is?

A. Low
B. High CPU waste
C. Zero
D. DMA-like
Correct Answer: B

Q185. Interrupt latency is?

A. Request to service start
B. Service time
C. Transfer
D. Return
Correct Answer: A

Q186. Multiple interrupts handled by?

A. Priority + masking
B. Single vector
C. Poll only
D. Cache
Correct Answer: A

Q187. Software interrupt is?

A. Trap/syscall
B. Hardware
C. NMI
D. DMA
Correct Answer: A

Q188. Edge vs level triggered: level needs?

A. Ack to clear
B. Pulse only
C. No ack
D. Poll
Correct Answer: A

Q189. IOP offloads?

A. ALU
B. CPU I/O burden
C. Memory
D. Cache
Correct Answer: B

Q190. Spooling overlaps?

A. CPU-I/O via buffer
B. ALU ops
C. Cache
D. Pipeline
Correct Answer: B

Q191. Buffer vs cache in I/O: buffer?

A. Speed mismatch absorb
B. Locality
C. Coherence
D. Replacement
Correct Answer: A

Q192. Double buffering allows?

A. Overlap I/O-compute
B. Half speed
C. No overlap
D. Cache coherence
Correct Answer: A

Q193. Memory-mapped I/O advantage?

A. Separate instructions
B. Use load/store, flexible
C. Port saving
D. Faster DMA
Correct Answer: B

Q194. Isolated I/O advantage?

A. Large space
B. No memory space waste, protection
C. Load/store
D. Cacheable
Correct Answer: B

Q195. Bus mastership by DMA needs?

A. HOLD/HLDA
B. INT
C. NMI
D. READY
Correct Answer: B

Q196. Flyby DMA transfers?

A. Via controller
B. Direct mem-device
C. Via CPU
D. Via cache
Correct Answer: B

Q197. I/O mapped to cacheable memory causes?

A. Stale data
B. Faster always safe
C. No issue
D. Coherence ok
Correct Answer: A

Q198. Uncacheable I/O region avoids?

A. Speed
B. Stale/coherence issues
C. Latency
D. Interrupts
Correct Answer: B

Q199. Throughput vs latency in I/O: streaming needs?

A. Low latency
B. High throughput
C. Both low
D. Neither
Correct Answer: B

Q200. Best bulk disk backup interface choice?

A. Programmed PIO
B. DMA/channel + buffering
C. Polling
D. Single-byte interrupt
Correct Answer: B

ANSWER KEY — Topic 5: Computer Organization

```
1-C 2-B 3-C 4-B 5-B 6-B 7-B 8-A 9-C 10-C
11-A 12-B 13-A 14-B 15-A 16-B 17-A 18-B 19-A 20-B
21-B 22-B 23-A 24-C 25-B 26-B 27-B 28-A 29-C 30-C
31-B 32-B 33-A 34-A 35-B 36-A 37-B 38-B 39-A 40-B
41-A 42-A 43-A 44-A 45-A 46-A 47-B 48-C 49-A 50-B
51-A 52-B 53-B 54-A 55-A 56-A 57-A 58-A 59-B 60-B
61-A 62-B 63-A 64-B 65-B 66-A 67-B 68-B 69-A 70-B
71-A 72-A 73-B 74-B 75-A 76-B 77-B 78-A 79-B 80-A
81-A 82-B 83-B 84-A 85-B 86-A 87-A 88-A 89-B 90-A
91-A 92-A 93-A 94-B 95-B 96-A 97-B 98-B 99-A 100-A
101-C 102-A 103-B 104-B 105-A 106-B 107-A 108-A 109-B 110-B
111-A 112-B 113-B 114-A 115-B 116-B 117-B 118-B 119-A 120-B
121-B 122-B 123-B 124-B 125-A 126-A 127-A 128-B 129-B 130-B
131-B 132-B 133-B 134-B 135-B 136-B 137-B 138-B 139-B 140-B
141-A 142-B 143-B 144-B 145-B 146-A 147-B 148-B 149-B 150-B
151-B 152-A 153-B 154-B 155-B 156-A 157-B 158-B 159-B 160-B
161-B 162-B 163-B 164-B 165-A 166-B 167-B 168-B 169-A 170-B
171-B 172-B 173-B 174-A 175-B 176-B 177-C 178-B 179-B 180-B
181-A 182-A 183-B 184-B 185-A 186-A 187-A 188-A 189-B 190-B
191-A 192-A 193-B 194-B 195-B 196-B 197-A 198-B 199-B 200-B
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Topic 6: Relational Database Design and SQL

Syllabus: *E-R diagrams: E-R diagrams and their transformation to relational design, Normalization: 1NF, 2NF, 3NF, BCNF and 4NF, Limitations of 4NF and BCNF, SQL: Data Definition Language (DDL), Data Manipulation Language (DML), Data Control Language (DCL) commands, Database objects like Views, indexes, sequences, synonyms, data dictionary.*

Q1. Entity set is?

A. Collection of similar entities
B. Attribute
C. Relation
D. Tuple
Correct Answer: A

Q2. Strong entity has?

A. No key
B. Primary key
C. Partial key
D. Foreign only
Correct Answer: B

Q3. Weak entity depends on?

A. Itself
B. Owner/identifying
C. Attribute
D. Relationship
Correct Answer: B

Q4. Weak entity key is?

A. Primary
B. Partial discriminator + owner key
C. Candidate
D. Surrogate only
Correct Answer: B

Q5. Relationship degree 2 is?

A. Unary
B. Binary
C. Ternary
D. N-ary
Correct Answer: B

Q6. Recursive relationship example?

A. Student-Course
B. Employee supervises Employee
C. Customer-Order
D. Book-Author
Correct Answer: B

Q7. Cardinality 1:N means?

A. One to one
B. One to many
C. Many to many
D. None
Correct Answer: B

Q8. Participation total means?

A. Optional
B. Mandatory
C. Partial
D. Multivalued
Correct Answer: B

Q9. ER to tables: 1:1 total both sides merged into?

A. Two tables
B. Single table
C. Three
D. None
Correct Answer: B

Q10. M:N relationship transforms to?

A. New table with both FKs
B. Add FK one side
C. Merge
D. No table
Correct Answer: A

Q11. 1:N transforms by?

A. New table
B. FK on N side
C. FK on 1 side
D. Merge always
Correct Answer: B

Q12. Multivalued attribute maps to?

A. New table
B. Single column
C. Ignore
D. Primary key
Correct Answer: A

Q13. Composite attribute maps to?

A. Single column
B. Component columns
C. New table
D. FK
Correct Answer: A

Q14. Derived attribute stored?

A. Always
B. Usually not
C. As key
D. As FK
Correct Answer: B

Q15. Primary key constraint ensures?

A. Null allowed
B. Unique+not null
C. Duplicate
D. Foreign
Correct Answer: A

Q16. Candidate key is?

A. Minimal superkey
B. Non-key
C. FK
D. Duplicate
Correct Answer: B

Q17. Superkey is?

A. Minimal
B. Any unique-identifying set
C. Single attr
D. FK only
Correct Answer: A

Q18. Foreign key references?

A. Candidate
B. Primary of referenced
C. Any
D. Non-key
Correct Answer: B

Q19. Referential integrity prevents?

A. Duplicates
B. Dangling references
C. Null PK
D. Redundancy
Correct Answer: B

Q20. Entity integrity prevents?

A. Null PK
B. Duplicate FK
C. Null FK
D. Redundancy
Correct Answer: A

Q21. Functional dependency X->Y means?

A. X determines Y
B. Y determines X
C. Independent
D. Multivalued
Correct Answer: A

Q22. Trivial FD example?

A. AB->A
B. A->B
C. B->A
D. AB->C
Correct Answer: A

Q23. Armstrong axiom not included?

A. Reflexivity
B. Augmentation
C. Transitivity
D. Commutativity
Correct Answer: D

Q24. Closure of {A} with A->B,B->C is?

A. {A}
B. {A,B}
C. {A,B,C}
D. {C}
Correct Answer: C

Q25. R(A,B,C) F={A->B,B->C}. Candidate key?

A. A
B. B
C. C
D. AB
Correct Answer: C

Q26. R(A,B,C,D) F={AB->C,C->D,D->A}. Candidate key?

A. AB
B. C
C. D
D. B
Correct Answer: A

Q27. If A->B and B->A then?

A. Both keys
B. Equivalent
C. No key
D. Redundant
Correct Answer: B

Q28. Full FD vs partial: partial depends on?

A. Whole key
B. Part of key
C. Non-key
D. FK
Correct Answer: B

Q29. Transitive dependency is?

A. Non-key via another non-key
B. Key->key
C. Trivial
D. MVD
Correct Answer: A

Q30. 1NF requires?

A. No duplicates
B. Atomic values
C. No partial
D. No transitive
Correct Answer: B

Q31. Table with multivalued phone violates?

A. 1NF
B. 2NF
C. 3NF
D. BCNF
Correct Answer: A

Q32. 2NF removes?

A. Atomic
B. Partial dependency
C. Transitive
D. MVD
Correct Answer: B

Q33. Relation in 1NF with single-attr PK is automatically?

A. 1NF only
B. 2NF
C. 3NF
D. BCNF
Correct Answer: B

Q34. 3NF removes?

A. Partial
B. Transitive
C. MVD
D. Join dep
Correct Answer: B

Q35. BCNF condition: every determinant must be?

A. Non-key
B. Superkey
C. Prime
D. FK
Correct Answer: B

Q36. Every BCNF is in?

A. 1NF only
B. 3NF
C. 4NF
D. 5NF
Correct Answer: B

Q37. 3NF preserves dependencies but BCNF may not; true?

A. No
B. Yes
C. Both preserve
D. Neither
Correct Answer: B

Q38. 4NF removes?

A. Transitive
B. MVD
C. Partial
D. Join
Correct Answer: B

Q39. MVD X->->Y means?

A. FD
B. Independent multi-values
C. Key
D. Join
Correct Answer: B

Q40. 5NF deals with?

A. FD
B. Join dependency
C. MVD
D. Atomic
Correct Answer: B

Q41. Normalization goal?

A. More redundancy
B. Reduce redundancy/anomaly
C. Speed only
D. Indexing
Correct Answer: B

Q42. Insert anomaly due to?

A. Redundancy
B. Keys
C. Indexes
D. Views
Correct Answer: B

Q43. R(A,B,C) A->B, B->C. Highest normal form?

A. 1NF
B. 2NF
C. 3NF
D. BCNF
Correct Answer: B

Q44. R(A,B,C) A->B, A->C, key A. Normal form?

A. 1NF
B. 2NF
C. 3NF
D. BCNF
Correct Answer: D

Q45. R(A,B,C,D) AB->C, C->D. Violates?

A. 1NF
B. 2NF
C. 3NF/BCNF
D. 4NF
Correct Answer: B

Q46. Lossless join needs?

A. Common attr key in one
B. No common
C. All keys
D. MVD
Correct Answer: C

Q47. Lossy decomposition causes?

A. Extra tuples
B. Loss tuples
C. No change
D. Faster
Correct Answer: A

Q48. Dependency preservation means?

A. All FDs checkable without join
B. Lossless
C. BCNF
D. Keys preserved
Correct Answer: B

Q49. BCNF decomposition is always?

A. Lossless
B. Dependency preserving
C. Both
D. Neither
Correct Answer: A

Q50. 3NF synthesis is?

A. Lossless+preserving
B. Lossy
C. Not preserving
D. Redundant
Correct Answer: A

Q51. Canonical cover removes?

A. Redundant FDs/attrs
B. Keys
C. Tables
D. Indexes
Correct Answer: A

Q52. Extraneous attribute test uses?

A. Closure
B. Sampling
C. Index
D. View
Correct Answer: A

Q53. R(A,B) A->B, B->A. Normal form?

A. 1NF
B. 2NF
C. 3NF
D. BCNF
Correct Answer: D

Q54. Prime attribute is?

A. Part of any candidate key
B. Non-key
C. FK
D. Multivalued
Correct Answer: A

Q55. Non-prime determines prime violates?

A. 2NF possibly
B. 1NF
C. 4NF
D. 5NF
Correct Answer: A

Q56. Limitation of normalization: over-normalization causes?

A. More joins, slower queries
B. Faster always
C. Less storage always good
D. No joins
Correct Answer: B

Q57. Denormalization used for?

A. OLTP purity
B. OLAP/read performance
C. Security
D. Integrity
Correct Answer: B

Q58. SQL full form?

A. Structured Query Language
B. Simple Query
C. Sequential
D. Standard Question
Correct Answer: B

Q59. DDL includes?

A. SELECT
B. CREATE/ALTER/DROP
C. GRANT
D. INSERT
Correct Answer: B

Q60. DML includes?

A. CREATE
B. SELECT/INSERT/UPDATE/DELETE
C. GRANT
D. REVOKE
Correct Answer: B

Q61. DCL includes?

A. SELECT
B. GRANT/REVOKE
C. ALTER
D. DELETE
Correct Answer: B

Q62. TCL includes?

A. COMMIT/ROLLBACK
B. CREATE
C. GRANT
D. SELECT
Correct Answer: B

Q63. CREATE TABLE belongs to?

A. DML
B. DDL
C. DCL
D. TCL
Correct Answer: B

Q64. DROP vs TRUNCATE: TRUNCATE?

A. Removes structure
B. Removes rows, keeps structure, no rollback in some DBs
C. Same as DELETE
D. DCL
Correct Answer: B

Q65. DELETE vs TRUNCATE: DELETE?

A. DDL
B. DML with WHERE, logged
C. No log
D. Drops table
Correct Answer: B

Q66. ALTER ADD COLUMN is?

A. DML
B. DDL
C. DCL
D. TCL
Correct Answer: B

Q67. Data types: VARCHAR vs CHAR?

A. Same
B. VARCHAR variable, CHAR fixed padded
C. CHAR variable
D. Both numeric
Correct Answer: B

Q68. NUMBER(10,2) means?

A. 10 digits 2 decimal
B. 10 rows
C. 2 tables
D. Exponent
Correct Answer: A

Q69. DATE default includes?

A. Time only
B. Date (+time)
C. String
D. Integer
Correct Answer: B

Q70. Constraint NOT NULL is?

A. Column
B. Table
C. FK
D. Check
Correct Answer: A

Q71. UNIQUE allows?

A. Duplicates
B. One null (Oracle) / multiple (SQL Server), no duplicates
C. Null PK
D. FK
Correct Answer: B

Q72. CHECK constraint validates?

A. Uniqueness
B. Predicate
C. FK
D. PK
Correct Answer: B

Q73. DEFAULT sets?

A. Null always
B. Value if not provided
C. PK
D. Index
Correct Answer: A

Q74. CREATE VIEW syntax correct?

A. CREATE VIEW v AS SELECT...
B. CREATE TABLE v
C. MAKE VIEW
D. VIEW CREATE
Correct Answer: B

Q75. SELECT DISTINCT removes?

A. Nulls
B. Duplicates
C. Table
D. Index
Correct Answer: B

Q76. WHERE vs HAVING: HAVING filters?

A. Rows
B. Groups after GROUP BY
C. Tables
D. Columns
Correct Answer: A

Q77. GROUP BY used with?

A. Aggregates
B. ORDER only
C. JOIN only
D. Subquery only
Correct Answer: B

Q78. COUNT(*) vs COUNT(col): latter?

A. Counts nulls
B. Ignores nulls
C. Same
D. Counts tables
Correct Answer: A

Q79. Output of SELECT 'A'||'B' in Oracle?

A. A
B. B
C. AB
D. Error
Correct Answer: C

Q80. LIKE 'A%' matches?

A. Starts with A
B. Ends with A
C. Contains A
D. Exactly A
Correct Answer: A

Q81. LIKE '_abc' matches how many chars + abc?

A. 0
B. 1
C. Many
D. None
Correct Answer: B

Q82. BETWEEN 10 AND 20 inclusive?

A. Yes
B. Exclusive
C. Only 10
D. Only 20
Correct Answer: A

Q83. IN vs EXISTS: EXISTS checks?

A. List
B. Existence, correlated
C. Equality
D. Null
Correct Answer: B

Q84. ORDER BY 2 means?

A. Second column
B. Value 2
C. Two rows
D. Desc
Correct Answer: B

Q85. ORDER BY default?

A. DESC
B. ASC
C. Random
D. Grouped
Correct Answer: B

Q86. Aggregate without GROUP BY returns?

A. Many rows
B. Single row
C. Zero
D. Error
Correct Answer: B

Q87. WHERE age IS NULL correct?

A. = NULL
B. IS NULL
C. LIKE NULL
D. IN NULL
Correct Answer: B

Q88. SELECT NVL(NULL,5) returns?

A. NULL
B. 5
C. 0
D. Error
Correct Answer: B

Q89. JOIN without ON results in?

A. Inner
B. Cartesian
C. Outer
D. Error always
Correct Answer: B

Q90. INNER JOIN returns?

A. All left
B. Matching only
C. All both + nulls
D. Cross
Correct Answer: B

Q91. LEFT OUTER JOIN returns?

A. Matching + unmatched left
B. Right only
C. Matching only
D. None
Correct Answer: A

Q92. FULL OUTER JOIN returns?

A. Matching only
B. Both unmatched + matched
C. Left only
D. Cross
Correct Answer: B

Q93. SELF JOIN needs?

A. Two tables
B. Aliases
C. View
D. Index
Correct Answer: B

Q94. EQUIJOIN uses?

A. =
B. >
C. BETWEEN
D. LIKE
Correct Answer: A

Q95. NATURAL JOIN joins on?

A. All same-name cols
B. PK only
C. FK only
D. First col
Correct Answer: A

Q96. Subquery in WHERE with > ALL means?

A. Greater than max
B. Greater than min
C. Equal
D. In list
Correct Answer: B

Q97. > ANY means?

A. Greater than max
B. Greater than min (at least one)
C. Equal all
D. None
Correct Answer: A

Q98. Correlated subquery executes?

A. Once
B. Per outer row
C. Never
D. After
Correct Answer: B

Q99. Output: SELECT 10/3 in integer DB?

A. 3
B. 3.33
C. 3.0
D. Error
Correct Answer: A

Q100. Output: SELECT MOD(10,3)?

A. 3
B. 1
C. 0
D. 10
Correct Answer: B

Q101. UNION vs UNION ALL: UNION?

A. Keeps duplicates
B. Removes duplicates, sorts
C. No sort
D. Intersect
Correct Answer: B

Q102. INTERSECT returns?

A. Union
B. Common
C. Difference
D. Cartesian
Correct Answer: B

Q103. MINUS/EXCEPT returns?

A. Common
B. In first not second
C. Union
D. Join
Correct Answer: B

Q104. UNION column requirement?

A. Same number/type-compatible
B. Any
C. Same name
D. Same key
Correct Answer: A

Q105. ORDER BY in UNION placed?

A. Each query
B. At end
C. First only
D. Not allowed
Correct Answer: B

Q106. INSERT without column list needs?

A. Partial values
B. All columns order
C. No values
D. Defaults only
Correct Answer: B

Q107. INSERT INTO ... SELECT does?

A. Create table
B. Bulk copy
C. Delete
D. Update
Correct Answer: B

Q108. UPDATE without WHERE?

A. No rows
B. All rows
C. Error
D. First row
Correct Answer: B

Q109. DELETE without WHERE?

A. Drops table
B. Deletes all rows
C. Error
D. No effect
Correct Answer: B

Q110. MERGE statement does?

A. Insert only
B. Upsert (insert/update)
C. Delete only
D. Join only
Correct Answer: B

Q111. GRANT SELECT ON t TO user is?

A. DDL
B. DCL
C. DML
D. TCL
Correct Answer: B

Q112. REVOKE removes?

A. Table
B. Privilege
C. Row
D. Index
Correct Answer: B

Q113. Role is?

A. Table
B. Privilege bundle
C. User
D. View
Correct Answer: B

Q114. WITH GRANT OPTION allows?

A. Revoke
B. Further grant
C. Delete table
D. Create DB
Correct Answer: B

Q115. COMMIT makes?

A. Undo
B. Permanent
C. Temporary
D. Lock
Correct Answer: B

Q116. ROLLBACK undoes?

A. Committed
B. Uncommitted
C. DDL always
D. Grant
Correct Answer: B

Q117. SAVEPOINT allows?

A. Full commit
B. Partial rollback
C. Lock
D. Index
Correct Answer: B

Q118. ACID: Atomicity ensured by?

A. Log/undo-redo
B. Locks
C. Timestamp
D. Index
Correct Answer: A

Q119. Consistency ensured by?

A. Recovery only
B. Integrity constraints + app
C. Locks
D. Deadlock
Correct Answer: B

Q120. Isolation levels: lowest?

A. Serializable
B. Read uncommitted
C. Repeatable read
D. Read committed
Correct Answer: B

Q121. Dirty read allowed in?

A. Read uncommitted
B. Serializable
C. Repeatable
D. Read committed
Correct Answer: A

Q122. Phantom prevented in?

A. Read uncommitted
B. Serializable
C. Read committed
D. None
Correct Answer: B

Q123. View is?

A. Physical table
B. Virtual/stored query
C. Index
D. Sequence
Correct Answer: B

Q124. Updatable view requires?

A. Join/group
B. Single table, no aggregate/distinct
C. Aggregate
D. UNION
Correct Answer: B

Q125. WITH CHECK OPTION ensures?

A. Performance
B. DML via view satisfies view predicate
C. Read only
D. Index
Correct Answer: B

Q126. Materialized view stores?

A. Query only
B. Data, refreshed
C. Index
D. Log
Correct Answer: B

Q127. Index improves?

A. Write speed
B. Read/selective queries
C. Storage
D. Redundancy
Correct Answer: B

Q128. Index overhead?

A. None
B. Storage + slower writes
C. Faster writes
D. No maintenance
Correct Answer: B

Q129. B-tree index good for?

A. Range queries
B. Exact only
C. Hash only
D. Full scan
Correct Answer: A

Q130. Bitmap index good for?

A. High cardinality OLTP
B. Low cardinality DW
C. Range
D. Text
Correct Answer: B

Q131. Unique index ensures?

A. Duplicates
B. Uniqueness
C. Nulls
D. FK
Correct Answer: B

Q132. Composite index (A,B) helps WHERE B=? alone?

A. Yes fully
B. Typically not (leftmost)
C. Always
D. Never any
Correct Answer: B

Q133. Clustered index order?

A. Unordered
B. Physical row order
C. Logical only
D. Hash
Correct Answer: B

Q134. CREATE INDEX is?

A. DML
B. DDL
C. DCL
D. TCL
Correct Answer: B

Q135. Sequence generates?

A. Random
B. Unique numbers
C. Strings
D. Dates
Correct Answer: B

Q136. NEXTVAL vs CURRVAL: CURRVAL needs?

A. Nothing
B. NEXTVAL first in session
C. Commit
D. Rollback
Correct Answer: B

Q137. Sequence gap due to?

A. Rollback/cache
B. Commit
C. Index
D. View
Correct Answer: A

Q138. Synonym is?

A. Copy
B. Alias for object
C. Duplicate table
D. Index
Correct Answer: B

Q139. Data dictionary stores?

A. User data
B. Metadata
C. Logs
D. Indexes only
Correct Answer: B

Q140. USER_TABLES vs ALL_TABLES vs DBA_TABLES differ by?

A. Size
B. Scope/privilege
C. Speed
D. Type
Correct Answer: B

Q141. Primary key automatically creates?

A. View
B. Index
C. Sequence
D. Synonym
Correct Answer: B

Q142. Foreign key creates index automatically? (Oracle)

A. Yes always
B. No, must create
C. Yes in all DBs
D. Creates view
Correct Answer: B

Q143. Query: SELECT COUNT(*) FROM emp WHERE dept=10; with 0 rows returns?

A. 0 rows
B. 0 (single row)
C. NULL
D. Error
Correct Answer: B

Q144. Query: SELECT SUM(sal) with all NULL returns?

A. 0
B. NULL
C. Error
D. 1
Correct Answer: B

Q145. GROUP BY dept HAVING COUNT(*)>5 filters?

A. Rows
B. Groups
C. Columns
D. Tables
Correct Answer: B

Q146. WHERE sal > (SELECT AVG(sal) FROM emp) is?

A. Correlated
B. Non-correlated aggregate subquery
C. Join
D. Union
Correct Answer: B

Q147. Second highest salary query uses?

A. MAX
B. Subquery < MAX or DENSE_RANK
C. MIN
D. COUNT
Correct Answer: B

Q148. DENSE_RANK vs RANK: DENSE?

A. Gaps
B. No gaps
C. Same
D. Row_number
Correct Answer: B

Q149. DELETE duplicate rows technique?

A. DISTINCT in delete
B. ROWID/min(rowid)
C. Index
D. View
Correct Answer: A

Q150. Query flattening is?

A. Optimizer transform
B. Normalization
C. Index
D. View
Correct Answer: B

Q151. Explain plan shows?

A. Data
B. Execution plan
C. Dictionary
D. Locks
Correct Answer: B

Q152. Cost-based optimizer uses?

A. Rules only
B. Statistics
C. Random
D. User hint only
Correct Answer: B

Q153. Selectivity low means?

A. Many rows
B. Few rows, index useful
C. All rows
D. No rows
Correct Answer: B

Q154. Join order affects?

A. Result
B. Performance
C. Correctness
D. Keys
Correct Answer: B

Q155. Nested-loop join best for?

A. Large-large
B. Small-outer, indexed inner
C. No index
D. Cartesian
Correct Answer: B

Q156. Hash join best for?

A. Small tables
B. Large equijoin, no index
C. Range
D. Sorted
Correct Answer: B

Q157. Sort-merge needs?

A. Hash
B. Sorted inputs
C. Index only
D. Small
Correct Answer: B

Q158. Transaction schedule serializable if?

A. Conflict-equivalent to serial
B. Has dirty read
C. Concurrent
D. Locked
Correct Answer: A

Q159. Conflict operations are?

A. Read-read
B. Read-write/write-write same item
C. Write-read diff
D. Commit
Correct Answer: B

Q160. Precedence graph cycle means?

A. Serializable
B. Not serializable
C. Deadlock
D. Recoverable
Correct Answer: B

Q161. Two-phase locking ensures?

A. Deadlock free
B. Serializability
C. No starvation
D. Recovery
Correct Answer: B

Q162. Strict 2PL holds locks till?

A. Growing
B. Commit/abort
C. Shrinking early
D. Start
Correct Answer: B

Q163. Deadlock prevention: wait-die uses?

A. Age/priority
B. Timeout only
C. Detection
D. No timestamps
Correct Answer: A

Q164. Deadlock detection uses?

A. Wait-for graph
B. Precedence
C. Lock table only
D. Log
Correct Answer: A

Q165. Timestamp ordering uses?

A. Locks only
B. Timestamps for order
C. 2PL
D. Graph
Correct Answer: B

Q166. Thomas write rule ignores?

A. Late reads
B. Obsolete writes
C. Early writes
D. Commits
Correct Answer: B

Q167. Recoverable schedule: commit after?

A. Readers commit
B. Read-from writers commit
C. Abort
D. Start
Correct Answer: B

Q168. Cascadeless means?

A. No dirty reads
B. No aborts
C. No commits
D. No locks
Correct Answer: B

Q169. Log-based recovery uses?

A. Indexes
B. Undo/redo
C. Shadows only
D. Checkpoints only
Correct Answer: B

Q170. Checkpoint reduces?

A. Locks
B. Recovery time
C. Deadlock
D. Serializability
Correct Answer: B

Q171. Shadow paging uses?

A. Log
B. Two page tables
C. Locks
D. Timestamps
Correct Answer: B

Q172. SQL injection prevented by?

A. Concatenation
B. Bind variables/prepared
C. Dynamic
D. Grant all
Correct Answer: B

Q173. Privilege to create table?

A. CONNECT
B. RESOURCE/CREATE TABLE
C. DBA only
D. SELECT
Correct Answer: B

Q174. PUBLIC grant means?

A. One user
B. All users
C. Admin
D. Role
Correct Answer: B

Q175. View for security hides?

A. All data
B. Sensitive cols/rows
C. Indexes
D. Keys
Correct Answer: B

Q176. Assertion is?

A. Table constraint
B. DB-wide predicate
C. View
D. Index
Correct Answer: B

Q177. Trigger fires on?

A. SELECT always
B. Insert/update/delete events
C. Commit
D. Grant
Correct Answer: B

Q178. Row vs statement trigger: row fires?

A. Once per statement
B. Per row
C. Once per DB
D. On select
Correct Answer: B

Q179. BEFORE vs AFTER trigger: BEFORE for?

A. Logging
B. Validation/modify new
C. Audit
D. Commit
Correct Answer: B

Q180. Instead-of trigger on?

A. Tables
B. Views
C. Indexes
D. Sequences
Correct Answer: B

Q181. Stored procedure vs function: function?

A. No return
B. Returns value, callable in SQL
C. No params
D. TCL only
Correct Answer: B

Q182. Cursor is for?

A. Set processing
B. Row-by-row
C. DDL
D. Index
Correct Answer: B

Q183. Embedded SQL uses?

A. API
B. Precompiler/host vars
C. Trigger
D. View
Correct Answer: B

Q184. Closure test: X+ with F={AB->C,C->D}. (AB)+?

A. AB
B. ABC
C. ABCD
D. ABD
Correct Answer: C

Q185. Minimal cover of {A->B,B->C,A->C}?

A. Same
B. A->B,B->C
C. A->C
D. B->A
Correct Answer: B

Q186. R(A,B,C,D,E) F={A->B,B->C,C->D,D->E}. Key?

A. A
B. B
C. E
D. AB
Correct Answer: A

Q187. Decomposition R1(A,B),R2(B,C) from R(A,B,C) with A->B,B->C lossless?

A. Yes
B. No
C. Depends
D. Lossy? Check B key in R2 yes lossless
Correct Answer: A

Q188. FD preserved if union closures equal; tool is?

A. Chase
B. Closure comparison
C. Graph
D. Index
Correct Answer: B

Q189. R(A,B,C) F={AB->C,C->A}. Keys?

A. AB only
B. AB and BC
C. C
D. A
Correct Answer: B

Q190. Highest NF of Q189?

A. 1NF
B. 2NF
C. 3NF
D. BCNF? C->A with C not superkey violates BCNF but 3NF holds as A prime
Correct Answer: C

Q191. Query: SELECT * FROM t WHERE col=NULL returns?

A. Null rows
B. Zero (use IS NULL)
C. All
D. Error
Correct Answer: B

Q192. COUNT DISTINCT col ignores?

A. Duplicates+nulls
B. All
C. Only duplicates
D. None
Correct Answer: A

Q193. FK ON DELETE CASCADE does?

A. Block delete
B. Delete children
C. Nullify
D. Nothing
Correct Answer: B

Q194. ON DELETE SET NULL requires?

A. Not null FK
B. Nullable FK
C. PK
D. Unique
Correct Answer: B

Q195. Deferrable constraint checked at?

A. Row time
B. Commit
C. Start
D. Grant
Correct Answer: B

Q196. Domain constraint example?

A. PK
B. CHECK age>0
C. FK
D. Index
Correct Answer: B

Q197. NULL in UNIQUE + FK semantics tricky; true?

A. No
B. Yes, nulls not equal/comparable
C. Always error
D. Ignore
Correct Answer: B

Q198. Three-valued logic: NULL AND FALSE =?

A. TRUE
B. FALSE
C. UNKNOWN
D. NULL only
Correct Answer: B

Q199. NULL OR TRUE =?

A. FALSE
B. TRUE
C. UNKNOWN
D. NULL
Correct Answer: B

Q200. Best candidate key choice heuristic?

A. Longest
B. Minimal, stable, numeric/surrogate
C. Nullable
D. Composite always
Correct Answer: B

ANSWER KEY — Topic 6: Relational Database Design and SQL

```
1-A 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-A
11-B 12-A 13-A 14-B 15-A 16-B 17-A 18-B 19-B 20-A
21-A 22-A 23-D 24-C 25-C 26-A 27-B 28-B 29-A 30-B
31-A 32-B 33-B 34-B 35-B 36-B 37-B 38-B 39-B 40-B
41-B 42-B 43-B 44-D 45-B 46-C 47-A 48-B 49-A 50-A
51-A 52-A 53-D 54-A 55-A 56-B 57-B 58-B 59-B 60-B
61-B 62-B 63-B 64-B 65-B 66-B 67-B 68-A 69-B 70-A
71-B 72-B 73-A 74-B 75-B 76-A 77-B 78-A 79-C 80-A
81-B 82-A 83-B 84-B 85-B 86-B 87-B 88-B 89-B 90-B
91-A 92-B 93-B 94-A 95-A 96-B 97-A 98-B 99-A 100-B
101-B 102-B 103-B 104-A 105-B 106-B 107-B 108-B 109-B 110-B
111-B 112-B 113-B 114-B 115-B 116-B 117-B 118-A 119-B 120-B
121-A 122-B 123-B 124-B 125-B 126-B 127-B 128-B 129-A 130-B
131-B 132-B 133-B 134-B 135-B 136-B 137-A 138-B 139-B 140-B
141-B 142-B 143-B 144-B 145-B 146-B 147-B 148-B 149-A 150-B
151-B 152-B 153-B 154-B 155-B 156-B 157-B 158-A 159-B 160-B
161-B 162-B 163-A 164-A 165-B 166-B 167-B 168-B 169-B 170-B
171-B 172-B 173-B 174-B 175-B 176-B 177-B 178-B 179-B 180-B
181-B 182-B 183-B 184-C 185-B 186-A 187-A 188-B 189-B 190-C
191-B 192-A 193-B 194-B 195-B 196-B 197-B 198-B 199-B 200-B
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Topic 7: Data and File Structures

Syllabus: *Data structures: Data, Information, Definition of data structure, Arrays, stacks, queues, linked lists, trees, graphs, priority queues, and heaps; File Structures: Fields, records and files, Sequential, direct, index-sequential and relative files, Hashing, inverted lists and multi-lists, B trees and B+ trees.*

Q1. Array random access time?

A. O(1)
B. O(n)
C. O(log n)
D. O(n^2)
Correct Answer: A

Q2. Array insertion at front cost?

A. O(1)
B. O(n)
C. O(log n)
D. O(0)
Correct Answer: B

Q3. Row-major address: base 1000, int 4B, a[3][4], a[1][2]?

A. 1024
B. 1020
C. 1044
D. 1008
Correct Answer: A

Q4. Sparse matrix best stored as?

A. 2D array
B. Triplet/linked
C. Stack
D. Queue
Correct Answer: B

Q5. Stack discipline?

A. FIFO
B. LIFO
C. Priority
D. Random
Correct Answer: B

Q6. Stack operations?

A. Enqueue/dequeue
B. Push/pop
C. Insert/delete arbitrary
D. Sort
Correct Answer: B

Q7. Infix a+b*c precedence gives?

A. (a+b)*c
B. a+(b*c)
C. a*b+c
D. abc+*
Correct Answer: B

Q8. Postfix ab+c* evaluates a=2,b=3,c=4?

A. 20
B. 14
C. 10
D. 24
Correct Answer: A

Q9. Prefix +*234 equals?

A. 14
B. 20
C. 10
D. 24
Correct Answer: A

Q10. Stack for recursion stores?

A. Heap
B. Activation records
C. Globals
D. Code
Correct Answer: B

Q11. Balanced parentheses uses?

A. Queue
B. Stack
C. Tree
D. Graph
Correct Answer: B

Q12. Infix to postfix needs stack for?

A. Operands
B. Operators
C. Output
D. Numbers
Correct Answer: B

Q13. Postfix evaluation complexity?

A. O(n^2)
B. O(n)
C. O(log n)
D. O(2^n)
Correct Answer: B

Q14. Two stacks in one array: overflow when?

A. One full
B. Total size met
C. Both empty
D. Never
Correct Answer: B

Q15. Queue discipline?

A. LIFO
B. FIFO
C. FILO
D. Random
Correct Answer: B

Q16. Circular queue full condition (size N)?

A. front==rear
B. (rear+1)%N==front
C. rear==0
D. front==0
Correct Answer: B

Q17. Queue via two stacks: enqueue O?

A. O(n)
B. O(1) amortized
C. O(log n)
D. O(n^2)
Correct Answer: B

Q18. Deque allows?

A. One end
B. Both ends
C. Priority
D. Sorted
Correct Answer: B

Q19. Priority queue best impl?

A. Array
B. Heap
C. Stack
D. Linked unsorted
Correct Answer: B

Q20. BFS uses?

A. Stack
B. Queue
C. Heap
D. Tree
Correct Answer: B

Q21. DFS uses?

A. Queue
B. Stack/recursion
C. Heap
D. Array
Correct Answer: B

Q22. Singly linked list search O?

A. O(1)
B. O(n)
C. O(log n)
D. O(0)
Correct Answer: B

Q23. Insert at head DLL/SLL O?

A. O(n)
B. O(1)
C. O(log n)
D. O(n^2)
Correct Answer: B

Q24. Delete given node pointer (no head) SLL?

A. O(n)
B. O(1) copy next
C. O(log n)
D. Impossible
Correct Answer: B

Q25. Doubly linked allows?

A. Forward only
B. Both directions
C. Random
D. Indexed
Correct Answer: B

Q26. Circular list last->next points to?

A. Null
B. First
C. Itself only
D. Middle
Correct Answer: B

Q27. Linked list vs array: linked?

A. Random access
B. Dynamic, no shift
C. Cache friendly
D. Less memory
Correct Answer: B

Q28. Reverse SLL complexity?

A. O(n^2)
B. O(n) time O(1) space
C. O(log n)
D. O(n log n)
Correct Answer: B

Q29. Detect cycle uses?

A. Two stacks
B. Floyd slow/fast
C. BFS
D. Sorting
Correct Answer: B

Q30. Middle of list in one pass?

A. Two pointers
B. Count only
C. Reverse
D. Sort
Correct Answer: A

Q31. Stack via linked list push O?

A. O(n)
B. O(1)
C. O(log n)
D. O(n^2)
Correct Answer: B

Q32. Polynomial add via sorted LL O?

A. O(1)
B. O(n+m)
C. O(n*m)
D. O(log n)
Correct Answer: B

Q33. Tree height with n nodes min?

A. n
B. log n
C. n log n
D. 1
Correct Answer: B

Q34. Full binary tree n nodes: leaves?

A. n/2
B. (n+1)/2
C. n-1
D. log n
Correct Answer: B

Q35. Max nodes at level l (root 0)?

A. l
B. 2^l
C. 2l
D. l^2
Correct Answer: B

Q36. Max nodes height h (root 1)?

A. 2^h
B. 2^h-1
C. h^2
D. 2h
Correct Answer: B

Q37. Inorder of BST gives?

A. Unsorted
B. Sorted
C. Reverse
D. Level
Correct Answer: B

Q38. Preorder sequence root?

A. Last
B. First
C. Middle
D. Random
Correct Answer: C

Q39. Postorder root?

A. First
B. Last
C. Middle
D. None
Correct Answer: B

Q40. Inorder: left-root-right; preorder of 4,2,1,3,6,5,7?

A. 1,2,3,4,5,6,7
B. 4,2,1,3,6,5,7
C. 1,3,2,5,7,6,4
D. 4,2,6,1,3,5,7
Correct Answer: B

Q41. BST search worst O?

A. O(log n)
B. O(n) skewed
C. O(1)
D. O(n log n)
Correct Answer: B

Q42. BST insert 10,5,15,3,7: height?

A. 1
B. 2
C. 3
D. 4
Correct Answer: C

Q43. BST delete node with two children replace with?

A. Any
B. Inorder succ/pred
C. Root
D. Leaf
Correct Answer: B

Q44. AVL balance factor?

A. 0 only
B. -1,0,1
C. -2 to 2
D. Any
Correct Answer: B

Q45. AVL rotations for LL?

A. Left
B. Right
C. LR
D. RL
Correct Answer: B

Q46. AVL search O?

A. O(n)
B. O(log n)
C. O(1)
D. O(n^2)
Correct Answer: B

Q47. Insert sorted into AVL n items O?

A. O(n)
B. O(n log n)
C. O(n^2)
D. O(log n)
Correct Answer: B

Q48. Splay/Red-black height?

A. O(n)
B. O(log n) amortized/guaranteed
C. O(1)
D. O(n^2)
Correct Answer: B

Q49. B-tree order m: max children?

A. m-1
B. m
C. m+1
D. 2m
Correct Answer: B

Q50. B+ tree stores data?

A. All nodes
B. Leaves only (+ linked)
C. Root only
D. Internal
Correct Answer: B

Q51. B/B+ good for?

A. RAM
B. Disk/file systems
C. Cache only
D. Stack
Correct Answer: B

Q52. B-tree search O?

A. O(1)
B. O(log_t n)
C. O(n)
D. O(n log n)
Correct Answer: B

Q53. Heap property max-heap?

A. Parent<=child
B. Parent>=child
C. Sorted
D. Balanced BST
Correct Answer: B

Q54. Build heap O?

A. O(n log n)
B. O(n)
C. O(log n)
D. O(1)
Correct Answer: B

Q55. Insert heap O?

A. O(1)
B. O(log n)
C. O(n)
D. O(n log n)
Correct Answer: B

Q56. Extract-max O?

A. O(1)
B. O(log n)
C. O(n)
D. O(0)
Correct Answer: B

Q57. Heapsort complexity?

A. O(n)
B. O(n log n) worst
C. O(n^2)
D. O(log n)
Correct Answer: B

Q58. Heapsort space?

A. O(n)
B. O(1) auxiliary
C. O(log n)
D. O(n^2)
Correct Answer: B

Q59. Priority queue extract-min heap?

A. O(1)
B. O(log n)
C. O(n)
D. O(n log n)
Correct Answer: B

Q60. Array heap parent of i (0-index)?

A. i/2
B. (i-1)/2
C. 2i
D. 2i+1
Correct Answer: B

Q61. Graph representations: adjacency matrix space?

A. O(V)
B. O(V^2)
C. O(E)
D. O(1)
Correct Answer: B

Q62. Adjacency list space?

A. O(V^2)
B. O(V+E)
C. O(E^2)
D. O(1)
Correct Answer: B

Q63. BFS complexity?

A. O(V)
B. O(V+E)
C. O(E^2)
D. O(V^2) always
Correct Answer: B

Q64. DFS complexity?

A. O(V+E)
B. O(V^2)
C. O(E log V)
D. O(V log V)
Correct Answer: A

Q65. BFS shortest path (unweighted) from source?

A. No
B. Yes
C. Only DFS
D. Only Dijkstra
Correct Answer: B

Q66. DFS for cycle in directed uses?

A. Visited only
B. Colors/recursion stack
C. Queue
D. Heap
Correct Answer: B

Q67. Topological sort uses?

A. BFS Kahn / DFS finish
B. Dijkstra
C. MST
D. Heap
Correct Answer: A

Q68. Topological possible only for?

A. Cyclic
B. DAG
C. Undirected
D. Complete
Correct Answer: A

Q69. Dijkstra handles?

A. Negative weights
B. Non-negative
C. All
D. Unweighted only
Correct Answer: B

Q70. Dijkstra with binary heap O?

A. O(V^2)
B. O((V+E) log V)
C. O(V+E)
D. O(E log E) only
Correct Answer: B

Q71. Bellman-Ford handles negatives? Complexity?

A. No, O(V+E)
B. Yes (no neg cycle), O(VE)
C. Yes all, O(log)
D. No
Correct Answer: B

Q72. Floyd-Warshall O?

A. O(V+E)
B. O(V^3)
C. O(V^2)
D. O(E log V)
Correct Answer: B

Q73. MST Kruskal uses?

A. Heap only
B. Sort + Union-Find
C. BFS
D. DFS
Correct Answer: B

Q74. Prim with heap O?

A. O(V^2)
B. O(E log V)
C. O(VE)
D. O(V^3)
Correct Answer: B

Q75. Union-Find with path compression ~?

A. O(n)
B. O(alpha(n))
C. O(log n) always
D. O(n^2)
Correct Answer: B

Q76. Pseudocode: for i=1..n for j=1..i print: complexity?

A. O(n)
B. O(n^2)
C. O(log n)
D. O(n log n)
Correct Answer: A

Q77. Pseudocode: while n>1 n=n/2: complexity?

A. O(n)
B. O(log n)
C. O(n log n)
D. O(1)
Correct Answer: B

Q78. T(n)=2T(n/2)+n gives?

A. O(n)
B. O(n log n)
C. O(n^2)
D. O(log n)
Correct Answer: B

Q79. T(n)=T(n-1)+n gives?

A. O(n)
B. O(n^2)
C. O(log n)
D. O(2^n)
Correct Answer: B

Q80. Binary search O?

A. O(n)
B. O(log n)
C. O(1)
D. O(n log n)
Correct Answer: B

Q81. Linear search worst?

A. O(log n)
B. O(n)
C. O(1)
D. O(n^2)
Correct Answer: B

Q82. Quick sort avg/worst?

A. O(n)/O(n^2)
B. O(n log n) avg, O(n^2) worst
C. O(n^2)/O(n)
D. O(log n)
Correct Answer: B

Q83. Merge sort?

A. O(n^2)
B. O(n log n) all
C. O(n)
D. O(log n)
Correct Answer: B

Q84. Output: stack push 1,2,3 pop, push 4 pop,pop: order?

A. 3,4,2
B. 3,2,1
C. 1,2,3
D. 4,3,2
Correct Answer: A

Q85. Queue enqueue 1,2,3 dequeue, enqueue 4 dequeue,dequeue: order?

A. 1,2,3
B. 2,3
C. 1,2
D. 1,4,2
Correct Answer: A

Q86. Postfix 5 1 2 + 4 * + 3 - =?

A. 14
B. 12
C. 11
D. 10
Correct Answer: A

Q87. Prefix - + 2 3 * 4 5 =? ((2+3)-(4*5))

A. -15
B. 15
C. 25
D. 5
Correct Answer: A

Q88. Infix (a+b)*(c-d) postfix?

A. ab+cd-*
B. ab+cd*-
C. abc+d-*
D. ab*cd+-
Correct Answer: A

Q89. Ackermann/recurrence depth uses?

A. Heap
B. Stack
C. Queue
D. Graph
Correct Answer: B

Q90. Tail recursion■■ to?

A. Loop
B. More stack
C. Heap
D. Tree
Correct Answer: A

Q91. BST inorder 1,2,3,4,5; preorder 3,1,2,4,5 root?

A. 1
B. 3
C. 5
D. 2
Correct Answer: B

Q92. AVL after insert balance 2 at root left-heavy, left child balanced: rotation?

A. L
B. R
C. LR
D. RL
Correct Answer: B

Q93. Min heap [10,20,30,40] insert 5: new root?

A. 10
B. 5
C. 20
D. 30
Correct Answer: B

Q94. Max heap extract root [20,15,10]: next root?

A. 20
B. 15
C. 10
D. 5
Correct Answer: B

Q95. Height of heap n nodes?

A. O(n)
B. O(log n)
C. O(1)
D. O(n log n)
Correct Answer: B

Q96. Level-order traversal uses?

A. Stack
B. Queue
C. Recursion
D. Heap
Correct Answer: B

Q97. Inorder without recursion uses?

A. Explicit stack
B. Queue
C. Heap
D. Array
Correct Answer: B

Q98. Expression tree leaves are?

A. Operators
B. Operands
C. Root
D. Null
Correct Answer: B

Q99. Huffman tree is for?

A. Sorting
B. Compression
C. Searching
D. MST
Correct Answer: B

Q100. Huffman greedy uses?

A. Stack
B. Min-heap
C. Queue
D. BST
Correct Answer: B

Q101. File: sequential access best for?

A. Random
B. Tape/batch
C. Indexed
D. Direct
Correct Answer: B

Q102. Direct/hash file gives?

A. Sequential
B. Random O(1) avg
C. Sorted
D. Slow
Correct Answer: B

Q103. Indexed sequential (ISAM) combines?

A. Sequential+index
B. Hash only
C. Random only
D. Linked
Correct Answer: A

Q104. Relative file record address via?

A. Key->offset calc
B. Index
C. Scan
D. Hash chain
Correct Answer: A

Q105. Master vs transaction file: master?

A. Temporary
B. Permanent base
C. Log
D. Index
Correct Answer: B

Q106. Sorting large file uses?

A. Quick in RAM
B. External k-way merge
C. Bubble
D. Heap only
Correct Answer: B

Q107. Blocking factor =?

A. File/block
B. Block/record
C. Records per block
D. Blocks per file
Correct Answer: C

Q108. Unblocked vs blocked: blocked?

A. Wasteful
B. Reduces I/O
C. Slower
D. Fragmented
Correct Answer: B

Q109. Overflow in ISAM handled by?

A. Rehash
B. Overflow area/chain
C. Split always
D. Ignore
Correct Answer: B

Q110. B+ tree vs ISAM: B+?

A. Static
B. Dynamic balanced
C. Sequential only
D. Hash
Correct Answer: B

Q111. Hashing: h(k) maps to?

A. Record
B. Bucket/address
C. File
D. Index
Correct Answer: B

Q112. Collision occurs when?

A. Different keys same address
B. Same key
C. Full file
D. Empty
Correct Answer: A

Q113. Chaining resolves by?

A. Rehash
B. Linked list per bucket
C. Probing
D. Double file
Correct Answer: B

Q114. Open addressing stores?

A. Elsewhere linked
B. In table itself
C. Overflow file
D. Index
Correct Answer: B

Q115. Linear probing suffers?

A. No clustering
B. Primary clustering
C. Secondary
D. No probe
Correct Answer: B

Q116. Double hashing uses?

A. One function
B. Two functions
C. Chaining
D. Sorting
Correct Answer: B

Q117. Load factor =?

A. n/m
B. m/n
C. collisions
D. buckets/records
Correct Answer: A

Q118. Rehash when load?

A. Low
B. High (>0.7-0.8)
C. Zero
D. Never
Correct Answer: B

Q119. Perfect hashing for?

A. Dynamic
B. Static set, no collisions
C. All
D. Overflow
Correct Answer: B

Q120. Extendible hashing uses?

A. Fixed array
B. Directory + global depth
C. Chain only
D. B-tree
Correct Answer: B

Q121. Inverted list is?

A. Forward index
B. Key -> list of records
C. Sequential
D. Heap
Correct Answer: B

Q122. Inverted file good for?

A. Range primary
B. Secondary key queries
C. Sequential batch
D. Sorting
Correct Answer: B

Q123. Multi-list links records with?

A. Arrays
B. Linked pointers per attribute
C. Hash
D. Tree
Correct Answer: B

Q124. Multi-list vs inverted: multi-list?

A. Separate index
B. Embedded links, slower traverse
C. Faster
D. No links
Correct Answer: B

Q125. Cellular multilist partitions into?

A. Buckets
B. Cells/subfiles
C. Blocks
D. Trees
Correct Answer: B

Q126. Index dense vs sparse: dense?

A. One per block
B. One per record
C. One per file
D. None
Correct Answer: B

Q127. Primary index on?

A. Non-key
B. Ordered key
C. Unordered
D. Secondary
Correct Answer: B

Q128. Secondary index needs?

A. Nothing
B. Extra level/bucket pointers
C. Sorted file
D. Hash only
Correct Answer: B

Q129. Clustering index file?

A. Unordered
B. Ordered on non-key
C. Hash
D. Direct
Correct Answer: B

Q130. B-tree node split when?

A. Underflow
B. Overflow/full
C. Empty
D. Delete only
Correct Answer: B

Q131. B-tree min keys (order m)?

A. m
B. ceil(m/2)-1
C. m-1
D. m/2+1
Correct Answer: B

Q132. B+ internal vs leaf: internal?

A. Data
B. Keys+pointers only
C. Records
D. Bitmap
Correct Answer: B

Q133. B+ range scan via?

A. Root only
B. Leaf linked list
C. Hash
D. Stack
Correct Answer: B

Q134. Trie is for?

A. Numbers
B. Strings/prefixes
C. Graphs
D. Heaps
Correct Answer: B

Q135. Trie search O?

A. O(n)
B. O(L) length
C. O(log n)
D. O(1)
Correct Answer: B

Q136. ISAM static vs B+ dynamic: ISAM degrades with?

A. Queries
B. Inserts/overflow
C. Reads
D. Deletes only
Correct Answer: A

Q137. VSAM is?

A. IBM dynamic B-tree-like
B. Hash
C. Sequential only
D. Inverted
Correct Answer: A

Q138. File directory holds?

A. Data
B. Metadata/extents
C. Indexes
D. Records
Correct Answer: B

Q139. Extent is?

A. Record
B. Contiguous blocks
C. File
D. Bit
Correct Answer: B

Q140. Fragmentation internal vs external: external?

A. Wasted inside block
B. Scattered free space
C. No waste
D. Inode
Correct Answer: B

Q141. Linked allocation file suffers?

A. Random access
B. Sequential
C. Sharing
D. Growth
Correct Answer: A

Q142. Indexed allocation (inode) supports?

A. Sequential only
B. Random + dynamic
C. Tape
D. No growth
Correct Answer: B

Q143. FAT vs inode: FAT?

A. Per-file index
B. Central table chain
C. Extents
D. B-tree
Correct Answer: B

Q144. Journaling FS for?

A. Speed only
B. Crash recovery
C. Compression
D. Encryption
Correct Answer: B

Q145. Pseudocode binary search mid=(l+h)/2; overflow safe?

A. Yes always
B. Use l+(h-l)/2
C. No fix
D. Use float
Correct Answer: B

Q146. Output: fun(n){if n<=1 return 1; return n*fun(n-1);} fun(4)?

A. 12
B. 24
C. 16
D. 4
Correct Answer: B

Q147. Output: fib iterative n=6 (0,1...)?

A. 5
B. 8
C. 13
D. 6
Correct Answer: B

Q148. Tower of Hanoi moves n=3?

A. 6
B. 7
C. 8
D. 9
Correct Answer: B

Q149. Complexity Hanoi?

A. O(n)
B. O(2^n)
C. O(n^2)
D. O(n log n)
Correct Answer: B

Q150. Output queue BFS from A: A->B,C; B->D; C->D. Order?

A. A,B,C,D
B. A,C,B,D
C. A,D,B,C
D. A,B,D,C
Correct Answer: A

Q151. DFS from A same graph (B first) order?

A. A,B,D,C
B. A,B,C,D
C. A,C,D,B
D. A,D,C,B
Correct Answer: A

Q152. Dijkstra A->D weights AB1,AC4,BD2,CD1 shortest?

A. 5
B. 3
C. 4
D. 2
Correct Answer: B

Q153. Kruskal picks edges sorted; first?

A. Largest
B. Smallest acyclic
C. Random
D. Last
Correct Answer: B

Q154. Stable sort preserves?

A. Speed
B. Equal order
C. Memory
D. Complexity
Correct Answer: C

Q155. Which stable O(n log n)?

A. Quick
B. Heap
C. Merge
D. Selection
Correct Answer: B

Q156. Counting sort O?

A. O(n log n)
B. O(n+k)
C. O(n^2)
D. O(log n)
Correct Answer: B

Q157. Radix sort base?

A. Comparisons
B. Digit passes
C. Heap
D. Divide
Correct Answer: B

Q158. Bucket sort avg?

A. O(n^2)
B. O(n+k) avg linear uniform
C. O(n log n) always
D. O(log n)
Correct Answer: B

Q159. Lower bound comparison sort?

A. O(n)
B. Omega(n log n)
C. O(log n)
D. O(1)
Correct Answer: B

Q160. Best for nearly sorted?

A. Merge
B. Insertion O(n)
C. Quick
D. Heap
Correct Answer: B

Q161. Best for linked list sort?

A. Quick array
B. Merge (no random)
C. Heap
D. Counting
Correct Answer: B

Q162. Recursion depth quick worst?

A. O(log n)
B. O(n)
C. O(1)
D. O(n log n)
Correct Answer: B

Q163. Tail recursion factorial space without opt?

A. O(1)
B. O(n)
C. O(log n)
D. O(n^2)
Correct Answer: B

Q164. Graph bipartite check via?

A. Dijkstra
B. BFS coloring
C. MST
D. Topo
Correct Answer: B

Q165. Articulation point via?

A. BFS only
B. DFS low/disc
C. Dijkstra
D. Union
Correct Answer: B

Q166. Strongly connected (directed) via?

A. BFS once
B. Kosaraju/Tarjan
C. MST
D. Topo
Correct Answer: B

Q167. Cycle undirected detection via?

A. Union-Find
B. Topo
C. Dijkstra
D. Heap
Correct Answer: B

Q168. AVL delete O?

A. O(1)
B. O(log n)
C. O(n)
D. O(n log n)
Correct Answer: B

Q169. Splay amortized O?

A. O(1)
B. O(log n)
C. O(n)
D. O(n^2)
Correct Answer: B

Q170. 2-3 tree is?

A. Binary
B. Balanced multiway, precursor B-tree
C. Heap
D. Trie
Correct Answer: B

Q171. Red-black vs AVL: RB?

A. Stricter
B. Looser, fewer rotations
C. Unbalanced
D. Array
Correct Answer: B

Q172. Segment tree query O?

A. O(1)
B. O(log n)
C. O(n)
D. O(n log n)
Correct Answer: B

Q173. Fenwick tree for?

A. Prefix sums
B. Sorting
C. MST
D. Strings
Correct Answer: B

Q174. Disjoint set union by rank keeps?

A. Height log
B. List
C. Stack
D. Queue
Correct Answer: B

Q175. Open hashing load >1 possible?

A. No
B. Yes (chaining)
C. Never
D. Only closed
Correct Answer: B

Q176. Closed hashing load must be?

A. >1
B. <=1
C. Any
D. 0
Correct Answer: B

Q177. Expected probe linear load 0.5?

A. 1.5
B. 2.5
C. 1
D. 5
Correct Answer: A

Q178. Double hashing probe sequence depends on?

A. Key
B. Table size only
C. Load only
D. Random
Correct Answer: B

Q179. Cuckoo hashing uses?

A. One table
B. Two functions/displacement
C. Chaining
D. Sorting
Correct Answer: B

Q180. Bloom filter may give?

A. False negative
B. False positive, no false negative
C. Exact
D. No error
Correct Answer: B

Q181. B+ fanout 100, height 3 capacity ~?

A. 1M
B. 1M? 100^3=1M
C. 300
D. 10000
Correct Answer: B

Q182. Height balanced ensures disk accesses?

A. Many
B. Few/logarithmic
C. Linear
D. Random
Correct Answer: B

Q183. Index sequential search steps?

A. Direct only
B. Index + data block
C. Full scan
D. Hash
Correct Answer: B

Q184. Cylinder/track/sector locality: place related on?

A. Random cylinders
B. Same cylinder
C. Different disks
D. Tape
Correct Answer: B

Q185. RAID + B-tree: RAID helps?

A. CPU
B. I/O parallelism
C. Height
D. Keys
Correct Answer: B

Q186. Log file for recovery is?

A. Random
B. Sequential append
C. Hash
D. Indexed
Correct Answer: B

Q187. Checkpoint file truncates?

A. Data
B. Log redo
C. Index
D. Directory
Correct Answer: B

Q188. Two-phase commit for?

A. Single DB
B. Distributed atomic
C. Sorting
D. Indexing
Correct Answer: B

Q189. Shadowing vs logging: shadow?

A. Log
B. Copy-on-write
C. Lock
D. Timestamp
Correct Answer: B

Q190. Pseudocode: sum=0; for i=1..n sum+=i; sum=?

A. n(n+1)/2
B. n^2
C. 2^n
D. n log n
Correct Answer: A

Q191. Output: p=1; for i=1..4 p*=i; p=?

A. 6
B. 24
C. 12
D. 10
Correct Answer: B

Q192. Output: x=0; for i=0..3 for j=0..3 x++; x=?

A. 12
B. 16
C. 9
D. 7
Correct Answer: B

Q193. Recursion T(n)=T(n/2)+1 -> iterative?

A. Linear
B. Binary search-like log
C. Quadratic
D. Exponential
Correct Answer: B

Q194. Master theorem needs?

A. Any recurrence
B. Divide-conquer aT(n/b)+f
C. Linear
D. Tail
Correct Answer: B

Q195. Amortized stack multi-pop analysis O per op?

A. O(n)
B. O(1)
C. O(log n)
D. O(n^2)
Correct Answer: B

Q196. Dynamic array double on full amortized push?

A. O(n^2)
B. O(1)
C. O(log n)
D. O(n)
Correct Answer: B

Q197. Circular buffer queue full vs empty distinguish by?

A. Size/count or sacrifice slot
B. Values
C. Pointers equal always full
D. Flags only data
Correct Answer: A

Q198. Priority queue via sorted array insert/extract?

A. O(1)/O(n)
B. O(n)/O(1) or vice versa
C. O(log n) both
D. O(1) both
Correct Answer: B

Q199. Best structure for LRU cache?

A. Array
B. Hash + DLL
C. Stack
D. BST
Correct Answer: B

Q200. Best for phone prefix autocomplete?

A. Heap
B. Trie
C. Graph
D. Queue
Correct Answer: B

ANSWER KEY — Topic 7: Data and File Structures

```
1-A 2-B 3-A 4-B 5-B 6-B 7-B 8-A 9-A 10-B
11-B 12-B 13-B 14-B 15-B 16-B 17-B 18-B 19-B 20-B
21-B 22-B 23-B 24-B 25-B 26-B 27-B 28-B 29-B 30-A
31-B 32-B 33-B 34-B 35-B 36-B 37-B 38-C 39-B 40-B
41-B 42-C 43-B 44-B 45-B 46-B 47-B 48-B 49-B 50-B
51-B 52-B 53-B 54-B 55-B 56-B 57-B 58-B 59-B 60-B
61-B 62-B 63-B 64-A 65-B 66-B 67-A 68-A 69-B 70-B
71-B 72-B 73-B 74-B 75-B 76-A 77-B 78-B 79-B 80-B
81-B 82-B 83-B 84-A 85-A 86-A 87-A 88-A 89-B 90-A
91-B 92-B 93-B 94-B 95-B 96-B 97-B 98-B 99-B 100-B
101-B 102-B 103-A 104-A 105-B 106-B 107-C 108-B 109-B 110-B
111-B 112-A 113-B 114-B 115-B 116-B 117-A 118-B 119-B 120-B
121-B 122-B 123-B 124-B 125-B 126-B 127-B 128-B 129-B 130-B
131-B 132-B 133-B 134-B 135-B 136-A 137-A 138-B 139-B 140-B
141-A 142-B 143-B 144-B 145-B 146-B 147-B 148-B 149-B 150-A
151-A 152-B 153-B 154-C 155-B 156-B 157-B 158-B 159-B 160-B
161-B 162-B 163-B 164-B 165-B 166-B 167-B 168-B 169-B 170-B
171-B 172-B 173-B 174-B 175-B 176-B 177-A 178-B 179-B 180-B
181-B 182-B 183-B 184-B 185-B 186-B 187-B 188-B 189-B 190-A
191-B 192-B 193-B 194-B 195-B 196-B 197-A 198-B 199-B 200-B
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Topic 8: Computer Networks

Syllabus: *Network fundamentals: Local Area Networks (LAN), Metropolitan Area Networks (MAN), Wide Area Networks (WAN), Wireless Networks, Inter Networks; Reference Models: The OSI model, TCP/IP model; Data Communication: Channel capacity, Transmission media - twisted pair, coaxial cables, fibre-optic cables, wireless transmission - radio, microwave, infrared and millimeter waves, Lightwave transmission, Telephones - local loop, trunks, multiplexing, switching, narrowband ISDN, broadband ISDN, ATM, High speed LANS, Cellular Radio, Communication satellites - geosynchronous and low-orbit; Internetworking: Switch / Hub, Bridge, Router, Gateways, Concatenated virtual circuits, Tunnelling, Fragmentation, Firewalls; Routing: Virtual circuits and datagrams, Routing algorithms, Congestion control; Network Security: Cryptography - public key, secret key, Domain Name System (DNS), Electronic Mail and Worldwide Web (WWW), The DNS, Resource Records, Name servers, E-mail architecture and Servers.*

Q1. LAN scope?

A. City
B. Building/campus
C. Country
D. Planet
Correct Answer: B

Q2. MAN scope?

A. Room
B. City/metro
C. Country
D. Personal
Correct Answer: B

Q3. WAN scope?

A. Office
B. Large geo/country
C. Chip
D. Home
Correct Answer: B

Q4. Wireless ad-hoc has?

A. AP
B. No infra
C. Base station
D. Switch
Correct Answer: B

Q5. Internetwork connects?

A. Hosts only
B. Networks via routers
C. Cables
D. Processes
Correct Answer: B

Q6. PAN example?

A. Internet
B. Bluetooth
C. MAN
D. WAN
Correct Answer: B

Q7. ARPANET is ancestor of?

A. LAN
B. Internet
C. ATM
D. ISDN
Correct Answer: B

Q8. Number of OSI layers?

A. 4
B. 5
C. 7
D. 9
Correct Answer: C

Q9. OSI order top to bottom?

A. P,S,T,N,D,P,Ph
B. App,Pres,Sess,Trans,Net,DL,Phy
C. Reverse
D. Mixed
Correct Answer: B

Q10. TCP/IP layers count (typical)?

A. 7
B. 4/5
C. 2
D. 8
Correct Answer: B

Q11. Physical layer deals with?

A. Routing
B. Bits/signals
C. Sessions
D. Encryption
Correct Answer: B

Q12. Data link handles?

A. Routing
B. Framing/MAC
C. Process
D. Compression
Correct Answer: B

Q13. Network layer?

A. Framing
B. Routing/IP
C. TCP
D. HTTP
Correct Answer: B

Q14. Transport examples?

A. IP,ARP
B. TCP/UDP
C. HTTP/FTP
D. Ethernet
Correct Answer: B

Q15. Session layer?

A. Bits
B. Dialog/sync
C. Routing
D. MAC
Correct Answer: B

Q16. Presentation does?

A. Routing
B. Encryption/compression/format
C. Framing
D. IP
Correct Answer: B

Q17. Application examples?

A. TCP
B. HTTP/DNS/FTP
C. IP
D. MAC
Correct Answer: B

Q18. Encapsulation order from app?

A. Segment-packet-frame
B. Frame-packet-segment
C. Packet-segment-frame
D. Bits first
Correct Answer: A

Q19. Port numbers at which layer?

A. Network
B. Transport
C. Data link
D. Physical
Correct Answer: B

Q20. IP address at?

A. Transport
B. Network
C. Data link
D. App
Correct Answer: B

Q21. MAC address at?

A. Network
B. Data link
C. Transport
D. Session
Correct Answer: B

Q22. Nyquist capacity 2B log2M: B=3kHz,M=8 gives?

A. 6000
B. 9000
C. 18000
D. 24000 bps? 2*3000*3=18000
Correct Answer: B

Q23. Shannon C=B log2(1+SNR): B=1kHz,SNR=15 gives?

A. 1000
B. 2000
C. 4000
D. 8000
Correct Answer: B

Q24. SNR 10dB means ratio?

A. 10
B. 100
C. 1
D. 1000? 10dB=10
Correct Answer: C

Q25. Baud vs bit rate: 4 bits/baud, 1000 baud?

A. 1000 bps
B. 4000 bps
C. 250 bps
D. 400 bps
Correct Answer: D

Q26. Guided media example?

A. Radio
B. Twisted pair/coax/fiber
C. Microwave
D. Satellite
Correct Answer: B

Q27. Unguided example?

A. Fiber
B. Radio/microwave
C. Coax
D. STP
Correct Answer: B

Q28. Fiber advantage?

A. EMI immune, high BW
B. Cheap tap
C. Slow
D. Short
Correct Answer: B

Q29. Twisted pair cancels?

A. Attenuation
B. EMI/crosstalk
C. Delay
D. Loss
Correct Answer: B

Q30. Coax vs UTP: coax?

A. Less BW
B. Higher BW/shielded
C. No shield
D. Wireless
Correct Answer: B

Q31. Telephone POTS bandwidth?

A. 20kHz
B. 3-4kHz
C. 1MHz
D. 100kHz
Correct Answer: B

Q32. Modem modulates?

A. Digital-digital
B. Digital-analog
C. Analog-digital only
D. Light
Correct Answer: B

Q33. PCM sampling voice 4kHz needs?

A. 4k samples
B. 8k samples/s
C. 16k
D. 2k
Correct Answer: B

Q34. PCM bits 8 per sample bitrate?

A. 32kbps
B. 64kbps
C. 16kbps
D. 8kbps
Correct Answer: B

Q35. ISDN BRI channels?

A. 23B+D
B. 2B+D
C. 30B+D
D. 1B+D
Correct Answer: B

Q36. ISDN B/D rates?

A. 16/64
B. 64/16 kbps
C. 1.5M/64k
D. Equal
Correct Answer: B

Q37. ATM cell size?

A. 48B
B. 53B (5+48)
C. 64B
D. 1500B
Correct Answer: B

Q38. ATM is?

A. Packet
B. Cell-switched, connection-oriented
C. Circuit only
D. Broadcast
Correct Answer: B

Q39. SONET base STS-1?

A. 51.84 Mbps
B. 155 Mbps
C. 622 Mbps
D. 2 Mbps
Correct Answer: B

Q40. DSL uses?

A. Fiber only
B. Phone line high freq
C. Coax only
D. Satellite
Correct Answer: B

Q41. Hub operates at?

A. Network
B. Physical
C. Transport
D. App
Correct Answer: B

Q42. Switch operates at (typical LAN)?

A. Physical
B. Data link
C. Network
D. App
Correct Answer: B

Q43. Bridge operates at?

A. Physical
B. Data link
C. Network
D. Transport
Correct Answer: B

Q44. Router operates at?

A. Data link
B. Network
C. Physical
D. Session
Correct Answer: B

Q45. Gateway operates up to?

A. Physical
B. Application
C. Data link
D. Network only
Correct Answer: B

Q46. Repeater function?

A. Filter
B. Amplify/regenerate
C. Route
D. Translate
Correct Answer: B

Q47. Hub vs switch: hub?

A. Intelligent
B. Broadcasts all ports, collision domain shared
C. Learns MAC
D. Routes IP
Correct Answer: B

Q48. Switch learns via?

A. Routing table
B. MAC table/self-learning
C. DNS
D. ARP only
Correct Answer: B

Q49. Collision domain per port on switch?

A. Shared
B. Separate
C. None
D. Single global
Correct Answer: B

Q50. Broadcast domain separated by?

A. Hub
B. Switch (no VLAN)
C. Router/VLAN
D. Repeater
Correct Answer: C

Q51. Store-and-forward switch?

A. No check
B. Full frame + CRC
C. Cut-through only
D. No buffer
Correct Answer: B

Q52. Cut-through advantage?

A. Reliable
B. Low latency
C. No errors
D. Large buffer
Correct Answer: B

Q53. Bridge filtering via?

A. IP
B. MAC
C. Port
D. URL
Correct Answer: B

Q54. Spanning Tree Protocol prevents?

A. Congestion
B. Loops
C. Errors
D. Delay
Correct Answer: B

Q55. Router forwarding via?

A. MAC table
B. Routing table/longest prefix
C. DNS
D. Ports only
Correct Answer: B

Q56. Tunneling encapsulates?

A. Same protocol
B. One packet inside another for transit
C. Encrypts only
D. Fragments only
Correct Answer: B

Q57. Fragmentation needed when?

A. Small packet
B. MTU smaller than datagram
C. Large MTU
D. Congestion only
Correct Answer: B

Q58. IP MTU Ethernet typical?

A. 576
B. 1500
C. 9000
D. 64
Correct Answer: B

Q59. Fragment offset units?

A. 1B
B. 8B
C. 1500B
D. 1 bit
Correct Answer: B

Q60. DF flag means?

A. Fragment
B. Don't fragment
C. More fragments
D. Reserved
Correct Answer: B

Q61. MF flag means?

A. Last
B. More fragments follow
C. No frag
D. Error
Correct Answer: B

Q62. Reassembly at?

A. Each router
B. Destination host
C. Source
D. Switch
Correct Answer: B

Q63. Firewall filters by?

A. Content always deep
B. IP/port/rules
C. MAC only
D. Voltage
Correct Answer: B

Q64. Stateful firewall tracks?

A. Nothing
B. Connection state
C. Routes
D. DNS
Correct Answer: B

Q65. Packet filter vs proxy: proxy?

A. Faster
B. App-level, breaks connection
C. Stateless
D. Physical
Correct Answer: B

Q66. NAT translates?

A. MAC
B. Private-public IP
C. Ports only
D. DNS
Correct Answer: B

Q67. PAT uses?

A. IP only
B. Ports to multiplex
C. MAC
D. DNS
Correct Answer: B

Q68. Routing algorithm: distance-vector example?

A. OSPF
B. RIP/BGP (DV)
C. IS-IS
D. Dijkstra link-state
Correct Answer: B

Q69. Link-state example?

A. RIP
B. OSPF
C. BGP path-vector
D. EIGRP only
Correct Answer: B

Q70. Dijkstra in link-state computes?

A. Random
B. Shortest path
C. Max flow
D. Spanning
Correct Answer: B

Q71. Distance-vector suffers?

A. Fast converge
B. Count-to-infinity
C. No loops
D. Scalable
Correct Answer: B

Q72. Split horizon + poison reverse mitigate?

A. Congestion
B. Loops/count-to-infinity
C. Errors
D. Fragmentation
Correct Answer: B

Q73. BGP is?

A. Interior DV
B. Exterior path-vector
C. Link-state
D. Static
Correct Answer: B

Q74. OSPF uses?

A. Hop count
B. Cost + flooding LSAs
C. Path vector
D. Manual
Correct Answer: B

Q75. RIP max hops?

A. 16 (15 usable)
B. 255
C. 64
D. 32
Correct Answer: A

Q76. Convergence faster in?

A. DV
B. Link-state
C. Static
D. Flooding only
Correct Answer: B

Q77. Hierarchical routing via?

A. Flat
B. Areas/autonomous systems
C. Single router
D. Hub
Correct Answer: B

Q78. Anycast delivers to?

A. All
B. Nearest/one of group
C. Broadcast
D. Multicast all
Correct Answer: B

Q79. Multicast address IPv4 range?

A. 192.168/16
B. 224.0.0.0/4
C. 10/8
D. 172.16/12
Correct Answer: B

Q80. IGMP is for?

A. Routing
B. Group membership
C. Error
D. Address assign
Correct Answer: B

Q81. Congestion vs flow control: congestion is?

A. Receiver overload
B. Network overload
C. Sender only
D. Link failure
Correct Answer: B

Q82. Flow control example?

A. OSPF
B. Stop-and-wait/sliding window
C. BGP
D. NAT
Correct Answer: B

Q83. Sliding window allows?

A. 1 packet
B. Multiple unacked
C. No ack
D. Fixed route
Correct Answer: B

Q84. Go-Back-N on error retransmits?

A. Only lost
B. Window from loss
C. All sent ever
D. None
Correct Answer: B

Q85. Selective Repeat retransmits?

A. Window
B. Only lost/NACKed, buffers
C. Go back
D. All
Correct Answer: B

Q86. TCP window is for?

A. Routing
B. Flow+congestion
C. Framing
D. Addressing
Correct Answer: B

Q87. Slow start grows?

A. Linear
B. Exponential (cwnd double)
C. Fixed
D. Decreases
Correct Answer: B

Q88. AIMD in congestion avoidance: on loss?

A. Double
B. Halve (multiplicative decrease)
C. Zero
D. Same
Correct Answer: B

Q89. Fast retransmit on?

A. Timeout only
B. 3 duplicate ACKs
C. 1 ACK
D. NACK
Correct Answer: B

Q90. Leaky bucket shapes to?

A. Bursty
B. Constant rate
C. Variable
D. Priority
Correct Answer: B

Q91. Token bucket allows?

A. No burst
B. Burst up to bucket
C. Constant only
D. Drops all
Correct Answer: B

Q92. RED drops?

A. Tail only
B. Early random on avg queue
C. Head
D. None
Correct Answer: B

Q93. Choke packets signal?

A. Route
B. Congestion source quench
C. Error
D. Ack
Correct Answer: B

Q94. QoS needs?

A. Best effort
B. Reservation/traffic classes
C. Hubs
D. Firewalls
Correct Answer: B

Q95. Secret-key crypto uses?

A. Two keys
B. Single shared key
C. Public only
D. Hash only
Correct Answer: B

Q96. Public-key uses?

A. Shared
B. Public+private pair
C. One key
D. No key
Correct Answer: B

Q97. Symmetric example?

A. RSA
B. AES/DES
C. Diffie-Hellman key only
D. DSA sign only
Correct Answer: B

Q98. Asymmetric example?

A. AES
B. RSA/ECC
C. DES
D. RC4
Correct Answer: A

Q99. Hash function for?

A. Encryption
B. Integrity/fingerprint
C. Routing
D. Compression
Correct Answer: B

Q100. Digital signature provides?

A. Confidentiality
B. Authentication/integrity/nonrepudiation
C. Routing
D. Speed
Correct Answer: B

Q101. Sign with which key?

A. Public
B. Private
C. Shared
D. Session
Correct Answer: B

Q102. Verify with?

A. Private
B. Public
C. Shared
D. Hash only
Correct Answer: B

Q103. Diffie-Hellman for?

A. Encryption
B. Key exchange
C. Signing
D. Hashing
Correct Answer: B

Q104. Man-in-middle defeats DH without?

A. Encryption
B. Authentication
C. Hash
D. Speed
Correct Answer: B

Q105. Certificate binds?

A. Key-route
B. Identity-public key via CA
C. IP-MAC
D. Port-app
Correct Answer: B

Q106. CA is?

A. Router
B. Trusted authority
C. Firewall
D. Host
Correct Answer: B

Q107. SSL/TLS operates above?

A. IP
B. TCP
C. Ethernet
D. HTTP only? Actually TLS over TCP
Correct Answer: B

Q108. IPsec modes: transport vs tunnel?

A. Same
B. Host-host vs gateway-gateway encapsulation
C. Reverse
D. No encryption
Correct Answer: B

Q109. DNS maps?

A. MAC-IP
B. Name-IP
C. Port-app
D. URL-content
Correct Answer: B

Q110. DNS hierarchy root?

A. Bottom
B. Top dot
C. Leaf
D. TLD only
Correct Answer: B

Q111. Iterative vs recursive: recursive server?

A. Refers
B. Resolves fully
C. Client does
D. No cache
Correct Answer: B

Q112. DNS record A?

A. IPv6
B. IPv4
C. Mail
D. Name server
Correct Answer: B

Q113. MX record for?

A. Host
B. Mail exchange
C. Alias
D. Reverse
Correct Answer: B

Q114. CNAME is?

A. Mail
B. Canonical/alias
C. Pointer
D. Text
Correct Answer: B

Q115. DNS uses transport?

A. TCP only
B. UDP mostly, TCP for zone/large
C. IP directly
D. ICMP
Correct Answer: B

Q116. DHCP assigns?

A. Names
B. IP dynamically
C. MAC
D. Routes only
Correct Answer: B

Q117. DHCP DORA order?

A. Request-offer-discover-ack
B. Discover-Offer-Request-Ack
C. Offer-Discover-Ack-Req
D. Ack first
Correct Answer: B

Q118. ARP resolves?

A. Name-IP
B. IP-MAC
C. MAC-IP (RARP)
D. Port
Correct Answer: B

Q119. RARP/BOOTP predecessor of?

A. DNS
B. DHCP
C. ARP
D. ICMP
Correct Answer: B

Q120. ICMP for?

A. Routing
B. Error/report (ping)
C. Address
D. Mail
Correct Answer: B

Q121. Ping uses?

A. TCP
B. ICMP echo
C. UDP
D. ARP
Correct Answer: B

Q122. Traceroute uses?

A. TTL expiry
B. Flooding
C. BGP
D. DNS
Correct Answer: B

Q123. Email protocols: SMTP for?

A. Retrieve
B. Send/transfer
C. Mailbox access
D. Directory
Correct Answer: B

Q124. POP3 vs IMAP: IMAP?

A. Download-delete
B. Server-side management
C. Send
D. No folders
Correct Answer: B

Q125. SMTP port?

A. 110
B. 25/587
C. 143
D. 80
Correct Answer: B

Q126. POP3 port?

A. 25
B. 110
C. 143
D. 443
Correct Answer: B

Q127. IMAP port?

A. 25
B. 110
C. 143
D. 21
Correct Answer: B

Q128. MIME for?

A. Routing
B. Non-ASCII/attachments
C. Encryption
D. Compression
Correct Answer: B

Q129. WWW protocol?

A. FTP
B. HTTP
C. SMTP
D. DNS
Correct Answer: B

Q130. HTTP port?

A. 21
B. 80
C. 25
D. 53
Correct Answer: B

Q131. HTTPS port?

A. 80
B. 443
C. 21
D. 110
Correct Answer: B

Q132. HTTP stateless; session via?

A. IP
B. Cookies/sessions
C. TCP
D. DNS
Correct Answer: B

Q133. Persistent vs non-persistent: persistent?

A. New conn each
B. Reuse
C. No TCP
D. UDP
Correct Answer: B

Q134. Proxy cache reduces?

A. Security
B. Latency/bandwidth
C. Reliability
D. Routing
Correct Answer: B

Q135. FTP uses how many connections?

A. 1
B. 2 (control+data)
C. 0
D. 3
Correct Answer: B

Q136. FTP ports?

A. 80/443
B. 20/21
C. 25/110
D. 53/67
Correct Answer: B

Q137. Telnet vs SSH: SSH?

A. Plaintext
B. Encrypted
C. Faster
D. UDP
Correct Answer: B

Q138. SNMP for?

A. Mail
B. Network management
C. Routing
D. Transfer
Correct Answer: B

Q139. CIDR /24 mask?

A. 255.0.0.0
B. 255.255.255.0
C. 255.255.0.0
D. 255.255.255.255
Correct Answer: B

Q140. Class C default mask?

A. /8
B. /16
C. /24
D. /32
Correct Answer: B

Q141. Private IP example?

A. 8.8.8.8
B. 192.168.1.1
C. 11.0.0.1 public
D. 1.1.1.1
Correct Answer: B

Q142. Loopback?

A. 0.0.0.0
B. 127.0.0.1
C. 255.255.255.255
D. 10.0.0.1
Correct Answer: B

Q143. Number hosts /26?

A. 256
B. 64 (62 usable)
C. 128
D. 32
Correct Answer: B

Q144. Subnet 192.168.1.0/26 subnets from /24?

A. 2
B. 4
C. 8
D. 16
Correct Answer: B

Q145. Supernet aggregates to?

A. Smaller prefix
B. Larger block/shorter prefix
C. More routes
D. Subnets
Correct Answer: B

Q146. Longest prefix match picks?

A. Shortest
B. Most specific
C. First
D. Random
Correct Answer: B

Q147. IPv6 length?

A. 32b
B. 64b
C. 128b
D. 48b
Correct Answer: C

Q148. IPv6 header vs IPv4: no?

A. Addresses
B. Checksum/fragmentation in routers
C. Version
D. Length
Correct Answer: B

Q149. IPv6 loopback?

A. ::1
B. 127.0.0.1
C. ::
D. fe80
Correct Answer: A

Q150. Transition: dual-stack/tunneling? Tunneling?

A. Both stacks
B. v6 over v4 encapsulation
C. Translation only
D. Native
Correct Answer: B

Q151. CSMA/CD used in?

A. Token ring
B. Classic Ethernet
C. Wifi
D. ATM
Correct Answer: B

Q152. CSMA/CA used in?

A. Ethernet
B. 802.11 wireless
C. Token
D. FDDI
Correct Answer: B

Q153. Why CA not CD in wireless?

A. Faster
B. Can't detect collision while sending
C. No collisions
D. Full duplex always
Correct Answer: B

Q154. Exponential backoff after?

A. Success
B. Collision
C. Idle
D. Ack
Correct Answer: B

Q155. Token ring controls via?

A. Contention
B. Token
C. Reservation only
D. Aloha
Correct Answer: B

Q156. Aloha efficiency slotted?

A. 18%
B. 36%
C. 90%
D. 100%
Correct Answer: B

Q157. Pure Aloha efficiency?

A. 36%
B. 18%
C. 50%
D. 90%
Correct Answer: B

Q158. Flow: stop-and-wait efficiency with long RTT?

A. High
B. Low
C. 100%
D. Unaffected
Correct Answer: B

Q159. Piggybacking combines?

A. Data+data
B. Ack+data
C. SYN+FIN
D. IP+TCP
Correct Answer: B

Q160. CRC detects?

A. All errors
B. Burst up to degree
C. Corrects all
D. Nothing
Correct Answer: B

Q161. Hamming corrects?

A. 0
B. Single-bit
C. All
D. Burst only
Correct Answer: B

Q162. Parity detects?

A. Even-bit
B. Odd/single-bit
C. All
D. Burst
Correct Answer: B

Q163. Checksum in IP covers?

A. Data only
B. Header only
C. Both
D. Nothing
Correct Answer: B

Q164. TCP checksum covers?

A. Header only
B. Pseudo+header+data
C. Nothing
D. IP header
Correct Answer: B

Q165. Three-way handshake?

A. SYN, SYN-ACK, ACK
B. SYN, ACK, FIN
C. REQ, REP, ACK
D. Hello only
Correct Answer: B

Q166. Why 3-way?

A. Speed
B. Sync seq both sides
C. Routing
D. Encryption
Correct Answer: A

Q167. TCP close FIN+ACK needs?

A. 1 packet
B. 4-way (or 3)
C. 2-way only
D. No ack
Correct Answer: B

Q168. Well-known ports range?

A. 0-1023
B. 1024-49151
C. 49152-65535
D. All
Correct Answer: B

Q169. Ephemeral ports?

A. 0-1023
B. 49152-65535 (dynamic)
C. 80
D. 25
Correct Answer: B

Q170. UDP header size?

A. 20B
B. 8B
C. 40B
D. 16B
Correct Answer: B

Q171. TCP header min?

A. 8B
B. 20B
C. 40B
D. 16B
Correct Answer: B

Q172. IP header min?

A. 8B
B. 20B
C. 40B
D. 60B
Correct Answer: B

Q173. Sequence numbers for?

A. Routing
B. Reliability/ordering
C. Flow only
D. Addressing
Correct Answer: B

Q174. Cumulative ACK means?

A. Per packet
B. Ack up to N
C. Selective
D. Negative
Correct Answer: B

Q175. Timeout RTT estimation uses?

A. Fixed
B. EWMA + deviation
C. Hop count
D. Random
Correct Answer: B

Q176. Congestion window vs advertised: effective =?

A. Sum
B. min(cwnd,rwnd)
C. max
D. Product
Correct Answer: B

Q177. 802.3 is?

A. Wireless
B. Ethernet
C. Token ring
D. Bluetooth
Correct Answer: B

Q178. 802.11 is?

A. Ethernet
B. WLAN
C. Bluetooth (802.15)
D. WiMAX (802.16)
Correct Answer: B

Q179. Bluetooth range?

A. MAN
B. PAN (~10m)
C. WAN
D. LAN large
Correct Answer: B

Q180. AP in infrastructure mode?

A. No AP
B. Central coordinator
C. Router only
D. Switch
Correct Answer: B

Q181. Handoff in cellular/wireless?

A. Frequency change
B. Switch base/AP
C. Routing
D. Fragmentation
Correct Answer: B

Q182. FDMA/TDMA/CDMA: CDMA uses?

A. Time slots
B. Codes
C. Frequencies
D. Tokens
Correct Answer: B

Q183. Hidden terminal problem solved by?

A. CSMA/CD
B. RTS/CTS
C. Aloha
D. Token
Correct Answer: B

Q184. Exposed terminal unnecessarily defers; RTS/CTS?

A. Helps fully
B. Partially wastes
C. No effect
D. Solves fully
Correct Answer: B

Q185. VLAN separates?

A. Physically
B. Logically broadcast domains
C. Routes
D. Collisions only
Correct Answer: B

Q186. Trunk carries?

A. Single VLAN
B. Multiple tagged (802.1Q)
C. Untagged only
D. IP
Correct Answer: B

Q187. STP blocks to?

A. Fastest
B. Loop-free tree
C. Mesh
D. Shortest only
Correct Answer: B

Q188. Data center: fat-tree/Clos for?

A. Loops
B. Bisection BW
C. Routing
D. Power
Correct Answer: B

Q189. SDN separates?

A. Data-data
B. Control-data plane
C. App-transport
D. MAC-IP
Correct Answer: B

Q190. OpenFlow controller?

A. Data plane
B. Centralized control
C. Physical
D. Host
Correct Answer: B

Q191. CDN replicates to?

A. Origin only
B. Edge near users
C. Core
D. Client
Correct Answer: B

Q192. P2P vs client-server: P2P?

A. Central
B. Decentral, peers serve
C. No sharing
D. Single point failure
Correct Answer: B

Q193. BitTorrent pieces via?

A. Central only
B. Tit-for-tat swarming
C. Broadcast
D. Multicast only
Correct Answer: B

Q194. Overlay network is?

A. Physical
B. Logical atop physical
C. Underlay
D. Link
Correct Answer: B

Q195. Tunneling IPv6 over IPv4 example?

A. NAT
B. 6to4/Teredo
C. DHCP
D. DNS
Correct Answer: B

Q196. Firewall DMZ places?

A. Internal only
B. Public servers isolated
C. Outside
D. No firewall
Correct Answer: B

Q197. IDS vs IPS: IPS?

A. Passive only
B. Active block
C. Firewall
D. Antivirus only
Correct Answer: B

Q198. VPN provides?

A. Public access
B. Private tunnel over public
C. Speed
D. No encryption
Correct Answer: B

Q199. MPLS labels for?

A. IP lookup each hop
B. Fast switching path
C. Encryption
D. Fragmentation
Correct Answer: B

Q200. Best routing for mobile ad-hoc?

A. Static
B. Dynamic/on-demand (AODV/DSR)
C. Fixed
D. Central
Correct Answer: B

ANSWER KEY — Topic 8: Computer Networks

```
1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-C 9-B 10-B
11-B 12-B 13-B 14-B 15-B 16-B 17-B 18-A 19-B 20-B
21-B 22-B 23-B 24-C 25-D 26-B 27-B 28-B 29-B 30-B
31-B 32-B 33-B 34-B 35-B 36-B 37-B 38-B 39-B 40-B
41-B 42-B 43-B 44-B 45-B 46-B 47-B 48-B 49-B 50-C
51-B 52-B 53-B 54-B 55-B 56-B 57-B 58-B 59-B 60-B
61-B 62-B 63-B 64-B 65-B 66-B 67-B 68-B 69-B 70-B
71-B 72-B 73-B 74-B 75-A 76-B 77-B 78-B 79-B 80-B
81-B 82-B 83-B 84-B 85-B 86-B 87-B 88-B 89-B 90-B
91-B 92-B 93-B 94-B 95-B 96-B 97-B 98-A 99-B 100-B
101-B 102-B 103-B 104-B 105-B 106-B 107-B 108-B 109-B 110-B
111-B 112-B 113-B 114-B 115-B 116-B 117-B 118-B 119-B 120-B
121-B 122-B 123-B 124-B 125-B 126-B 127-B 128-B 129-B 130-B
131-B 132-B 133-B 134-B 135-B 136-B 137-B 138-B 139-B 140-B
141-B 142-B 143-B 144-B 145-B 146-B 147-C 148-B 149-A 150-B
151-B 152-B 153-B 154-B 155-B 156-B 157-B 158-B 159-B 160-B
161-B 162-B 163-B 164-B 165-B 166-A 167-B 168-B 169-B 170-B
171-B 172-B 173-B 174-B 175-B 176-B 177-B 178-B 179-B 180-B
181-B 182-B 183-B 184-B 185-B 186-B 187-B 188-B 189-B 190-B
191-B 192-B 193-B 194-B 195-B 196-B 197-B 198-B 199-B 200-B
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Topic 9: System Software & Compilers

Syllabus: *8085 assembly fundamentals, assemblers (2-pass/single-pass), macros, loader/linker (loading/linking/relocation/relocatability/linkage editing), text editors/debuggers, compilation vs interpretation, bootstrap, phases of compilation, lexical analysis/Lex, CFG/parsing/parse trees/leftmost-rightmost derivations, bottom-up (shift-reduce/operator precedence/LR)/YACC, top-down (left recursion removal/recursive descent/predictive), intermediate codes (quadruples/triples), code generation & optimization.*

Q1. Which register in 8085 is used as accumulator?

A. A
B. B
C. H
D. SP
Correct Answer: A

Q2. Size of accumulator in 8085 is:

A. 4-bit
B. 8-bit
C. 16-bit
D. 32-bit
Correct Answer: B

Q3. 8085 has how many general purpose 8-bit registers?

A. 4
B. 5
C. 6
D. 7
Correct Answer: A

Q4. Which pair forms the 16-bit HL register pair?

A. H and L
B. B and C
C. D and E
D. A and F
Correct Answer: A

Q5. Program Counter in 8085 is:

A. 8-bit
B. 16-bit
C. 20-bit
D. 32-bit
Correct Answer: B

Q6. Stack Pointer in 8085 is:

A. 8-bit
B. 16-bit
C. 6-bit
D. 4-bit
Correct Answer: B

Q7. Number of flags in 8085 flag register used is:

A. 4
B. 5
C. 8
D. 16
Correct Answer: B

Q8. Which is NOT a flag in 8085?

A. Overflow
B. Zero
C. Carry
D. Parity
Correct Answer: A

Q9. HLT instruction in 8085 means:

A. Halt processor
B. Hold bus
C. High level transfer
D. Hardware trap
Correct Answer: A

Q10. NOP instruction performs:

A. No operation
B. Normal output pulse
C. Next operation
D. Null pointer
Correct Answer: A

Q11. LDA instruction in 8085 is:

A. Load accumulator direct
B. Load accumulator indirect
C. Logical AND
D. Long address
Correct Answer: A

Q12. STA instruction stores:

A. Accumulator to memory
B. Stack to accumulator
C. Status to accumulator
D. Stack to memory
Correct Answer: A

Q13. MOV A,B means:

A. Copy B to A
B. Copy A to B
C. Add B to A
D. Move memory to bus
Correct Answer: A

Q14. ADD B instruction adds:

A. B to accumulator
B. B to H
C. B to stack
D. B to PC
Correct Answer: A

Q15. INR M means:

A. Increment memory pointed by HL
B. Increment microprocessor
C. Input from memory
D. Interrupt memory
Correct Answer: A

Q16. DCR C means:

A. Decrement C by 1
B. Decode C
C. Divide by C
D. Direct call
Correct Answer: A

Q17. JMP instruction is:

A. Unconditional jump
B. Jump if zero
C. Jump if carry
D. Jump subroutine
Correct Answer: A

Q18. JZ jumps when:

A. Zero flag=1
B. Carry=1
C. Sign=0
D. Parity=1
Correct Answer: A

Q19. CALL instruction:

A. Calls subroutine, saves return address on stack
B. Calls interrupt only
C. Copies accumulator to stack
D. Clears accumulator
Correct Answer: A

Q20. RET returns from:

A. Subroutine
B. Interrupt only
C. Halt
D. Reset
Correct Answer: A

Q21. PUSH B pushes:

A. BC pair onto stack
B. B only
C. Accumulator only
D. PC only
Correct Answer: A

Q22. POP H pops:

A. Top of stack to HL
B. HL to stack
C. H only
D. Hold state
Correct Answer: A

Q23. XCHG exchanges:

A. HL with DE
B. A with B
C. PC with SP
D. H with L
Correct Answer: A

Q24. 8085 address bus is:

A. 8-bit
B. 16-bit
C. 20-bit
D. 32-bit
Correct Answer: B

Q25. 8085 data bus is:

A. 8-bit
B. 16-bit
C. 32-bit
D. 4-bit
Correct Answer: A

Q26. 8085 can address memory:

A. 64KB
B. 4KB
C. 1MB
D. 16KB
Correct Answer: A

Q27. Which addressing mode is MVI A,05H?

A. Immediate
B. Direct
C. Register indirect
D. Implied
Correct Answer: A

Q28. Which addressing mode is LDA 2000H?

A. Direct
B. Immediate
C. Register
D. Implicit
Correct Answer: A

Q29. Which is register indirect?

A. MOV A,M
B. MVI A,05
C. LDA 2000
D. CMA
Correct Answer: A

Q30. CMA instruction:

A. Complement accumulator
B. Clear memory
C. Compare accumulator
D. Call memory
Correct Answer: A

Q31. Main function of assembler is to:

A. Translate assembly to machine code
B. Translate high-level to assembly
C. Load program into memory
D. Debug program
Correct Answer: A

Q32. Two-pass assembler performs:

A. Pass1 builds symbol table, Pass2 generates code
B. Both passes generate code
C. Pass1 executes, Pass2 assembles
D. Pass1 loads, Pass2 links
Correct Answer: A

Q33. In Pass1 of two-pass assembler, MOT stands for:

A. Mnemonic Operation Table
B. Memory Operation Table
C. Machine Output Table
D. Macro Operation Table
Correct Answer: A

Q34. POT stands for:

A. Pseudo-Operation Table
B. Program Operation Table
C. Pass Operation Table
D. Print Operation Table
Correct Answer: A

Q35. Symbol Table stores:

A. Symbols and addresses/values
B. Only opcodes
C. Only macros
D. Object code
Correct Answer: A

Q36. Literal Table stores:

A. Literals and addresses
B. Symbols only
C. Macros only
D. Errors
Correct Answer: A

Q37. Forward reference problem is solved by:

A. Two-pass assembler
B. Single-pass with backpatching only always fails
C. Compiler
D. Loader
Correct Answer: A

Q38. Single-pass assembler handles forward reference by:

A. Backpatching / fix-ups
B. Ignoring
C. Two symbol tables
D. Halting
Correct Answer: A

Q39. Pseudo-ops are:

A. Assembler directives, no machine code
B. Machine instructions
C. Macros
D. Executable instructions
Correct Answer: A

Q40. Example of pseudo-op:

A. START, END, EQU, DS
B. ADD, SUB
C. LDA, STA
D. JMP, CALL
Correct Answer: A

Q41. EQU pseudo-op is used to:

A. Equate symbol to value
B. Equalize registers
C. End program
D. Erase memory
Correct Answer: A

Q42. ORIGIN/START directive specifies:

A. Starting address
B. End address
C. Stack size
D. Macro start
Correct Answer: A

Q43. Base table in assembler is used for:

A. Relocation
B. Macro expansion
C. Lexical analysis
D. Debugging
Correct Answer: A

Q44. Cross assembler is one that:

A. Runs on one machine, generates code for another
B. Runs cross-platform UI
C. Assembles two programs
D. Needs two passes on cross memory
Correct Answer: A

Q45. One-pass assembler is suitable for:

A. Systems with limited memory / single scan
B. Only high-level languages
C. No symbols
D. No literals
Correct Answer: A

Q46. Pass2 of assembler uses:

A. Symbol/Literal tables + MOT/POT to generate object code
B. Only source
C. Only object
D. Linker output
Correct Answer: A

Q47. Intermediate file between Pass1 and Pass2 contains:

A. Intermediate code with addresses
B. Final executable
C. High-level code
D. Debug info only
Correct Answer: A

Q48. Load-and-go assembler:

A. Generates code directly in memory for execution, no object file
B. Needs loader always
C. Generates high-level code
D. Requires linkage editor
Correct Answer: A

Q49. Which error is detected in Pass1?

A. Undefined symbol / duplicate symbol
B. Runtime divide by zero
C. Logic error
D. Deadlock
Correct Answer: A

Q50. Which error is detected in Pass2?

A. Invalid opcode / addressability
B. Syntax of macro definition always
C. Deadlock
D. Page fault
Correct Answer: A

Q51. Macro definition uses directives:

A. MACRO and MEND
B. START and END
C. CALL and RETURN
D. PUSH and POP
Correct Answer: A

Q52. Macro call causes:

A. Expansion
B. Condensation
C. Compilation
D. Loading
Correct Answer: A

Q53. Macro processor expands:

A. Before assembly
B. After loading
C. During execution
D. After linking
Correct Answer: A

Q54. Positional parameters in macros are distinguished by:

A. Position
B. Keyword name
C. Type
D. Value range
Correct Answer: A

Q55. Keyword parameters are distinguished by:

A. Name
B. Position only
C. Address
D. Opcode
Correct Answer: A

Q56. Nested macro calls mean:

A. Macro calls another macro
B. Macro calls itself only
C. Two assemblers
D. Recursive loader
Correct Answer: A

Q57. MDT stands for:

A. Macro Definition Table
B. Memory Definition Table
C. Mnemonic Definition Table
D. Machine Data Table
Correct Answer: A

Q58. MNT stands for:

A. Macro Name Table
B. Memory Name Table
C. Machine Name Table
D. Mnemonic Name Table
Correct Answer: A

Q59. ALA stands for in macro processing:

A. Argument List Array
B. Address List Array
C. Assembler List Array
D. Allocation Array
Correct Answer: A

Q60. Conditional macro expansion uses:

A. IF/ELSE/ENDIF in macro
B. JMP only
C. Loader condition
D. Interrupt
Correct Answer: A

Q61. Function of loader is to:

A. Load program into memory for execution
B. Translate source
C. Edit text
D. Optimize code
Correct Answer: A

Q62. Absolute loader loads program at:

A. Fixed address, no relocation
B. Any address with relocation
C. Virtual address only
D. Cache only
Correct Answer: A

Q63. Relocating loader allows:

A. Loading at different addresses via relocation
B. Only fixed address
C. No address translation
D. Only overlays
Correct Answer: A

Q64. Relocation is needed when:

A. Program loaded at address different from assembled
B. Program has no symbols
C. No linker used
D. Macro used
Correct Answer: A

Q65. Relocatability is ability to:

A. Execute from different memory locations
B. Relocate files on disk
C. Move compiler
D. Reboot OS
Correct Answer: A

Q66. Linker combines:

A. Object modules into executable
B. Source files into assembly
C. Macros into tables
D. Pages into frames
Correct Answer: A

Q67. Linkage editing includes:

A. Resolving external symbols and address adjustment
B. Editing source text
C. Compiling functions
D. Debugging
Correct Answer: A

Q68. External symbol resolution is done by:

A. Linker
B. Text editor
C. Debugger
D. Lexer
Correct Answer: A

Q69. Static linking occurs:

A. Before execution
B. During execution on demand
C. At runtime by OS
D. In CPU
Correct Answer: A

Q70. Dynamic linking occurs:

A. At runtime / load time
B. During assembly
C. During macro expansion
D. During editing
Correct Answer: A

Q71. Binder is another name for:

A. Linker
B. Loader
C. Assembler
D. Compiler
Correct Answer: A

Q72. Overlay structure is used when:

A. Program larger than memory
B. Program smaller than memory
C. No I/O
D. No linker
Correct Answer: A

Q73. Bootstrap loader is:

A. Small program to load OS kernel
B. Compiler starter
C. Macro loader
D. Text formatter
Correct Answer: A

Q74. Text editor provides:

A. Create/modify source files
B. Execute programs
C. Allocate memory
D. Schedule CPU
Correct Answer: A

Q75. Debugger allows:

A. Breakpoints, step execution, watch variables
B. Only editing
C. Only compiling
D. Only loading
Correct Answer: A

Q76. Compiler translates:

A. High-level to machine/object code in one go before execution
B. Line by line during execution
C. Assembly to high-level
D. Object to source
Correct Answer: A

Q77. Interpreter:

A. Executes source line by line without producing object file
B. Produces exe before execution
C. Links modules
D. Loads OS
Correct Answer: A

Q78. Compiled programs generally:

A. Have faster execution
B. Have slower execution
C. No optimization possible
D. Always portable
Correct Answer: A

Q79. Bootstrap in compilers means:

A. Writing compiler in its own language using minimal initial compiler
B. Booting OS
C. Loading assembler
D. Starting debugger
Correct Answer: A

Q80. T-diagrams represent:

A. Compiler bootstrapping relationships
B. Memory layout
C. Parse trees
D. Deadlock
Correct Answer: A

Q81. Cross-compiler generates code for:

A. Different machine than host
B. Same machine only
C. Human reading
D. Interpreter only
Correct Answer: A

Q82. Decompiler converts:

A. Machine/object to high-level
B. High-level to machine
C. Assembly to macro
D. Text to binary
Correct Answer: A

Q83. First phase of compilation is:

A. Lexical analysis
B. Code generation
C. Optimization
D. Syntax analysis
Correct Answer: A

Q84. Last phase of compilation is:

A. Code generation
B. Lexical analysis
C. Parsing
D. Semantic analysis
Correct Answer: A

Q85. Correct order of phases is:

A. Lexical-Syntax-Semantic-ICG-Optimization-CodeGen
B. Syntax-Lexical-CodeGen-Optimization
C. Semantic-Lexical-Syntax-CodeGen
D. CodeGen-Lexical-Syntax-Semantic
Correct Answer: A

Q86. Lexical analysis converts:

A. Source characters to tokens
B. Tokens to parse tree
C. Parse tree to IC
D. IC to object
Correct Answer: A

Q87. Tool Lex generates:

A. Lexical analyzer
B. Parser
C. Code generator
D. Loader
Correct Answer: A

Q88. Input to Lex is:

A. Regular expressions / patterns with actions
B. CFG
C. Quadruples
D. Object code
Correct Answer: A

Q89. Output of Lex is typically:

A. C program for scanner (lex.yy.c)
B. Assembly file
C. EXE
D. Parse table
Correct Answer: A

Q90. Token is:

A. Sequence of characters with collective meaning (identifier, keyword)
B. Single character only
C. Parse tree
D. Grammar rule
Correct Answer: A

Q91. Lexeme is:

A. Actual character sequence forming token
B. Token type
C. Grammar
D. Symbol table entry only
Correct Answer: A

Q92. Pattern describes:

A. Rule for lexemes of a token (via regex)
B. Parse tree shape
C. Object layout
D. Memory map
Correct Answer: A

Q93. Which handles whitespace/comments?

A. Lexical analyzer (usually skips)
B. Code generator
C. Optimizer
D. Linker
Correct Answer: A

Q94. Longest match rule in Lex means:

A. Longest possible lexeme preferred
B. Shortest preferred
C. First rule always
D. Random
Correct Answer: A

Q95. Symbol table is used in:

A. All phases
B. Only code gen
C. Only linking
D. Only editing
Correct Answer: A

Q96. CFG stands for:

A. Context-Free Grammar
B. Combined Formal Grammar
C. Code Flow Graph
D. Control Flow Grammar
Correct Answer: A

Q97. CFG consists of:

A. Terminals, Non-terminals, Productions, Start symbol
B. Only terminals
C. Only tokens
D. Frames and pages
Correct Answer: A

Q98. Parse tree shows:

A. Derivation structure of string per grammar
B. Memory allocation
C. Symbol table
D. Flowchart
Correct Answer: A

Q99. Leftmost derivation expands:

A. Leftmost non-terminal first
B. Rightmost first
C. Random
D. All at once
Correct Answer: A

Q100. Rightmost derivation expands:

A. Rightmost non-terminal first
B. Leftmost first
C. Terminals first
D. All at once
Correct Answer: A

Q101. Rightmost derivation in reverse corresponds to:

A. Bottom-up parsing
B. Top-down parsing
C. Lexical analysis
D. Code optimization
Correct Answer: A

Q102. Leftmost derivation corresponds to:

A. Top-down parsing
B. Bottom-up parsing
C. Code gen
D. Linking
Correct Answer: A

Q103. Ambiguous grammar has:

A. More than one parse tree for a string
B. No parse tree
C. Exactly one leftmost derivation
D. No terminals
Correct Answer: A

Q104. Which grammar is suitable for deterministic parsing?

A. Unambiguous
B. Ambiguous
C. Regular only never CFG
D. Type-0 only
Correct Answer: A

Q105. Sentence of grammar is:

A. String of terminals derivable from start
B. Any production
C. Any non-terminal
D. Token stream only
Correct Answer: A

Q106. Sentential form may contain:

A. Terminals and non-terminals
B. Only terminals
C. Only start symbol
D. Object code
Correct Answer: A

Q107. Handle in bottom-up parsing is:

A. Substring matching RHS to reduce
B. Leftmost token
C. Symbol table handle
D. File handle
Correct Answer: A

Q108. Shift-reduce parsing actions are:

A. Shift, Reduce, Accept, Error
B. Push, Pop only
C. Lex, Yacc
D. Load, Store
Correct Answer: A

Q109. Shift means:

A. Push input symbol onto stack
B. Pop stack to output
C. Replace RHS with LHS
D. Discard stack
Correct Answer: A

Q110. Reduce means:

A. Replace RHS on stack with LHS
B. Push input
C. Accept input
D. Delete input
Correct Answer: A

Q111. Operator precedence parsing handles:

A. Operators with precedence/associativity, no consecutive non-terminals
B. All CFGs
C. Left recursion automatically
D. Semantic actions only
Correct Answer: A

Q112. Limitation of operator precedence:

A. Cannot handle epsilon / consecutive non-terminals well
B. Too powerful for all grammars
C. Needs LR table always
D. Cannot handle expressions
Correct Answer: A

Q113. LR parsing is:

A. Bottom-up, L=left-to-right scan, R=rightmost derivation reverse
B. Top-down
C. Left-recursive only
D. Lexical only
Correct Answer: A

Q114. LR(0) items have:

A. No lookahead
B. 1 lookahead
C. k lookaheads always >1
D. Infinite lookahead
Correct Answer: A

Q115. SLR uses:

A. FOLLOW sets for reduce decisions
B. FIRST only
C. No sets
D. Precedence only
Correct Answer: A

Q116. CLR(1)/LALR differ in:

A. Power and table size; LALR merges states
B. Both same power as SLR
C. Top-down vs bottom-up
D. Lex vs Yacc
Correct Answer: A

Q117. Most powerful among SLR, LALR, CLR:

A. Canonical LR
B. SLR
C. Operator precedence
D. Recursive descent
Correct Answer: A

Q118. YACC generates:

A. LALR parser (bottom-up)
B. Lexical analyzer
C. Top-down parser only
D. Code optimizer
Correct Answer: A

Q119. Input to YACC is:

A. CFG with actions/semantic routines
B. Regular expressions only
C. Object code
D. Quadruples
Correct Answer: A

Q120. YACC output is typically:

A. y.tab.c parser in C
B. lex.yy.c scanner
C. EXE
D. Symbol table only
Correct Answer: A

Q121. YACC works with Lex by:

A. Parser calls yylex() for tokens
B. Lex calls yyparse()
C. No interaction
D. Linker merges
Correct Answer: A

Q122. Conflict when both shift and reduce possible is:

A. Shift-reduce conflict
B. Reduce-reduce resolved automatically
C. Lex error
D. Link error
Correct Answer: A

Q123. Reduce-reduce conflict means:

A. Two reductions possible, needs grammar revision
B. Shift and reduce possible
C. No action
D. Accept
Correct Answer: A

Q124. Dangling else causes:

A. Shift-reduce conflict
B. Reduce-reduce conflict
C. Lex error
D. No conflict
Correct Answer: A

Q125. Viable prefix in LR parsing is:

A. Prefix of sentential form that can appear on stack
B. Any prefix of input
C. Suffix of grammar
D. Token prefix
Correct Answer: A

Q126. Augmented grammar adds:

A. New start S'->S
B. New terminal
C. Epsilon removal
D. Left factoring
Correct Answer: A

Q127. Top-down parsing builds parse tree from:

A. Root to leaves (start symbol)
B. Leaves to root
C. Middle out
D. Random
Correct Answer: A

Q128. Challenge for top-down:

A. Left recursion and backtracking
B. Shift-reduce conflicts
C. Relocation
D. Paging
Correct Answer: A

Q129. Left recursion example:

A. A->Aalph|beta
B. A->betaA|beta
C. A->aB
D. A->epsilon only
Correct Answer: A

Q130. Left recursion removal for A->Aalph|beta gives:

A. A->betaA', A'->alphA'|epsilon
B. A->alphaA|beta
C. A->beta only
D. A->Abeta
Correct Answer: A

Q131. Indirect left recursion is removed by:

A. Ordering and substitution algorithm
B. Direct substitution only one step
C. Left factoring
D. Precedence
Correct Answer: A

Q132. Recursive descent parsing is:

A. Top-down with recursive procedures for non-terminals
B. Bottom-up table driven
C. Operator precedence
D. LR
Correct Answer: A

Q133. Predictive parsing uses:

A. Lookahead to choose production without backtracking
B. Backtracking always
C. Stack of handles
D. Precedence table only
Correct Answer: A

Q134. LL(1) means:

A. Left-to-right, Leftmost derivation, 1 lookahead
B. Left-to-right, Left recursion, 1 pass
C. Leftmost rightmost 1 grammar
D. Lex lookahead 1
Correct Answer: A

Q135. FIRST(alpha) is set of:

A. Terminals beginning strings derived from alpha
B. Terminals following alpha
C. Non-terminals of alpha
D. Last symbols
Correct Answer: A

Q136. FOLLOW(A) is set of:

A. Terminals that can appear immediately after A
B. First of A
C. All productions of A
D. Start symbols
Correct Answer: A

Q137. LL(1) table M[A,a] uses:

A. FIRST and FOLLOW
B. LR items
C. Precedence
D. Symbol table
Correct Answer: A

Q138. Condition for LL(1) (no conflict):

A. Disjoint SELECT sets / FIRST-FOLLOW disjoint for epsilon-productions
B. Ambiguous grammar
C. Left recursive grammar
D. No FIRST needed
Correct Answer: A

Q139. Left factoring removes:

A. Common prefixes to enable predictive parsing
B. Left recursion
C. Ambiguity fully
D. Epsilon
Correct Answer: A

Q140. Example left factored: A->alpha beta1 | alpha beta2 becomes:

A. A->alpha A', A'->beta1|beta2
B. A->beta A'
C. A->alpha only
D. A->epsilon
Correct Answer: A

Q141. Backtracking in top-down is avoided by:

A. Predictive LL(1) tables
B. More backtracking
C. LR tables
D. Macros
Correct Answer: A

Q142. Which is top-down?

A. Recursive descent, LL, predictive
B. SLR, LALR, CLR
C. Operator precedence
D. Shift-reduce
Correct Answer: A

Q143. Which is bottom-up?

A. LR, SLR, LALR, operator precedence
B. LL, recursive descent
C. Predictive
D. Lex
Correct Answer: A

Q144. Semantic analysis checks:

A. Type consistency, declarations, scope
B. Token spelling
C. Whitespace
D. Address relocation
Correct Answer: A

Q145. Syntax-directed translation attaches:

A. Semantic actions/attributes to productions
B. Tokens to Lex only
C. Frames to pages
D. Macros to loader
Correct Answer: A

Q146. S-attributed definitions use:

A. Synthesized attributes only
B. Inherited only
C. Both unrestricted
D. No attributes
Correct Answer: A

Q147. L-attributed definitions allow:

A. Left-to-right evaluation (synthesized + inherited with restrictions)
B. Right-to-left only
C. No inherited
D. Random
Correct Answer: A

Q148. Three-address code has at most:

A. Three addresses (x=y op z)
B. One address
C. Five addresses
D. No addresses
Correct Answer: A

Q149. Quadruple fields are:

A. op, arg1, arg2, result
B. op, arg, result only
C. Three fields only
D. Two fields
Correct Answer: A

Q150. Triple fields are:

A. op, arg1, arg2 (result implicit by position)
B. op, arg1, arg2, result
C. Only op
D. Label only
Correct Answer: A

Q151. Indirect triples use:

A. Pointer list to triples
B. Direct result field
C. Quadruples
D. Symbol table only
Correct Answer: A

Q152. Advantage of triples over quadruples:

A. Saves space (no explicit temp names)
B. Faster code gen always
C. Easier optimization movement
D. No disadvantage
Correct Answer: A

Q153. Disadvantage of triples:

A. Moving code requires pointer updates / less flexible
B. Uses more space
C. Cannot represent expressions
D. Needs more temps
Correct Answer: A

Q154. Example a=b+c in quadruple:

A. (+,b,c,t1) then (=,t1,-,a)
B. (+,b,c)
C. (=,a,b)
D. (+,a,b,c,d,e)
Correct Answer: A

Q155. DAG for basic block helps in:

A. Detecting common subexpressions
B. Parsing
C. Lexing
D. Linking
Correct Answer: A

Q156. Basic block is:

A. Straight-line code with single entry/exit
B. Block with jumps inside
C. Function
D. Macro
Correct Answer: A

Q157. Flow graph nodes are:

A. Basic blocks
B. Tokens
C. Symbols
D. Frames
Correct Answer: A

Q158. Next-use information is used for:

A. Register allocation
B. Parsing
C. Lexing
D. Loading
Correct Answer: A

Q159. Peephole optimization examines:

A. Small window of instructions
B. Whole program
C. Grammar
D. Symbol table only
Correct Answer: A

Q160. Examples of peephole:

A. Redundant loads, strength reduction, dead code elimination locally
B. LR table compaction
C. Macro expansion
D. Deadlock avoidance
Correct Answer: A

Q161. Loop optimization includes:

A. Invariant hoisting, induction variable elimination, unrolling
B. Left factoring
C. Backpatching
D. Paging
Correct Answer: A

Q162. Common subexpression elimination is:

A. Global/local optimization avoiding recomputation
B. Parsing technique
C. Macro technique
D. Loading technique
Correct Answer: A

Q163. Copy propagation replaces:

A. Use of copy variable with original
B. Constants with variables
C. Loops with recursion
D. Tokens with lexemes
Correct Answer: A

Q164. Constant folding evaluates:

A. Constants at compile time
B. Variables at runtime
C. Grammar at parse time
D. Addresses at load time
Correct Answer: A

Q165. Strength reduction replaces:

A. Expensive ops with cheaper (e.g., *2 by shift)
B. Cheap with expensive
C. Loads with stores
D. Branches with calls
Correct Answer: A

Q166. Dead code elimination removes:

A. Unreachable / unused computation
B. Live code
C. Entry block
D. Symbol table
Correct Answer: A

Q167. Code motion moves:

A. Loop-invariant code out of loop
B. Code into loop
C. Tokens
D. Macros
Correct Answer: A

Q168. Register allocation aims to:

A. Minimize loads/stores, keep values in registers
B. Maximize memory access
C. Increase spills
D. Avoid registers
Correct Answer: A

Q169. Graph coloring is used for:

A. Register allocation
B. Parsing
C. Lexing
D. Scheduling
Correct Answer: A

Q170. Spilling occurs when:

A. Insufficient registers, value stored to memory
B. Sufficient registers
C. No interference
D. No variables
Correct Answer: A

Q171. Activation record contains:

A. Return address, locals, params, dynamic/static links
B. Only opcodes
C. Only tokens
D. Only macros
Correct Answer: A

Q172. Static link points to:

A. Enclosing scope activation for non-locals
B. Caller activation
C. Heap top
D. Code segment
Correct Answer: A

Q173. Dynamic link points to:

A. Caller activation record
B. Global data
C. Heap
D. Symbol table
Correct Answer: A

Q174. Parameter passing by value:

A. Copies actual to formal, callee changes not reflected
B. Reflects changes
C. Passes address
D. Passes name textually
Correct Answer: A

Q175. Call by reference passes:

A. Address, changes reflected
B. Copy only
C. Name string
D. Value only
Correct Answer: A

Q176. Display is used for:

A. Fast access to non-locals in nested scopes
B. Monitor display
C. Lex display
D. Loader display
Correct Answer: A

Q177. Heap allocation is needed for:

A. Dynamic data / recursion with variable lifetime
B. Static locals only
C. Compile-time constants only
D. Register allocation
Correct Answer: A

Q178. Duff's / Sethi-Ullman relates to:

A. Code generation / register number ordering
B. Parsing
C. Lexing
D. Linking
Correct Answer: A

Q179. Object code generation from IR must consider:

A. Target instruction selection, addressing modes
B. Grammar ambiguity only
C. Macro names only
D. Editor commands
Correct Answer: A

Q180. Cross-optimization concerns:

A. Machine-dependent vs independent levels
B. Only peephole
C. Only loops
D. Only macros
Correct Answer: A

Q181. Machine-independent optimization acts on:

A. Intermediate code
B. Target assembly only
C. Source text layout
D. Object layout on disk
Correct Answer: A

Q182. Machine-dependent optimization acts on:

A. Target code (registers, pipelines)
B. Tokens
C. Grammar
D. Symbol names
Correct Answer: A

Q183. Data-flow analysis includes:

A. Reaching definitions, live variables, available expressions
B. FIRST/FOLLOW only
C. Symbol table hashing only
D. Paging only
Correct Answer: A

Q184. Dominators in flow graphs are used for:

A. Finding loops / natural loops
B. Lexing
C. Macro expansion
D. Relocation
Correct Answer: A

Q185. Back edge (v->u where u dominates v) indicates:

A. Loop
B. Straight code
C. Dead code
D. Entry
Correct Answer: A

Q186. Which parsing is most general but slow?

A. CYK / Earley for general CFG
B. LL(1)
C. LR(1)
D. Operator precedence
Correct Answer: A

Q187. Handle pruning corresponds to:

A. Bottom-up parsing steps
B. Top-down expansion
C. Lex scanning
D. Optimization
Correct Answer: A

Q188. Kernel vs non-kernel LR items:

A. Kernel includes initial item and those with dot not at left except S'->.S
B. Same
C. Kernel is for LL
D. Non-kernel for Lex
Correct Answer: A

Q189. SLR table construction uses:

A. LR(0) automaton + FOLLOW
B. Canonical collection + lookaheads
C. Precedence matrix only
D. FIRST only
Correct Answer: A

Q190. LALR table size vs CLR:

A. Smaller (merged) but slightly less powerful
B. Larger
C. Same power always
D. Unrelated
Correct Answer: A

Q191. Error recovery in LR often uses:

A. Error productions / panic mode with synchronizing tokens
B. Ignoring all input
C. Reboot
D. Relocation
Correct Answer: A

Q192. Panic-mode recovery skips to:

A. Synchronizing token (e.g., ;, end)
B. Start of file always
C. End of program halt
D. Random token
Correct Answer: A

Q193. Phrase-level recovery:

A. Local correction (insert/delete) on error
B. Skip whole program
C. Reload OS
D. Reassemble
Correct Answer: A

Q194. Bootstrapping a compiler for new machine needs:

A. Cross-compiler then self-hosting
B. Only interpreter
C. Only loader
D. Only editor
Correct Answer: A

Q195. Self-compilation validates:

A. Compiler correctness via fixed point
B. Loader speed
C. Editor features
D. Macro power
Correct Answer: A

Q196. One-address machine uses:

A. Accumulator (e.g., ADD M)
B. Stack
C. Three registers
D. No addresses
Correct Answer: A

Q197. Zero-address machine uses:

A. Stack (PUSH/POP, ADD with no address)
B. Accumulator
C. Three addresses
D. Two addresses
Correct Answer: A

Q198. Assembly directive DB defines:

A. Byte(s) of data
B. Double word always 32-bit
C. Debugger breakpoint
D. Data bus
Correct Answer: A

Q199. DAD instruction in 8085:

A. Add HL with register pair (16-bit add)
B. Add 8-bit
C. Decimal adjust
D. Decrement
Correct Answer: A

Q200. DAA instruction is used for:

A. Decimal adjust after BCD addition
B. Data access
C. Direct addressing
D. Dynamic allocation
Correct Answer: A

ANSWER KEY — Topic 9: System Software & Compilers

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1-A 2-B 3-A 4-A 5-B 6-B 7-B 8-A 9-A 10-A
11-A 12-A 13-A 14-A 15-A 16-A 17-A 18-A 19-A 20-A
21-A 22-A 23-A 24-B 25-A 26-A 27-A 28-A 29-A 30-A
31-A 32-A 33-A 34-A 35-A 36-A 37-A 38-A 39-A 40-A
41-A 42-A 43-A 44-A 45-A 46-A 47-A 48-A 49-A 50-A
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181-A 182-A 183-A 184-A 185-A 186-A 187-A 188-A 189-A 190-A
191-A 192-A 193-A 194-A 195-A 196-A 197-A 198-A 199-A 200-A
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Topic 10: Operating Systems & Unix

Syllabus: *OS functions, multiprogramming/multiprocessing/multitasking, virtual memory/paging/fragmentation, mutual exclusion/critical regions/lock-unlock, CPU/IO/resource scheduling, deadlock & Banker's, UNIX file system/process/bourne shell/variables/command-line, filters (pr/head/tail/cut/paste/sort/uniq/tr/join/grep/egrep/fgrep/sed/awk), syscalls (create/open/close/read/write/lseek/link/unlink/stat/fstat/umask/chmod/exec/fork/wait/system).*

Q1. Operating System primary function is:

A. Resource management and coordination of hardware
B. Only user interface
C. Only memory management
D. Only file handling
Correct Answer: A

Q2. Multiprogramming aims to:

A. Keep CPU busy by having multiple programs in memory
B. Run only one program
C. Reduce memory to zero
D. Increase I/O wait
Correct Answer: A

Q3. Multiprocessing uses:

A. Multiple processors/cores
B. Single processor
C. No CPU
D. Only I/O devices
Correct Answer: A

Q4. Multitasking allows:

A. Multiple tasks to share CPU time
B. One task at a time
C. No task switching
D. Only batch processing
Correct Answer: A

Q5. Virtual memory provides:

A. Illusion of larger main memory than physically available
B. Only disk storage
C. Only faster CPU
D. No memory at all
Correct Answer: A

Q6. Paging divides memory into:

A. Fixed-size blocks called frames/pages
B. Variable-size segments
C. Only one block
D. No blocks
Correct Answer: A

Q7. Fragmentation in memory occurs when:

A. Memory is wasted in small unusable chunks
B. Memory is perfectly organized
C. No memory exists
D. Only disk fragmentation
Correct Answer: A

Q8. Internal fragmentation happens when:

A. Allocated memory is slightly larger than needed
B. Memory is perfectly allocated
C. No memory assigned
D. Only external
Correct Answer: A

Q9. External fragmentation happens when:

A. Free memory is scattered in small blocks
B. Memory is contiguous
C. No free memory
D. Only internal
Correct Answer: A

Q10. Mutual exclusion requires:

A. Only one process accesses critical resource at a time
B. All processes access simultaneously
C. No resource sharing
D. Only one thread
Correct Answer: A

Q11. Critical region is:

A. Section of code accessing shared resources
B. Part of CPU only
C. Memory address only
D. I/O port only
Correct Answer: A

Q12. Lock operation:

A. Acquires a mutex/semaphore
B. Releases a mutex/semaphore
C. Deletes a process
D. Creates a thread
Correct Answer: A

Q13. Unlock operation:

A. Releases a mutex/semaphore
B. Acquires a mutex/semaphore
C. Locks a file
D. Deletes a thread
Correct Answer: A

Q14. CPU scheduling aims to:

A. Maximize CPU utilization and throughput
B. Minimize memory only
C. Only increase I/O
D. Reduce processes to zero
Correct Answer: A

Q15. Shortest Job First (SJF) scheduling:

A. Selects process with smallest burst time
B. Selects first arrived
C. Selects longest
D. Selects random
Correct Answer: A

Q16. Round Robin scheduling uses:

A. Time quantum for each process
B. Priority only
C. First come only
D. Shortest burst
Correct Answer: A

Q17. Priority scheduling may cause:

A. Starvation of low-priority processes
B. No starvation ever
C. Only fast execution
D. CPU idle
Correct Answer: A

Q18. Deadlock occurs when:

A. Processes are blocked waiting for each other
B. CPU is fast
C. Memory is full
D. Only one process
Correct Answer: A

Q19. Necessary conditions for deadlock:

A. Mutual exclusion, hold and wait, no preemption, circular wait
B. Only mutual exclusion
C. Only circular wait
D. Only preemption
Correct Answer: A

Q20. Banker's Algorithm is used for:

A. Deadlock avoidance by checking safe states
B. Deadlock detection only
C. Process creation
D. Memory allocation
Correct Answer: A

Q21. Resource allocation graph is used for:

A. Detecting and avoiding deadlock
B. Only scheduling
C. Only memory
D. Only I/O
Correct Answer: A

Q22. UNIX file system is hierarchical:

A. Tree structure rooted at /
B. Flat structure
C. Circular structure
D. No structure
Correct Answer: A

Q23. In UNIX, everything is treated as:

A. A file
B. Only a directory
C. Only a process
D. Only a device
Correct Answer: A

Q24. Shell is:

A. Command interpreter that mediates between user and OS kernel
B. Part of CPU
C. Only a file
D. A hardware component
Correct Answer: A

Q25. Bourne Shell is:

A. Original UNIX command-line interpreter
B. Windows shell
C. Graphical interface
D. A compiler
Correct Answer: A

Q26. UNIX variable PATH specifies:

A. Directories to search for commands
B. User home directory
C. Current directory only
D. File permissions
Correct Answer: A

Q27. In UNIX, $HOME is a:

A. Shell variable pointing to home directory
B. System error code
C. File name
D. Shell command
Correct Answer: A

Q28. Command-line arguments in UNIX are:

A. Passed to commands/shell scripts
B. Ignored by shell
C. Only for system calls
D. Not used
Correct Answer: A

Q29. UNIX pipe (|) connects:

A. Output of one command to input of another
B. Two files
C. Two users
D. Two CPUs
Correct Answer: A

Q30. UNIX filter processes:

A. Input and produces output
B. Only stores files
C. Only manages processes
D. Only creates users
Correct Answer: A

Q31. pr command in UNIX:

A. Prints file contents
B. Deletes files
C. Copies files
D. Compiles files
Correct Answer: A

Q32. head command in UNIX:

A. Displays first few lines of file
B. Last few lines
C. Middle lines
D. All lines
Correct Answer: A

Q33. tail command in UNIX:

A. Displays last few lines of file
B. First few lines
C. Middle lines
D. All lines
Correct Answer: A

Q34. cut command in UNIX:

A. Extracts columns/fields from lines
B. Combines files
C. Sorts data
D. Deletes files
Correct Answer: A

Q35. paste command in UNIX:

A. Merges lines of files side by side
B. Sorts files
C. Deletes lines
D. Prints only
Correct Answer: A

Q36. sort command in UNIX:

A. Sorts lines of a file
B. Copies files
C. Deletes duplicates
D. Prints heads
Correct Answer: A

Q37. uniq command in UNIX:

A. Filters out repeated adjacent lines
B. Sorts all lines
C. Joins files
D. Translates characters
Correct Answer: A

Q38. tr command in UNIX:

A. Translates or deletes characters
B. Sorts files
C. Cuts columns
D. Prints files
Correct Answer: A

Q39. join command in UNIX:

A. Joins lines of two files on a common field
B. Sorts files
C. Prints lines
D. Deletes lines
Correct Answer: A

Q40. grep command in UNIX:

A. Searches for patterns in files
B. Joins files
C. Sorts lines
D. Deletes files
Correct Answer: A

Q41. egrep is:

A. Extended grep supporting additional regex metacharacters
B. Graphical grep
C. Only searches directories
D. A compiler
Correct Answer: A

Q42. fgrep is:

A. Fixed-string grep (fast pattern matching without regex)
B. File grep
C. Floating grep
D. Fast compiler
Correct Answer: A

Q43. sed is a:

A. Stream editor for filtering/transformation
B. System editor
C. Shell
D. Compiler
Correct Answer: A

Q44. awk is a:

A. Pattern scanning and processing language
B. Shell command
C. File system
D. Compiler
Correct Answer: A

Q45. System call create() is used to:

A. Create a new file
B. Create a directory
C. Create a process
D. Delete a file
Correct Answer: A

Q46. System call open() is used to:

A. Open an existing file
B. Create a file
C. Delete a file
D. Read only
Correct Answer: A

Q47. System call close() is used to:

A. Close an open file descriptor
B. Open a file
C. Create a file
D. Delete a file
Correct Answer: A

Q48. System call read() is used to:

A. Read data from a file descriptor
B. Write data
C. Delete data
D. Close a file
Correct Answer: A

Q49. System call write() is used to:

A. Write data to a file descriptor
B. Read data
C. Close a file
D. Open a file
Correct Answer: A

Q50. System call lseek() is used to:

A. Reposition the file offset
B. Create a file
C. Delete a file
D. Open a file
Correct Answer: A

Q51. System call link() is used to:

A. Create a new directory entry (hard link) for a file
B. Delete a file
C. Copy a file
D. Move a file
Correct Answer: A

Q52. System call unlink() is used to:

A. Remove a directory entry (delete a file)
B. Create a file
C. Link files
D. Open a file
Correct Answer: A

Q53. System call stat() is used to:

A. Get file status/metadata
B. Create a file
C. Delete a file
D. Open a file
Correct Answer: A

Q54. System call fstat() is used to:

A. Get file status from an open file descriptor
B. Create a file
C. Delete a file
D. Close a file
Correct Answer: A

Q55. System call umask() is used to:

A. Set file mode creation mask
B. Set file size
C. Delete a file
D. Read a file
Correct Answer: A

Q56. System call chmod() is used to:

A. Change file permissions
B. Change file owner
C. Change file size
D. Move a file
Correct Answer: A

Q57. System call exec() is used to:

A. Replace current process image with a new program
B. Create a new file
C. Delete a file
D. Copy data
Correct Answer: A

Q58. System call fork() is used to:

A. Create a new child process
B. Delete a process
C. Wait for a process
D. Read a file
Correct Answer: A

Q59. System call wait() is used to:

A. Wait for a child process to terminate
B. Create a process
C. Read a file
D. Write a file
Correct Answer: A

Q60. System call system() is used to:

A. Execute a shell command from a program
B. Create a file
C. Delete a file
D. Copy data
Correct Answer: A

Q61. Process in UNIX is identified by:

A. Process ID (PID)
B. File name
C. User name only
D. Directory name
Correct Answer: A

Q62. Parent process in UNIX:

A. Has a child process
B. Is a kernel thread
C. Is a system call
D. Has no PID
Correct Answer: A

Q63. init process in UNIX has PID:

A. 0 or 1
B. 100
C. 999
D. Any random
Correct Answer: A

Q64. Daemon in UNIX is:

A. Background process running without user interaction
B. Foreground process
C. Shell command
D. System file
Correct Answer: A

Q65. Signal in UNIX is used to:

A. Notify a process of an event
B. Create a file
C. Delete a process
D. Read data
Correct Answer: A

Q66. SIGKILL signal:

A. Terminates a process immediately
B. Stops a process
C. Ignored by process
D. Pauses process
Correct Answer: A

Q67. SIGINT signal:

A. Interrupts a process (e.g., Ctrl+C)
B. Kills a process permanently
C. Ignores a process
D. Creates a process
Correct Answer: A

Q68. Context switch in OS:

A. Saves state of current process and loads state of another
B. Deletes a process
C. Creates a file
D. Reads data
Correct Answer: A

Q69. Process state ready means:

A. Process is waiting to be assigned to CPU
B. Process is running
C. Process is blocked
D. Process is terminated
Correct Answer: A

Q70. Process state blocked means:

A. Process is waiting for an event
B. Process is running
C. Process is ready
D. Process is terminated
Correct Answer: A

Q71. Semaphore is used for:

A. Process synchronization and mutual exclusion
B. Memory allocation
C. File creation
D. CPU scheduling
Correct Answer: A

Q72. Mutex is:

A. Mutual exclusion lock for processes/threads
B. A file type
C. A system call
D. A memory area
Correct Answer: A

Q73. Starvation in OS:

A. A process never gets resources it needs
B. A process runs too fast
C. A process uses too much memory
D. A process is killed
Correct Answer: A

Q74. Thrashing in OS:

A. Excessive paging causing system to spend more time paging than executing
B. Fast CPU execution
C. No memory needed
D. Efficient processing
Correct Answer: A

Q75. Page fault is:

A. An interrupt when a page is not in memory
B. A file system error
C. A CPU error
D. A network error
Correct Answer: A

Q76. TLB (Translation Lookaside Buffer) is:

A. Cache for page table entries
B. A disk cache
C. A CPU register
D. A network buffer
Correct Answer: A

Q77. Demand paging loads a page:

A. Only when it is referenced
B. At process start always
C. At system boot
D. In background always
Correct Answer: A

Q78. Page replacement algorithm FIFO:

A. Replaces the oldest page in memory
B. Replaces the most used
C. Replaces random
D. Replaces next
Correct Answer: A

Q79. Page replacement algorithm LRU:

A. Replaces the least recently used page
B. Replaces oldest
C. Replaces random
D. Replaces most used
Correct Answer: A

Q80. Page replacement algorithm Optimal:

A. Replaces the page that will not be used for longest time
B. Replaces oldest
C. Replaces random
D. Replaces first
Correct Answer: A

Q81. Working set model is used for:

A. Keeping track of pages referenced recently by a process
B. Only memory allocation
C. Only scheduling
D. Only I/O
Correct Answer: A

Q82. Thrashing can be avoided by:

A. Proper page/frame ratio and working set
B. Adding more disk
C. Removing all processes
D. Increasing CPU speed
Correct Answer: A

Q83. Shared memory IPC:

A. Allows processes to share a region of memory
B. Uses only messages
C. Uses only signals
D. Uses only pipes
Correct Answer: A

Q84. Message passing IPC:

A. Processes communicate by sending/receiving messages
B. Shares memory
C. Uses shared files
D. Uses sockets only
Correct Answer: A

Q85. Deadlock detection uses:

A. Resource allocation graphs and cycle detection
B. Only time stamps
C. Only memory
D. Only CPU
Correct Answer: A

Q86. Deadlock recovery can involve:

A. Killing one or more processes
B. Always restarting
C. Only adding memory
D. Only deleting files
Correct Answer: A

Q87. Preemption can resolve deadlock by:

A. Taking resources from a process
B. Killing all processes
C. Ignoring the deadlock
D. Adding more CPU
Correct Answer: A

Q88. Banker's Algorithm requires:

A. Advance knowledge of maximum resource needs
B. No knowledge at all
C. Only current usage
D. Only future requests
Correct Answer: A

Q89. Safe state in Banker's Algorithm means:

A. System can allocate resources without deadlock
B. System is always deadlocked
C. System has no resources
D. System has only one process
Correct Answer: A

Q90. Critical section problem requires:

A. Mutual exclusion, progress, bounded waiting
B. Only mutual exclusion
C. Only progress
D. Only bounded waiting
Correct Answer: A

Q91. Peterson's solution is for:

A. Two-process mutual exclusion
B. Three-process scheduling
C. Memory allocation
D. Deadlock detection
Correct Answer: A

Q92. Test-and-set instruction:

A. Atomically reads and sets a memory location
B. Reads two locations
C. Writes to a file
D. Deletes a variable
Correct Answer: A

Q93. Compare-and-swap instruction:

A. Atomically compares and swaps values
B. Reads and writes files
C. Sends messages
D. Deletes processes
Correct Answer: A

Q94. Semaphores were introduced by:

A. Dijkstra
B. Bellman
C. Turing
D. Kernighan
Correct Answer: A

Q95. Binary semaphore can have values:

A. 0 and 1
B. Any integer
C. Only 0
D. Only 1
Correct Answer: A

Q96. Counting semaphore can have:

A. Non-negative integer values
B. Only 0 or 1
C. Negative values only
D. Only positive
Correct Answer: A

Q97. Bounded buffer problem is also called:

A. Producer-consumer problem
B. Reader-writer problem
C. Dining philosophers
D. Sleeping barber
Correct Answer: A

Q98. Readers-writer problem allows:

A. Multiple readers or one writer
B. Only one reader
C. Multiple writers
D. No readers
Correct Answer: A

Q99. Starvation vs livelock:

A. Starvation is no progress; livelock is active but no useful progress
B. They are the same
C. Starvation is fast execution
D. Livelock uses no CPU
Correct Answer: A

Q100. Belady's anomaly relates to:

A. FIFO page replacement can increase page faults with more frames
B. LRU always works
C. Optimal always fails
D. No anomaly exists
Correct Answer: A

Q101. Swapping moves a process:

A. Between main memory and disk
B. Between CPU and memory
C. Between disk and network
D. Between files
Correct Answer: A

Q102. Virtual address space is:

A. The address space as seen by the process
B. Physical memory address
C. Disk address
D. Network address
Correct Answer: A

Q103. Page table maps:

A. Virtual pages to physical frames
B. Physical to virtual
C. Files to processes
D. Processes to users
Correct Answer: A

Q104. Inverted page table has:

A. One entry per physical frame
B. One entry per process
C. One per user
D. One per file
Correct Answer: A

Q105. Multi-level page table reduces:

A. Size of page table needed by a process
B. CPU speed
C. Disk size
D. Memory size
Correct Answer: A

Q106. Copy-on-write is used in:

A. Forking to share pages until modified
B. Memory allocation
C. File copying
D. Process scheduling
Correct Answer: A

Q107. Memory-mapped files allow:

A. File I/O using memory accesses
B. Only disk I/O
C. Only network I/O
D. Only CPU operations
Correct Answer: A

Q108. TLB miss leads to:

A. Page table walk in memory
B. Page fault always
C. Program crash
D. CPU reset
Correct Answer: A

Q109. Large pages (huge pages) reduce:

A. TLB misses
B. CPU speed
C. Memory size
D. Disk I/O
Correct Answer: A

Q110. Preemptive scheduling:

A. OS forcibly interrupts running process
B. Process voluntarily yields
C. Only batch
D. No preemption possible
Correct Answer: A

Q111. Non-preemptive scheduling:

A. Process keeps CPU until it voluntarily yields or terminates
B. OS interrupts anytime
C. Only priority
D. No scheduling
Correct Answer: A

Q112. Convoy effect occurs when:

A. Short processes wait behind long ones
B. Processes run too fast
C. No waiting
D. Only I/O
Correct Answer: A

Q113. Context switching overhead includes:

A. Saving and restoring registers, PC, PSW
B. Only memory access
C. Only disk access
D. Only network
Correct Answer: A

Q114. CPU burst time is:

A. Amount of time a process spends executing on CPU
B. Amount of time waiting for I/O
C. Amount of memory used
D. Amount of disk access
Correct Answer: A

Q115. I/O burst time is:

A. Amount of time a process spends doing I/O
B. Amount of CPU time
C. Amount of memory
D. Amount of disk space
Correct Answer: A

Q116. Process control block (PCB) contains:

A. Process state, program counter, CPU registers, memory management info
B. Only process name
C. Only memory address
D. Only user ID
Correct Answer: A

Q117. Long-term scheduler selects:

A. Jobs from the job pool to be brought into memory
B. Processes from ready queue
C. Processes from disk
D. Threads only
Correct Answer: A

Q118. Short-term scheduler selects:

A. Process from ready queue to execute on CPU
B. Jobs from pool
C. Processes from disk
D. Threads only
Correct Answer: A

Q119. Throughput is:

A. Number of processes completed per time unit
B. Time to complete one process
C. Time spent waiting
D. Number of processes in ready queue
Correct Answer: A

Q120. Turnaround time is:

A. Total time from submission to completion
B. Time spent in ready queue
C. Time spent in running state only
D. Time spent waiting for I/O
Correct Answer: A

Q121. Waiting time is:

A. Total time a process has been in the ready queue
B. Time spent executing
C. Time spent in I/O
D. Time from creation
Correct Answer: A

Q122. Response time is:

A. Time from submission until first response
B. Time to complete the process
C. Time spent in waiting only
D. Time spent in I/O only
Correct Answer: A

Q123. Preemption occurs when:

A. OS interrupts a running process to schedule another
B. Process voluntarily yields
C. System shuts down
D. Only in batch
Correct Answer: A

Q124. Dispatch latency is:

A. Time to stop one process and start another
B. Time to create a process
C. Time to load a file
D. Time to read memory
Correct Answer: A

Q125. Deterministic scheduling is:

A. All processes have known execution times
B. All processes have random times
C. Only one process
D. Only batch jobs
Correct Answer: A

Q126. Guaranteed scheduling uses:

A. Formula based on fair share of CPU
B. Priority only
C. Round Robin only
D. First come only
Correct Answer: A

Q127. Lottery scheduling uses:

A. Tickets for CPU time allocation
B. Priority only
C. Fixed time slices
D. Random selection
Correct Answer: A

Q128. In UNIX, the file extension is:

A. Not required/meaningful for file type
B. Always .txt
C. Always .sh
D. Always .c
Correct Answer: A

Q129. chmod command changes:

A. File permissions
B. File owner
C. File size
D. File content
Correct Answer: A

Q130. umask sets:

A. Default permissions for newly created files
B. Current directory
C. Shell variable
D. File name
Correct Answer: A

Q131. The /etc/passwd file contains:

A. User account information
B. Passwords in plain text
C. Only usernames
D. Only group names
Correct Answer: A

Q132. The /etc/shadow file stores:

A. Encrypted password information
B. User home directories
C. Shell configuration
D. File permissions
Correct Answer: A

Q133. su command in UNIX:

A. Switch user identity
B. Shut down system
C. Start a service
D. Change directory
Correct Answer: A

Q134. sudo command allows:

A. Execute commands as another user (usually root)
B. Shut down without password
C. Create a new user
D. Delete a file
Correct Answer: A

Q135. cron daemon in UNIX:

A. Schedules periodic execution of commands
B. Manages memory
C. Controls network
D. Handles files
Correct Answer: A

Q136. crontab file contains:

A. Scheduled commands and their timings
B. User passwords
C. File contents
D. Network addresses
Correct Answer: A

Q137. In UNIX, stdin refers to:

A. Standard input (usually keyboard)
B. Standard output
C. Standard error
D. Standard file
Correct Answer: A

Q138. In UNIX, stdout refers to:

A. Standard output (usually screen)
B. Standard input
C. Standard error
D. A file
Correct Answer: A

Q139. In UNIX, stderr refers to:

A. Standard error output
B. Standard input
C. Standard output
D. A device
Correct Answer: A

Q140. The pipe symbol | in UNIX:

A. Connects stdout of one command to stdin of another
B. Creates a file
C. Deletes a file
D. Moves a file
Correct Answer: A

Q141. Redirection operator > in UNIX:

A. Sends output to a file
B. Reads input from a file
C. Appends to a file
D. Prints to screen
Correct Answer: A

Q142. Redirection operator >> in UNIX:

A. Appends output to a file
B. Overwrites a file
C. Reads from a file
D. Deletes a file
Correct Answer: A

Q143. The cat command in UNIX:

A. Concatenates and displays file contents
B. Creates a directory
C. Deletes a file
D. Copies a file
Correct Answer: A

Q144. The more/less command in UNIX:

A. Displays file contents one screen at a time
B. Creates files
C. Deletes files
D. Copies files
Correct Answer: A

Q145. The wc command in UNIX:

A. Counts lines, words, and characters
B. Changes permissions
C. Moves files
D. Finds files
Correct Answer: A

Q146. The who command in UNIX:

A. Displays currently logged-in users
B. Shows file contents
C. Displays processes
D. Shows network connections
Correct Answer: A

Q147. The whoami command in UNIX:

A. Displays current username
B. Displays home directory
C. Displays shell
D. Displays IP address
Correct Answer: A

Q148. The hostname command in UNIX:

A. Displays system hostname
B. Shows current user
C. Shows memory
D. Shows disk space
Correct Answer: A

Q149. The date command in UNIX:

A. Displays current date and time
B. Changes file date
C. Shows system uptime
D. Displays calendar
Correct Answer: A

Q150. The cal command in UNIX:

A. Displays a calendar
B. Shows system time
C. Shows users
D. Shows processes
Correct Answer: A

Q151. The alias command in UNIX:

A. Creates shortcuts for commands
B. Changes file names
C. Sets permissions
D. Creates variables
Correct Answer: A

Q152. The unalias command in UNIX:

A. Removes command aliases
B. Creates new aliases
C. Lists all aliases
D. Resets shell
Correct Answer: A

Q153. The history command in UNIX:

A. Displays command history
B. Shows file contents
C. Shows users
D. Shows memory
Correct Answer: A

Q154. The echo command in UNIX:

A. Displays a line of text
B. Reads a file
C. Deletes a file
D. Copies a file
Correct Answer: A

Q155. The printf command in UNIX:

A. Formats and prints data
B. Creates files
C. Deletes files
D. Changes directories
Correct Answer: A

Q156. The read command in UNIX shell:

A. Reads input from user
B. Prints output
C. Deletes variables
D. Creates files
Correct Answer: A

Q157. In shell scripting, $? contains:

A. Exit status of last command
B. PID of last command
C. Home directory
D. Shell version
Correct Answer: A

Q158. In shell scripting, $$ is:

A. PID of the current shell
B. Exit status
C. Home directory
D. Script name
Correct Answer: A

Q159. In shell scripting, $! is:

A. PID of last background process
B. Exit status
C. Script name
D. Home directory
Correct Answer: A

Q160. In shell scripting, $0 is:

A. Name of the shell script
B. First argument
C. Exit status
D. Home directory
Correct Answer: A

Q161. In shell scripting, $1, $2... refer to:

A. Positional parameters (arguments)
B. Environment variables
C. Shell functions
D. Process IDs
Correct Answer: A

Q162. The test command in shell checks:

A. File attributes and values
B. Network connections
C. CPU usage
D. Memory usage
Correct Answer: A

Q163. The [[ ]] construct in bash is:

A. Extended test command with more features
B. A loop construct
C. A function definition
D. A file operation
Correct Answer: A

Q164. The case statement in shell:

A. Matches patterns and executes corresponding commands
B. A loop
C. A function
D. A variable
Correct Answer: A

Q165. The for loop in shell:

A. Iterates over a list of values
B. Executes conditionally
C. Defines functions
D. Reads files
Correct Answer: A

Q166. The while loop in shell:

A. Executes commands while condition is true
B. Executes once
C. Defines functions
D. Reads user input
Correct Answer: A

Q167. The until loop in shell:

A. Executes commands until condition becomes true
B. Executes forever
C. Defines functions
D. Exits immediately
Correct Answer: A

Q168. The break command in shell:

A. Exits a loop
B. Defines a function
C. Reads input
D. Creates files
Correct Answer: A

Q169. The continue command in shell:

A. Skips to next iteration of a loop
B. Exits the shell
C. Deletes a file
D. Prints output
Correct Answer: A

Q170. A shell function is defined with:

A. function_name() { ... }
B. def function_name():
C. func function_name()
D. create function_name()
Correct Answer: A

Q171. The source/dot command in shell:

A. Executes commands from a file in current shell
B. Runs a script in subshell
C. Compiles a file
D. Creates a file
Correct Answer: A

Q172. Here document in shell uses:

A. << to redirect input until a delimiter
B. A for loop
C. A while condition
D. A function call
Correct Answer: A

Q173. The trap command in shell:

A. Catches signals and executes commands
B. Creates traps
C. Deletes files
D. Compiles code
Correct Answer: A

Q174. The getopts command in shell:

A. Parses command-line options
B. Gets process options
C. Reads files
D. Displays help
Correct Answer: A

Q175. The shift command in shell:

A. Shifts positional parameters
B. Moves files
C. Changes directory
D. Deletes variables
Correct Answer: A

Q176. In shell, export makes a variable:

A. Available to child processes
B. Available only in current script
C. Temporary
D. System-wide
Correct Answer: A

Q177. The type command in shell:

A. Shows how a command is interpreted
B. Changes file type
C. Shows file contents
D. Lists users
Correct Answer: A

Q178. The which command in shell:

A. Finds the path of a command
B. Displays help
C. Shows history
D. Lists files
Correct Answer: A

Q179. The find command in UNIX:

A. Searches for files based on criteria
B. Compiles programs
C. Deletes processes
D. Shows users
Correct Answer: A

Q180. The locate command in UNIX:

A. Quickly finds files using a database
B. Searches memory
C. Shows network
D. Displays time
Correct Answer: A

Q181. The grep -r option:

A. Recursively searches directories
B. Reverses search
C. Counts lines
D. Shows file size
Correct Answer: A

Q182. The tar command in UNIX:

A. Archives files (tape archive)
B. Compiles programs
C. Deletes files
D. Shows disk space
Correct Answer: A

Q183. The gzip command in UNIX:

A. Compresses files
B. Decompresses files
C. Both A and depends on mode
D. Only archives
Correct Answer: C

Q184. The bzip2 command:

A. Compresses files with better compression than gzip
B. Only archives
C. Only decompresses
D. Only extracts
Correct Answer: A

Q185. The zip command in UNIX:

A. Creates zip archives
B. Only compresses
C. Only extracts
D. Only lists
Correct Answer: A

Q186. The chmod symbolic mode u+x means:

A. Adds execute permission for owner
B. Removes write permission
C. Adds read for group
D. Removes all permissions
Correct Answer: A

Q187. The chown command changes:

A. File owner and/or group
B. File permissions
C. File content
D. File name
Correct Answer: A

Q188. The ln command creates:

A. Links (hard or symbolic)
B. Files
C. Directories only
D. User accounts
Correct Answer: A

Q189. A symbolic link (soft link) in UNIX:

A. Points to the path of another file
B. Is an exact copy of the file
C. Is a hard link
D. Is a directory
Correct Answer: A

Q190. A hard link in UNIX:

A. Shares the same inode as the original file
B. Points to file path
C. Is a copy of the file
D. Is a symbolic link
Correct Answer: A

Q191. The dd command in UNIX:

A. Converts and copies files
B. Deletes files
C. Compiles programs
D. Shows users
Correct Answer: A

Q192. The diff command in UNIX:

A. Compares files line by line
B. Deletes files
C. Compiles programs
D. Shows processes
Correct Answer: A

Q193. The patch command in UNIX:

A. Applies differences to files
B. Creates patches
C. Deletes files
D. Compiles files
Correct Answer: A

Q194. The split command in UNIX:

A. Splits a file into smaller parts
B. Joins files
C. Deletes files
D. Compiles files
Correct Answer: A

Q195. The tee command in UNIX:

A. Reads from stdin and writes to stdout and files
B. Deletes files
C. Compiles code
D. Shows history
Correct Answer: A

Q196. The xargs command in UNIX:

A. Builds and executes command lines from stdin
B. Deletes files
C. Compiles programs
D. Shows users
Correct Answer: A

Q197. The screen command in UNIX:

A. Allows multiple terminal sessions
B. Deletes files
C. Compiles programs
D. Shows users
Correct Answer: A

Q198. The tmux command in UNIX:

A. Terminal multiplexer for multiple sessions
B. File manager
C. Compiler
D. Shell script
Correct Answer: A

Q199. The sort -r option in UNIX:

A. Sorts in reverse order
B. Sorts numerically
C. Sorts by file size
D. Sorts by date
Correct Answer: A

Q200. The uniq -c option:

A. Counts repeated lines
B. Removes duplicates only
C. Sorts lines
D. Joins files
Correct Answer: A

ANSWER KEY — Topic 10: Operating Systems & Unix

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1-A 2-A 3-A 4-A 5-A 6-A 7-A 8-A 9-A 10-A
11-A 12-A 13-A 14-A 15-A 16-A 17-A 18-A 19-A 20-A
21-A 22-A 23-A 24-A 25-A 26-A 27-A 28-A 29-A 30-A
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61-A 62-A 63-A 64-A 65-A 66-A 67-A 68-A 69-A 70-A
71-A 72-A 73-A 74-A 75-A 76-A 77-A 78-A 79-A 80-A
81-A 82-A 83-A 84-A 85-A 86-A 87-A 88-A 89-A 90-A
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101-A 102-A 103-A 104-A 105-A 106-A 107-A 108-A 109-A 110-A
111-A 112-A 113-A 114-A 115-A 116-A 117-A 118-A 119-A 120-A
121-A 122-A 123-A 124-A 125-A 126-A 127-A 128-A 129-A 130-A
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141-A 142-A 143-A 144-A 145-A 146-A 147-A 148-A 149-A 150-A
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191-A 192-A 193-A 194-A 195-A 196-A 197-A 198-A 199-A 200-A
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Topic 11: Software Engineering

Syllabus: *SDLC/waterfall/prototype/spiral, software metrics/project management (COCOMO, FP, CPM/PERT), system/detailed/function-oriented/OO/UI design + design metrics, coding/testing levels, quality/reliability, clean room, re-engineering. Include McCabe cyclomatic, testing types.*

Q1. SDLC stands for:

A. Software Development Life Cycle
B. System Design Life Cycle
C. Software Deployment Life Cycle
D. System Development Loop
Correct Answer: A

Q2. Waterfall model is:

A. Sequential and linear model
B. Iterative model
C. Incremental model
D. Spiral model
Correct Answer: A

Q3. Waterfall model phases are:

A. Requirements, Design, Implementation, Testing, Maintenance
B. Design, Code, Test, Deploy
C. Plan, Do, Check, Act
D. Requirements, Build, Test, Deploy
Correct Answer: A

Q4. The waterfall model has which limitation:

A. Cannot go back to previous phase easily
B. Too iterative
C. Too flexible
D. No phases
Correct Answer: A

Q5. Prototyping model is:

A. Build a prototype to clarify requirements
B. Sequential model
C. Spiral model
D. Waterfall variant
Correct Answer: A

Q6. Incremental model:

A. Delivers product in increments/parts
B. One big delivery
C. No delivery
D. Only testing
Correct Answer: A

Q7. Spiral model combines:

A. Development and risk analysis in repeated cycles
B. Only design phases
C. Only testing
D. Only coding
Correct Answer: A

Q8. Spiral model has which key feature:

A. Risk analysis at each iteration
B. No planning
C. No design
D. No coding
Correct Answer: A

Q9. Radial model is also known as:

A. Spiral model
B. Waterfall
C. Prototype
D. Incremental
Correct Answer: A

Q10. V-Model is:

A. Verification and validation at each development stage
B. Only testing model
C. Only design model
D. Only coding model
Correct Answer: A

Q11. Agile methodology emphasizes:

A. Iterative delivery, collaboration, adaptability
B. Heavy documentation
C. Sequential phases
D. Fixed requirements
Correct Answer: A

Q12. Scrum is a:

A. Agile framework with sprints and roles
B. Waterfall variant
C. Testing framework
D. Design tool
Correct Answer: A

Q13. Scrum roles include:

A. Product Owner, Scrum Master, Development Team
B. Manager, Developer, Tester
C. Analyst, Designer, Coder
D. PM, Dev, QA
Correct Answer: A

Q14. Sprint in Scrum is:

A. Time-boxed iteration
B. A document
C. A meeting type
D. A testing tool
Correct Answer: A

Q15. Kanban is a:

A. Visual workflow management method
B. Coding language
C. Testing framework
D. Design tool
Correct Answer: A

Q16. Extreme Programming (XP) emphasizes:

A. Technical excellence and customer involvement
B. Documentation only
C. Sequential development
D. Heavy planning
Correct Answer: A

Q17. Pair programming is a practice of:

A. Two programmers working at one workstation
B. One programmer coding alone
C. Three programmers coding
D. Programmers reviewing code separately
Correct Answer: A

Q18. Test-Driven Development (TDD):

A. Write tests before code
B. Write code before tests
C. Write no tests
D. Write tests after deployment
Correct Answer: A

Q19. Software metrics measure:

A. Properties of software products and processes
B. Only code lines
C. Only cost
D. Only time
Correct Answer: A

Q20. COCOMO stands for:

A. Constructive Cost Model
B. Cost of Code Optimization Method
C. Coding Cost Output Measure
D. Cumulative Cost Model
Correct Answer: A

Q21. COCOMO estimates:

A. Effort, cost, and schedule for software projects
B. Only code complexity
C. Only testing effort
D. Only team size
Correct Answer: A

Q22. COCOMO Basic model uses:

A. Lines of Code (KLOC) as primary driver
B. Only team experience
C. Only project type
D. Only development language
Correct Answer: A

Q23. COCOMO Intermediate model adds:

A. Product, hardware, personnel, project attributes
B. Only cost factors
C. Only schedule factors
D. Only effort factors
Correct Answer: A

Q24. COCOMO Advanced model includes:

A. All factors including risk and quality
B. Only basic COCOMO
C. Only intermediate COCOMO
D. No factors
Correct Answer: A

Q25. Function Point (FP) analysis measures:

A. Functional size of software from user perspective
B. Lines of code
C. Number of modules
D. Number of bugs
Correct Answer: A

Q26. FP analysis counts:

A. External inputs, outputs, files, interfaces
B. Only code lines
C. Only modules
D. Only functions
Correct Answer: A

Q27. COCOMO effort equation is:

A. Effort = a * (KLOC)^b
B. Effort = a + b * KLOC
C. Effort = a * KLOC / b
D. Effort = a - b * KLOC
Correct Answer: A

Q28. CPM (Critical Path Method) is used for:

A. Finding the longest path determining minimum project duration
B. Finding shortest path
C. Finding average path
D. Finding random path
Correct Answer: A

Q29. PERT uses:

A. Three time estimates (optimistic, most likely, pessimistic)
B. One time estimate
C. Two time estimates
D. Four time estimates
Correct Answer: A

Q30. Critical path in CPM/PERT:

A. Longest path through the network diagram
B. Shortest path
C. Average path
D. Any path
Correct Answer: A

Q31. System design phase produces:

A. High-level architecture and module definitions
B. Only code
C. Only tests
D. Only documentation
Correct Answer: A

Q32. Detailed design specifies:

A. Internal logic of each module
B. Only system architecture
C. Only data flow
D. Only interfaces
Correct Answer: A

Q33. Function-oriented design uses:

A. Data flow diagrams to derive module structure
B. Only class diagrams
C. Only sequence diagrams
D. Only activity diagrams
Correct Answer: A

Q34. Object-Oriented Design uses:

A. Classes, objects, inheritance, polymorphism
B. Only functions
C. Only data structures
D. Only modules
Correct Answer: A

Q35. UI design focuses on:

A. User interface layout, navigation, interaction
B. System architecture
C. Database schema
D. Network topology
Correct Answer: A

Q36. Design metrics include:

A. Coupling, cohesion, complexity measures
B. Only lines of code
C. Only module count
D. Only function count
Correct Answer: A

Q37. Cohesion is highest when:

A. All elements work toward single well-defined purpose
B. Elements do different things
C. Elements are unrelated
D. Elements are loosely connected
Correct Answer: A

Q38. Coupling is lowest when:

A. Modules are independent with minimal dependencies
B. Modules share data heavily
C. Modules are tightly coupled
D. Modules are inseparable
Correct Answer: A

Q39. Internal coherence/high cohesion is desirable because:

A. Easier maintenance and reuse
B. Harder to understand
C. More code
D. More bugs
Correct Answer: A

Q40. Low coupling is desirable because:

A. Changes in one module less likely to affect others
B. Changes affect all modules
C. Modules become bigger
D. Modules become slower
Correct Answer: A

Q41. Coding phase converts:

A. Design into source code
B. Design into documentation
C. Requirements into code
D. Tests into code
Correct Answer: A

Q42. Coding standards include:

A. Naming conventions, indentation, commenting rules
B. Only indentation rules
C. Only naming rules
D. Only commenting rules
Correct Answer: A

Q43. Unit testing tests:

A. Individual modules/functions
B. Integration of modules
C. Whole system
D. User acceptance
Correct Answer: A

Q44. Integration testing tests:

A. Interfaces between integrated modules
B. Individual functions
C. User acceptance
D. System performance
Correct Answer: A

Q45. System testing tests:

A. Complete integrated system against requirements
B. Individual modules
C. Code logic
D. Performance only
Correct Answer: A

Q46. Acceptance testing tests:

A. System meets business/user requirements
B. Code correctness
C. Module interfaces
D. Performance only
Correct Answer: A

Q47. Regression testing ensures:

A. New changes don't break existing functionality
B. Only new features work
C. Only old features work
D. No features break
Correct Answer: A

Q48. Smoke testing is:

A. Preliminary testing to check if build is stable
B. Deep testing
C. Only integration testing
D. Only unit testing
Correct Answer: A

Q49. Alpha testing is performed by:

A. Users at developer's site
B. Users at customer's site
C. Independent testers
D. Developers only
Correct Answer: A

Q50. Beta testing is performed by:

A. End users at customer's site
B. Developers only
C. Testers at developer site
D. No users
Correct Answer: A

Q51. White-box testing:

A. Tests internal logic/structure of code
B. Tests functionality only
C. Tests performance only
D. Tests usability only
Correct Answer: A

Q52. Black-box testing:

A. Tests functionality without knowledge of internal structure
B. Tests code structure
C. Tests performance
D. Tests security
Correct Answer: A

Q53. Glass-box testing is:

A. Another name for white-box testing
B. Black-box testing
C. Gray-box testing
D. Performance testing
Correct Answer: A

Q54. Gray-box testing combines:

A. White-box and black-box techniques
B. Only white-box
C. Only black-box
D. Only gray-box
Correct Answer: A

Q55. Statement coverage tests:

A. Each executable statement executed at least once
B. Each branch executed
C. Each path executed
D. Each condition executed
Correct Answer: A

Q56. Branch/Decision coverage tests:

A. Each boolean expression outcome evaluated
B. Each statement executed
C. Each path executed
D. Each condition executed
Correct Answer: A

Q57. Path coverage tests:

A. Every possible path through the program
B. Only statements
C. Only branches
D. Only conditions
Correct Answer: A

Q58. McCabe's cyclomatic complexity V(G) =:

A. E - N + 2P (edges - nodes + 2*components)
B. Nodes - edges
C. Edges + nodes
D. Edges * nodes
Correct Answer: A

Q59. McCabe's cyclomatic complexity determines:

A. Number of independent paths / minimum test cases
B. Only code lines
C. Only modules
D. Only functions
Correct Answer: A

Q60. A program with cyclomatic complexity 5 has:

A. 5 independent paths to test
B. 5 functions
C. 5 modules
D. 5 lines of code
Correct Answer: A

Q61. Software quality assurance includes:

A. Process, product, and project quality assurance
B. Only code testing
C. Only documentation
D. Only reviews
Correct Answer: A

Q62. CMMI stands for:

A. Capability Maturity Model Integration
B. Code Maturity Model Integration
C. Coding Model Integration
D. Capacity Model Integration
Correct Answer: A

Q63. ISO 9001 relates to:

A. Quality management systems
B. Only software
C. Only hardware
D. Only processes
Correct Answer: A

Q64. ISO/IEC 12207 is about:

A. Software lifecycle processes
B. Only coding
C. Only testing
D. Only maintenance
Correct Answer: A

Q65. Six Sigma aims for:

A. Near-zero defect rate (3.4 defects per million)
B. High defect rate
C. No quality goals
D. Only testing
Correct Answer: A

Q66. Reliability is:

A. Probability of failure-free operation for specified time
B. Software speed
C. Software size
D. Software complexity
Correct Answer: A

Q67. MTBF stands for:

A. Mean Time Between Failures
B. Mean Time Before Failure
C. Maximum Time Between Failures
D. Minimum Time Between Failures
Correct Answer: A

Q68. MTTR stands for:

A. Mean Time To Repair
B. Maximum Time To Repair
C. Mean Time To Replace
D. Minimum Time To Repair
Correct Answer: A

Q69. Cleanroom software engineering emphasizes:

A. Correctness by construction, formal verification
B. Rapid prototyping
C. Heavy testing
D. Agile methods
Correct Answer: A

Q70. Cleanroom uses:

A. Statistical quality assurance and formal methods
B. Only testing
C. Only prototyping
D. Only Agile
Correct Answer: A

Q71. Formal verification uses:

A. Mathematical proof of correctness
B. Testing only
C. Inspection only
D. Walkthrough only
Correct Answer: A

Q72. Software re-engineering:

A. Revising and updating existing software
B. Writing new code from scratch
C. Deleting old software
D. Ignoring old software
Correct Answer: A

Q73. Reverse engineering is:

A. Analyzing a system to identify components and their relationships
B. Writing code
C. Testing software
D. Deploying software
Correct Answer: A

Q74. Re-engineering includes:

A. Understanding, restructuring, and forward engineering legacy systems
B. Only rewriting
C. Only testing
D. Only documenting
Correct Answer: A

Q75. Legacy system challenges include:

A. Difficulty understanding, outdated technology, documentation gaps
B. Modern technology
C. Clear documentation
D. Easy modification
Correct Answer: A

Q76. Software maintenance types include:

A. Corrective, adaptive, perfective, preventive
B. Only corrective
C. Only adaptive
D. Only perfective
Correct Answer: A

Q77. Adaptive maintenance:

A. Modifying software to work in changed environment
B. Fixing bugs
C. Improving performance
D. Preventing future issues
Correct Answer: A

Q78. Perfective maintenance:

A. Enhancing features and performance
B. Fixing bugs
C. Adapting to new environment
D. Preventing issues
Correct Answer: A

Q79. Preventive maintenance:

A. Improving future maintainability
B. Fixing bugs
C. Adding features
D. Adapting environment
Correct Answer: A

Q80. Configuration management tracks:

A. Changes to software artifacts over time
B. Only code
C. Only documentation
D. Only tests
Correct Answer: A

Q81. Version control systems like Git:

A. Track and manage changes to source code
B. Compile code
C. Test software
D. Deploy software
Correct Answer: A

Q82. CMMI maturity levels:

A. Initial, Managed, Defined, Quantitatively Managed, Optimizing
B. Initial, Intermediate, Advanced
C. Level 1-5 only
D. Basic, Intermediate, Expert
Correct Answer: A

Q83. Requirements engineering includes:

A. Elicitation, analysis, specification, validation
B. Only coding
C. Only testing
D. Only design
Correct Answer: A

Q84. Use case diagram in UML represents:

A. Actors and their interactions with the system
B. Only data flow
C. Only class structure
D. Only sequence
Correct Answer: A

Q85. Class diagram in UML shows:

A. Classes, attributes, methods, and relationships
B. Only sequences
C. Only activities
D. Only use cases
Correct Answer: A

Q86. Sequence diagram shows:

A. Object interactions over time
B. Class structure
C. Use cases
D. Data flow
Correct Answer: A

Q87. Activity diagram is:

A. Workflow of activities and actions
B. Class diagram
C. Sequence diagram
D. Use case diagram
Correct Answer: A

Q88. Software design patterns like Singleton ensure:

A. Only one instance of a class exists
B. Multiple instances
C. No instances
D. Only static methods
Correct Answer: A

Q89. Factory pattern provides:

A. An interface for creating objects, letting subclasses decide which class to instantiate
B. Only one object
C. Only static objects
D. Only abstract objects
Correct Answer: A

Q90. Observer pattern:

A. Notifies dependent objects of state changes
B. Creates objects
C. Manages database
D. Handles UI only
Correct Answer: A

Q91. Software project estimation techniques include:

A. Expert judgment, analogy, parametric models
B. Only COCOMO
C. Only PERT
D. Only CPM
Correct Answer: A

Q92. Risk management includes:

A. Identification, analysis, planning, monitoring
B. Only identification
C. Only analysis
D. Only monitoring
Correct Answer: A

Q93. Risk analysis assesses:

A. Probability and impact of risks
B. Only cost
C. Only schedule
D. Only quality
Correct Answer: A

Q94. Software configuration audit verifies:

A. Conformance to configured items
B. Only code quality
C. Only test results
D. Only user satisfaction
Correct Answer: A

Q95. Functional Point Analysis was developed by:

A. Albrecht
B. Dijkstra
C. Boehm
D. Cocke
Correct Answer: A

Q96. Code review is a type of:

A. Static analysis / inspection
B. Dynamic testing
C. Performance testing
D. Integration testing
Correct Answer: A

Q97. Fagan inspection is:

A. Formal structured inspection process
B. Pair programming
C. Code review without checklist
D. Random testing
Correct Answer: A

Q98. Walkthrough is:

A. Developer-led review of work product
B. Automated testing
C. Code compilation
D. User acceptance test
Correct Answer: A

Q99. Software reliability is measured by:

A. Failure rate, MTBF, availability
B. Lines of code
C. Number of modules
D. Testing hours
Correct Answer: A

Q100. Availability =:

A. MTBF / (MTBF + MTTR)
B. MTTR / MTBF
C. MTBF + MTTR
D. MTBF * MTTR
Correct Answer: A

Q101. Software product lines are:

A. A set of software systems sharing a common core domain
B. Only one product
C. Only one company
D. Only one language
Correct Answer: A

Q102. Domain engineering produces:

A. Reusable components and architectures
B. Only code
C. Only tests
D. Only documentation
Correct Answer: A

Q103. Application engineering produces:

A. Specific products from product line assets
B. Only domain assets
C. Only reusable code
D. Only documentation
Correct Answer: A

Q104. Model Driven Engineering (MDE) uses:

A. Models as primary artifacts
B. Only code
C. Only documentation
D. Only tests
Correct Answer: A

Q105. UML stands for:

A. Unified Modeling Language
B. Universal Modeling Language
C. Unified Module Language
D. Universal Module Language
Correct Answer: A

Q106. Software reuse includes:

A. Code reuse, design reuse, analysis reuse
B. Only code reuse
C. Only design reuse
D. Only test reuse
Correct Answer: A

Q107. Component-based development:

A. Assembling software from pre-built components
B. Writing all code from scratch
C. Only using libraries
D. Only using frameworks
Correct Answer: A

Q108. Service-Oriented Architecture (SOA):

A. Uses loosely coupled services communicating over network
B. Tightly coupled modules
C. Only client-server
D. Only peer-to-peer
Correct Answer: A

Q109. Microservices architecture:

A. Small, independent services communicating via APIs
B. Monolithic architecture
C. Only client-server
D. Only database systems
Correct Answer: A

Q110. DevOps integrates:

A. Development and operations for continuous delivery
B. Only development
C. Only operations
D. Only testing
Correct Answer: A

Q111. CI/CD stands for:

A. Continuous Integration / Continuous Deployment
B. Code Integration / Code Deployment
C. Continuous Inspection / Continuous Debugging
D. Code Inspection / Code Debugging
Correct Answer: A

Q112. Technical debt refers to:

A. Implied cost of future reworking due to shortcuts
B. Only budget debt
C. Only code debt
D. Only time debt
Correct Answer: A

Q113. Software architecture styles include:

A. Client-server, layered, pipe-filter, event-driven
B. Only client-server
C. Only layered
D. Only pipe-filter
Correct Answer: A

Q114. Layered architecture pattern:

A. Organizes system into layers with specific responsibilities
B. Only one layer
C. No layers
D. All layers same
Correct Answer: A

Q115. Pipe-filter pattern is used in:

A. Unix shells, compilers
B. Database systems
C. Network protocols
D. UI design
Correct Answer: A

Q116. MVC pattern separates:

A. Model, View, Controller
B. Input, Output, Processing
C. Database, Logic, UI
D. Frontend, Backend, Database
Correct Answer: A

Q117. Software testing at integration level uses:

A. Integration test cases
B. Only unit tests
C. Only system tests
D. Only acceptance tests
Correct Answer: A

Q118. Stubs are used in:

A. Top-down integration testing
B. Bottom-up integration testing
C. System testing
D. Acceptance testing
Correct Answer: A

Q119. Drivers are used in:

A. Bottom-up integration testing
B. Top-down integration testing
C. System testing
D. Unit testing
Correct Answer: A

Q120. Big-bang integration:

A. All modules integrated at once
B. Incremental integration
C. Top-down only
D. Bottom-up only
Correct Answer: A

Q121. Code coverage tools measure:

A. Percentage of code exercised by tests
B. Only code lines
C. Only branches
D. Only functions
Correct Answer: A

Q122. Mutation testing:

A. Introducing small changes to check if tests catch them
B. Only black-box testing
C. Only white-box testing
D. Only integration testing
Correct Answer: A

Q123. Boundary value analysis tests:

A. Values at boundaries of equivalence classes
B. Only random values
C. Only normal values
D. Only extreme values
Correct Answer: A

Q124. Equivalence partitioning:

A. Divides input domain into classes
B. Only boundary testing
C. Only random testing
D. Only code testing
Correct Answer: A

Q125. State transition testing uses:

A. State diagrams to design test cases
B. Only code paths
C. Only boundary values
D. Only equivalence classes
Correct Answer: A

Q126. Decision table testing:

A. Tests combinations of conditions
B. Only single conditions
C. Only random conditions
D. Only boundary conditions
Correct Answer: A

Q127. Use case testing:

A. Tests from use case specifications
B. Only code testing
C. Only performance testing
D. Only security testing
Correct Answer: A

Q128. User acceptance testing (UAT):

A. Final testing by end users before release
B. Developer testing
C. Tester testing
D. Automated testing
Correct Answer: A

Q129. Performance testing includes:

A. Load testing, stress testing, scalability testing
B. Only unit testing
C. Only integration testing
D. Only code review
Correct Answer: A

Q130. Security testing verifies:

A. Protection against unauthorized access and attacks
B. Only functionality
C. Only performance
D. Only usability
Correct Answer: A

Q131. Compatibility testing checks:

A. Software works across different environments
B. Only functionality
C. Only performance
D. Only security
Correct Answer: A

Q132. Installation testing:

A. Tests installation and uninstallation process
B. Only code testing
C. Only integration testing
D. Only performance testing
Correct Answer: A

Q133. Recovery testing:

A. Tests system recovery after crashes
B. Only functionality
C. Only performance
D. Only compatibility
Correct Answer: A

Q134. Exploratory testing:

A. Simultaneous learning, design, and execution
B. Scripted testing only
C. Automated testing only
D. Only regression testing
Correct Answer: A

Q135. Ad-hoc testing:

A. Informal, unplanned testing
B. Scripted testing
C. Automated testing
D. Only formal testing
Correct Answer: A

Q136. Software inspection formal steps include:

A. Planning, Overview, Preparation, Inspection, Rework
B. Only review
C. Only walkthrough
D. Only testing
Correct Answer: A

Q137. Software product quality is measured by:

A. Internal, external, and use-case quality measures
B. Only code lines
C. Only user satisfaction
D. Only testing results
Correct Answer: A

Q138. ISO/IEC 25010 quality model includes:

A. Functional suitability, performance efficiency, usability, reliability, etc.
B. Only functionality
C. Only performance
D. Only usability
Correct Answer: A

Q139. Software project management includes:

A. Planning, organizing, staffing, directing, controlling
B. Only coding
C. Only testing
D. Only design
Correct Answer: A

Q140. Work Breakdown Structure (WBS):

A. Hierarchical decomposition of project tasks
B. Only code modules
C. Only test cases
D. Only documents
Correct Answer: A

Q141. Gantt chart shows:

A. Tasks and their timelines
B. Only code
C. Only test results
D. Only costs
Correct Answer: A

Q142. PERT chart shows:

A. Activity dependencies and timelines
B. Only costs
C. Only resources
D. Only quality
Correct Answer: A

Q143. Resource leveling is:

A. Adjusting schedule to balance resource demand
B. Increasing resources
C. Reducing scope
D. Adding time
Correct Answer: A

Q144. Crashing a schedule:

A. Adding resources to shorten duration at increased cost
B. Reducing quality
C. Adding scope
D. Removing tasks
Correct Answer: A

Q145. Fast tracking:

A. Performing activities in parallel that were planned sequentially
B. Adding resources
C. Reducing scope
D. Extending timeline
Correct Answer: A

Q146. Cost estimation includes:

A. Direct costs, indirect costs, and contingency
B. Only salaries
C. Only hardware
D. Only software
Correct Answer: A

Q147. Earned Value Management (EVM) uses:

A. PV, EV, AC to measure project performance
B. Only cost
C. Only schedule
D. Only scope
Correct Answer: A

Q148. SPI (Schedule Performance Index) =:

A. EV / PV
B. PV / EV
C. AC / EV
D. EV / AC
Correct Answer: A

Q149. CPI (Cost Performance Index) =:

A. EV / AC
B. AC / EV
C. PV / EV
D. EV / PV
Correct Answer: A

Q150. Risk mitigation:

A. Reducing probability or impact of risks
B. Ignoring risks
C. Transferring risks only
D. Avoiding all activities
Correct Answer: A

Q151. Risk transfer involves:

A. Shifting risk impact to a third party
B. Eliminating risks
C. Accepting all risks
D. Ignoring risks
Correct Answer: A

Q152. Software contracts types include:

A. Fixed-price, cost-reimbursable, time-and-materials
B. Only fixed-price
C. Only cost-reimbursable
D. Only time-and-materials
Correct Answer: A

Q153. Function Point Analysis counts:

A. External Inputs, External Outputs, External Inquiries, Internal Logical Files, External Interface Files
B. Only code lines
C. Only modules
D. Only functions
Correct Answer: A

Q154. Object-Oriented metrics include:

A. LCOM, CBO, RFC, WMC
B. Only lines of code
C. Only module count
D. Only function count
Correct Answer: A

Q155. WMC (Weighted Methods per Class):

A. Sum of complexity of all methods in a class
B. Number of methods
C. Number of classes
D. Number of attributes
Correct Answer: A

Q156. CBO (Coupling Between Objects):

A. Number of other classes a class is coupled to
B. Number of methods
C. Number of attributes
D. Number of packages
Correct Answer: A

Q157. RFC (Response for a Class):

A. Number of methods that can respond to a message sent to the class
B. Number of classes
C. Number of methods only
D. Number of attributes only
Correct Answer: A

Q158. LCOM (Lack of Cohesion in Methods):

A. Measures degree of lack of cohesion in a class
B. Measures coupling
C. Measures complexity
D. Measures inheritance
Correct Answer: A

Q159. Software reliability models include:

A. Markov, exponential, Weibull models
B. Only linear models
C. Only polynomial models
D. Only constant models
Correct Answer: A

Q160. Cleanroom inspection levels:

A. Structure, Algorithm, Functional, Domain inspection
B. Only code inspection
C. Only design inspection
D. Only testing
Correct Answer: A

Q161. Fagan inspection phases:

A. Planning, Overview, Preparation, Inspection, Rework
B. Only inspection
C. Only review
D. Only testing
Correct Answer: A

Q162. Software re-engineering steps:

A. Reverse engineering, restructuring, forward engineering
B. Only forward engineering
C. Only reverse engineering
D. Only restructuring
Correct Answer: A

Q163. Legacy system re-engineering approaches:

A. Wrap, migrate, encapsulate, replace
B. Only delete
C. Only rewrite
D. Only ignore
Correct Answer: A

Q164. Software migration involves:

A. Moving software to new platform/environment
B. Only code changes
C. Only documentation
D. Only testing
Correct Answer: A

Q165. Component-based software engineering focuses on:

A. Assembly of pre-built reusable components
B. Writing all from scratch
C. Only using frameworks
D. Only using libraries
Correct Answer: A

Q166. Software product line engineering:

A. Domain engineering and application engineering
B. Only application engineering
C. Only domain engineering
D. Only component engineering
Correct Answer: A

Q167. Model-driven architecture (MDA):

A. Platform-independent models transformed to platform-specific models
B. Only coding
C. Only testing
D. Only documentation
Correct Answer: A

Q168. Aspect-Oriented Programming (AOP):

A. Separating cross-cutting concerns
B. Only OOP
C. Only functional programming
D. Only procedural programming
Correct Answer: A

Q169. Software architecture description uses:

A. ADLs (Architecture Description Languages)
B. Only natural language
C. Only diagrams
D. Only tables
Correct Answer: A

Q170. ATAM (Architecture Tradeoff Analysis Method):

A. Evaluating software architecture quality attributes
B. Only testing
C. Only coding
D. Only design
Correct Answer: A

Q171. Software supply chain security:

A. Managing risks in third-party components
B. Only internal code
C. Only testing
D. Only deployment
Correct Answer: A

Q172. DevSecOps adds:

A. Security practices to DevOps pipeline
B. Only development
C. Only operations
D. Only testing
Correct Answer: A

Q173. Infrastructure as Code (IaC):

A. Managing infrastructure through configuration files
B. Only manual setup
C. Only hardware
D. Only software
Correct Answer: A

Q174. Microservices communication patterns:

A. Synchronous (REST/gRPC) and asynchronous (message queues)
B. Only synchronous
C. Only asynchronous
D. Only shared memory
Correct Answer: A

Q175. Circuit breaker pattern:

A. Prevents cascading failures in distributed systems
B. Only load balancing
C. Only caching
D. Only database optimization
Correct Answer: A

Q176. Event sourcing stores:

A. State as a sequence of events
B. Only current state
C. Only snapshots
D. Only logs
Correct Answer: A

Q177. CQRS pattern separates:

A. Command and query responsibilities
B. Only reading
C. Only writing
D. Only database
Correct Answer: A

Q178. Software architecture evaluation uses:

A. SAAM, ATAM, ARID methods
B. Only code review
C. Only testing
D. Only documentation
Correct Answer: A

Q179. Software reuse levels include:

A. Code, design, analysis, domain reuse
B. Only code
C. Only design
D. Only testing
Correct Answer: A

Q180. Domain analysis produces:

A. Domain models and reusable assets
B. Only code
C. Only tests
D. Only documentation
Correct Answer: A

Q181. Software factoring:

A. Refactoring code for better structure
B. Only adding features
C. Only fixing bugs
D. Only deleting code
Correct Answer: A

Q182. Code smell detection:

A. Identifying patterns indicating deeper problems
B. Only code metrics
C. Only testing
D. Only documentation
Correct Answer: A

Q183. Technical review is:

A. Formal evaluation of work products by stakeholders
B. Informal discussion
C. Peer programming
D. Code compilation
Correct Answer: A

Q184. Software audit checks:

A. Conformance to standards and procedures
B. Only code quality
C. Only user satisfaction
D. Only performance
Correct Answer: A

Q185. Capability Level 2 in CMMI:

A. Managed - processes are planned and executed
B. Optimizing
C. Defining
D. Quantitatively managed
Correct Answer: A

Q186. Capability Level 3 in CMMI:

A. Defined - processes are described and understood
B. Initial
C. Managed
D. Optimizing
Correct Answer: A

Q187. Software process improvement uses:

A. Process models, metrics, and feedback
B. Only coding standards
C. Only testing
D. Only documentation
Correct Answer: A

Q188. Software quality assurance roles include:

A. QA auditor, process engineer, tester
B. Only developer
C. Only manager
D. Only analyst
Correct Answer: A

Q189. Software configuration management plan includes:

A. Identification, control, status accounting, audit
B. Only coding
C. Only testing
D. Only deployment
Correct Answer: A

Q190. Change control board (CCB):

A. Approves or rejects change requests
B. Only implements changes
C. Only tests changes
D. Only documents changes
Correct Answer: A

Q191. Software product line management involves:

A. Managing domain assets and application engineering
B. Only product development
C. Only testing
D. Only deployment
Correct Answer: A

Q192. Software architecture evaluation uses:

A. SAAM, ATAM, ARID methods
B. Only code review
C. Only testing
D. Only documentation
Correct Answer: A

Q193. Architecture Level (AL) in CMMI:

A. Architecture is evaluated and managed
B. Only developed
C. Only documented
D. Only tested
Correct Answer: A

Q194. Software security engineering focuses on:

A. Building security into the software lifecycle
B. Only testing
C. Only deployment
D. Only coding
Correct Answer: A

Q195. Threat modeling identifies:

A. Potential threats and vulnerabilities
B. Only code bugs
C. Only performance issues
D. Only design flaws
Correct Answer: A

Q196. Software assurance includes:

A. Ensuring software meets security and quality requirements
B. Only functionality
C. Only performance
D. Only usability
Correct Answer: A

Q197. Model-based testing uses:

A. Models to generate test cases
B. Only code
C. Only manual testing
D. Only documentation
Correct Answer: A

Q198. Automated test generation:

A. Tools generate test cases from models/code
B. Only manual test cases
C. Only test scripts
D. Only test reports
Correct Answer: A

Q199. Software testing trends include:

A. Shift-left testing, AI-based testing, continuous testing
B. Only manual testing
C. Only unit testing
D. Only integration testing
Correct Answer: A

Q200. DevOps culture emphasizes:

A. Collaboration, automation, continuous feedback
B. Only development
C. Only operations
D. Only testing
Correct Answer: A

ANSWER KEY — Topic 11: Software Engineering

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1-A 2-A 3-A 4-A 5-A 6-A 7-A 8-A 9-A 10-A
11-A 12-A 13-A 14-A 15-A 16-A 17-A 18-A 19-A 20-A
21-A 22-A 23-A 24-A 25-A 26-A 27-A 28-A 29-A 30-A
31-A 32-A 33-A 34-A 35-A 36-A 37-A 38-A 39-A 40-A
41-A 42-A 43-A 44-A 45-A 46-A 47-A 48-A 49-A 50-A
51-A 52-A 53-A 54-A 55-A 56-A 57-A 58-A 59-A 60-A
61-A 62-A 63-A 64-A 65-A 66-A 67-A 68-A 69-A 70-A
71-A 72-A 73-A 74-A 75-A 76-A 77-A 78-A 79-A 80-A
81-A 82-A 83-A 84-A 85-A 86-A 87-A 88-A 89-A 90-A
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141-A 142-A 143-A 144-A 145-A 146-A 147-A 148-A 149-A 150-A
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181-A 182-A 183-A 184-A 185-A 186-A 187-A 188-A 189-A 190-A
191-A 192-A 193-A 194-A 195-A 196-A 197-A 198-A 199-A 200-A
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Topic 12a: Parallel Computing

Syllabus: *Parallel virtual machine (PVM) and message passing interface (MPI) libraries and calls, Advanced architectures, Today's fastest computers.*

Q1. PVM stands for:

A. Parallel Virtual Memory
B. Parallel Virtual Machine
C. Private Virtual Machine
D. Programmed Virtual Module
Correct Answer: B

Q2. MPI stands for:

A. Message Passing Interface
B. Multiple Processing Integration
C. Memory Passing Interconnect
D. Message Programming Instruction
Correct Answer: A

Q3. Which of the following is a distributed-memory parallel programming standard?

A. OpenMP
B. Pthreads
C. MPI
D. CUDA (only shared-memory view)
Correct Answer: C

Q4. In Flynn's taxonomy, SISD refers to:

A. Single Instruction Single Data
B. Single Instruction Multiple Data
C. Multiple Instruction Single Data
D. Multiple Instruction Multiple Data
Correct Answer: A

Q5. GPUs are best classified as:

A. SISD
B. SIMD
C. MISD
D. SISD and MISD
Correct Answer: B

Q6. Which MPI routine must be called first in every MPI program?

A. MPI_Comm_size
B. MPI_Init
C. MPI_Finalize
D. MPI_Start
Correct Answer: B

Q7. Which MPI routine must be called last to cleanly exit?

A. MPI_Init
B. MPI_Finalize
C. MPI_Abort
D. MPI_Stop
Correct Answer: B

Q8. In MPI, MPI_Comm_rank returns:

A. Total number of processes
B. Rank (ID) of calling process in communicator
C. Processor clock speed
D. Message tag
Correct Answer: B

Q9. In MPI, MPI_Comm_size returns:

A. Rank of process
B. Size of message in bytes
C. Number of processes in communicator
D. Size of communicator buffer
Correct Answer: C

Q10. Default communicator that includes all processes is:

A. MPI_COMM_SELF
B. MPI_COMM_ALL
C. MPI_COMM_WORLD
D. MPI_COMM_NULL
Correct Answer: C

Q11. Which MPI call sends a message (blocking)?

A. MPI_Recv
B. MPI_Send
C. MPI_Bcast
D. MPI_Barrier
Correct Answer: B

Q12. Which MPI call receives a message (blocking)?

A. MPI_Send
B. MPI_Recv
C. MPI_Scatter
D. MPI_Gather
Correct Answer: B

Q13. Non-blocking send in MPI is:

A. MPI_Send
B. MPI_Isend
C. MPI_Ssend
D. MPI_Bsend
Correct Answer: B

Q14. MPI_Bcast is used for:

A. Point-to-point communication
B. One-to-all broadcast within communicator
C. Process termination
D. Mutual exclusion
Correct Answer: B

Q15. MPI_Reduce is used to:

A. Reduce number of processes
B. Combine values from all processes to one root using an operation (e.g., SUM, MAX)
C. Decrease message size
D. Remove deadlock
Correct Answer: B

Q16. MPI_Barrier does:

A. Encrypts messages
B. Synchronizes all processes in a communicator; no process proceeds until all reach it
C. Starts MPI
D. Splits communicator
Correct Answer: B

Q17. MPI_Scatter:

A. Gathers data to root
B. Distributes distinct chunks from root to all processes
C. Broadcasts identical data
D. Scatters processes across nodes
Correct Answer: B

Q18. MPI_Gather:

A. Collects distinct data from all processes to root
B. Sends identical data to all
C. Terminates processes
D. Creates new communicator
Correct Answer: A

Q19. PVM was developed primarily at:

A. Microsoft
B. Oak Ridge National Laboratory / University of Tennessee
C. IBM Zurich
D. IIT Bombay
Correct Answer: B

Q20. Which is TRUE for PVM vs MPI?

A. PVM supports dynamic process spawning and fault tolerance better; MPI is standardized for high performance static SPMD
B. MPI supports only shared memory
C. PVM is only for GPUs
D. MPI cannot do point-to-point
Correct Answer: A

Q21. In PVM, the daemon that must run on each host is called:

A. pvmd
B. mpichd
C. sshd
D. httpd
Correct Answer: A

Q22. PVM function pvm_spawn is used to:

A. Terminate tasks
B. Start new tasks/processes on virtual machine
C. Send mail
D. Allocate shared memory
Correct Answer: B

Q23. PVM function pvm_send / pvm_recv are for:

A. File I/O
B. Message send/receive between tasks
C. Memory allocation
D. Thread creation
Correct Answer: B

Q24. Amdahl's Law gives:

A. Maximum theoretical speedup limited by sequential fraction
B. Minimum memory required
C. Network bandwidth
D. Cache hit ratio
Correct Answer: A

Q25. If 90% of a program is parallelizable, with infinite processors, maximum speedup per Amdahl is:

A. 5x
B. 9x
C. 10x
D. 90x
Correct Answer: C

Q26. Gustafson's Law (scaled speedup) argues:

A. Speedup scales with problem size; parallel portion grows with larger problems
B. Parallel computing is impossible
C. Sequential time always dominates
D. Cache is irrelevant
Correct Answer: A

Q27. Speedup S(p) is defined as:

A. T_parallel / T_serial
B. T_serial / T_parallel
C. T_serial + T_parallel
D. p / T_serial
Correct Answer: B

Q28. Efficiency E(p) is defined as:

A. S(p) / p
B. S(p) * p
C. p / S(p)
D. T(p) / p
Correct Answer: A

Q29. Which is an example of shared-memory API?

A. MPI
B. OpenMP / Pthreads
C. PVM
D. TCP sockets only
Correct Answer: B

Q30. UMA stands for:

A. Uniform Memory Access
B. Unified Message Architecture
C. Universal Multi Agent
D. Uniform Message Passing
Correct Answer: A

Q31. NUMA stands for:

A. Non-Uniform Memory Access
B. New Unified Memory Array
C. Non-Unified Message Application
D. Neutral Memory Access
Correct Answer: A

Q32. In a cluster, nodes are connected via:

A. High-speed interconnect (e.g., InfiniBand, Ethernet)
B. Only dial-up modem
C. Single CPU bus only
D. Printer cable
Correct Answer: A

Q33. Beowulf clusters are built using:

A. Off-the-shelf commodity PCs + open-source software + LAN
B. Only superconductor CPUs
C. Only mainframes
D. Mobile phones only
Correct Answer: A

Q34. Flynn's MIMD includes:

A. Multicore CPUs, clusters, MPPs
B. Only vector processors
C. Only SISD machines
D. Only pipelined CPUs
Correct Answer: A

Q35. SIMD machines include:

A. Vector processors, array processors, GPUs
B. Only MIMD clusters
C. Uniprocessors only
D. Distributed databases only
Correct Answer: A

Q36. Which topology has diameter log2(N) for N nodes?

A. Linear array
B. Hypercube
C. Ring
D. Star (center bottleneck)
Correct Answer: B

Q37. Deadlock in message passing can occur when:

A. Two processes both do blocking Send waiting for matching Recv with no buffering
B. Using MPI_Finalize correctly
C. Using broadcast only
D. Single process program
Correct Answer: A

Q38. MPI_TAG is used to:

A. Identify / filter messages
B. Terminate process
C. Measure time
D. Allocate memory
Correct Answer: A

Q39. MPI_Status object after MPI_Recv contains:

A. Source, tag, error fields
B. Only process rank
C. Only clock speed
D. Compiler version
Correct Answer: A

Q40. Collective communications in MPI require:

A. All processes in communicator to call the routine
B. Only root to call
C. No communicator
D. Only two processes
Correct Answer: A

Q41. Which MPI operation finds maximum value across processes?

A. MPI_Reduce with MPI_MAX
B. MPI_Send with MPI_MAX
C. MPI_Barrier with MPI_MAX
D. MPI_Init with MPI_MAX
Correct Answer: A

Q42. Granularity in parallel computing refers to:

A. Ratio of computation to communication; fine vs coarse grain
B. Size of hard disk
C. Screen resolution
D. Clock frequency only
Correct Answer: A

Q43. Data parallelism means:

A. Same operation on different data partitions across processors
B. Different operations on same data
C. Only task queue
D. Sequential execution
Correct Answer: A

Q44. Task parallelism means:

A. Different tasks/threads execute different operations, possibly on same or different data
B. Same instruction on all data
C. No parallelism
D. Only pipelining in ALU
Correct Answer: A

Q45. Advanced architecture: Multi-core is:

A. Multiple processing cores on single chip sharing/dedicated caches
B. Multiple computers in different countries only
C. Single ALU only
D. Only I/O device
Correct Answer: A

Q46. GPGPU computing commonly uses:

A. CUDA, OpenCL
B. Only COBOL
C. Only HTML
D. Only PVM daemons
Correct Answer: A

Q47. Today's fastest supercomputers (exascale era, e.g., Frontier, Aurora) achieve:

A. ExaFLOPS (~10^18 floating-point ops/sec)
B. Only kiloFLOPS
C. Only megaFLOPS
D. 10 FLOPS
Correct Answer: A

Q48. TOP500 list ranks supercomputers based on:

A. LINPACK benchmark (solving dense linear systems)
B. Number of printers attached
C. Power consumption only
D. Disk capacity only
Correct Answer: A

Q49. Which interconnect is most common in TOP supercomputers for low latency?

A. InfiniBand / proprietary high-speed interconnects (e.g., Slingshot, Tofu)
B. Dial-up 56kbps
C. Bluetooth Low Energy only
D. RS-232 serial
Correct Answer: A

Q50. PVM supports:

A. Heterogeneous network computing - virtual machine hides architecture differences (XDR for data conversion)
B. Only homogeneous Intel machines
C. No network support
D. Only shared memory threads
Correct Answer: A

ANSWER KEY — Topic 12a: Parallel Computing

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1-B 2-A 3-C 4-A 5-B 6-B 7-B 8-B 9-C 10-C
11-B 12-B 13-B 14-B 15-B 16-B 17-B 18-A 19-B 20-A
21-A 22-B 23-B 24-A 25-C 26-A 27-B 28-A 29-B 30-A
31-A 32-A 33-A 34-A 35-A 36-B 37-A 38-A 39-A 40-A
41-A 42-A 43-A 44-A 45-A 46-A 47-A 48-A 49-A 50-A
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Topic 12b: Mobile Computing

Syllabus: *Mobile connectivity - Cells, Framework, wireless delivery technology and switching methods, mobile information access devices, mobile data internetworking standards, cellular data communication protocols, mobile computing applications, Mobile databases - protocols, scope, tools and technology, M-business.*

Q1. Mobile Computing framework primarily enables?

A. Fixed desktop access
B. Anytime anywhere access
C. Only wired access
D. Batch processing
Correct Answer: B

Q2. Cell in cellular network is?

A. Geographic service area
B. Battery unit
C. SIM card
D. Antenna tower
Correct Answer: A

Q3. Base Station (BS) function is?

A. Billing only
B. Radio communication with mobiles
C. Manufacturing phones
D. Satellite launch
Correct Answer: B

Q4. MSC stands for?

A. Mobile Switching Center
B. Mobile Service Cell
C. Main Signal Controller
D. Message Switching Code
Correct Answer: A

Q5. HLR stores?

A. Permanent subscriber info
B. Temporary visitor info
C. Signal strength
D. Battery level
Correct Answer: A

Q6. VLR stores?

A. Permanent data
B. Visiting subscriber temporary data
C. Cell size
D. Frequency plan
Correct Answer: B

Q7. Frequency reuse means?

A. Same frequency in distant cells
B. One frequency per country
C. No reuse
D. Wired reuse
Correct Answer: A

Q8. Cell splitting is done to?

A. Increase capacity
B. Decrease capacity
C. Stop handoff
D. Remove MSC
Correct Answer: A

Q9. Sectoring uses?

A. Directional antennas
B. Omnidirectional only
C. No antennas
D. Satellites only
Correct Answer: A

Q10. Cluster size N=7 means?

A. 7-cell reuse pattern
B. 7 users
C. 7 MSCs
D. 7 SIMs
Correct Answer: A

Q11. FDMA divides by?

A. Frequency
B. Time
C. Code
D. Space only
Correct Answer: A

Q12. TDMA divides by?

A. Time slots
B. Frequency only
C. Codes
D. Power
Correct Answer: A

Q13. CDMA uses?

A. Spread spectrum codes
B. Time slots only
C. Frequency only
D. Wires
Correct Answer: A

Q14. GSM uses combination of?

A. FDMA+TDMA
B. CDMA only
C. SDMA only
D. PDMA
Correct Answer: A

Q15. GSM operates in which band (primary)?

A. 900/1800 MHz
B. 10 GHz
C. 100 Hz
D. Infrared only
Correct Answer: A

Q16. CDMA capacity compared to GSM is generally?

A. Higher
B. Zero
C. Same as wired
D. Lower always
Correct Answer: A

Q17. Handoff means?

A. Transfer call between cells
B. Switch off phone
C. Recharge battery
D. Change SIM
Correct Answer: A

Q18. Hard handoff is?

A. Break-before-make
B. Make-before-break
C. No break
D. Wired transfer
Correct Answer: A

Q19. Soft handoff used in?

A. CDMA
B. Wired LAN
C. FDMA only
D. Paging only
Correct Answer: A

Q20. MAHO stands for?

A. Mobile Assisted Handoff
B. Manual Handoff
C. Master Handoff
D. Message Handoff
Correct Answer: A

Q21. Wireless delivery framework includes?

A. Air interface + network
B. Only cable
C. Only printer
D. Only desktop
Correct Answer: A

Q22. Circuit switching is?

A. Dedicated path
B. No path
C. Packet only
D. Message only
Correct Answer: A

Q23. Packet switching is best for?

A. Bursty data
B. Continuous voice only
C. Fixed line
D. No data
Correct Answer: A

Q24. Switching in mobile networks handled by?

A. MSC/GMSC
B. SIM
C. Charger
D. Display
Correct Answer: A

Q25. GMSC connects mobile network to?

A. PSTN
B. Printer
C. Keyboard
D. Monitor
Correct Answer: A

Q26. Wireless transmission medium is?

A. Air/radio waves
B. Fiber only
C. Coaxial only
D. Paper
Correct Answer: A

Q27. Path loss means?

A. Signal weakens with distance
B. Battery loss
C. Data gain
D. No loss
Correct Answer: A

Q28. Multipath fading caused by?

A. Reflections/refraction
B. Single path
C. Wired link
D. Battery
Correct Answer: A

Q29. Doppler effect in mobile is due to?

A. Mobility
B. Static user
C. Wired phone
D. SMS length
Correct Answer: A

Q30. Co-channel interference is from?

A. Same frequency cells
B. Different services
C. Wired phones
D. Battery
Correct Answer: A

Q31. Adjacent channel interference due to?

A. Neighbor frequencies leakage
B. Same cell ID
C. SIM
D. OS
Correct Answer: A

Q32. Power control important in?

A. CDMA
B. Wired LAN
C. Paging
D. Printing
Correct Answer: A

Q33. Near-far problem occurs in?

A. CDMA
B. Wired Ethernet
C. FDMA only
D. Bluetooth only
Correct Answer: A

Q34. Authentication Center (AuC) is for?

A. Security/authentication
B. Billing print
C. Tower paint
D. Phone cover
Correct Answer: A

Q35. EIR stores?

A. Equipment identity/IMEI
B. User voice
C. SMS text
D. Battery
Correct Answer: A

Q36. Mobile access device example is?

A. Smartphone/PDA
B. Desktop server
C. Mainframe
D. Printer
Correct Answer: A

Q37. PDA stands for?

A. Personal Digital Assistant
B. Public Data Access
C. Private Device Allocation
D. Personal Dialing Agent
Correct Answer: A

Q38. Smartphone OS example is?

A. Android
B. MS-DOS
C. UNIX server only
D. BIOS
Correct Answer: A

Q39. SIM stands for?

A. Subscriber Identity Module
B. Simple Identity Machine
C. Signal Meter
D. System Mode
Correct Answer: A

Q40. IMEI identifies?

A. Mobile equipment
B. Subscriber
C. Network
D. SMS
Correct Answer: A

Q41. IMSI identifies?

A. Subscriber
B. Phone model
C. Tower
D. Battery
Correct Answer: A

Q42. Key constraint of mobile device is?

A. Limited battery/memory
B. Unlimited power
C. Fixed location
D. Large screen always
Correct Answer: A

Q43. Mobile device portability requires?

A. Low power/lightweight
B. Heavy mainframe
C. Wired power always
D. No wireless
Correct Answer: A

Q44. Pager is?

A. One-way messaging device
B. Laptop
C. Router
D. Satellite
Correct Answer: A

Q45. Laptop as mobile device uses?

A. Wireless NIC/Cellular modem
B. No network
C. Only floppy
D. Only serial
Correct Answer: A

Q46. Sensors in mobiles include?

A. GPS/accelerometer
B. Printer
C. Scanner
D. Projector only
Correct Answer: A

Q47. Mobile device connectivity via?

A. Cellular/Wi-Fi/Bluetooth
B. Only Ethernet
C. Only serial
D. No connectivity
Correct Answer: A

Q48. Limited display implies need for?

A. Content adaptation
B. No adaptation
C. Larger servers
D. More cables
Correct Answer: A

Q49. Mobile file system must handle?

A. Disconnection/mobility
B. Only fixed disk
C. No caching
D. No sync
Correct Answer: A

Q50. Mobile OS must support?

A. Power management/wireless
B. Only batch
C. No multitasking
D. Mainframe jobs
Correct Answer: A

Q51. Internetworking standards ensure?

A. Interoperability
B. Incompatibility
C. Isolation
D. Single vendor lock
Correct Answer: A

Q52. TCP/IP in mobile needs adaptation due to?

A. Mobility/errors/disconnection
B. Wired stability
C. No errors
D. Fixed IP always
Correct Answer: A

Q53. Mobile IP home agent does?

A. Forward to mobile's current location
B. Block calls
C. Charge battery
D. Print bills
Correct Answer: A

Q54. Care-of-address is?

A. Temporary mobile location address
B. Permanent home
C. Email ID
D. SIM number
Correct Answer: A

Q55. Foreign agent in Mobile IP?

A. Serves visitor in foreign network
B. Blocks roaming
C. Makes phones
D. Sells SIM
Correct Answer: A

Q56. Tunneling in Mobile IP is?

A. Encapsulate to forward
B. Digging
C. Encryption only
D. Antenna setup
Correct Answer: A

Q57. IPv6 advantage for mobiles?

A. Large address space
B. Smaller addresses
C. No mobility
D. No security
Correct Answer: A

Q58. IEEE 802.11 is for?

A. WLAN/Wi-Fi
B. GSM
C. Bluetooth only
D. 3G
Correct Answer: A

Q59. Bluetooth range is approx?

A. 10m
B. 10km
C. 100km
D. 1000km
Correct Answer: A

Q60. Infrared (IrDA) requires?

A. Line of sight
B. Long range
C. Satellite
D. Tower
Correct Answer: A

Q61. OSI layer for routing?

A. Network
B. Physical
C. Session
D. Presentation
Correct Answer: A

Q62. TCP suffers in wireless due to?

A. Mistaking loss as congestion
B. Too fast
C. No loss
D. Wired speed
Correct Answer: A

Q63. Indirect-TCP (I-TCP) splits connection at?

A. Base station
B. Mobile only
C. Satellite
D. Printer
Correct Answer: A

Q64. Snooping TCP advantage?

A. Local retransmission
B. End-to-end only
C. No wireless
D. Wired only
Correct Answer: A

Q65. WAP stands for?

A. Wireless Application Protocol
B. Wired Access Program
C. Wide Area Printer
D. Web And Phone
Correct Answer: A

Q66. WAP gateway converts between?

A. WAP and HTTP/Internet
B. Voice and printer
C. SMS and fax only
D. GSM and CDMA hardware
Correct Answer: A

Q67. WML stands for?

A. Wireless Markup Language
B. Wired Markup Language
C. Wide Mode Language
D. Web Master Language
Correct Answer: A

Q68. WAP protocol stack includes WSP/WTP/WTLS/WDP. WTLS provides?

A. Security
B. Routing only
C. Billing
D. Display
Correct Answer: A

Q69. WDP corresponds to which layer?

A. Transport/datagram
B. Physical only
C. Application
D. Session
Correct Answer: A

Q70. WAP microbrowser is?

A. Small browser for mobiles
B. Desktop Chrome
C. Server
D. Router
Correct Answer: A

Q71. SMS stands for?

A. Short Message Service
B. System Message Service
C. Signal Mode Service
D. Subscriber Mode Service
Correct Answer: A

Q72. SMS length limit is?

A. 160 chars
B. 1000 chars
C. 10 chars
D. Unlimited
Correct Answer: A

Q73. SMS uses which channel?

A. Signaling/control channel
B. Voice only
C. Data cable
D. Infrared
Correct Answer: A

Q74. SMS Center (SMSC) function?

A. Store-and-forward SMS
B. Voice call
C. Charge phone
D. Make tower
Correct Answer: A

Q75. Cell broadcast SMS is?

A. One-to-many in cell
B. One-to-one only
C. Wired mail
D. Voice call
Correct Answer: A

Q76. GPRS stands for?

A. General Packet Radio Service
B. Global Packet Relay System
C. Grouped Phone Radio Service
D. General Phone Routing
Correct Answer: A

Q77. GPRS is which generation?

A. 2.5G
B. 1G
C. 5G
D. 0G
Correct Answer: A

Q78. GPRS charging is based on?

A. Data volume
B. Call duration only
C. Battery
D. Distance
Correct Answer: A

Q79. SGSN in GPRS is?

A. Serving GPRS Support Node
B. Signal Node
C. Simple Node
D. Satellite Node
Correct Answer: A

Q80. GGSN connects GPRS to?

A. External packet networks/Internet
B. Printer
C. Voice only
D. SIM
Correct Answer: A

Q81. EDGE stands for?

A. Enhanced Data rates for GSM Evolution
B. Extended Data Group
C. Emergency Data Gateway
D. European Data Group
Correct Answer: A

Q82. EDGE uses modulation?

A. 8-PSK
B. FM only
C. AM only
D. QPSK only
Correct Answer: A

Q83. EDGE speed compared to GPRS is?

A. Higher
B. Lower
C. Equal
D. Zero
Correct Answer: A

Q84. 3G primary goal?

A. High-speed data+multimedia
B. Voice only
C. SMS only
D. Paging only
Correct Answer: A

Q85. UMTS stands for?

A. Universal Mobile Telecommunications System
B. United Mobile Telephone Service
C. Universal Message Transfer System
D. User Mode Telecom
Correct Answer: A

Q86. W-CDMA used in?

A. 3G
B. 1G
C. Pager
D. Wired LAN
Correct Answer: A

Q87. IMT-2000 relates to?

A. 3G standard
B. 1G
C. Printer
D. Bluetooth
Correct Answer: A

Q88. 4G technology example?

A. LTE
B. GSM
C. Pager
D. AMPS
Correct Answer: A

Q89. LTE stands for?

A. Long Term Evolution
B. Low Telecom Evolution
C. Local Transfer Exchange
D. Long Telephone Exchange
Correct Answer: A

Q90. 4G offers?

A. All-IP high broadband
B. Analog voice only
C. Circuit only
D. SMS only
Correct Answer: A

Q91. AMPS is?

A. 1G analog
B. 4G
C. 5G
D. Packet data
Correct Answer: A

Q92. D-AMPS (IS-136) uses?

A. TDMA
B. CDMA only
C. FDMA only
D. Wired
Correct Answer: A

Q93. IS-95 uses?

A. CDMA
B. TDMA only
C. FDMA only
D. Analog
Correct Answer: A

Q94. GSM channel bandwidth?

A. 200 kHz
B. 1.25 MHz
C. 5 MHz
D. 10 MHz
Correct Answer: A

Q95. IS-95 channel bandwidth?

A. 1.25 MHz
B. 200 kHz
C. 30 kHz
D. 10 kHz
Correct Answer: A

Q96. Multiple access in GSM?

A. TDMA with 8 slots
B. 64 slots
C. 1 slot
D. 100 slots
Correct Answer: A

Q97. Authentication in GSM uses?

A. Ki + A3/A8
B. Password only
C. No auth
D. IPsec
Correct Answer: A

Q98. GSM cipher A5 is for?

A. Encryption
B. Compression
C. Billing
D. Routing
Correct Answer: A

Q99. Location update in GSM happens when?

A. Crossing location area
B. Switching off only
C. Never
D. Charging
Correct Answer: A

Q100. Paging is used to?

A. Locate mobile for incoming call
B. Charge battery
C. Send email
D. Print
Correct Answer: A

Q101. Cellular data protocols include?

A. GSM/CDMA/GPRS
B. HTTP only
C. FTP only
D. SMTP only
Correct Answer: A

Q102. Circuit-switched data in GSM gives?

A. 9.6 kbps
B. 100 Mbps
C. 1 Gbps
D. 10 Gbps
Correct Answer: A

Q103. HSCSD stands for?

A. High Speed Circuit Switched Data
B. High System Code Division
C. Home Subscriber Data
D. Hybrid Switching Data
Correct Answer: A

Q104. Mobile apps categories include?

A. Native/web/hybrid
B. Only native
C. Only mainframe
D. Only DOS
Correct Answer: A

Q105. Native app is?

A. Platform-specific installed
B. Browser only
C. Server only
D. Printer driver
Correct Answer: A

Q106. Mobile middleware provides?

A. Connectivity/sync/adaptation
B. Hardware only
C. Tower only
D. Battery
Correct Answer: A

Q107. J2ME was used for?

A. Mobile Java apps
B. Desktop only
C. Server only
D. Routing
Correct Answer: A

Q108. Android apps written mainly in?

A. Java/Kotlin
B. COBOL
C. Fortran
D. Pascal
Correct Answer: A

Q109. iOS apps written in?

A. Swift/Objective-C
B. COBOL
C. BASIC only
D. Assembly only
Correct Answer: A

Q110. Mobile database means?

A. DB accessible via mobile/distributed
B. Only central DB
C. No DB
D. Paper DB
Correct Answer: A

Q111. Mobile database issue is?

A. Disconnection/data sync
B. Unlimited bandwidth
C. Fixed location
D. No security need
Correct Answer: A

Q112. Replication in mobile DB is for?

A. Availability/offline access
B. Delete data
C. Stop sync
D. Increase cost
Correct Answer: A

Q113. Hoarding/caching in mobile DB helps during?

A. Disconnection
B. Charging only
C. Printing
D. Booting
Correct Answer: A

Q114. Data broadcast (push) in mobile is?

A. Server broadcasts to many
B. Client pulls only
C. Wired only
D. No broadcast
Correct Answer: A

Q115. Pull model is?

A. Client requests
B. Server pushes always
C. No request
D. Tower pushes
Correct Answer: A

Q116. Conflict resolution needed due to?

A. Concurrent updates offline
B. Single user
C. Wired lock
D. No updates
Correct Answer: A

Q117. Two-tier replication involves?

A. Mobile + server DB
B. Two printers
C. Two towers
D. Two SIMs
Correct Answer: A

Q118. Mobile transaction ACID is hard due to?

A. Mobility/failures
B. Fixed network
C. No transactions
D. Large servers
Correct Answer: A

Q119. Timeout-based commit used because?

A. Mobiles may disconnect
B. Servers fast
C. No disconnect
D. Wired stable
Correct Answer: A

Q120. Mobile DB tools include?

A. SQLite/Oracle Lite/DB2 Everyplace
B. Only Oracle server
C. MS Word
D. Photoshop
Correct Answer: A

Q121. SQLite is?

A. Lightweight embedded DB
B. Server DB only
C. Network protocol
D. Browser
Correct Answer: A

Q122. Scope of mobile DB protocol includes?

A. Sync/conflict/security
B. Only display
C. Only battery
D. Only tower
Correct Answer: A

Q123. SyncML is for?

A. Data synchronization
B. Voice coding
C. Tower setup
D. Billing
Correct Answer: A

Q124. Data dissemination modes are push/pull/hybrid. Hybrid means?

A. Combination
B. Push only
C. Pull only
D. Neither
Correct Answer: A

Q125. Location-dependent data means?

A. Data varies by location
B. Same everywhere
C. No location
D. Fixed data
Correct Answer: A

Q126. Mobile query must consider?

A. Location/mobility
B. Only syntax
C. Only server speed
D. Printer
Correct Answer: A

Q127. M-business stands for?

A. Mobile business/commerce
B. Manual business
C. Mainframe business
D. Morning business
Correct Answer: A

Q128. M-commerce enables?

A. Transactions via mobile
B. Only shop visit
C. Cash only
D. No payment
Correct Answer: A

Q129. M-business applications include?

A. Banking/ticketing/shopping
B. Only farming
C. Only printing
D. Only teaching
Correct Answer: A

Q130. WAP banking uses?

A. Secure mobile browser
B. Desktop only
C. Paper
D. Voice only
Correct Answer: A

Q131. Mobile payment modes include?

A. Micropayment/macropayment
B. Cash only
C. Cheque only
D. No payment
Correct Answer: A

Q132. Security in M-commerce needs?

A. Authentication/encryption
B. No security
C. Open data
D. Shared PIN
Correct Answer: A

Q133. Mobile advertising advantage?

A. Personalized/location-based
B. Generic only
C. No reach
D. Paper only
Correct Answer: A

Q134. LBS stands for?

A. Location Based Services
B. Local Billing System
C. Large Base Station
D. Low Bandwidth Service
Correct Answer: A

Q135. GPS helps M-business via?

A. Positioning/navigation
B. Billing only
C. Display only
D. Battery
Correct Answer: A

Q136. Supply chain M-business benefit?

A. Real-time tracking
B. Delay
C. Paper only
D. No tracking
Correct Answer: A

Q137. Mobile ticketing uses?

A. SMS/QR/barcode
B. Cash only
C. Paper ledger
D. Voice only
Correct Answer: A

Q138. M-business challenge?

A. Security/small screen/bandwidth
B. Unlimited screen
C. No mobility
D. Fixed users
Correct Answer: A

Q139. B2C in M-commerce means?

A. Business to Consumer
B. Business to Code
C. Battery to Consumer
D. Base to Cell
Correct Answer: A

Q140. B2B in M-commerce means?

A. Business to Business
B. Battery to Business
C. Base to Base
D. Browser to Business
Correct Answer: A

Q141. P2P mobile payment example?

A. User-to-user transfer
B. Printer to phone
C. Tower to tower
D. SIM to SIM hardware
Correct Answer: A

Q142. Mobile wallet stores?

A. Digital payment credentials
B. Clothes
C. Photos only
D. Towers
Correct Answer: A

Q143. NFC used for?

A. Contactless payment
B. Long-range calls
C. Satellite
D. Tower setup
Correct Answer: A

Q144. Roaming enables?

A. Service outside home network
B. No service outside
C. Free battery
D. Extra SIM
Correct Answer: A

Q145. International roaming needs?

A. Agreements/standards
B. Same tower
C. No network
D. New phone always
Correct Answer: A

Q146. Handoff failure leads to?

A. Dropped call
B. Better signal
C. Free call
D. Longer battery
Correct Answer: A

Q147. Dropped call rate measures?

A. QoS
B. Battery
C. Price
D. Screen size
Correct Answer: A

Q148. Blocking probability is when?

A. New call denied no channel
B. Call connected
C. SMS delivered
D. Phone off
Correct Answer: A

Q149. Erlang measures?

A. Traffic intensity
B. Signal power
C. Battery
D. Distance
Correct Answer: A

Q150. Spectrum is?

A. Scarce resource
B. Unlimited
C. Free infinite
D. Wired
Correct Answer: A

Q151. 900 MHz propagation vs 1800 MHz?

A. Better coverage lower freq
B. Same
C. Worse always
D. No coverage
Correct Answer: A

Q152. Microcell is for?

A. Dense urban/high capacity
B. Rural large
C. Satellite
D. Ocean
Correct Answer: A

Q153. Macrocell covers?

A. Large rural area
B. Room only
C. Desk only
D. Building floor
Correct Answer: A

Q154. Picocell covers?

A. Small indoor
B. Country
C. State
D. Continent
Correct Answer: A

Q155. Femtocell is?

A. Home small base station
B. Large tower
C. Satellite
D. MSC
Correct Answer: A

Q156. Reuse distance larger gives?

A. Less interference
B. More interference
C. No effect
D. More capacity always
Correct Answer: A

Q157. Capacity increase via?

A. Cell splitting/sectoring
B. Larger cells only
C. Less spectrum
D. Fewer towers
Correct Answer: A

Q158. Mobile agent in apps is?

A. Migrating software component
B. Human agent
C. SIM seller
D. Tower worker
Correct Answer: A

Q159. Client-server mobile app requires?

A. Wireless connectivity
B. No network
C. Mainframe only
D. Paper
Correct Answer: A

Q160. Offline mobile app uses?

A. Local storage/sync later
B. No storage
C. Server always
D. Printer
Correct Answer: A

Q161. Mobile Web Services use?

A. HTTP/SOAP/REST
B. SMS only
C. Voice only
D. Pager only
Correct Answer: A

Q162. XML/JSON in mobile for?

A. Data exchange
B. Voice coding
C. Tower control
D. Battery
Correct Answer: A

Q163. Mobile VPN provides?

A. Secure remote access
B. Free internet
C. Faster battery
D. Larger screen
Correct Answer: A

Q164. Mobile security threats include?

A. Eavesdropping/theft/malware
B. Only battery drain
C. Tower paint
D. SIM color
Correct Answer: A

Q165. Eavesdropping prevented by?

A. Encryption
B. Louder speaker
C. Larger phone
D. More towers
Correct Answer: A

Q166. Mobile virus spreads via?

A. Apps/messages/Bluetooth
B. Battery
C. Charger
D. Cover
Correct Answer: A

Q167. Firewalls on mobile for?

A. Network protection
B. Heating
C. Display
D. Sound
Correct Answer: A

Q168. PIN in mobile for?

A. User authentication
B. Tower ID
C. Signal boost
D. Battery save
Correct Answer: A

Q169. PUK is used to?

A. Unblock SIM
B. Boost signal
C. Increase memory
D. Change OS
Correct Answer: A

Q170. Mobile backup is needed due to?

A. Loss/theft/failure
B. Unlimited safety
C. Fixed location
D. No data
Correct Answer: A

Q171. OTA stands for?

A. Over The Air provisioning
B. Office Telecom Access
C. On Time Arrival
D. Open Transfer Area
Correct Answer: A

Q172. OTA updates allow?

A. Remote config/software update
B. Hardware repair only
C. Battery replace
D. Tower paint
Correct Answer: A

Q173. Mobile portal is?

A. Gateway to services
B. Physical gate
C. Tower
D. Cable
Correct Answer: A

Q174. Personalization in mobile apps means?

A. Customized per user/context
B. Same for all
C. No profile
D. Fixed ads
Correct Answer: A

Q175. Context-aware app uses?

A. Location/time/profile
B. Only time
C. No context
D. Only battery
Correct Answer: A

Q176. Push notification is?

A. Server-initiated alert
B. Client request only
C. Email only
D. SMS only voice
Correct Answer: A

Q177. Mobile database recovery must handle?

A. Frequent disconnects
B. Stable power always
C. No failures
D. Wired safety
Correct Answer: A

Q178. Versioning in sync is for?

A. Detect conflicts
B. Delete all
C. Stop sync
D. Charge battery
Correct Answer: A

Q179. Serializability in mobile transactions is?

A. Correctness criterion
B. Display mode
C. Battery mode
D. Signal type
Correct Answer: A

Q180. Mobile learning (M-learning) is?

A. Learning via mobiles
B. Tower learning
C. Manual learning
D. Machine only
Correct Answer: A

Q181. M-health uses mobiles for?

A. Telemedicine/monitoring
B. Tower repair
C. Billing only
D. Printing
Correct Answer: A

Q182. Mobile inventory management benefits?

A. Real-time stock update
B. Paper delay
C. No update
D. Annual check
Correct Answer: A

Q183. Field service M-business uses?

A. Remote dispatch/update
B. Office only
C. No mobility
D. Paper only
Correct Answer: A

Q184. Mobile CRM helps?

A. On-field customer service
B. No customers
C. Fixed desk only
D. Printer service
Correct Answer: A

Q185. Disadvantage of M-business?

A. Security/device limits
B. Unlimited bandwidth
C. Large screens
D. No cost
Correct Answer: A

Q186. SMS banking is?

A. Balance/alerts via SMS
B. Branch visit only
C. ATM only
D. Voice only
Correct Answer: A

Q187. USSD in mobile banking is?

A. Session-based menu service
B. Voice call
C. Video call
D. Email
Correct Answer: A

Q188. WAP vs SMS banking: WAP needs?

A. Data bearer/browser
B. No phone
C. Paper
D. Tower ownership
Correct Answer: A

Q189. Mobile top-up/recharge is example of?

A. M-commerce
B. E-mail
C. Printing
D. Fax
Correct Answer: A

Q190. Auction via mobile is?

A. M-business app
B. Tower app
C. Battery app
D. SIM app
Correct Answer: A

Q191. Reservation systems on mobile use?

A. Online booking/payment
B. Counter only
C. Paper only
D. No network
Correct Answer: A

Q192. Entertainment M-business includes?

A. Games/music/video
B. Only voice
C. Only SMS
D. Only banking
Correct Answer: A

Q193. Mobile gaming needs?

A. Graphics/low latency
B. No display
C. No network
D. Paper
Correct Answer: A

Q194. Location tracking in logistics uses?

A. GPS + cellular
B. Paper map
C. No tech
D. Pager only voice
Correct Answer: A

Q195. Privacy concern in LBS?

A. Location disclosure
B. Battery color
C. Phone weight
D. Screen size
Correct Answer: A

Q196. Opt-in for LBS means?

A. User consent
B. Forced tracking
C. No consent
D. Tower consent
Correct Answer: A

Q197. Future mobile (old syllabus view 4G) promised?

A. 100 Mbps+ broadband
B. 9.6 kbps only
C. Analog only
D. No data
Correct Answer: A

Q198. Seamless mobility means?

A. Uninterrupted across networks
B. Disconnect always
C. Manual reconnect
D. No roaming
Correct Answer: A

Q199. Convergence in mobile means?

A. Voice/data/video integrated
B. Separate only
C. Voice only
D. Data only
Correct Answer: A

Q200. Main M-business driver?

A. Ubiquity/personalization
B. Fixed lines
C. Paper
D. Desktops only
Correct Answer: A

ANSWER KEY — Topic 12b: Mobile Computing

1-B 2-A 3-B 4-A 5-A 6-B 7-A 8-A 9-A 10-A
11-A 12-A 13-A 14-A 15-A 16-A 17-A 18-A 19-A 20-A
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101-A 102-A 103-A 104-A 105-A 106-A 107-A 108-A 109-A 110-A
111-A 112-A 113-A 114-A 115-A 116-A 117-A 118-A 119-A 120-A
121-A 122-A 123-A 124-A 125-A 126-A 127-A 128-A 129-A 130-A
131-A 132-A 133-A 134-A 135-A 136-A 137-A 138-A 139-A 140-A
141-A 142-A 143-A 144-A 145-A 146-A 147-A 148-A 149-A 150-A
151-A 152-A 153-A 154-A 155-A 156-A 157-A 158-A 159-A 160-A
161-A 162-A 163-A 164-A 165-A 166-A 167-A 168-A 169-A 170-A
171-A 172-A 173-A 174-A 175-A 176-A 177-A 178-A 179-A 180-A
181-A 182-A 183-A 184-A 185-A 186-A 187-A 188-A 189-A 190-A
191-A 192-A 193-A 194-A 195-A 196-A 197-A 198-A 199-A 200-A

Topic 12c: Data Warehousing and Mining

Syllabus: *Data Warehousing: Data Warehouse environment, architecture of a data warehouse methodology, analysis, design, construction and administration; Data Mining: Extracting models and patterns from large databases, data mining techniques, classification, regression, clustering, summarization, dependency modeling, link analysis, sequencing analysis, mining scientific and business data.*

Q1. Data warehouse is?

A. Subject-oriented integrated time-variant nonvolatile
B. Transactional only
C. Temporary cache
D. Operational file
Correct Answer: A

Q2. Data warehouse environment includes?

A. Source/staging/warehouse/marts
B. Printer only
C. Keyboard only
D. CPU only
Correct Answer: A

Q3. OLTP focuses on?

A. Day-to-day transactions
B. Analysis only
C. Mining only
D. Archiving only
Correct Answer: A

Q4. OLAP focuses on?

A. Analysis/decision support
B. Transactions only
C. Data entry only
D. Billing only
Correct Answer: A

Q5. OLTP vs OLAP: data in OLTP is?

A. Current detailed normalized
B. Historical multidimensional
C. Aggregated only
D. External only
Correct Answer: A

Q6. OLAP query type is?

A. Complex ad-hoc multidimensional
B. Simple short update
C. Delete only
D. Insert only
Correct Answer: A

Q7. Warehouse architecture tiers commonly?

A. 3-tier
B. 1-tier only
C. No tier
D. 10-tier
Correct Answer: A

Q8. Bottom tier is?

A. Warehouse server/RDBMS
B. Client browser
C. Printer
D. OLAP server
Correct Answer: A

Q9. Middle tier is?

A. OLAP server
B. Data source
C. Monitor
D. Keyboard
Correct Answer: A

Q10. Top tier is?

A. Client/query tools
B. Disk only
C. Tape only
D. Source DB
Correct Answer: A

Q11. Data warehouse methodology steps include?

A. Analysis/design/construction/admin
B. Coding only
C. Testing only
D. Purchase only
Correct Answer: A

Q12. Requirement analysis in warehouse collects?

A. Business needs/KPIs
B. Hardware color
C. Staff age
D. Office size
Correct Answer: A

Q13. Dimensional modeling designs?

A. Facts/dimensions
B. ER only
C. Network only
D. Files only
Correct Answer: A

Q14. Fact table contains?

A. Measures/metrics
B. Only text
C. Only keys
D. Only names
Correct Answer: A

Q15. Dimension table contains?

A. Descriptive attributes
B. Only measures
C. Only facts
D. No data
Correct Answer: A

Q16. Star schema has?

A. Central fact + denormalized dims
B. Fully normalized
C. No fact
D. No dims
Correct Answer: A

Q17. Snowflake schema is?

A. Normalized dimensions
B. Denormalized only
C. No fact
D. Star with no center
Correct Answer: A

Q18. Star vs snowflake: query speed?

A. Star faster (fewer joins)
B. Snowflake always faster
C. Equal always
D. No queries
Correct Answer: A

Q19. Fact constellation is?

A. Multiple facts sharing dims
B. Single fact only
C. No facts
D. Only dims
Correct Answer: A

Q20. Slowly changing dimension Type 2?

A. New row per change
B. Overwrite
C. No history
D. Delete
Correct Answer: A

Q21. ETL stands for?

A. Extract Transform Load
B. Enter Table Load
C. Extract Table Link
D. Event Time Log
Correct Answer: A

Q22. Staging area is for?

A. Cleaning/transform before load
B. User queries
C. Final reports
D. Archiving only
Correct Answer: A

Q23. Data cleaning handles?

A. Missing/noisy/inconsistent
B. Clean data only
C. Hardware
D. Software install
Correct Answer: A

Q24. Data construction phase includes?

A. ETL + load + test
B. Requirement only
C. Design only
D. Admin only
Correct Answer: A

Q25. Metadata in warehouse is?

A. Data about data
B. User data
C. Fact data
D. Backup data
Correct Answer: A

Q26. Operational data store (ODS) is?

A. Current integrated for ops
B. Historical only
C. Archive only
D. External only
Correct Answer: A

Q27. Data mart is?

A. Departmental subset
B. Entire enterprise only
C. Source system
D. Staging only
Correct Answer: A

Q28. Dependent mart is fed from?

A. Warehouse
B. Source directly
C. Printer
D. User
Correct Answer: A

Q29. Independent mart is fed from?

A. Source directly
B. Warehouse only
C. OLAP
D. Browser
Correct Answer: A

Q30. Warehouse administration includes?

A. Security/backup/tuning
B. Coding only
C. Design only
D. Purchase only
Correct Answer: A

Q31. Roll-up operation?

A. Aggregate to higher level
B. Drill to detail
C. Slice only
D. Delete
Correct Answer: A

Q32. Drill-down operation?

A. Detail from aggregate
B. Aggregate only
C. Filter only
D. Delete
Correct Answer: A

Q33. Slice operation?

A. One dimension selection
B. All dims
C. No filter
D. Aggregation only
Correct Answer: A

Q34. Dice operation?

A. Multi-dim subcube
B. Single value
C. Delete cube
D. Create fact
Correct Answer: A

Q35. Pivot/rotate is?

A. Reorient cube view
B. Delete cube
C. Load data
D. Clean data
Correct Answer: A

Q36. MOLAP stores?

A. Multidimensional cubes
B. Relational only
C. Files only
D. No storage
Correct Answer: A

Q37. ROLAP stores?

A. Relational tables
B. Cubes only
C. XML only
D. Flat files
Correct Answer: A

Q38. HOLAP is?

A. Hybrid MOLAP+ROLAP
B. History only
C. High OLAP printer
D. Home OLAP
Correct Answer: A

Q39. Data cube lattice includes?

A. All cuboids/group-bys
B. Single table
C. No aggregation
D. Source only
Correct Answer: A

Q40. Iceberg cube stores only?

A. Cells meeting threshold
B. All cells
C. No cells
D. Dims only
Correct Answer: A

Q41. Warehouse design: top-down by?

A. Inmon
B. Kimball
C. Codd
D. Gray
Correct Answer: A

Q42. Bottom-up marts-first by?

A. Kimball
B. Inmon
C. Turing
D. Shannon
Correct Answer: A

Q43. Granularity means?

A. Level of detail
B. Data size in GB
C. User count
D. Table count
Correct Answer: A

Q44. Partitioning in warehouse helps?

A. Performance/manageability
B. Slow queries
C. More storage only
D. No use
Correct Answer: A

Q45. Indexing in warehouse e.g.?

A. Bitmap/join index
B. No index
C. Primary only
D. Sequential only
Correct Answer: A

Q46. Bitmap index best for?

A. Low-cardinality cols
B. High-cardinality
C. Text blobs
D. Images
Correct Answer: A

Q47. Materialized view stores?

A. Precomputed aggregates
B. Query text only
C. Source only
D. Metadata only
Correct Answer: A

Q48. View maintenance is?

A. Keep aggregates consistent
B. Delete views
C. Create users
D. Backup only
Correct Answer: A

Q49. Data warehouse analysis task?

A. OLAP/reporting/mining
B. Data entry
C. Transaction update
D. Deletion
Correct Answer: A

Q50. Backup/recovery in admin ensures?

A. Availability
B. Deletion
C. Slowness
D. Redundancy removal
Correct Answer: A

Q51. Data mining is?

A. Knowledge discovery from data
B. Digging coal
C. Data entry
D. Data deletion
Correct Answer: A

Q52. KDD steps order?

A. Selection/preprocess/transform/mine/evaluate
B. Mine first only
C. Evaluate first
D. No steps
Correct Answer: A

Q53. Data mining models include predictive/descriptive. Predictive is?

A. Classification/regression
B. Clustering only
C. Summarization only
D. Association only
Correct Answer: A

Q54. Descriptive models include?

A. Clustering/association
B. Classification only
C. Regression only
D. Prediction only
Correct Answer: A

Q55. Classification predicts?

A. Categorical label
B. Continuous value
C. Clusters
D. Sequences
Correct Answer: A

Q56. Regression predicts?

A. Numeric continuous
B. Class label
C. Cluster ID
D. Rule only
Correct Answer: A

Q57. Clustering is?

A. Unsupervised grouping
B. Supervised labeling
C. Regression
D. Classification
Correct Answer: A

Q58. Summarization gives?

A. Compact description
B. Full detail
C. Raw data
D. No output
Correct Answer: A

Q59. Dependency modeling finds?

A. Relationships/causality
B. No relation
C. Single attribute
D. Random data
Correct Answer: A

Q60. Link analysis finds?

A. Associations/links
B. No links
C. Single row
D. Sorting only
Correct Answer: A

Q61. Sequence analysis finds?

A. Ordered patterns over time
B. Unordered only
C. Static only
D. No order
Correct Answer: A

Q62. Association rule X=>Y support is?

A. P(X union Y)
B. P(Y|X) only
C. Count X only
D. Lift only
Correct Answer: A

Q63. Confidence of X=>Y is?

A. P(Y|X)
B. P(X)
C. P(Y)
D. Lift+support
Correct Answer: A

Q64. Lift >1 means?

A. Positive correlation
B. Negative
C. Independent
D. No rule
Correct Answer: A

Q65. Apriori principle?

A. Subset of frequent is frequent
B. Superset always frequent
C. No pruning
D. Random search
Correct Answer: A

Q66. Apriori join+prune uses?

A. L(k-1) to make Ck
B. Full scan always
C. No candidates
D. Decision tree
Correct Answer: A

Q67. Frequent itemset needs min?

A. Support threshold
B. Confidence only
C. Lift only
D. No threshold
Correct Answer: A

Q68. Strong rule needs min support + min?

A. Confidence
B. Lift only
C. Leverage only
D. Chi-square only
Correct Answer: A

Q69. FP-Growth avoids?

A. Candidate generation
B. Tree building
C. Scanning
D. Mining
Correct Answer: A

Q70. Decision tree internal node is?

A. Attribute test
B. Class label
C. Root only
D. Leaf only
Correct Answer: A

Q71. Decision tree leaf is?

A. Class label
B. Test
C. Split
D. Branch
Correct Answer: A

Q72. ID3 uses which measure?

A. Information gain
B. Gini only
C. Distance
D. Lift
Correct Answer: A

Q73. Entropy of pure set?

A. 0
B. 1
C. Max
D. Infinite
Correct Answer: A

Q74. Information gain is?

A. Entropy(parent)-weighted child
B. Child-parent
C. Gini+gain
D. Support
Correct Answer: A

Q75. Gini index used in?

A. CART
B. Apriori
C. K-means
D. PageRank
Correct Answer: A

Q76. Overfitting in trees handled by?

A. Pruning
B. Growing more
C. No validation
D. More depth always
Correct Answer: A

Q77. Prepruning stops?

A. Early by threshold
B. After full
C. Never
D. At leaves only
Correct Answer: A

Q78. Postpruning?

A. Grow full then trim
B. Stop early
C. No tree
D. Random cut
Correct Answer: A

Q79. Naive Bayes assumes?

A. Conditional independence
B. Dependence always
C. No probability
D. No prior
Correct Answer: A

Q80. Bayes theorem P(H|X)=?

A. P(X|H)P(H)/P(X)
B. P(X)/P(H)
C. P(H)+P(X)
D. P(H)-P(X)
Correct Answer: A

Q81. k-NN is?

A. Lazy instance-based
B. Eager tree
C. Rule-based
D. Neural only
Correct Answer: A

Q82. k-means is for?

A. Clustering
B. Classification
C. Regression
D. Association
Correct Answer: A

Q83. k-means steps?

A. Init assign update repeat
B. Sort only
C. Prune only
D. Split only
Correct Answer: A

Q84. k-means distance commonly?

A. Euclidean
B. Support
C. Confidence
D. Entropy
Correct Answer: A

Q85. k-means drawback?

A. Sensitive to k/init/outliers
B. No drawback
C. Supervised
D. Handles categorical well
Correct Answer: A

Q86. k-medoids uses?

A. Actual medoid object
B. Mean always
C. No center
D. Mode only
Correct Answer: A

Q87. PAM stands for?

A. Partitioning Around Medoids
B. Pattern Mining
C. Principal Mining
D. Probability Model
Correct Answer: A

Q88. Hierarchical clustering types?

A. Agglomerative/divisive
B. Flat only
C. Density only
D. Grid only
Correct Answer: A

Q89. AGNES is?

A. Agglomerative
B. Divisive
C. Density
D. Grid
Correct Answer: A

Q90. DIANA is?

A. Divisive
B. Agglomerative
C. Partitioning
D. Model-based
Correct Answer: A

Q91. Dendrogram shows?

A. Merge/split hierarchy
B. Decision boundary
C. ROC
D. Lift chart
Correct Answer: A

Q92. DBSCAN groups by?

A. Density
B. k centers
C. Hierarchy
D. Grid only
Correct Answer: A

Q93. DBSCAN core point needs minPts within?

A. Eps
B. k
C. Support
D. Confidence
Correct Answer: A

Q94. Outlier in DBSCAN is?

A. Noise point
B. Core always
C. Border always
D. Cluster center
Correct Answer: A

Q95. Regression linear model y=?

A. a+bx
B. a/b
C. Cluster
D. Rule
Correct Answer: A

Q96. Least squares minimizes?

A. Sum squared errors
B. Support
C. Entropy
D. Gain
Correct Answer: A

Q97. Multiple regression has?

A. Multiple predictors
B. One predictor
C. No predictor
D. Class label
Correct Answer: A

Q98. Logistic regression predicts?

A. Probability/class
B. Continuous only
C. Clusters
D. Rules
Correct Answer: A

Q99. Neural network learning via?

A. Backpropagation
B. Apriori
C. K-means
D. Pruning
Correct Answer: A

Q100. SVM finds?

A. Max-margin hyperplane
B. Decision tree
C. Clusters
D. Rules
Correct Answer: A

Q101. Mining scientific data challenge?

A. High-dim/noisy/experiments
B. Simple transactions only
C. No noise
D. Small data
Correct Answer: A

Q102. Mining business data uses?

A. RFM/churn/basket
B. DNA only
C. Spectra only
D. Images only
Correct Answer: A

Q103. Market basket data is?

A. Transactional items
B. Time series only
C. Spatial only
D. Text only
Correct Answer: A

Q104. Churn prediction is?

A. Classification
B. Clustering only
C. Association only
D. Sorting
Correct Answer: A

Q105. Credit scoring is?

A. Classification/risk
B. Clustering only
C. Summarization
D. Sequencing
Correct Answer: A

Q106. Fraud detection uses?

A. Outlier/anomaly
B. Sorting only
C. Reporting only
D. OLTP only
Correct Answer: A

Q107. Web mining types?

A. Content/structure/usage
B. Only content
C. Only hardware
D. Only cables
Correct Answer: A

Q108. Text mining steps?

A. Tokenize/stem/mine
B. Mine without clean
C. Hardware setup
D. No preprocessing
Correct Answer: A

Q109. TF-IDF measures?

A. Term importance
B. Support
C. Confidence
D. Distance
Correct Answer: A

Q110. Sequential pattern e.g.?

A. Buys A then B
B. Buys A and B unordered
C. Cluster
D. Regression
Correct Answer: A

Q111. GSP algorithm mines?

A. Sequential patterns
B. Decision trees
C. Clusters
D. Outliers
Correct Answer: A

Q112. Time-series mining finds?

A. Trends/seasonality
B. Static rules
C. No time
D. Spatial only
Correct Answer: A

Q113. Spatial mining considers?

A. Location/adjacency
B. Time only
C. No space
D. Transactions only
Correct Answer: A

Q114. Multimedia mining mines?

A. Image/audio/video
B. Numbers only
C. Text only
D. Transactions only
Correct Answer: A

Q115. Feature selection helps by?

A. Reduce dims/overfit
B. Increase dims
C. Add noise
D. Remove labels
Correct Answer: A

Q116. Data discretization converts?

A. Continuous to categorical
B. Categorical to continuous
C. Text to image
D. Numbers to noise
Correct Answer: A

Q117. Normalization range 0-1 is?

A. Min-max
B. Z-score
C. Decimal
D. No norm
Correct Answer: A

Q118. Z-score uses?

A. Mean/stddev
B. Min/max
C. Median only
D. Mode
Correct Answer: A

Q119. Missing values handled by?

A. Ignore/fill/mean
B. Delete DB
C. Stop mining
D. Ignore mining
Correct Answer: A

Q120. Noisy data smoothed by?

A. Binning/regression
B. No smoothing
C. Deletion only
D. Encryption
Correct Answer: A

Q121. Binning methods?

A. Equal-width/equal-freq
B. Random only
C. No bins
D. Single bin
Correct Answer: A

Q122. Correlation analysis detects?

A. Redundant attributes
B. Useful always keep all
C. No relation
D. Outliers only
Correct Answer: A

Q123. Chi-square for?

A. Categorical correlation
B. Numeric regression
C. Clustering
D. Classification label
Correct Answer: A

Q124. Evaluation: accuracy is?

A. Correct/total
B. TP/FP
C. FP only
D. TN only
Correct Answer: A

Q125. Precision is?

A. TP/(TP+FP)
B. TN/total
C. TP/FN
D. Accuracy
Correct Answer: A

Q126. Recall is?

A. TP/(TP+FN)
B. TP/(TP+FP)
C. TN/FP
D. Accuracy
Correct Answer: A

Q127. F1 is?

A. Harmonic mean P/R
B. Mean accuracy
C. Sum P+R
D. Max P,R
Correct Answer: A

Q128. Confusion matrix shows?

A. Actual vs predicted
B. Support only
C. Distance
D. Lift
Correct Answer: A

Q129. Cross-validation is for?

A. Reliable estimate
B. Faster training
C. More overfit
D. Less data use
Correct Answer: A

Q130. Holdout method splits?

A. Train/test
B. Train only
C. Test only
D. No split
Correct Answer: A

Q131. ROC plots?

A. TPR vs FPR
B. Precision vs support
C. Recall vs accuracy
D. Lift vs time
Correct Answer: A

Q132. AUC higher means?

A. Better classifier
B. Worse
C. Random
D. No model
Correct Answer: A

Q133. Underfitting means?

A. Too simple/high bias
B. Too complex
C. Perfect fit
D. No bias
Correct Answer: A

Q134. Overfitting means?

A. Too complex/high variance
B. Too simple
C. Perfect generalize
D. No variance
Correct Answer: A

Q135. Ensemble combines?

A. Multiple models
B. Single model
C. No models
D. Single tree only
Correct Answer: A

Q136. Bagging e.g.?

A. Random Forest
B. Single tree
C. K-means
D. Apriori
Correct Answer: A

Q137. Boosting e.g.?

A. AdaBoost
B. K-means
C. Apriori
D. DBSCAN
Correct Answer: A

Q138. Random Forest uses?

A. Bagged trees+random features
B. Single tree
C. Rules only
D. Linear only
Correct Answer: A

Q139. Warehouse vs database: warehouse is?

A. Read-only analytical
B. Update-heavy
C. Real-time writes
D. Volatile
Correct Answer: A

Q140. Nonvolatile means?

A. Stable no updates
B. Frequent updates
C. Deleted often
D. Temporary
Correct Answer: A

Q141. Time-variant means?

A. Historical time-stamped
B. Current only
C. No time
D. Real-time only
Correct Answer: A

Q142. Integrated means?

A. Unified schemas/formats
B. Separate formats
C. Isolated
D. Redundant conflicts kept
Correct Answer: A

Q143. Subject-oriented means?

A. Around themes (sales/customer)
B. Around apps
C. Around hardware
D. Around users login
Correct Answer: A

Q144. ETL tool example?

A. Informatica
B. MS Word
C. Photoshop
D. Browser
Correct Answer: A

Q145. OLAP tool example?

A. Cognos/MicroStrategy
B. Notepad
C. Paint
D. Compiler
Correct Answer: A

Q146. Data cube operation slice+dice helps?

A. Business analysis
B. Data entry
C. Transaction
D. Backup
Correct Answer: A

Q147. Dashboard shows?

A. KPIs/visuals
B. Raw logs
C. Code
D. Hardware
Correct Answer: A

Q148. Reports in warehouse are?

A. Predefined/ad-hoc
B. No reports
C. Code listings
D. Logs only
Correct Answer: A

Q149. Security in warehouse via?

A. Roles/access control
B. Open access
C. No passwords
D. Shared login
Correct Answer: A

Q150. Scalability needed due to?

A. Volume growth
B. Small data
C. Fixed size
D. No growth
Correct Answer: A

Q151. Apriori example: milk=>bread s=30% c=60% means?

A. 30% transactions both, 60% milk buyers buy bread
B. 60% all buy
C. 30% bread only
D. No meaning
Correct Answer: A

Q152. Closed frequent itemset?

A. No superset same support
B. All subsets
C. Max length only
D. Empty set
Correct Answer: A

Q153. Max frequent itemset?

A. No superset frequent
B. All frequent
C. Closed only
D. Single item
Correct Answer: A

Q154. Rule pruning by?

A. Min confidence/lift
B. Max support only
C. No pruning
D. Random delete
Correct Answer: A

Q155. Decision tree split for continuous?

A. Threshold/binary split
B. No split
C. Categorical only
D. Random
Correct Answer: A

Q156. C4.5 uses?

A. Gain ratio
B. Gini only
C. Distance
D. Support
Correct Answer: A

Q157. CART produces?

A. Binary trees
B. N-ary always
C. No tree
D. Forest only
Correct Answer: A

Q158. Tree induction complexity due to?

A. Attribute selection/partition
B. No computation
C. Sorting only
D. Printing
Correct Answer: A

Q159. Rule extraction from tree by?

A. Root-to-leaf paths
B. Leaf only
C. Random
D. Prune only
Correct Answer: A

Q160. k-means k chosen by?

A. Elbow/domain
B. Random fixed 1
C. Always 2
D. No choice
Correct Answer: A

Q161. Empty cluster in k-means handled by?

A. Reinit/split largest
B. Stop
C. Delete all
D. Ignore
Correct Answer: A

Q162. Categorical clustering uses?

A. K-modes
B. K-means Euclidean
C. Regression
D. SVM
Correct Answer: A

Q163. Outlier detection methods?

A. Statistical/distance/density
B. Sorting only
C. Reporting only
D. OLTP
Correct Answer: A

Q164. Isolation Forest isolates?

A. Anomalies
B. Normal only
C. Clusters
D. Rules
Correct Answer: A

Q165. Mining DNA/sequential bio uses?

A. Sequence alignment/patterns
B. Basket only
C. Sales only
D. No mining
Correct Answer: A

Q166. Scientific visualization helps?

A. Interpret patterns
B. Hide patterns
C. Store data
D. Delete noise
Correct Answer: A

Q167. Business KPI example?

A. Sales/profit/churn
B. Gene length
C. Spectra
D. Temperature only
Correct Answer: A

Q168. CRM mining uses?

A. Segmentation/cross-sell
B. Gene mining
C. Spectra
D. No CRM
Correct Answer: A

Q169. Segmentation uses?

A. Clustering
B. Classification only
C. Regression only
D. Sorting
Correct Answer: A

Q170. Target marketing uses?

A. Classification/profiles
B. Random mail
C. No data
D. Mass only
Correct Answer: A

Q171. Risk management uses?

A. Regression/scoring
B. Clustering only
C. No model
D. Reports only
Correct Answer: A

Q172. Demand forecasting uses?

A. Regression/time series
B. Association only
C. Clustering only
D. Sorting
Correct Answer: A

Q173. Inventory optimization uses?

A. Association/forecast
B. Classification only
C. No mining
D. Manual only
Correct Answer: A

Q174. Web usage mining uses?

A. Logs/clickstreams
B. Content only
C. Hardware
D. Cables
Correct Answer: A

Q175. PageRank mines?

A. Web structure
B. Content only
C. Usage only
D. Hardware
Correct Answer: A

Q176. Collaborative filtering for?

A. Recommendations
B. Clustering only
C. Sorting
D. Indexing
Correct Answer: A

Q177. Content-based filtering uses?

A. Item features
B. User ratings only
C. No features
D. Random
Correct Answer: A

Q178. Data quality issues?

A. Incomplete/noisy/inconsistent
B. Perfect data
C. No issues
D. Large data only
Correct Answer: A

Q179. Concept hierarchy enables?

A. Multi-level drill
B. Single level
C. No aggregation
D. Flat analysis
Correct Answer: A

Q180. Star-net query model is for?

A. Multidimensional queries
B. Transactions
C. Backups
D. ETL code
Correct Answer: A

Q181. Data warehouse bus architecture shares?

A. Conformed dims
B. Facts only
C. Staging only
D. Tools only
Correct Answer: A

Q182. Lifecycle: operations phase monitors?

A. Usage/performance
B. Requirements only
C. Design only
D. Build only
Correct Answer: A

Q183. Change management in warehouse handles?

A. Source/schema evolution
B. No changes
C. Hardware color
D. Staff only
Correct Answer: A

Q184. Growth management plans for?

A. Volume/users
B. Shrinkage
C. Deletion
D. Downtime
Correct Answer: A

Q185. Cost-benefit of warehouse measured by?

A. ROI/decision value
B. Lines of code
C. Tables count
D. Disk weight
Correct Answer: A

Q186. Big data vs warehouse: big data is?

A. Volume/velocity/variety
B. Small structured only
C. Relational only
D. Static only
Correct Answer: A

Q187. Hadoop complements warehouse by?

A. Unstructured processing
B. Replace all BI
C. No use
D. Storage only small
Correct Answer: A

Q188. Cloud warehouse e.g.?

A. Snowflake/BigQuery/Redshift
B. Notepad
C. Paint
D. DOS
Correct Answer: A

Q189. Real-time warehouse enables?

A. Near-real-time BI
B. Batch only
C. Annual reports
D. No BI
Correct Answer: A

Q190. Active warehouse supports?

A. Tactical decisions/automation
B. Archive only
C. Static reports
D. No decisions
Correct Answer: A

Q191. Data lake vs warehouse: lake stores?

A. Raw multi-format
B. Modeled only
C. Aggregates only
D. Reports only
Correct Answer: A

Q192. ETT vs ETL: ELT loads then?

A. Transforms in target
B. No transform
C. Transforms before
D. Deletes
Correct Answer: A

Q193. Data governance ensures?

A. Quality/compliance
B. No rules
C. Open chaos
D. Deletion
Correct Answer: A

Q194. Master data management provides?

A. Single truth for entities
B. Multiple truths
C. No entities
D. Transactions only
Correct Answer: A

Q195. Slow query in warehouse solved by?

A. Aggregates/index/partition
B. More transactions
C. Delete warehouse
D. Smaller RAM
Correct Answer: A

Q196. Concurrency in warehouse is?

A. Many readers/few writers
B. Many writers
C. No readers
D. Single user
Correct Answer: A

Q197. Data mining languages include?

A. DMQL extensions
B. Only SQL
C. Only Java
D. Only HTML
Correct Answer: A

Q198. Privacy-preserving mining uses?

A. Anonymization/perturbation
B. Open publish
C. No privacy
D. Delete mining
Correct Answer: A

Q199. K-anonymity ensures?

A. Indistinguishable k-records
B. Unique IDs
C. No anonymity
D. Full disclosure
Correct Answer: A

Q200. Future trend in mining?

A. Stream/big data/AI integration
B. Manual ledgers
C. No automation
D. Paper reports
Correct Answer: A

ANSWER KEY — Topic 12c: Data Warehousing and Mining

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1-A 2-A 3-A 4-A 5-A 6-A 7-A 8-A 9-A 10-A
11-A 12-A 13-A 14-A 15-A 16-A 17-A 18-A 19-A 20-A
21-A 22-A 23-A 24-A 25-A 26-A 27-A 28-A 29-A 30-A
31-A 32-A 33-A 34-A 35-A 36-A 37-A 38-A 39-A 40-A
41-A 42-A 43-A 44-A 45-A 46-A 47-A 48-A 49-A 50-A
51-A 52-A 53-A 54-A 55-A 56-A 57-A 58-A 59-A 60-A
61-A 62-A 63-A 64-A 65-A 66-A 67-A 68-A 69-A 70-A
71-A 72-A 73-A 74-A 75-A 76-A 77-A 78-A 79-A 80-A
81-A 82-A 83-A 84-A 85-A 86-A 87-A 88-A 89-A 90-A
91-A 92-A 93-A 94-A 95-A 96-A 97-A 98-A 99-A 100-A
101-A 102-A 103-A 104-A 105-A 106-A 107-A 108-A 109-A 110-A
111-A 112-A 113-A 114-A 115-A 116-A 117-A 118-A 119-A 120-A
121-A 122-A 123-A 124-A 125-A 126-A 127-A 128-A 129-A 130-A
131-A 132-A 133-A 134-A 135-A 136-A 137-A 138-A 139-A 140-A
141-A 142-A 143-A 144-A 145-A 146-A 147-A 148-A 149-A 150-A
151-A 152-A 153-A 154-A 155-A 156-A 157-A 158-A 159-A 160-A
161-A 162-A 163-A 164-A 165-A 166-A 167-A 168-A 169-A 170-A
171-A 172-A 173-A 174-A 175-A 176-A 177-A 178-A 179-A 180-A
181-A 182-A 183-A 184-A 185-A 186-A 187-A 188-A 189-A 190-A
191-A 192-A 193-A 194-A 195-A 196-A 197-A 198-A 199-A 200-A
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Topic 12d: Artificial Intelligence

Syllabus: *Reasoning, problem-solving, Knowledge Representation, Learning, Natural language processing.*

Q1. AI aims to?

A. Build intelligent agents
B. Stop computers
C. Manual work only
D. Replace data only
Correct Answer: A

Q2. Intelligent agent perceives via?

A. Sensors/actuators
B. Printer only
C. Keyboard only
D. Monitor only
Correct Answer: A

Q3. PEAS stands for?

A. Performance Environment Actuators Sensors
B. Program Effect Action State
C. Performance Entity Agent System
D. Percept Action State
Correct Answer: A

Q4. Rational agent maximizes?

A. Performance measure
B. Cost always
C. Errors
D. Time only
Correct Answer: A

Q5. Types of agents: reflex, model, goal, utility. Simple reflex uses?

A. Condition-action rules
B. Full history
C. Goals
D. Utility
Correct Answer: A

Q6. Problem-solving as search needs?

A. State/space/goal/actions
B. Only goal
C. Only state
D. No actions
Correct Answer: A

Q7. Search: initial state + actions + goal test +?

A. Path cost
B. Printer
C. Battery
D. Screen
Correct Answer: A

Q8. State space is?

A. All reachable states
B. Single state
C. Goal only
D. Action only
Correct Answer: A

Q9. BFS uses?

A. Queue FIFO
B. Stack LIFO
C. Priority cost
D. Random
Correct Answer: A

Q10. BFS complete/optimal for?

A. Uniform unweighted shallow
B. Deep infinite
C. Weighted cost
D. Heuristic
Correct Answer: A

Q11. BFS time complexity b^d where b=?

A. Branching factor
B. Depth
C. Cost
D. Heuristic
Correct Answer: A

Q12. DFS uses?

A. Stack LIFO
B. Queue
C. Priority
D. Random
Correct Answer: A

Q13. DFS drawback?

A. Incomplete infinite depth
B. Always optimal
C. High memory
D. No recursion
Correct Answer: A

Q14. DFS space complexity?

A. O(bm)
B. O(b^d)
C. O(bd) always high
D. Exponential always
Correct Answer: A

Q15. Depth-limited search limits?

A. Depth bound
B. Breadth
C. Cost
D. Heuristic
Correct Answer: A

Q16. Iterative deepening combines?

A. BFS completeness + DFS memory
B. BFS+Heuristic
C. DFS+Cost
D. Random+Greedy
Correct Answer: A

Q17. Uniform-cost search expands?

A. Lowest path cost
B. Shallowest
C. Deepest
D. Random
Correct Answer: A

Q18. Best-first search uses?

A. Evaluation function
B. FIFO
C. LIFO
D. No function
Correct Answer: A

Q19. Greedy search uses?

A. h(n) only
B. g(n) only
C. g+h
D. No heuristic
Correct Answer: A

Q20. A* evaluation f(n)=?

A. g(n)+h(n)
B. g-h
C. h/g
D. g*h
Correct Answer: A

Q21. A* optimal if h is?

A. Admissible (never overestimates)
B. Overestimates
C. Zero always
D. Random
Correct Answer: A

Q22. Consistent heuristic means?

A. Monotone triangle inequality
B. Overestimates
C. No property
D. Inconsistent always
Correct Answer: A

Q23. Hill climbing is?

A. Local greedy no backtrack
B. Global exhaustive
C. BFS variant
D. DFS variant
Correct Answer: A

Q24. Hill climbing problem?

A. Local maxima/plateau/ridge
B. No problems
C. High memory
D. Slow complete
Correct Answer: A

Q25. Simulated annealing allows?

A. Occasional downhill moves
B. Only uphill
C. No moves
D. Exhaustive
Correct Answer: A

Q26. Genetic algorithm operators?

A. Selection/crossover/mutation
B. Queue/stack
C. BFS/DFS
D. Sort/search
Correct Answer: A

Q27. Minimax is for?

A. Adversarial games
B. Single agent
C. No opponent
D. Sorting
Correct Answer: A

Q28. Minimax MAX/MIN levels represent?

A. Players
B. Costs
C. Heuristics
D. Depths only
Correct Answer: A

Q29. Alpha-beta pruning?

A. Cutoff irrelevant branches
B. Add branches
C. No pruning
D. Expand all
Correct Answer: A

Q30. Alpha value is?

A. Best for MAX
B. Best for MIN
C. Worst
D. Random
Correct Answer: A

Q31. Beta value is?

A. Best for MIN
B. Best for MAX
C. Average
D. Heuristic
Correct Answer: A

Q32. Tic-tac-toe game tree size manageable by?

A. Search+pruning
B. No search
C. Random only
D. Manual only
Correct Answer: A

Q33. Constraint satisfaction problem has variables/domains/?

A. Constraints
B. Costs only
C. Heuristics only
D. Goals only
Correct Answer: A

Q34. Backtracking search for CSP?

A. Depth-first assignment
B. BFS only
C. A* only
D. Greedy only
Correct Answer: A

Q35. Forward checking does?

A. Prune inconsistent values early
B. Expand all
C. No pruning
D. Random assign
Correct Answer: A

Q36. Arc consistency AC-3 ensures?

A. Binary constraints consistent
B. No constraints
C. Unary only
D. Global only
Correct Answer: A

Q37. Knowledge Representation (KR) is?

A. Encode knowledge for reasoning
B. Store images only
C. Display only
D. Print only
Correct Answer: A

Q38. Propositional logic deals with?

A. True/false propositions
B. Objects/relations
C. Probabilities
D. Fuzzy only
Correct Answer: A

Q39. First-order logic adds?

A. Quantifiers/objects/predicates
B. Propositions only
C. No variables
D. No inference
Correct Answer: A

Q40. Universal quantifier symbol?

A. ∀
B. ∃
C. ¬
D. ∧
Correct Answer: A

Q41. Existential quantifier symbol?

A. ∃
B. ∀
C. →
D. ↔
Correct Answer: A

Q42. Inference by Modus Ponens: P, P→Q gives?

A. Q
B. P
C. ¬Q
D. ¬P
Correct Answer: A

Q43. Resolution in logic needs?

A. CNF clauses
B. DNF only
C. No clauses
D. Facts only
Correct Answer: A

Q44. Unification finds?

A. Substitution to match
B. Difference
C. Distance
D. Cost
Correct Answer: A

Q45. Semantic network uses?

A. Nodes/links for concepts
B. Tables only
C. Trees only
D. Rules only
Correct Answer: A

Q46. IS-A link represents?

A. Inheritance
B. Instance only
C. Action
D. Time
Correct Answer: A

Q47. Frames have slots/fillers for?

A. Attributes/values
B. Actions only
C. Goals only
D. Costs
Correct Answer: A

Q48. Frame default value is?

A. Assumed if unknown
B. Always empty
C. No default
D. Deleted
Correct Answer: A

Q49. Ontology defines?

A. Shared conceptualization
B. Single fact
C. Algorithm
D. Hardware
Correct Answer: A

Q50. OWL is for?

A. Web ontology
B. Programming only
C. Hardware
D. OS
Correct Answer: A

Q51. Inheritance in KR allows?

A. Reuse properties
B. No reuse
C. Delete all
D. Duplicate all
Correct Answer: A

Q52. Nonmonotonic reasoning allows?

A. Retract conclusions
B. Only add
C. No reasoning
D. Monotonic only
Correct Answer: A

Q53. Default reasoning example: birds fly except?

A. Penguins
B. Sparrows
C. Pigeons
D. Eagles
Correct Answer: A

Q54. Fuzzy logic truth values?

A. Degree 0-1
B. Only 0/1
C. No truth
D. Binary only
Correct Answer: A

Q55. Expert system components: KB/inference/?

A. User interface/explanation
B. Printer only
C. CPU only
D. Monitor only
Correct Answer: A

Q56. Forward chaining is?

A. Data-driven
B. Goal-driven
C. No data
D. Backward only
Correct Answer: A

Q57. Backward chaining is?

A. Goal-driven
B. Data-driven
C. No goal
D. Forward only
Correct Answer: A

Q58. Rete algorithm speeds?

A. Pattern matching
B. Sorting
C. Searching
D. Parsing
Correct Answer: A

Q59. Uncertainty handled by?

A. Probability/Bayes/fuzzy
B. Logic only
C. No uncertainty
D. Determinism only
Correct Answer: A

Q60. Bayes net nodes/edges represent variables/?

A. Dependencies
B. Costs
C. Actions
D. States only
Correct Answer: A

Q61. Conditional independence in Bayes net simplifies?

A. Joint distribution
B. No simplification
C. Adds complexity
D. Removes nodes
Correct Answer: A

Q62. Learning in AI is?

A. Improve via experience
B. Fixed program
C. No data
D. Manual only
Correct Answer: A

Q63. Supervised learning uses?

A. Labeled examples
B. Unlabeled only
C. No examples
D. Rewards only
Correct Answer: A

Q64. Unsupervised learning uses?

A. Unlabeled pattern discovery
B. Labels always
C. Teacher always
D. Rewards only
Correct Answer: A

Q65. Reinforcement learning uses?

A. Rewards/punishments
B. Labels only
C. No feedback
D. Supervised pairs
Correct Answer: A

Q66. Classification example?

A. Spam vs ham
B. Clustering
C. Grouping
D. Density
Correct Answer: A

Q67. Regression example?

A. House price predict
B. Spam detect
C. Cluster docs
D. Group users
Correct Answer: A

Q68. Clustering example?

A. Customer segmentation
B. Spam label
C. Price predict
D. Next word
Correct Answer: A

Q69. Decision tree learning splits by?

A. Info gain/gini
B. Distance
C. Support
D. Lift
Correct Answer: A

Q70. Overfitting in ML solved by?

A. Prune/regularize/validate
B. More complexity
C. No validation
D. Memorize
Correct Answer: A

Q71. Training/validation/test split for?

A. Generalization estimate
B. Faster code
C. More overfit
D. Less data
Correct Answer: A

Q72. Cross-validation k-fold averages over?

A. k folds
B. 1 fold
C. No folds
D. Test only
Correct Answer: A

Q73. Linear regression minimizes?

A. MSE
B. Accuracy
C. Entropy
D. Gain
Correct Answer: A

Q74. Logistic regression output?

A. Probability 0-1
B. Any number
C. Cluster
D. Rule
Correct Answer: A

Q75. Perceptron is?

A. Single neuron classifier
B. Multi-layer only
C. Unsupervised
D. Clustering
Correct Answer: A

Q76. Activation sigmoid outputs?

A. 0-1
B. -inf to inf
C. Binary only
D. -1 to 1 only tanh
Correct Answer: A

Q77. Backpropagation computes?

A. Gradients via chain rule
B. Info gain
C. Support
D. Distance
Correct Answer: A

Q78. Gradient descent updates by?

A. Opposite gradient step
B. Random jump
C. Increase loss
D. No update
Correct Answer: A

Q79. Learning rate too high causes?

A. Divergence/oscillation
B. Slow converge
C. Perfect fit
D. No effect
Correct Answer: A

Q80. CNN best for?

A. Images
B. Text only
C. Tables only
D. Audio only
Correct Answer: A

Q81. RNN best for?

A. Sequences
B. Images only
C. Tables
D. Static vectors
Correct Answer: A

Q82. SVM margin maximizes?

A. Separation
B. Errors
C. Support vectors count
D. Overfit
Correct Answer: A

Q83. Kernel trick allows?

A. Non-linear separation
B. Linear only
C. No separation
D. Clustering
Correct Answer: A

Q84. Naive Bayes spam filter uses?

A. Word probabilities
B. Distance
C. Rules only
D. Clustering
Correct Answer: A

Q85. k-NN classifies by?

A. Majority neighbors
B. Distant points
C. Global mean
D. Tree path
Correct Answer: A

Q86. Dimensionality curse means?

A. Sparse/high-dim difficulty
B. Easy high-dim
C. No issue
D. Low-dim hard
Correct Answer: A

Q87. PCA is for?

A. Dimensionality reduction
B. Classification
C. Clustering
D. Regression
Correct Answer: A

Q88. Feature engineering creates?

A. Informative inputs
B. Noise
C. Labels
D. Duplicates
Correct Answer: A

Q89. Bias-variance tradeoff balances?

A. Under vs overfit
B. Speed only
C. Memory only
D. Cost only
Correct Answer: A

Q90. Regularization L2 adds?

A. Penalty weights
B. More features
C. No penalty
D. Delete model
Correct Answer: A

Q91. NLP stands for?

A. Natural Language Processing
B. Neural Logic Program
C. Network Layer Protocol
D. New Language Program
Correct Answer: A

Q92. NLP stages order?

A. Morph/lex/syntax/semantic/pragmatic
B. Semantic first only
C. No stages
D. Pragmatic only
Correct Answer: A

Q93. Morphological analysis studies?

A. Word structure/roots
B. Sentences only
C. Meaning only
D. Speech only
Correct Answer: A

Q94. Lexical analysis does?

A. Tokenize/POS
B. Parse only
C. Generate only
D. Translate only
Correct Answer: A

Q95. Syntax analysis does?

A. Parsing/grammar
B. Meaning only
C. Speech only
D. Context only
Correct Answer: A

Q96. CFG stands for?

A. Context-Free Grammar
B. Combined Function Grammar
C. Central Form Grammar
D. Context Form Group
Correct Answer: A

Q97. Parse tree shows?

A. Syntactic structure
B. Meaning only
C. Sounds only
D. Words only
Correct Answer: A

Q98. Ambiguity in NLP means?

A. Multiple parses/meanings
B. Single meaning
C. No meaning
D. Clear text
Correct Answer: A

Q99. Semantic analysis extracts?

A. Meaning
B. Sounds
C. Letters
D. Pixels
Correct Answer: A

Q100. Word sense disambiguation resolves?

A. Correct meaning
B. Spelling only
C. Grammar only
D. Speech only
Correct Answer: A

Q101. Pragmatic analysis uses?

A. Context/intent
B. Letters only
C. Sounds only
D. Syntax only
Correct Answer: A

Q102. POS tagging labels?

A. Noun/verb/etc
B. Pixels
C. Phonemes
D. Files
Correct Answer: A

Q103. NER finds?

A. Persons/orgs/locations
B. Verbs only
C. Nouns only
D. Adjectives
Correct Answer: A

Q104. Stop words are?

A. Common low-value (the/is)
B. Keywords only
C. Rare words
D. Nouns only
Correct Answer: A

Q105. Stemming reduces to?

A. Root (running→run)
B. Full form
C. Synonym
D. Antonym
Correct Answer: A

Q106. Lemmatization vs stemming: lemma is?

A. Valid base form
B. Crude chop
C. Synonym
D. Uppercase
Correct Answer: A

Q107. Bag-of-words ignores?

A. Order
B. Words
C. Frequency
D. Vocabulary
Correct Answer: A

Q108. TF-IDF in NLP weights?

A. Important terms
B. Stop words high
C. All equal
D. Random
Correct Answer: A

Q109. N-gram is?

A. n consecutive words
B. Single letter
C. Single sentence
D. Random words
Correct Answer: A

Q110. Bigram example?

A. Natural language
B. Natural
C. Language processing system
D. N
Correct Answer: A

Q111. Language model predicts?

A. Next word probability
B. Previous image
C. POS only
D. Parse only
Correct Answer: A

Q112. Machine translation translates?

A. Language to language
B. Speech to image
C. Text to code only
D. Code to image
Correct Answer: A

Q113. Sentiment analysis detects?

A. Opinion/positive-negative
B. Grammar only
C. Length only
D. Author only
Correct Answer: A

Q114. Question answering needs IR +?

A. Comprehension/inference
B. Sorting only
C. Printing
D. Storage only
Correct Answer: A

Q115. Speech recognition converts?

A. Audio to text
B. Text to audio
C. Image to text
D. Video to image
Correct Answer: A

Q116. Speech synthesis converts?

A. Text to speech
B. Speech to text
C. Image to speech
D. Text to image
Correct Answer: A

Q117. Phoneme is?

A. Smallest sound unit
B. Word
C. Sentence
D. Paragraph
Correct Answer: A

Q118. HMM in speech/NLP models?

A. Sequential states
B. Images
C. Trees
D. Rules
Correct Answer: A

Q119. Information retrieval ranks by?

A. Relevance
B. Date only
C. Length only
D. Random
Correct Answer: A

Q120. Precision/recall in IR measure?

A. Retrieval quality
B. Speed only
C. Memory
D. Cost
Correct Answer: A

Q121. Turing Test tests?

A. Machine intelligence indistinguishability
B. Speed
C. Memory
D. Power
Correct Answer: A

Q122. Chinese Room argues?

A. Syntax not semantics
B. Machines understand fully
C. No argument
D. Pro-AI proof
Correct Answer: A

Q123. Strong AI claims?

A. True understanding/consciousness
B. Simulate only
C. No intelligence
D. Weak tools only
Correct Answer: A

Q124. Weak AI claims?

A. Simulate intelligent behavior
B. Consciousness
C. True understanding
D. Human mind
Correct Answer: A

Q125. Reasoning types: deductive gives?

A. Guaranteed conclusion
B. Probable only
C. Guess
D. Analogy only
Correct Answer: A

Q126. Inductive reasoning gives?

A. Generalize from examples
B. Guarantee
C. Deduce only
D. Prove
Correct Answer: A

Q127. Abductive reasoning gives?

A. Best explanation
B. Guarantee
C. Deduction
D. Induction only
Correct Answer: A

Q128. Problem-solving by planning needs operators +?

A. Initial/goal states
B. No states
C. Only operators
D. Only goals
Correct Answer: A

Q129. STRIPS plans with?

A. Preconditions/effects
B. No effects
C. Only goals
D. Only states
Correct Answer: A

Q130. Heuristic function estimates?

A. Cost to goal
B. Past cost
C. Total path
D. Branching
Correct Answer: A

Q131. Admissible example: straight-line distance for?

A. Route finding
B. Chess
C. Poker
D. Sorting
Correct Answer: A

Q132. Means-ends analysis reduces?

A. Difference current-goal
B. Search space to zero
C. Goals
D. Operators
Correct Answer: A

Q133. Production system: WM/rules/control. WM is?

A. Working memory
B. World map
C. Web module
D. Weight matrix
Correct Answer: A

Q134. Rote learning memorizes?

A. Facts directly
B. Generalizes
C. Discovers
D. Reasons
Correct Answer: A

Q135. Explanation-based learning uses?

A. Domain theory/examples
B. No theory
C. Random
D. Rewards only
Correct Answer: A

Q136. Version space contains?

A. Consistent hypotheses
B. Single hypothesis
C. No hypotheses
D. Data only
Correct Answer: A

Q137. Find-S vs Candidate-Elimination: Find-S gives?

A. Most specific
B. Most general
C. All versions
D. None
Correct Answer: A

Q138. Inductive bias is?

A. Assumptions to generalize
B. No assumptions
C. Data only
D. Noise
Correct Answer: A

Q139. Decision tree pruning in ML avoids?

A. Overfit
B. Underfit
C. Bias
D. Data
Correct Answer: A

Q140. Ensemble improves by?

A. Combine diverse models
B. Single model
C. Less data
D. More bias
Correct Answer: A

Q141. Reasoning under uncertainty uses Bayes. Prior is?

A. Before evidence
B. After evidence
C. Likelihood
D. Posterior
Correct Answer: A

Q142. Posterior is?

A. After evidence
B. Before evidence
C. Likelihood only
D. Prior only
Correct Answer: A

Q143. MAP chooses?

A. Max posterior hypothesis
B. Max likelihood only
C. Min prior
D. Random
Correct Answer: A

Q144. MLE chooses?

A. Max likelihood
B. Max posterior
C. Min likelihood
D. Random
Correct Answer: A

Q145. EM algorithm for?

A. Missing/latent vars
B. Complete data only
C. Supervised only
D. Sorting
Correct Answer: A

Q146. Markov Decision Process has states/actions/rewards +?

A. Transition probs/discount
B. No probs
C. Only rewards
D. Only states
Correct Answer: A

Q147. Policy in RL maps?

A. State to action
B. Reward to state
C. Action to reward
D. State to reward only
Correct Answer: A

Q148. Q-learning learns?

A. Action-values
B. Models only
C. Policies directly only
D. Supervised labels
Correct Answer: A

Q149. Exploration vs exploitation tradeoff balances?

A. Try new vs best known
B. Exploit only
C. Explore only
D. No tradeoff
Correct Answer: A

Q150. Epsilon-greedy does?

A. Random with prob epsilon
B. Greedy always
C. Random always
D. No action
Correct Answer: A

Q151. Knowledge engineering builds?

A. Knowledge bases
B. Hardware
C. Networks
D. Databases only ops
Correct Answer: A

Q152. Frame problem in KR?

A. Represent change efficiently
B. Picture frames
C. Window frames
D. No problem
Correct Answer: A

Q153. Qualification problem is?

A. List all preconditions impossible
B. Quantify always easy
C. No preconditions
D. Single precondition
Correct Answer: A

Q154. Common-sense reasoning needs?

A. Defaults/context
B. Logic only strict
C. No defaults
D. Math only
Correct Answer: A

Q155. Truth maintenance system maintains?

A. Belief consistency
B. Truth always fixed
C. No beliefs
D. Delete all
Correct Answer: A

Q156. Modal logic adds?

A. Necessity/possibility
B. Quantifiers only
C. Propositions only
D. Fuzzy only
Correct Answer: A

Q157. Temporal logic reasons about?

A. Time
B. Space only
C. Beliefs
D. Necessity
Correct Answer: A

Q158. Description logics underpin?

A. Ontologies/OWL
B. Games
C. Planning
D. Learning only
Correct Answer: A

Q159. Prolog uses?

A. Horn clauses/backtracking
B. Objects only
C. Functions only
D. Loops only
Correct Answer: A

Q160. LISP in AI for?

A. Symbolic processing
B. Numeric only
C. Hardware
D. OS
Correct Answer: A

Q161. Breadth vs depth: completeness needs?

A. Finite branching + BFS
B. DFS always
C. Depth always
D. Random
Correct Answer: A

Q162. Optimality needs uniform cost +?

A. Admissible/BFS/UCS
B. DFS
C. Greedy
D. Hill climbing
Correct Answer: A

Q163. Time vs space tradeoff: IDDFS saves?

A. Space
B. Time always
C. Neither
D. Completeness
Correct Answer: A

Q164. Bidirectional search meets?

A. Middle
B. Start only
C. Goal only
D. Random
Correct Answer: A

Q165. Constraint propagation reduces?

A. Search
B. Constraints
C. Variables
D. Domains to larger
Correct Answer: A

Q166. Game: perfect info e.g.?

A. Chess
B. Poker
C. Bridge
D. Dice
Correct Answer: A

Q167. Game: imperfect info e.g.?

A. Poker
B. Chess
C. Tic-tac-toe
D. Checkers
Correct Answer: A

Q168. Expectiminimax for?

A. Chance nodes
B. Deterministic only
C. No chance
D. Single player
Correct Answer: A

Q169. Evaluation function in games estimates?

A. Position utility
B. Exact win always
C. Moves count
D. Time
Correct Answer: A

Q170. Horizon effect is?

A. Delayed consequence beyond depth
B. No effect
C. Infinite search
D. Perfect play
Correct Answer: A

Q171. Learning curve plots?

A. Performance vs experience
B. Cost vs time
C. Error vs nodes
D. Space vs time
Correct Answer: A

Q172. Training error vs test error gap shows?

A. Overfit
B. Underfit
C. Perfect fit
D. No fit
Correct Answer: A

Q173. Hyperparameter tuning via?

A. Grid/random search
B. Learning weights
C. No tuning
D. Test set training
Correct Answer: A

Q174. Feature scaling helps?

A. Gradient/distance methods
B. Trees always
C. No effect
D. Rules
Correct Answer: A

Q175. One-hot encoding for?

A. Categorical
B. Numeric
C. Images
D. Audio
Correct Answer: A

Q176. Word embedding (Word2Vec) captures?

A. Semantic similarity
B. Spelling only
C. Length
D. Frequency only
Correct Answer: A

Q177. Cosine similarity measures?

A. Vector angle
B. Euclidean only
C. Support
D. Confidence
Correct Answer: A

Q178. POS HMM/Viterbi finds?

A. Best tag sequence
B. Single tag
C. Parse tree
D. Meaning
Correct Answer: A

Q179. CKY parsing needs grammar in?

A. CNF
B. DNF
C. No form
D. English only
Correct Answer: A

Q180. Dependency parsing shows?

A. Head-dependent relations
B. Constituents only
C. Meaning only
D. Sounds
Correct Answer: A

Q181. Coreference resolves?

A. Pronoun antecedents
B. POS only
C. NER only
D. Parsing only
Correct Answer: A

Q182. Discourse analysis studies?

A. Coherent text structure
B. Single sentence
C. Words only
D. Sounds only
Correct Answer: A

Q183. Dialogue system parts: NLU/DM/NLG. DM is?

A. Dialogue management
B. Data mining
C. Deep model
D. Disk manager
Correct Answer: A

Q184. NLG generates?

A. Natural text
B. Speech signals only
C. Parse trees
D. Tags only
Correct Answer: A

Q185. Chatbot ELIZA used?

A. Pattern matching
B. Deep learning
C. Neural nets
D. Logic only full
Correct Answer: A

Q186. BLEU score evaluates?

A. Translation quality
B. Classification
C. Clustering
D. Parsing speed
Correct Answer: A

Q187. Perplexity lower means?

A. Better language model
B. Worse
C. Random
D. No model
Correct Answer: A

Q188. Smoothing in LM handles?

A. Zero probabilities
B. High probs
C. No probs
D. Parsing
Correct Answer: A

Q189. Spam filter Naive Bayes feature?

A. Word presence
B. Length only
C. Sender IP only
D. Time only
Correct Answer: A

Q190. Face recognition pipeline: detect/align/extract/?

A. Match/classify
B. Print
C. Store only
D. Delete
Correct Answer: A

Q191. Expert system DENDRAL for?

A. Chemistry
B. Medicine MYCIN
C. Geology
D. Finance
Correct Answer: A

Q192. MYCIN for?

A. Medical diagnosis
B. Chemistry
C. Chess
D. Translation
Correct Answer: A

Q193. AI ethics includes bias/fairness/?

A. Transparency/accountability
B. Speed only
C. Memory only
D. Cost only
Correct Answer: A

Q194. Explainable AI (XAI) provides?

A. Interpretable reasons
B. Black box only
C. No reasons
D. Faster only
Correct Answer: A

Q195. AI planning vs search: planning uses?

A. Structured representations
B. Atomic states only
C. No actions
D. No goals
Correct Answer: A

Q196. Subsymbolic AI e.g.?

A. Neural nets
B. Logic only
C. Rules only
D. Frames only
Correct Answer: A

Q197. Symbolic AI e.g.?

A. Logic/rules
B. Neural nets only
C. Genetic only
D. Fuzzy only
Correct Answer: A

Q198. Hybrid AI combines?

A. Symbolic+subsymbolic
B. Single approach
C. Hardware only
D. Data only
Correct Answer: A

Q199. Current AI trend?

A. Deep learning+LLMs
B. Manual rules only
C. No learning
D. Symbolic only
Correct Answer: A

Q200. AI future challenge?

A. Generalization/safety/ethics
B. Speed only
C. Memory only
D. No challenges
Correct Answer: A

ANSWER KEY — Topic 12d: Artificial Intelligence

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1-A 2-A 3-A 4-A 5-A 6-A 7-A 8-A 9-A 10-A
11-A 12-A 13-A 14-A 15-A 16-A 17-A 18-A 19-A 20-A
21-A 22-A 23-A 24-A 25-A 26-A 27-A 28-A 29-A 30-A
31-A 32-A 33-A 34-A 35-A 36-A 37-A 38-A 39-A 40-A
41-A 42-A 43-A 44-A 45-A 46-A 47-A 48-A 49-A 50-A
51-A 52-A 53-A 54-A 55-A 56-A 57-A 58-A 59-A 60-A
61-A 62-A 63-A 64-A 65-A 66-A 67-A 68-A 69-A 70-A
71-A 72-A 73-A 74-A 75-A 76-A 77-A 78-A 79-A 80-A
81-A 82-A 83-A 84-A 85-A 86-A 87-A 88-A 89-A 90-A
91-A 92-A 93-A 94-A 95-A 96-A 97-A 98-A 99-A 100-A
101-A 102-A 103-A 104-A 105-A 106-A 107-A 108-A 109-A 110-A
111-A 112-A 113-A 114-A 115-A 116-A 117-A 118-A 119-A 120-A
121-A 122-A 123-A 124-A 125-A 126-A 127-A 128-A 129-A 130-A
131-A 132-A 133-A 134-A 135-A 136-A 137-A 138-A 139-A 140-A
141-A 142-A 143-A 144-A 145-A 146-A 147-A 148-A 149-A 150-A
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161-A 162-A 163-A 164-A 165-A 166-A 167-A 168-A 169-A 170-A
171-A 172-A 173-A 174-A 175-A 176-A 177-A 178-A 179-A 180-A
181-A 182-A 183-A 184-A 185-A 186-A 187-A 188-A 189-A 190-A
191-A 192-A 193-A 194-A 195-A 196-A 197-A 198-A 199-A 200-A
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Topic 12e: E-Technologies

Syllabus: *Electronic Commerce: Framework, Media Convergence of Applications, Consumer Applications, Organisation Applications; Electronic Payment Systems: Digital Token, Smart Cards, Credit Cards, Risks in Electronic Payment System, Designing Electronic Payment Systems; Electronic Data Interchange (EDI): Concepts, Applications, (Legal, Security and Privacy) issues, EDI and Electronic Commerce, Standardization and EDI, EDI Software Implementation, EDI Envelope for Message Transport, Internet-Based EDI; Digital Libraries and Data Warehousing: Concepts, Types of Digital documents, Issues behind document Infrastructure, Corporate Data Warehouses; Software Agents: Characteristics and Properties of Agents, Technology behind Software Agents (Applets, Browsers and Software Agents); Broadband Telecommunications: Concepts, Frame Relay, Cell Relay, Switched Multimegabit Data Service, Asynchronous Transfer Mode, Main concepts in Geographical Information System (GIS), E-cash, E-Business, ERP packages.*

Q1. In E-commerce framework, which layer provides multimedia content creation and distribution infrastructure?

A. Network infrastructure
B. Multimedia content & network publishing
C. Messaging & information distribution
D. Common business services
Correct Answer: B

Q2. Media convergence in E-commerce mainly refers to convergence of:

A. Price and product
B. Computing, communications & content
C. Buyers and sellers
D. Cash and credit
Correct Answer: B

Q3. Which is NOT a pillar of E-commerce framework (Kalakota & Whinston)?

A. Public policy
B. Technical standards
C. Network infrastructure
D. Cash on delivery logistics
Correct Answer: D

Q4. Information superhighway foundation for E-commerce is:

A. Multimedia content
B. Network infrastructure (telecom, cable, wireless)
C. Business services
D. Public policy
Correct Answer: B

Q5. Messaging layer in E-commerce framework typically includes:

A. EDI, e-mail, HTTP
B. HTML only
C. ERP only
D. GIS only
Correct Answer: A

Q6. Structured vs unstructured information distribution is handled at:

A. Network layer
B. Messaging & information distribution layer
C. Policy layer
D. Hardware layer
Correct Answer: B

Q7. Which supports secure messaging in E-commerce?

A. SET, SSL, S/MIME
B. JPEG only
C. MPEG only
D. FTP anonymous only
Correct Answer: A

Q8. Common business services layer includes:

A. Encryption, authentication, e-payment
B. Fibre cable laying
C. TV broadcasting
D. Paper ledger
Correct Answer: A

Q9. Media convergence enables:

A. Same digital content over TV, PC, mobile
B. Only print media
C. Only voice calls
D. Separation of data and video
Correct Answer: A

Q10. Digital convergence example:

A. Smart TV streaming e-shop + payment
B. Typewriter
C. Postal mail
D. Fax only
Correct Answer: A

Q11. Which network is backbone for E-commerce information superhighway?

A. Internet, intranet, extranet
B. Only LAN
C. Only PAN
D. Pigeon post
Correct Answer: A

Q12. Technical standards for E-commerce documents include:

A. EDI, XML, HTTP, SET
B. Only Morse code
C. Only ASCII art
D. Only voice
Correct Answer: A

Q13. Public policy issues in E-commerce framework include:

A. Privacy, universal access, taxation
B. Colour of website only
C. Shop timing only
D. Paper size
Correct Answer: A

Q14. Multimedia content layer deals with:

A. Creation of text, audio, video, graphics
B. Road transport
C. Cash counting
D. Warehouse locks
Correct Answer: A

Q15. Network publishing in E-commerce means:

A. Publishing catalogues / content on web for global access
B. Printing newspapers only
C. Publishing books offline
D. Notice board
Correct Answer: A

Q16. E-commerce framework's top layer for applications is:

A. E-commerce applications (retail, banking, trading)
B. Copper wire
C. Modem hardware
D. Operating system kernel
Correct Answer: A

Q17. Convergence of media reduces:

A. Need for separate devices/networks for voice, video, data
B. Internet speed
C. Choice
D. Competition
Correct Answer: A

Q18. Which is an I-way component?

A. Fibre, cellular, satellite access
B. Bullock cart
C. Telegraph only
D. Steam engine
Correct Answer: A

Q19. Interoperability in E-commerce is ensured by:

A. Standards & public policy
B. Monopoly
C. Secrecy
D. Manual registers
Correct Answer: A

Q20. E-commerce success depends on integration of:

A. Technology + business services + policy
B. Only technology
C. Only advertising
D. Only logistics
Correct Answer: A

Q21. Consumer-oriented E-commerce application includes:

A. Online shopping, e-banking, e-auction
B. Factory automation only
C. Payroll only
D. Inventory coding only
Correct Answer: A

Q22. Organisation-oriented (intra-organizational) application is:

A. Supply chain, ERP, workflow
B. Online games only
C. Movie streaming only
D. Personal blog
Correct Answer: A

Q23. Mercantile process from consumer perspective includes:

A. Search, compare, order, pay, delivery, service
B. Only manufacturing
C. Only mining
D. Only farming
Correct Answer: A

Q24. B2C example:

A. Amazon retail to customer
B. Company to supplier
C. Govt to govt
D. Employee to employee
Correct Answer: A

Q25. B2B example:

A. Manufacturer ordering raw material via extranet
B. Customer buying book
C. Citizen paying tax
D. Student reading blog
Correct Answer: A

Q26. C2C example:

A. eBay auction between consumers
B. RBI to bank
C. Company payroll
D. ERP module
Correct Answer: A

Q27. Intra-organizational E-commerce supports:

A. Workgroup communication, document workflow
B. Public ads only
C. TV shows
D. Radio
Correct Answer: A

Q28. Consumer search & discovery tool:

A. Search engines, catalogs, directories
B. Hammer
C. Ledger
D. Typewriter
Correct Answer: A

Q29. Organisation apps use mainly:

A. Intranet & extranet + EDI
B. Only public internet chat
C. Only FM radio
D. Only SMS jokes
Correct Answer: A

Q30. E-tailing■■:

A. Consumer-oriented retail online
B. Mining
C. Fishing
D. Construction
Correct Answer: A

Q31. E-banking service for consumer:

A. Balance check, fund transfer, bill pay
B. Ore smelting
C. Crop harvesting
D. Road laying
Correct Answer: A

Q32. On-demand services (video, information) are:

A. Consumer apps
B. Core manufacturing
C. Legal courts
D. Military drills
Correct Answer: A

Q33. Supply-chain integration is:

A. Organisation-to-organisation app
B. Personal diary
C. Movie review
D. Greeting card
Correct Answer: A

Q34. E-procurement benefits organisation by:

A. Lower cost, faster sourcing
B. Higher paper use
C. Slower approval
D. More travel
Correct Answer: A

Q35. Workflow automation in organisation handles:

A. Document routing & approvals
B. Cooking
C. Driving
D. Painting
Correct Answer: A

Q36. Group buying / auctions are:

A. Consumer mercantile models
B. Mining models
C. Textile looms
D. Railway signals
Correct Answer: A

Q37. Personalized marketing (recommendation) targets:

A. Individual consumer behaviour
B. Factory machines
C. Office furniture
D. Farm animals
Correct Answer: A

Q38. Customer service online includes:

A. Tracking, returns, helpdesk
B. Smelting
C. Weaving
D. Plumbing
Correct Answer: A

Q39. B2G example:

A. Firm filing tax / tender online
B. Friend chatting
C. Farmer ploughing
D. Child playing
Correct Answer: A

Q40. G2C example:

A. Citizen downloading certificates / paying bills
B. Shop to supplier
C. Factory to factory
D. Student to student notes
Correct Answer: A

Q41. E-payment system essential property:

A. Atomicity, privacy, anonymity option, convertibility
B. Only colour
C. Only size
D. Only weight
Correct Answer: A

Q42. Digital token-based payment uses:

A. Electronic tokens representing cash value
B. Paper cheques only
C. Barter goods
D. Gold bars physically mailed
Correct Answer: A

Q43. Smart card contains:

A. Embedded microprocessor/memory chip
B. Only magnetic stripe with no chip
C. Only photo
D. Only paper
Correct Answer: A

Q44. Credit card online processing involves:

A. Merchant, acquirer, issuer, cardholder, gateway
B. Only buyer
C. Only postman
D. Only printer
Correct Answer: A

Q45. Debit card differs from credit in that:

A. Directly debits bank balance
B. Gives unlimited loan
C. No authentication
D. No bank involved
Correct Answer: A

Q46. E-cheque is:

A. Digital version of paper cheque with signature/encryption
B. Physical cash
C. Free gift
D. Coupon only
Correct Answer: A

Q47. Risk in e-payment includes:

A. Fraud, chargeback, replay, double-spending
B. Only paper tear
C. Only ink fade
D. Only stamp loss
Correct Answer: A

Q48. Double-spending problem mainly in:

A. Digital cash without proper verification
B. Paper cash with watermark
C. Barter
D. Demand draft in person
Correct Answer: A

Q49. E-payment design requires:

A. Integrity, authentication, non-repudiation, scalability
B. Only attractive logo
C. Only font style
D. Only background music
Correct Answer: A

Q50. SET protocol was designed for:

A. Secure card payments over internet
B. Video streaming
C. Email jokes
D. File compression
Correct Answer: A

Q51. SSL/TLS in payment provides:

A. Encrypted channel between browser & server
B. Printed receipt
C. Cash counting
D. Coin sorting
Correct Answer: A

Q52. 3-D Secure (Verified by Visa) adds:

A. Extra authentication via issuer
B. Extra delivery charge
C. Extra packaging
D. Extra advertisement
Correct Answer: A

Q53. E-wallet stores:

A. Card details, addresses for fast checkout
B. Vegetables
C. Clothes
D. Fuel
Correct Answer: A

Q54. Micropayments are for:

A. Very small value (articles, game items)
B. Buying houses
C. Buying aircraft
D. Buying factories
Correct Answer: A

Q55. Stored-value card example:

A. Metro card, gift card
B. Visiting card
C. Greeting card
D. Playing card
Correct Answer: A

Q56. Contactless smart card uses:

A. NFC/RFID
B. Manual typing only
C. Fax
D. Telegram
Correct Answer: A

Q57. Credit card risk for merchant:

A. Chargeback & stolen cards
B. Too much cash
C. Heavy coins
D. Paper overload
Correct Answer: A

Q58. Credit card risk for buyer:

A. Identity theft, phishing
B. Getting discount
C. Getting cashback
D. Getting loyalty
Correct Answer: A

Q59. Digital cash must be:

A. Unforgeable, transferable, divisible
B. Easily copied
C. Non-verifiable
D. Non-portable
Correct Answer: A

Q60. Anonymity in e-cash means:

A. Payer identity not linked to transaction (like cash)
B. Payee unknown to all including bank always
C. No records at all
D. No encryption
Correct Answer: A

Q61. EDI full form:

A. Electronic Data Interchange
B. Electronic Device Interface
C. Electrical Data Internet
D. Encrypted Disk Image
Correct Answer: A

Q62. EDI is:

A. Structured computer-to-computer exchange of business documents
B. Human email chat
C. Video call
D. Manual fax reading
Correct Answer: A

Q63. EDI document example:

A. Purchase order, invoice, ASN
B. Movie file
C. Song MP3
D. Family photo
Correct Answer: A

Q64. EDI benefits:

A. Reduced paper, errors, faster cycle
B. More postage
C. More delays
D. More manual entry
Correct Answer: A

Q65. EDI application in retail:

A. Auto replenishment, POS to supplier
B. Film editing
C. Music composing
D. Painting
Correct Answer: A

Q66. EDI in manufacturing:

A. JIT scheduling, supplier coordination
B. Canteen menu
C. Garden maintenance
D. Festival planning
Correct Answer: A

Q67. EDI in trade / customs:

A. Manifest, clearance declarations
B. Love letters
C. Greetings
D. Jokes
Correct Answer: A

Q68. EDI vs e-mail:

A. EDI structured/machine-readable; email unstructured/human-readable
B. Same
C. Email structured, EDI unstructured
D. Both video
Correct Answer: A

Q69. Direct EDI uses:

A. Point-to-point private links / VAN
B. Only public chat rooms
C. Only social media
D. Only radio
Correct Answer: A

Q70. VAN in EDI is:

A. Value Added Network for mailboxing, audit
B. Van vehicle
C. Vacation network
D. Voice area network
Correct Answer: A

Q71. Legal issue in EDI:

A. Validity of electronic contracts & signatures
B. Colour of paper
C. Font size only
D. Stamp price
Correct Answer: A

Q72. Security in EDI needs:

A. Authentication, integrity, confidentiality, non-repudiation
B. Only speed
C. Only cost
D. Only volume
Correct Answer: A

Q73. Privacy in EDI means:

A. Protect trading partner data from disclosure
B. Publish all data
C. Share passwords
D. Open mailbox to all
Correct Answer: A

Q74. Non-repudiation ensures:

A. Sender cannot deny sending
B. Receiver cannot deny receiving
C. Only compression
D. Only backup (combined: both cannot deny)
Correct Answer: A

Q75. EDI standardization need arises to:

A. Allow different systems to interoperate
B. Increase incompatibility
C. Promote proprietary lock-in
D. Avoid trade
Correct Answer: A

Q76. UN/EDIFACT is:

A. International EDI standard by UN
B. Video codec
C. Audio format
D. Image format
Correct Answer: A

Q77. ANSI X12 is:

A. US national EDI standard
B. European video standard
C. Asian food standard
D. African telecom standard
Correct Answer: A

Q78. EDIFACT vs X12:

A. EDIFACT international, X12 mainly North America
B. Both same body
C. X12 for video, EDIFACT for audio
D. No difference at all
Correct Answer: A

Q79. EDI software functions:

A. Mapping, translation, communication, audit
B. Only gaming
C. Only browsing
D. Only printing
Correct Answer: A

Q80. Mapping in EDI software:

A. Convert in-house format to standard & vice versa
B. Draw road maps
C. Draw building plans
D. Draw charts only
Correct Answer: A

Q81. EDI envelope concept includes:

A. Interchange / group / transaction set wrappers with control headers
B. Paper envelope only
C. Email attachment only
D. Gift wrap
Correct Answer: A

Q82. ISA/IEA segments in X12 denote:

A. Interchange control header/trailer
B. Image size
C. Audio track
D. Video frame
Correct Answer: A

Q83. GS/GE segments denote:

A. Functional group header/trailer
B. General store
C. Gas station
D. Guest entry
Correct Answer: A

Q84. ST/SE segments denote:

A. Transaction set header/trailer
B. Street address
C. State entry
D. Stock exchange
Correct Answer: A

Q85. Control numbers in envelope ensure:

A. Tracking, sequencing, duplicate detection
B. Decoration
C. Advertisement
D. Entertainment
Correct Answer: A

Q86. Internet-EDI transports via:

A. AS1/AS2/AS3 over SMTP/HTTP/FTP
B. Only courier
C. Only pigeon
D. Only telex
Correct Answer: A

Q87. AS2 provides:

A. Secure HTTP + encryption + signed MDN receipt
B. Video chat
C. Voice mail
D. Fax tone
Correct Answer: A

Q88. MDN in AS2 is:

A. Message Disposition Notification (receipt)
B. Mobile number
C. Movie database
D. Mail delivery noodle
Correct Answer: A

Q89. Web-EDI is for:

A. SMEs via browser forms to join EDI network
B. Only large firms with mainframes
C. Only gamers
D. Only film makers
Correct Answer: A

Q90. EDI over internet advantage vs VAN:

A. Lower cost, ubiquity
B. Higher cost, less reach
C. No security possible
D. No standards
Correct Answer: A

Q91. EDIINT stands for:

A. EDI over Internet (secure)
B. Entertainment interface
C. Energy Intel
D. Engine internal
Correct Answer: A

Q92. Security for Internet-EDI uses:

A. Encryption, digital signatures, certificates
B. Plain postcard
C. Open broadcast
D. No password
Correct Answer: A

Q93. EDI audit trail helps:

A. Dispute resolution & compliance
B. Movie ratings
C. Song lyrics
D. Food recipes
Correct Answer: A

Q94. Trading partner agreement covers:

A. Technical, legal, operational rules
B. Movie tickets
C. Picnic plan
D. Birthday party
Correct Answer: A

Q95. EDI implementation steps:

A. Analysis, mapping, partner setup, pilot, rollout
B. Only purchase hardware
C. Only hire driver
D. Only paint office
Correct Answer: A

Q96. EDI failure risk includes:

A. Mapping errors, version mismatch, downtime
B. Too much paper saved
C. Too fast
D. Too accurate
Correct Answer: A

Q97. Hybrid EDI + XML is used because:

A. XML offers readability/web integration while EDI offers maturity
B. Neither useful
C. Only for games
D. Only for movies
Correct Answer: A

Q98. ebXML purpose:

A. XML-based B2B framework for SMEs/global trade
B. Music album
C. Photo editor
D. Video game
Correct Answer: A

Q99. Legal validity of EDI messages often needs:

A. Digital signature law (e.g., IT Act) support
B. Witness dance
C. Paper backup always mandatory
D. No law
Correct Answer: A

Q100. EDI vs E-commerce:

A. EDI subset focused B2B doc exchange; E-commerce broader including B2C
B. Identical
C. Unrelated
D. EDI is B2C only
Correct Answer: A

Q101. Digital library is:

A. Organized electronic collection with metadata & retrieval
B. Pile of CDs unsorted
C. Computer shop
D. Printing press
Correct Answer: A

Q102. Digital library functions:

A. Capture, index, store, search, deliver
B. Only sell hardware
C. Only repair PCs
D. Only provide power
Correct Answer: A

Q103. Metadata standard for digital libraries:

A. Dublin Core, MARC
B. JPEG only
C. MP3 only
D. AVI only
Correct Answer: A

Q104. OAI-PMH is for:

A. Metadata harvesting/interoperability
B. Video gaming
C. Email spam
D. Virus spread
Correct Answer: A

Q105. Digital library retrieval uses:

A. Indexing, full-text search, relevance ranking
B. Random pick
C. Alphabetical shelf only
D. Guessing
Correct Answer: A

Q106. Digitization challenge:

A. Copyright, preservation, quality
B. Only electricity bill
C. Only room size
D. Only chair comfort
Correct Answer: A

Q107. Data warehousing is:

A. Subject-oriented, integrated, time-variant, non-volatile store for analysis
B. Transactional register
C. Cache memory
D. Dustbin
Correct Answer: A

Q108. Data warehouse vs OLTP:

A. Warehouse for analysis/history; OLTP for daily transactions
B. Same
C. Warehouse for typing, OLTP for movies
D. No difference
Correct Answer: A

Q109. OLAP operations include:

A. Roll-up, drill-down, slice, dice, pivot
B. Copy-paste only
C. Print-scan only
D. Boot-shutdown only
Correct Answer: A

Q110. Star schema has:

A. Central fact table + denormalized dimensions
B. Only one table
C. No facts
D. Only images
Correct Answer: A

Q111. Snowflake schema:

A. Normalized dimensions branching
B. Snow weather data only
C. Circular tables
D. No keys
Correct Answer: A

Q112. ETL stands for:

A. Extract, Transform, Load
B. Email, Transfer, Link
C. Enter, Type, Leave
D. Edit, Test, Launch
Correct Answer: A

Q113. Data mart is:

A. Subset of warehouse for department
B. Shopping mall
C. Open market
D. Flea market
Correct Answer: A

Q114. Data cleaning in warehousing:

A. Remove duplicates, fix inconsistencies
B. Dust shelves
C. Wash CDs
D. Sweep floor
Correct Answer: A

Q115. Metadata in warehouse describes:

A. Source, transformation, lineage
B. Movie story
C. Song tune
D. Game cheat
Correct Answer: A

Q116. Archival in digital library needs:

A. Long-term preservation, migration, emulation
B. Throw away old
C. Burn books
D. Delete all
Correct Answer: A

Q117. Search in digital library vs OPAC:

A. Full-text + multimedia vs catalogue only
B. Same as attendance register
C. No search
D. Only by colour
Correct Answer: A

Q118. Data mining on warehouse finds:

A. Patterns, trends, associations
B. Coal
C. Gold physically
D. Oil drilling
Correct Answer: A

Q119. Warehouse non-volatile means:

A. Data stable, not updated in place; appended
B. Easily deleted
C. Changes every second
D. Volatile RAM
Correct Answer: A

Q120. Time-variant means:

A. Keeps history over time
B. Shows current time only
C. No dates
D. Clock display
Correct Answer: A

Q121. Software agent characteristic:

A. Autonomous, reactive, pro-active, social/communicative
B. Passive file only
C. Manual tool only
D. Static poster
Correct Answer: A

Q122. Agent autonomy means:

A. Operates without constant human control
B. Needs keystroke for every step
C. Cannot sense
D. Cannot act
Correct Answer: A

Q123. Reactivity of agent:

A. Perceives environment & responds timely
B. Ignores environment
C. Sleeps always
D. Crashes
Correct Answer: A

Q124. Pro-activeness means:

A. Goal-directed initiative
B. Only reactive
C. No goals
D. Random
Correct Answer: A

Q125. Social ability means:

A. Communicates with other agents/users (ACL/KQML)
B. Isolated
C. No language
D. No interaction
Correct Answer: A

Q126. Mobility property (mobile agent):

A. Migrates across network to execute near data
B. Fixed to one CPU
C. Cannot move code
D. Only human moves
Correct Answer: A

Q127. Learning property:

A. Adapts from experience
B. Never changes
C. Forgets all
D. Hardcoded forever
Correct Answer: A

Q128. Rational agent acts to:

A. Maximize expected performance measure
B. Minimize performance
C. Act randomly
D. Harm user
Correct Answer: A

Q129. Intelligent agent components (PEAS):

A. Performance, Environment, Actuators, Sensors
B. People, Entertainment, Art, Sport
C. Price, Expense, Audit, Sale
D. Power, Energy, Atom, Soil
Correct Answer: A

Q130. Applet is:

A. Small Java program embedded in webpage, runs in sandbox
B. Large OS
C. Device driver
D. Full database
Correct Answer: A

Q131. Applet security model:

A. Sandbox restricts file/network access
B. Full disk access
C. No restriction
D. Kernel access
Correct Answer: A

Q132. Browser role for agents/applets:

A. Executes, renders, provides JVM/plug-in, manages trust
B. Only prints
C. Only scans
D. Only stores power
Correct Answer: A

Q133. Web browser components:

A. UI, engine, rendering, JS interpreter, networking, storage
B. Only screen
C. Only keyboard
D. Only mouse
Correct Answer: A

Q134. Search agent (bot) does:

A. Crawls/indexes web automatically
B. Cooks food
C. Drives car
D. Paints house
Correct Answer: A

Q135. Shopping agent compares:

A. Prices/features across merchants
B. Weather
C. Road traffic only
D. Movie ratings only
Correct Answer: A

Q136. Agent communication language:

A. KQML / FIPA-ACL
B. Hindi only
C. Morse only
D. Binary music
Correct Answer: A

Q137. Multi-agent system needs:

A. Coordination, cooperation/negotiation
B. Isolation only
C. No communication
D. Single thread only
Correct Answer: A

Q138. Interface agent learns:

A. User habits to assist (mail filter, recommender)
B. Machine repair
C. Road rules
D. Cooking
Correct Answer: A

Q139. Java applets vs servlets:

A. Applet client-side, servlet server-side
B. Same side
C. Both hardware
D. Both networks cables
Correct Answer: A

Q140. Browser cookies help e-commerce/agents by:

A. Session, personalization (with privacy concerns)
B. Cooking
C. Heating
D. Lighting
Correct Answer: A

Q141. Frame Relay is:

A. Packet-switched WAN with variable frames, no per-hop error correction
B. Circuit voice only
C. Broadcast TV
D. Paging only
Correct Answer: A

Q142. Frame Relay DLCI is:

A. Identifier for virtual circuit
B. Movie title
C. Phone number
D. IP address version
Correct Answer: A

Q143. CIR in Frame Relay means:

A. Committed Information Rate (guaranteed avg)
B. Central Intelligence Report
C. Circuit Idle Ratio
D. Current Internet Rate
Correct Answer: A

Q144. Frame Relay virtual circuits can be:

A. PVC or SVC
B. Only physical wires
C. Only satellite
D. Only fibre cut
Correct Answer: A

Q145. Cell Relay uses:

A. Fixed-size cells (e.g., 53-byte ATM)
B. Variable long frames
C. Only voice waves
D. Only letters
Correct Answer: A

Q146. ATM cell size:

A. 53 bytes (5 header + 48 payload)
B. 1500 bytes
C. 64KB
D. 1MB
Correct Answer: A

Q147. ATM VPI/VCI are for:

A. Virtual path/channel routing
B. Video play indicators
C. Voice pitch
D. Vendor IDs
Correct Answer: A

Q148. ATM service classes include:

A. CBR, VBR, ABR, UBR
B. Only best-effort
C. Only voice
D. Only text
Correct Answer: A

Q149. SMDS full form:

A. Switched Multimegabit Data Service
B. Simple Mail Delivery System
C. Single Mode Digital Signal
D. Shared Media Distribution Service
Correct Answer: A

Q150. SMDS technology based on:

A. DQDB (IEEE 802.6), connectionless, 53-byte cells
B. Token ring only
C. Dial-up only
D. Pigeon post
Correct Answer: A

Q151. Broadband telecom need for E-commerce:

A. High-speed multimedia (video, voice, data) support
B. Slow telegram sufficient
C. No network
D. Only voice
Correct Answer: A

Q152. Frame vs Cell relay difference:

A. Variable vs fixed size; cell better for voice/video QoS
B. No difference
C. Frame for voice only
D. Cell for text only
Correct Answer: A

Q153. ATM adaptation layers (AAL):

A. Interface higher layers to ATM (AAL1-5)
B. Movie layers
C. Cake layers
D. Paint layers
Correct Answer: A

Q154. AAL5 commonly for:

A. Data (IP over ATM)
B. Only voice calls
C. Only TV
D. Only radio
Correct Answer: A

Q155. Congestion in Frame Relay indicated by:

A. FECN, BECN, DE bits
B. Colour lights
C. Sound alarm
D. Smell
Correct Answer: A

Q156. ATM is:

A. Connection-oriented with virtual circuits
B. Connectionless broadcast
C. Postal service
D. Courier on foot
Correct Answer: A

Q157. SONET/SDH supports broadband by:

A. High-speed optical transmission for ATM/B-ISDN
B. Copper telegraph
C. Smoke signals
D. Drum beats
Correct Answer: A

Q158. B-ISDN uses:

A. ATM as transfer mode
B. Only narrowband voice
C. Only Morse
D. Only fax
Correct Answer: A

Q159. SMDS addressing uses:

A. E.164 telephone-like numbers
B. House colours
C. Nicknames
D. Star signs
Correct Answer: A

Q160. DQDB topology is:

A. Dual bus
B. Star only
C. Ring only
D. Mesh only
Correct Answer: A

Q161. GIS full form:

A. Geographic Information System
B. General Internet Service
C. Global Information Speed
D. Graphical Interface Software
Correct Answer: A

Q162. GIS captures:

A. Spatial + attribute data (maps, coordinates)
B. Only songs
C. Only movies
D. Only bills
Correct Answer: A

Q163. GIS data types:

A. Raster & vector
B. Only text
C. Only audio
D. Only video
Correct Answer: A

Q164. Vector data uses:

A. Points, lines, polygons
B. Pixels only
C. Sound waves
D. Smell
Correct Answer: A

Q165. Raster data uses:

A. Grid of cells/pixels (satellite imagery)
B. Only lines
C. Only tables
D. Only charts
Correct Answer: A

Q166. GIS application in business:

A. Site selection, logistics, market analysis
B. Only gaming
C. Only chatting
D. Only typing
Correct Answer: A

Q167. GPS + GIS enables:

A. Tracking, navigation, fleet management
B. Cooking
C. Singing
D. Dancing
Correct Answer: A

Q168. Spatial query example:

A. Find stores within 5 km
B. Find rhyming words
C. Find song lyrics
D. Find jokes
Correct Answer: A

Q169. Map overlay in GIS:

A. Combine layers to find relationships
B. Place tablecloth
C. Cover book
D. Wrap gift
Correct Answer: A

Q170. GIS software example:

A. ArcGIS, QGIS
B. MS Paint only
C. Notepad only
D. Calculator only
Correct Answer: A

Q171. E-cash properties include:

A. Divisibility, transferability, anonymity option
B. Only divisibility of paper
C. Only weight
D. Only colour
Correct Answer: A

Q172. E-business is broader than E-commerce as it includes:

A. Internal processes + CRM + SCM + collaboration
B. Only online selling
C. Only buying
D. Only advertising
Correct Answer: A

Q173. ERP full form:

A. Enterprise Resource Planning
B. Electronic Retail Price
C. Emergency Response Plan
D. Entertainment Radio Program
Correct Answer: A

Q174. ERP modules include:

A. Finance, HR, inventory, sales, production
B. Only games
C. Only movies
D. Only songs
Correct Answer: A

Q175. ERP benefits:

A. Integration, data consistency, efficiency
B. More silos
C. More duplication
D. Slower reporting
Correct Answer: A

Q176. ERP vs SCM:

A. ERP internal resources; SCM external supply chain (can integrate)
B. Same
C. Unrelated
D. ERP for home use
Correct Answer: A

Q177. E-CRM focuses on:

A. Web-based customer interaction & retention
B. Machine repair
C. Road building
D. Crop yield
Correct Answer: A

Q178. E-business model B2B hub:

A. Marketplace connecting many buyers/sellers
B. Personal diary
C. Family album
D. School notebook
Correct Answer: A

Q179. E-cash offline vs online:

A. Offline no bank during payment; online needs bank verification
B. Same always needs bank
C. Both need courier
D. Both need cheque book
Correct Answer: A

Q180. Blind signature in e-cash provides:

A. Untraceability by bank while ensuring validity
B. Visible signature always linked
C. No security
D. No verification
Correct Answer: A

Q181. E-commerce threat:

A. Hacking, spoofing, sniffing
B. Good packaging
C. Fast delivery
D. Discounts
Correct Answer: A

Q182. Consumer protection in e-payment needs:

A. Refunds, grievance, secure disclosure
B. No receipts
C. No support
D. No warranty
Correct Answer: A

Q183. EDI + ERP integration allows:

A. Automatic order-to-invoice without re-entry
B. Manual typing more
C. Paper increase
D. Delay increase
Correct Answer: A

Q184. Data warehouse supports E-business by:

A. Customer analytics, sales trends
B. Power supply
C. Furniture
D. Stationery
Correct Answer: A

Q185. Broadband + GIS + agents together enable:

A. Location-based m-commerce (nearest ATM/offer)
B. Typewriter revival
C. Postal increase
D. Fax revival
Correct Answer: A

Q186. M-commerce is:

A. E-commerce via mobile/wireless
B. Only desktop trade
C. Only TV trade
D. Only paper trade
Correct Answer: A

Q187. Digital library + E-commerce example:

A. E-books sale, pay-per-article
B. Selling bricks only
C. Selling cement only
D. Selling sand only
Correct Answer: A

Q188. Software agent in E-commerce negotiates:

A. Price, compares, bids (auction bot)
B. Cooks
C. Cleans
D. Watches
Correct Answer: A

Q189. Frame Relay + EDI historically:

A. Cost-effective WAN for EDI before internet-EDI
B. Never used
C. Only for games
D. Only for movies
Correct Answer: A

Q190. ATM + E-commerce relevance:

A. Supported multimedia backbone for high-volume transactions/video catalog
B. Irrelevant
C. Only for cash machines confusion (ATM also Automated Teller Machine)
D. Only for traffic lights
Correct Answer: A

Q191. E-business infrastructure includes:

A. Networks, servers, middleware, security, payment gateways
B. Only chairs
C. Only tables
D. Only fans
Correct Answer: A

Q192. Payment gateway role:

A. Authorize & route card transactions securely
B. Open physical gate
C. Guard building
D. Direct traffic
Correct Answer: A

Q193. E-cash vs E-cheque:

A. Cash token-like immediate; cheque account-based with clearing
B. Same
C. Cash needs cheque book
D. Cheque needs no account
Correct Answer: A

Q194. ERP implementation challenge:

A. Change management, customization, cost
B. Only power plug
C. Only mouse pad
D. Only screen saver
Correct Answer: A

Q195. Success factor for E-commerce tech mix:

A. Interoperable standards + security + usability + broadband
B. Only graphics
C. Only slogans
D. Only discounts
Correct Answer: A

Q196. Media convergence + broadband enables:

A. Video catalog, live demo, interactive TV shopping
B. Only text bills
C. Only black-white images
D. Only voice bills
Correct Answer: A

Q197. EDI envelope + security ensures:

A. Correct routing + integrity for B2B orders
B. Pretty cover
C. Gift wrap
D. Movie cover
Correct Answer: A

Q198. Data warehousing + agents enable:

A. Personalized recommendations from history
B. Random ads
C. No targeting
D. Spam increase
Correct Answer: A

Q199. GIS + logistics in E-business optimizes:

A. Delivery routes, warehouse location
B. Web colours
C. Font sizes
D. Logo shapes
Correct Answer: A

Q200. Future E-technologies trend:

A. Mobile payments, AI agents, blockchain, IoT integration
B. Return to barter only
C. Abandon internet
D. Paper-only trade ---
Correct Answer: A

ANSWER KEY — Topic 12e: E-Technologies

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1-B 2-B 3-D 4-B 5-A 6-B 7-A 8-A 9-A 10-A
11-A 12-A 13-A 14-A 15-A 16-A 17-A 18-A 19-A 20-A
21-A 22-A 23-A 24-A 25-A 26-A 27-A 28-A 29-A 30-A
31-A 32-A 33-A 34-A 35-A 36-A 37-A 38-A 39-A 40-A
41-A 42-A 43-A 44-A 45-A 46-A 47-A 48-A 49-A 50-A
51-A 52-A 53-A 54-A 55-A 56-A 57-A 58-A 59-A 60-A
61-A 62-A 63-A 64-A 65-A 66-A 67-A 68-A 69-A 70-A
71-A 72-A 73-A 74-A 75-A 76-A 77-A 78-A 79-A 80-A
81-A 82-A 83-A 84-A 85-A 86-A 87-A 88-A 89-A 90-A
91-A 92-A 93-A 94-A 95-A 96-A 97-A 98-A 99-A 100-A
101-A 102-A 103-A 104-A 105-A 106-A 107-A 108-A 109-A 110-A
111-A 112-A 113-A 114-A 115-A 116-A 117-A 118-A 119-A 120-A
121-A 122-A 123-A 124-A 125-A 126-A 127-A 128-A 129-A 130-A
131-A 132-A 133-A 134-A 135-A 136-A 137-A 138-A 139-A 140-A
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171-A 172-A 173-A 174-A 175-A 176-A 177-A 178-A 179-A 180-A
181-A 182-A 183-A 184-A 185-A 186-A 187-A 188-A 189-A 190-A
191-A 192-A 193-A 194-A 195-A 196-A 197-A 198-A 199-A 200-A
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Topic 12f: Windows Programming

Syllabus: *Introduction to Windows programming - Win32, Microsoft Foundation Classes (MFC), Documents and views, Resources, Message handling in Windows; Simple Applications (in Windows): Scrolling, splitting views, docking toolbars, status bars, common dialogs; Advanced Windows Programming: Multiple Document Interface (MDI), Multithreading, Object linking and Embedding (OLE), Active X controls, Active Template Library (ATL), Network programming.*

Q1. WinMain signature in Win32 is:

A. int main()
B. int WINAPI WinMain(HINSTANCE,HINSTANCE,LPSTR,int)
C. void start()
D. DWORD ThreadProc()
Correct Answer: B

Q2. First parameter of WinMain (hInstance) denotes:

A. Window handle
B. Command line
C. Application instance handle
D. Message ID
Correct Answer: C

Q3. nCmdShow in WinMain specifies:

A. Thread ID
B. Icon
C. Cursor
D. How window is to be shown
Correct Answer: D

Q4. Core Win32 message loop uses:

A. GetMessage-TranslateMessage-DispatchMessage
B. printf-scanf
C. malloc-free
D. open-close only
Correct Answer: A

Q5. GetMessage returns FALSE when:

A. On WM_PAINT
B. On WM_QUIT
C. On WM_CREATE
D. On WM_MOUSEMOVE
Correct Answer: B

Q6. Window procedure signature is:

A. void func()
B. int main(int)
C. LRESULT CALLBACK WndProc(HWND,UINT,WPARAM,LPARAM)
D. DWORD APIENTRY DllMain()
Correct Answer: C

Q7. Registering window class uses:

A. RegisterClassEx + WNDCLASSEX
B. CreatePen
C. SelectObject
D. BitBlt
Correct Answer: A

Q8. Creating main window uses:

A. fopen
B. malloc
C. printf
D. CreateWindowEx
Correct Answer: D

Q9. Message map equivalent in raw Win32 vs MFC:

A. Identical syntax
B. Switch-case on msg vs BEGIN_MESSAGE_MAP macros
C. No handling needed
D. Only BIOS interrupts
Correct Answer: B

Q10. DefWindowProc is used to:

A. Destroy app
B. Create thread
C. Provide default handling for unprocessed messages
D. Compile resources
Correct Answer: C

Q11. WM_CREATE is sent:

A. At window creation
B. At window destroy
C. On key press
D. On timer only
Correct Answer: A

Q12. WM_PAINT should be handled with:

A. malloc
B. CreateThread
C. socket
D. BeginPaint / EndPaint + HDC drawing
Correct Answer: D

Q13. Invalidating window to trigger repaint uses:

A. Sleep
B. InvalidateRect
C. WaitForSingleObject
D. RegOpenKey
Correct Answer: B

Q14. MFC application class derived from:

A. CWnd
B. CDocument
C. CWinApp
D. CView
Correct Answer: C

Q15. CWinApp::InitInstance typically:

A. Creates document template & main frame
B. Deletes OS
C. Formats disk
D. Installs driver
Correct Answer: A

Q16. Root of MFC hierarchy:

A. CWnd
B. CView
C. CWinApp
D. CObject
Correct Answer: D

Q17. CCmdTarget enables:

A. Drawing only
B. Message routing / command handling
C. Memory chip
D. Network cable
Correct Answer: B

Q18. CWnd encapsulates:

A. HWND + window operations
B. Only files
C. Only threads
D. Only database
Correct Answer: A

Q19. CFrameWnd vs CView:

A. Same
B. Both documents
C. Frame is container window; View displays document data
D. View is frame
Correct Answer: C

Q20. CDocument stores:

A. Pixels only
B. HWND only
C. Threads only
D. Application data + serialization
Correct Answer: D

Q21. Document/View separation benefit:

A. Faster CPU
B. Multiple views on same data (splitter, MDI)
C. No benefit
D. Larger exe only
Correct Answer: B

Q22. CWinApp message pump is in:

A. Run() with PreTranslateMessage
B. main() of C
C. BIOS
D. Bootloader
Correct Answer: A

Q23. AfxWinMain internally calls:

A. printf
B. scanf
C. InitInstance, Run
D. Format
Correct Answer: C

Q24. hPrevInstance in Win32 is:

A. Used heavily in Win32
B. Thread handle
C. File handle
D. Obsolete/NULL in Win32
Correct Answer: D

Q25. LPSTR lpCmdLine gives:

A. Window style
B. Command-line string
C. Icon
D. Menu
Correct Answer: B

Q26. WNDCLASSEX::lpfnWndProc holds:

A. Pointer to window procedure
B. Brush colour only
C. Thread ID
D. File path
Correct Answer: A

Q27. WS_OVERLAPPEDWINDOW style means:

A. Invisible
B. Child only
C. Standard titled, sizable window with sysmenu/min-max
D. Button only
Correct Answer: C

Q28. ShowWindow + UpdateWindow after CreateWindow to:

A. Delete window
B. Hide forever
C. Start thread
D. Display and force initial paint
Correct Answer: D

Q29. PostQuitMessage posts:

A. WM_QUIT
B. WM_PAINT
C. WM_CREATE
D. WM_CLOSE only
Correct Answer: A

Q30. GetMessage vs PeekMessage:

A. Same
B. GetMessage blocks; PeekMessage non-blocking (for games/idle)
C. Peek blocks more
D. GetMessage for DOS only
Correct Answer: B

Q31. MFC class CObject provides:

A. Windows
B. Sockets
C. Serialization, RTTI, diagnostics (DECLARE/IMPLEMENT_DYNAMIC)
D. Printing only
Correct Answer: C

Q32. CArchive works with:

A. Serialization (<< / >> with CDocument::Serialize)
B. Audio only
C. Video only
D. BIOS
Correct Answer: A

Q33. DECLARE_DYNCREATE enables:

A. Drawing
B. Threading
C. Networking
D. Runtime object creation by class name
Correct Answer: D

Q34. CDocument::OnNewDocument called when:

A. On exit
B. New file
C. On print
D. On help
Correct Answer: B

Q35. CDocument::Serialize loads/saves via:

A. CArchive
B. CWnd
C. CMenu
D. CThread
Correct Answer: A

Q36. CView::OnDraw receives:

A. HANDLE file
B. SOCKET
C. CDC*
D. HMENU
Correct Answer: C

Q37. CDC encapsulates:

A. Window handle
B. File
C. Thread
D. Device context (HDC)
Correct Answer: D

Q38. CPaintDC used:

A. Anywhere
B. Only in OnDraw/WM_PAINT
C. In threads only
D. In DllMain
Correct Answer: B

Q39. CClientDC used:

A. Drawing outside paint (e.g., mouse handlers)
B. Only for printing
C. Only for files
D. Only for registry
Correct Answer: A

Q40. Resource script file extension:

A. .cpp
B. .h
C. .rc
D. .exe
Correct Answer: C

Q41. Resource header for IDs:

A. resource.h
B. windows.exe
C. stdio.h
D. winsock.dll
Correct Answer: A

Q42. Dialog resource editor creates:

A. .exe
B. .obj
C. .lib
D. Dialog template + DoModal() via CDialog
Correct Answer: D

Q43. Modal vs Modeless dialog:

A. Same
B. Modal blocks (DoModal); modeless (Create/ShowWindow) allows parallel
C. Modeless blocks
D. Modal has no return
Correct Answer: B

Q44. DDX/DDV stands for:

A. Disk defrag
B. DirectX
C. Dialog Data Exchange/Validation
D. Data delete
Correct Answer: C

Q45. Message box API:

A. MessageBox / AfxMessageBox
B. printf
C. cout
D. Beep only
Correct Answer: A

Q46. Menu resource attached via:

A. malloc
B. free
C. new
D. LoadMenu / SetMenu + COMMAND handlers
Correct Answer: D

Q47. Accelerator table provides:

A. Faster CPU
B. Keyboard shortcuts (LoadAccelerators, TranslateAccelerator)
C. Faster disk
D. Faster network
Correct Answer: B

Q48. Icon/cursor resources loaded by:

A. fopen
B. CreateFile
C. LoadIcon / LoadCursor
D. socket
Correct Answer: C

Q49. Version resource stores:

A. Version, copyright info
B. Pixels
C. Threads
D. Packets
Correct Answer: A

Q50. String Table resource helps:

A. Obfuscation
B. Hardcode
C. Speed
D. Localization (LoadString)
Correct Answer: D

Q51. WM_COMMAND originates from:

A. Mouse move
B. Menu, button, accelerator
C. Paint
D. Timer only
Correct Answer: B

Q52. WM_NOTIFY vs WM_COMMAND:

A. Same
B. NOTIFY older
C. NOTIFY carries NMHDR with richer info for common controls
D. COMMAND for listview only
Correct Answer: C

Q53. MFC message map macro block:

A. BEGIN_MESSAGE_MAP … END_MESSAGE_MAP + ON_COMMAND/ON_WM_PAINT
B. begin-end
C. if-else only
D. switch in main
Correct Answer: A

Q54. ON_WM_PAINT maps to:

A. OnCreate
B. OnClose
C. OnQuit
D. OnPaint
Correct Answer: D

Q55. Command routing order MFC (MDI):

A. App first
B. View → Document → Frame → App
C. OS first
D. Random
Correct Answer: B

Q56. ON_UPDATE_COMMAND_UI used for:

A. Drawing
B. Threading
C. Enabling/disabling/checking menu/toolbar items
D. Deleting files
Correct Answer: C

Q57. PreTranslateMessage allows:

A. Filtering/translating accelerator/tooltip before dispatch
B. Compiling
C. Linking
D. Formatting disk
Correct Answer: A

Q58. Window messages WM_LBUTTONDOWN params:

A. Only HWND
B. Only MSG
C. Only HDC
D. x,y in LPARAM, flags in WPARAM
Correct Answer: D

Q59. Capturing mouse uses:

A. printf
B. SetCapture / ReleaseCapture
C. malloc
D. free
Correct Answer: B

Q60. Keyboard messages:

A. WM_PAINT
B. WM_TIMER
C. WM_KEYDOWN/WM_CHAR/WM_KEYUP
D. WM_QUIT only
Correct Answer: C

Q61. Scroll bar messages:

A. WM_HSCROLL/WM_VSCROLL with SB_* codes
B. WM_CREATE only
C. WM_DESTROY only
D. WM_QUIT only
Correct Answer: A

Q62. CScrollView provides:

A. No scroll
B. Only dialogs
C. Only threads
D. Automatic scroll handling (SetScrollSizes, OnInitialUpdate)
Correct Answer: D

Q63. SetScrollSizes parameters include:

A. Only colour
B. Map mode + total size + page/line size
C. Only font
D. Only icon
Correct Answer: B

Q64. Mapping modes (MM_TEXT, MM_LOenzi etc.) control:

A. Threads
B. Sockets
C. Logical-to-device coordinate conversion
D. File paths
Correct Answer: C

Q65. Splitter window class:

A. CSplitterWnd
B. CDialog
C. CMenu
D. CFile
Correct Answer: A

Q66. Static vs Dynamic splitter:

A. Same
B. No difference
C. Static for MDI only
D. Static fixed panes; dynamic allows split/unsplit at runtime
Correct Answer: D

Q67. CreateStatic / CreateView used in:

A. Dialogs
B. Splitter pane creation
C. Threads
D. Sockets
Correct Answer: B

Q68. Docking toolbar base class:

A. CView
B. CDocument
C. CToolBar / CControlBar
D. CWinApp
Correct Answer: C

Q69. EnableDocking + DockControlBar allows:

A. Draggable dockable toolbars
B. Fixed menus
C. No UI
D. Fullscreen only
Correct Answer: A

Q70. CStatusBar panes set via:

A. printf
B. malloc
C. socket
D. SetIndicators + SetPaneInfo/Text
Correct Answer: D

Q71. Rebar / dialog bar are:

A. File types
B. Extended control bars for combos/bands
C. Threads
D. Sockets
Correct Answer: B

Q72. Common dialog for open file:

A. CColorDialog
B. CFontDialog
C. CFileDialog(TRUE)
D. CPrintDialog only
Correct Answer: C

Q73. Save dialog uses:

A. CFileDialog(FALSE)
B. MessageBox
C. CWinApp
D. CView
Correct Answer: A

Q74. CFontDialog + LOGFONT selects:

A. File
B. Colour only
C. Printer only
D. Font
Correct Answer: D

Q75. CColorDialog returns:

A. File path
B. Chosen COLORREF
C. Font
D. Brush handle only
Correct Answer: B

Q76. CPrintDialog / Page Setup enables:

A. Network config
B. Thread config
C. Printing + device context setup
D. File delete
Correct Answer: C

Q77. Print preview in MFC uses:

A. OnFilePrintPreview (CPreviewView)
B. WinExec
C. ShellExecute only
D. Beep
Correct Answer: A

Q78. MDI frame classes:

A. CFrameWnd only
B. CDialog only
C. CView only
D. CMDIFrameWnd + CMDIChildWnd
Correct Answer: D

Q79. CMultiDocTemplate manages:

A. Single window only
B. Multiple document instances + frame/view linkage
C. Threads only
D. Sockets only
Correct Answer: B

Q80. Active MDI child handling via:

A. printf
B. malloc
C. MDIGetActive / MDIMaximize / MDINext
D. fopen
Correct Answer: C

Q81. SDI vs MDI:

A. SDI one document; MDI multiple children in parent frame
B. Same
C. SDI multiple, MDI single
D. No difference
Correct Answer: A

Q82. Tiling/cascading MDI children uses:

A. DrawText
B. BitBlt
C. CreateThread
D. MDITile / MDICascade
Correct Answer: D

Q83. Worker thread entry in MFC/Win32:

A. WinMain
B. ThreadProc / AfxBeginThread (UINT func(LPVOID))
C. WndProc
D. DllMain only
Correct Answer: B

Q84. UI thread has:

A. No pump
B. Only files
C. Message pump (CWinThread with PreTranslateMessage)
D. Only console
Correct Answer: C

Q85. Creating thread via Win32:

A. CreateThread
B. CreateWindow
C. CreateFile
D. CreateMenu
Correct Answer: A

Q86. Creating thread via MFC:

A. new only
B. malloc
C. fopen
D. AfxBeginThread
Correct Answer: D

Q87. Thread sync object for mutual exclusion:

A. HDC
B. Mutex / CriticalSection (CCriticalSection)
C. HWND
D. HMENU
Correct Answer: B

Q88. CriticalSection vs Mutex:

A. Same
B. Mutex inter-process, lighter CS intra-process (Enter/LeaveCriticalSection)
C. CS faster intra-process; mutex cross-process + ownership
D. CS for files only
Correct Answer: C

Q89. Event objects (CEvent) used for:

A. Signaling (SetEvent/WaitForSingleObject)
B. Drawing
C. Printing
D. Filing
Correct Answer: A

Q90. Semaphore limits:

A. One thread only always
B. No limit
C. For UI only
D. Counted concurrent access (CSemaphore)
Correct Answer: D

Q91. Deadlock occurs when:

A. No sync
B. Circular wait on locks
C. Fast code
D. No threads
Correct Answer: B

Q92. Interlocked functions provide:

A. File IO
B. Window draw
C. Atomic ops (InterlockedIncrement)
D. Socket connect
Correct Answer: C

Q93. PostThreadMessage vs PostMessage:

A. To thread queue vs window queue
B. Same
C. To file
D. To printer
Correct Answer: A

Q94. Thread local storage (TLS) is for:

A. Shared global
B. Window text
C. Menu
D. Per-thread data (TlsAlloc/SetValue)
Correct Answer: D

Q95. Terminating thread safely:

A. TerminateThread always
B. Return / exit + WaitForSingleObject; avoid force-kill
C. Pull plug
D. Delete file
Correct Answer: B

Q96. Suspend/Resume thread APIs:

A. Show/Hide
B. Open/Close
C. SuspendThread / ResumeThread
D. Play/Pause media only
Correct Answer: C

Q97. OLE full form:

A. Object Linking and Embedding
B. Online Event
C. Open Link Engine
D. Object List Editor
Correct Answer: A

Q98. Linked vs Embedded OLE object:

A. Same
B. Both static
C. No difference
D. Linked references external; embedded stores copy inside container
Correct Answer: D

Q99. OLE container vs server:

A. Same
B. Container hosts; server provides object (e.g., Word in MFC app)
C. Server hosts
D. Container provides
Correct Answer: B

Q100. COM foundation for OLE/ActiveX uses:

A. Plain text
B. printf
C. IUnknown (QueryInterface/AddRef/Release) + GUIDs
D. Batch files
Correct Answer: C

Q101. COleDocument / COleClientItem support:

A. Compound documents with embedded items
B. Only text files
C. Only threads
D. Only sockets
Correct Answer: A

Q102. In-place activation allows:

A. Separate window only
B. No editing
C. Print only
D. Edit server object inside container UI (menus merge)
Correct Answer: D

Q103. ActiveX control is:

A. DOS program
B. COM-based reusable UI/control (OCX) with properties/methods/events
C. Image file
D. Text file
Correct Answer: B

Q104. ActiveX container (MFC dialog) hosts via:

A. printf
B. malloc
C. Wrapper class from typelibrary + properties/methods
D. fopen
Correct Answer: C

Q105. IDispatch enables:

A. Late binding automation (Invoke)
B. Early binding only
C. Static linking only
D. No scripting
Correct Answer: A

Q106. ATL stands for:

A. Advanced Text Library
B. Audio Tape Library
C. Auto Thread Locker
D. Active Template Library
Correct Answer: D

Q107. ATL advantage vs MFC for COM:

A. Larger
B. Lightweight template-based, no heavy runtime
C. No COM
D. Only for GUI
Correct Answer: B

Q108. ATL macros BEGIN_COM_MAP defines:

A. Message map
B. Thread map
C. Interface map (QueryInterface entries)
D. Menu map
Correct Answer: C

Q109. CComModule / CComPtr help with:

A. Module registration + smart COM pointers (ref counting)
B. Drawing
C. Filing
D. Printing
Correct Answer: A

Q110. IDL file defines:

A. Window layout
B. Menu
C. Bitmap
D. COM interfaces/type library (MIDL compiles)
Correct Answer: D

Q111. Registering COM server uses:

A. printf
B. regsvr32 / DllRegisterServer
C. copy/paste
D. reboot only
Correct Answer: B

Q112. Apartments (STA/MTA) relate to:

A. Housing
B. Files
C. COM threading model (CoInitialize/CoInitializeEx)
D. Windows size
Correct Answer: C

Q113. Marshaling in COM is for:

A. Cross-apartment/process parameter passing
B. Drawing lines
C. Filing papers
D. Printing pages
Correct Answer: A

Q114. Connection points in ActiveX provide:

A. Power plugs
B. Network cables
C. File handles
D. Events/callbacks to container (sinks)
Correct Answer: D

Q115. MFC OLE init requires:

A. printf
B. AfxOleInit()
C. malloc
D. fopen
Correct Answer: B

Q116. Network programming base API on Windows:

A. BIOS
B. DOS
C. Winsock (WSAStartup, sockets)
D. PRINT
Correct Answer: C

Q117. WSAStartup negotiates:

A. Winsock version (e.g., 2.2)
B. Window size
C. Font
D. Colour
Correct Answer: A

Q118. TCP socket sequence (server):

A. connect-send-close only
B. bind only
C. listen only
D. socket-bind-listen-accept-send/recv-closesocket
Correct Answer: D

Q119. TCP vs UDP:

A. Same
B. TCP reliable connection-oriented; UDP datagram unreliable/fast
C. UDP reliable
D. TCP datagram
Correct Answer: B

Q120. getaddrinfo / gethostbyname used for:

A. Drawing
B. Printing
C. Hostname→IP resolution
D. File delete
Correct Answer: C

Q121. htons/ntohl convert:

A. Host↔network byte order
B. Text to image
C. Font sizes
D. Colours
Correct Answer: A

Q122. CSocket / CAsyncSocket MFC classes wrap:

A. Files
B. Windows
C. Menus
D. Winsock with message-driven callbacks (OnReceive/OnAccept)
Correct Answer: D

Q123. Blocking vs Non-blocking sockets:

A. Same
B. Blocking waits; non-blocking + select/WSAAsyncSelect returns immediately
C. Non-blocking waits more
D. Blocking for UDP only
Correct Answer: B

Q124. select() allows:

A. Drawing choice
B. Menu choice
C. Multiplexing / waiting on multiple sockets
D. Font choice
Correct Answer: C

Q125. Overlapped IO / IOCP provides:

A. Scalable async network IO
B. Sync file delete
C. Sync draw
D. Sync print
Correct Answer: A

Q126. HTTP programming can use:

A. Only sockets raw always
B. Only MFC draw
C. Only BIOS
D. WinInet/WinHTTP (InternetOpen, HttpOpenRequest) or sockets
Correct Answer: D

Q127. WNDPROC hook vs subclassing:

A. Same as painting
B. Intercept/override window messages (SetWindowLong)
C. File hook
D. Thread hook only
Correct Answer: B

Q128. GDI object for lines:

A. CBrush
B. CFont
C. CPen
D. CBitmap
Correct Answer: C

Q129. GDI object for fills:

A. CBrush
B. CPen
C. CFont
D. CPalette only
Correct Answer: A

Q130. Device contexts memory (off-screen) uses:

A. Beep
B. Sleep
C. Wait
D. CreateCompatibleDC + CreateCompatibleBitmap (double buffering)
Correct Answer: D

Q131. BitBlt / StretchBlt do:

A. Delete files
B. Fast bitmap block transfer (with scaling for Stretch)
C. Send packets
D. Create threads
Correct Answer: B

Q132. WM_TIMER handling uses:

A. malloc
B. socket
C. SetTimer/KillTimer + OnTimer
D. fopen
Correct Answer: C

Q133. Modeless dialog creation:

A. Create + ShowWindow (+ DestroyWindow)
B. DoModal only
C. MessageBox only
D. WinExec
Correct Answer: A

Q134. EndDialog vs DestroyWindow:

A. Same
B. Both for modeless
C. Both for threads
D. EndDialog for modal; DestroyWindow for modeless
Correct Answer: D

Q135. CListCtrl view modes:

A. Only icon
B. Icon, Report, List, SmallIcon
C. Only text
D. Only bitmap
Correct Answer: B

Q136. CTreeCtrl nodes use:

A. Pixels
B. Packets
C. HTREEITEM + TVINSERTSTRUCT
D. Threads
Correct Answer: C

Q137. PropertySheet/Page (CPropertySheet) implements:

A. Tabbed dialogs
B. Menus only
C. Toolbars only
D. Sockets
Correct Answer: A

Q138. DIB vs DDB:

A. Same
B. Both printers
C. Both files
D. Device-independent vs device-dependent bitmap
Correct Answer: D

Q139. OnEraseBkgnd returning TRUE means:

A. Error
B. Background handled (avoid flicker)
C. Repaint always
D. Exit app
Correct Answer: B

Q140. Double buffering avoids:

A. Network lag
B. Disk lag
C. Flicker
D. Thread lag
Correct Answer: C

Q141. File open dialog filter string format:

A. "Text (*.txt)|*.txt||"
B. Plain sentence
C. Only "*.*"
D. No format
Correct Answer: A

Q142. CWinApp::ExitInstance cleans:

A. Nothing
B. Only windows
C. Only menus
D. App cleanup (GDI, sockets, threads)
Correct Answer: D

Q143. AfxGetApp / AfxGetMainWnd return:

A. HDC/HWND only
B. App object / main window pointers
C. Files
D. Sockets
Correct Answer: B

Q144. ASSERT vs VERIFY in MFC:

A. Same release
B. Both ignored
C. ASSERT debug-only; VERIFY evaluates in release too
D. Both for release
Correct Answer: C

Q145. Unicode (TCHAR, _T) handling:

A. Portable ANSI/Unicode builds (use LPCTSTR, _T)
B. Only ANSI
C. Only ASCII art
D. No text
Correct Answer: A

Q146. DLL entry for MFC extension DLL:

A. WinMain
B. main
C. ThreadProc
D. DllMain + AFX_MANAGE_STATE
Correct Answer: D

Q147. AFX_MANAGE_STATE needed to:

A. Draw faster
B. Switch module state for MFC DLL resources
C. Speed network
D. Manage threads only
Correct Answer: B

Q148. Mass virtual functions OnDraw/Serialize show:

A. No polymorphism
B. Overloading only
C. MFC document/view polymorphism
D. Templates only
Correct Answer: C

Q149. Message reflection (ON_NOTIFY_REFLECT) is for:

A. Let control handle its own notifications (e.g., custom draw)
B. Mirror display
C. Network echo
D. Thread echo
Correct Answer: A

Q150. Custom control drawing via:

A. Beep
B. Sleep
C. Wait
D. WM_DRAWITEM / OnDrawItem + owner-draw
Correct Answer: D

Q151. Timer vs Multithreading for background:

A. Same
B. Timer for light UI periodic; threads for heavy/parallel
C. Timer for heavy
D. Threads for UI only
Correct Answer: B

Q152. Worker thread must not directly:

A. Compute
B. Read file
C. Touch UI (use PostMessage)
D. Allocate memory
Correct Answer: C

Q153. SendMessage vs PostMessage:

A. Send synchronous (waits); Post async (queues)
B. Same
C. Post waits
D. Send queues only
Correct Answer: A

Q154. CriticalSection MFC wrapper:

A. CWinApp
B. CView
C. CDoc
D. CCriticalSection + CSingleLock
Correct Answer: D

Q155. CMutex vs CCriticalSection scope:

A. Same machine only both
B. Mutex named cross-process; CS intra-process
C. CS cross-process
D. Mutex intra only
Correct Answer: B

Q156. CEvent auto vs manual reset:

A. Same
B. No difference
C. Auto resets after one waiter; manual stays signaled till ResetEvent
D. Both manual
Correct Answer: C

Q157. WaitForMultipleObjects waits on:

A. Multiple handles
B. One pixel
C. One file
D. One menu
Correct Answer: A

Q158. UI update from worker via:

A. Direct CDC
B. Direct CWnd
C. malloc
D. PostMessage custom WM_USER+n
Correct Answer: D

Q159. Thread priority via:

A. ShowWindow
B. SetThreadPriority
C. SetWindowText
D. SetMenu
Correct Answer: B

Q160. OnIdle processing in MFC:

A. Never
B. Only for games
C. CWinApp::OnIdle for background UI updates
D. Only for drivers
Correct Answer: C

Q161. SDI template class:

A. CSingleDocTemplate
B. CMultiDocTemplate
C. CDialogTemplate
D. CThreadTemplate
Correct Answer: A

Q162. DocTemplate resources link:

A. Random
B. Only code
C. Only data
D. IDR_MAINFRAME (menu, icon, accel) + doc/view/frame classes
Correct Answer: D

Q163. RUNTIME_CLASS macro provides:

A. Compile-time only
B. Runtime class info for dynamic creation
C. Static text
D. Menu
Correct Answer: B

Q164. IMPLEMENT_DYNCREATE in .cpp pairs with:

A. BEGIN_MESSAGE_MAP
B. malloc
C. DECLARE_DYNCREATE in header
D. #include only
Correct Answer: C

Q165. View initial update via:

A. OnInitialUpdate
B. OnDraw only
C. OnCreate only
D. WinMain
Correct Answer: A

Q166. Document modified flag:

A. No flag
B. Only file date
C. Only read-only
D. SetModifiedFlag + IsModified for save prompt
Correct Answer: D

Q167. File MRU list handled by:

A. Manual code only
B. CWinApp MRU (LoadStdProfileSettings, AddToRecentFileList)
C. BIOS
D. Boot ini
Correct Answer: B

Q168. Registry vs INI for settings:

A. Same speed always
B. INI for large data
C. Registry (SetRegistryKey) structured; INI legacy files
D. No settings
Correct Answer: C

Q169. Help context (F1) uses:

A. Help IDs + WinHelp/HTML Help
B. printf
C. Beep
D. MessageBox only
Correct Answer: A

Q170. Drag-drop files (WM_DROPFILES) needs:

A. No setup
B. Only menu
C. Only thread
D. DragAcceptFiles + HDROP handling
Correct Answer: D

Q171. Clipboard (OpenClipboard/SetClipboardData) shares:

A. Threads
B. Data across apps (CF_TEXT, CF_BITMAP)
C. Files only
D. Sockets
Correct Answer: B

Q172. Drag-drop OLE (COleDropSource/Target) enables:

A. File delete
B. Window move only
C. Rich drag-drop with formats
D. Text only
Correct Answer: C

Q173. Shell execute to open doc:

A. ShellExecute / CreateProcess
B. malloc
C. free
D. BitBlt
Correct Answer: A

Q174. Process vs Thread:

A. Same
B. Thread has own address space
C. Process lightweight
D. Process isolated space; thread shares within process
Correct Answer: D

Q175. CreateProcess parameters include:

A. Only exe name
B. App name, cmdline, security, creation flags, STARTUPINFO
C. Only icon
D. Only menu
Correct Answer: B

Q176. Named pipes (CreateNamedPipe) for:

A. Drawing
B. Printing only
C. IPC (local/remote byte streams)
D. Filing only
Correct Answer: C

Q177. Mailslots vs Pipes:

A. Mailslots one-way broadcast; pipes two-way reliable
B. Same
C. Pipes broadcast
D. Mailslots two-way
Correct Answer: A

Q178. Winsock cleanup via:

A. closesocket only
B. close()
C. ExitProcess
D. WSACleanup
Correct Answer: D

Q179. socket() params:

A. HWND,MSG
B. af, type, protocol (e.g., AF_INET,SOCK_STREAM)
C. HDC only
D. HMENU only
Correct Answer: B

Q180. bind() associates:

A. Window to class
B. Thread to CPU
C. Socket to IP/port
D. Menu to window
Correct Answer: C

Q181. listen() + accept() do:

A. Queue + accept incoming TCP connections
B. Draw + paint
C. Open + save
D. Send + recv only
Correct Answer: A

Q182. connect() is used by:

A. Server only
B. Printer only
C. Plotter only
D. Client to reach server
Correct Answer: D

Q183. send()/recv() vs WSASend/Recv:

A. Same always
B. Basic vs overlapped-capable ex versions
C. No difference
D. Only for UDP
Correct Answer: B

Q184. closesocket() should be paired with:

A. CloseWindow
B. DeleteFile
C. Earlier socket() + shutdown() for graceful
D. DestroyMenu
Correct Answer: C

Q185. gethostname + inet_addr handle:

A. Name/IP conversion
B. Font conversion
C. Colour conversion
D. Menu conversion
Correct Answer: A

Q186. Firewall-friendly COM (DCOM) uses:

A. Only files
B. Only local
C. No network
D. RPC + CoCreateInstanceEx for remote objects
Correct Answer: D

Q187. Apartment init for UI thread with COM:

A. No init
B. CoInitialize (STA) + message pump
C. Only MTA
D. Only files
Correct Answer: B

Q188. BSTR vs CString:

A. Same
B. Both narrow
C. BSTR COM length-prefixed wide; CString MFC wrapper (convert via _bstr_t/CComBSTR)
D. Both for files
Correct Answer: C

Q189. HRESULT S_OK means:

A. Success
B. Failure
C. Pending
D. Cancelled
Correct Answer: A

Q190. ActiveX signing (Authenticode) ensures:

A. Faster load
B. Smaller size
C. Better UI
D. Trust/origin for safe download (reduce warnings)
Correct Answer: D

Q191. ATL vs MFC footprint reminder:

A. ATL heavier
B. ATL minimal (no document/view bloat) ideal for controls/servers
C. MFC minimal
D. Same
Correct Answer: B

Q192. Dual interface (IDispatch + vtable) allows:

A. Only C++
B. Only script
C. Both early (fast C++) + late (VB/script) binding
D. Neither
Correct Answer: C

Q193. Type library (.tlb) generated from:

A. IDL via MIDL
B. .rc via rc.exe
C. .cpp via cl.exe only
D. .bmp via paint
Correct Answer: A

Q194. Proxy/stub DLL handles:

A. Drawing
B. Printing
C. Filing
D. Marshaling for custom interfaces
Correct Answer: D

Q195. Common control init needs:

A. printf
B. InitCommonControlsEx
C. malloc
D. fopen
Correct Answer: B

Q196. Tooltip support via:

A. No code
B. Only menus
C. EnableToolTips + ON_NOTIFY(TTN_)
D. Only dialogs modal
Correct Answer: C

Q197. Win32 error via:

A. GetLastError + FormatMessage
B. errno only
C. printf only
D. Beep
Correct Answer: A

Q198. Debug output via:

A. printf to console always
B. cout only
C. MessageBox only
D. OutputDebugString + TRACE macro
Correct Answer: D

Q199. High-DPI awareness needs:

A. Ignore
B. Manifest + GetDpiForWindow / scaling
C. Larger fonts only
D. Smaller windows
Correct Answer: B

Q200. Best practice for MFC/Win32 app shutdown:

A. Kill process
B. Pull plug
C. Close documents (SaveModified), destroy windows, cleanup GDI/sockets/threads, PostQuitMessage
D. Delete exe ---
Correct Answer: C

ANSWER KEY — Topic 12f: Windows Programming

```
1-B 2-C 3-D 4-A 5-B 6-C 7-A 8-D 9-B 10-C
11-A 12-D 13-B 14-C 15-A 16-D 17-B 18-A 19-C 20-D
21-B 22-A 23-C 24-D 25-B 26-A 27-C 28-D 29-A 30-B
31-C 32-A 33-D 34-B 35-A 36-C 37-D 38-B 39-A 40-C
41-A 42-D 43-B 44-C 45-A 46-D 47-B 48-C 49-A 50-D
51-B 52-C 53-A 54-D 55-B 56-C 57-A 58-D 59-B 60-C
61-A 62-D 63-B 64-C 65-A 66-D 67-B 68-C 69-A 70-D
71-B 72-C 73-A 74-D 75-B 76-C 77-A 78-D 79-B 80-C
81-A 82-D 83-B 84-C 85-A 86-D 87-B 88-C 89-A 90-D
91-B 92-C 93-A 94-D 95-B 96-C 97-A 98-D 99-B 100-C
101-A 102-D 103-B 104-C 105-A 106-D 107-B 108-C 109-A 110-D
111-B 112-C 113-A 114-D 115-B 116-C 117-A 118-D 119-B 120-C
121-A 122-D 123-B 124-C 125-A 126-D 127-B 128-C 129-A 130-D
131-B 132-C 133-A 134-D 135-B 136-C 137-A 138-D 139-B 140-C
141-A 142-D 143-B 144-C 145-A 146-D 147-B 148-C 149-A 150-D
151-B 152-C 153-A 154-D 155-B 156-C 157-A 158-D 159-B 160-C
161-A 162-D 163-B 164-C 165-A 166-D 167-B 168-C 169-A 170-D
171-B 172-C 173-A 174-D 175-B 176-C 177-A 178-D 179-B 180-C
181-A 182-D 183-B 184-C 185-A 186-D 187-B 188-C 189-A 190-D
191-B 192-C 193-A 194-D 195-B 196-C 197-A 198-D 199-B 200-C
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